Learn Being Forward

COURSES · CAMPUS PARTNERSHIPS · PLACEMENTS

STUDY NOTES · BASIC TO ADVANCED

Quantitative Aptitude **Calculate Fast, Solve Right**

Complete notes for the Learn Being Forward Quantitative Aptitude syllabus: every topic explained with visuals, formulas and shortcut tricks, followed by basic, intermediate and advanced questions with fully worked solutions.

33	3	56	825
modules	levels	minimum days	solved questions

How to use these notes

These notes follow the **Learn Being Forward Quantitative Aptitude syllabus** exactly: the same 33 modules, in the same order, with every topic and subtopic explained step by step. Each subtopic number (for example **3.4.4**) matches the syllabus, so you can always find where you are.

Level	Modules	Minimum	Outcome
1. Beginner	1 to 12	19 days	Calculate quickly and master the number system, HCF and LCM, simplification, fractions, averages, percentages, ratios, partnership, mixtures and ages
2. Intermediate	13 to 24	18 days	Solve profit and loss, interest, time and work, pipes, speed and distance, boats, trains, races, algebra, progressions and logarithms with shortcuts
3. Advanced	25 to 33	19 days	Master permutations, probability, geometry, mensuration, coordinate geometry, data interpretation and data sufficiency, and perform in timed company-specific tests
Total	1 to 33	56 days	About 10 weeks at 6 study days per week

1 · Understand
Read the concept and study the diagram until you can explain the formula in your own words.

2 · Learn the trick
Every topic has shortcut tricks in gold boxes. Practise them on the worked examples.

3 · Solve with a timer
Attempt the basic, intermediate and advanced questions on paper, without a calculator, before looking at any solution.

4 · Check and log
Compare with the step-by-step solutions and write every mistake in your error log.

Every topic follows the same pattern: the idea in plain words, a visual model (number line, bar model, table, chart or figure), the standard formulas, **shortcut tricks** for exam speed, and worked examples. Each module then gives about 25 practice questions in three levels, **basic**, **intermediate** and **advanced**, in the styles used by campus tests (TCS NQT, Infosys, Wipro, Accenture, Cognizant, Capgemini, AMCAT, eLitmus) and bank and government exams. Every question is fully solved. Every numerical answer in these notes was computed and checked by a program while the notes were built.

HOW TO READ THE BOXES
Gold-bordered boxes hold **shortcut tricks**, worked examples and notes, red boxes warn about common traps, and navy boxes hold the key formula to remember. Questions are numbered in navy squares; each solution box repeats the question number with the answer in gold, followed by the full working.

Course plan at a glance

Module	Topic	Level	Min days
01	Introduction to quantitative aptitude	Beginner	1
02	Speed mathematics and calculation techniques	Beginner	3
03	Number system	Beginner	3
04	HCF and LCM	Beginner	1

Module	Topic	Level	Min days
05	Simplification, approximation, surds and indices	Beginner	2
06	Decimals and fractions	Beginner	1
07	Averages	Beginner	1
08	Percentages	Beginner	2
09	Ratio and proportion	Beginner	2
10	Partnership	Beginner	1
11	Mixtures and alligation	Beginner	1
12	Problems on ages	Beginner	1
13	Profit, loss and discount	Intermediate	2
14	Simple interest	Intermediate	1
15	Compound interest	Intermediate	2
16	Time and work	Intermediate	2
17	Pipes and cisterns	Intermediate	1
18	Time, speed and distance	Intermediate	2
19	Boats and streams	Intermediate	1
20	Problems on trains	Intermediate	1
21	Races and games	Intermediate	1
22	Algebra: equations and inequalities	Intermediate	2
23	Progressions	Intermediate	2
24	Logarithms	Intermediate	1
25	Permutations and combinations	Advanced	2
26	Probability	Advanced	2
27	Geometry	Advanced	3
28	Mensuration	Advanced	2
29	Coordinate geometry and trigonometry	Advanced	2
30	Data interpretation	Advanced	3
31	Data sufficiency	Advanced	1
32	Statistics, set theory and miscellaneous topics	Advanced	1
33	Exam strategy, mock tests and company-specific preparation	Advanced	3

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LEVEL 1

Beginner level

Calculate quickly and master the number system, HCF and LCM, simplification, fractions, averages, percentages, ratios, partnership, mixtures and ages

12 MODULES · 19 DAYS MINIMUM

Module 01 · Introduction to quantitative aptitude	1 day
Module 02 · Speed mathematics and calculation techniques	3 days
Module 03 · Number system	3 days
Module 04 · HCF and LCM	1 day
Module 05 · Simplification, approximation, surds and indices	2 days
Module 06 · Decimals and fractions	1 day
Module 07 · Averages	1 day
Module 08 · Percentages	2 days
Module 09 · Ratio and proportion	2 days
Module 10 · Partnership	1 day
Module 11 · Mixtures and alligation	1 day
Module 12 · Problems on ages	1 day

MODULE 01 · BEGINNER

Introduction to quantitative aptitude

Minimum duration **1 day** | Level **Beginner**

This module explains what quantitative aptitude is, why companies and exams test it, where it is tested, what test patterns look like, and how to study it with this course. It ends with a 25-question diagnostic test across the whole syllabus so you can measure your starting level.

IN THIS MODULE

1.1 What is quantitative aptitude?
1.2 Why do we need it?
1.3 Where is it tested?
1.4 Typical test patterns
1.5 How to study aptitude
1.6 Diagnostic test

1.1 What is quantitative aptitude?

1.1.1 Definition

Quantitative aptitude is the ability to solve numerical and mathematical problems **quickly and accurately**, usually without a calculator. It is not advanced mathematics: almost everything uses school-level arithmetic and algebra, applied cleverly under time pressure.

1.1.2 The five areas

Arithmetic
percentages, ratios, averages, profit and loss, interest, time and work, speed and distance

Number system
divisibility, factors, remainders, HCF and LCM, unit digits

Algebra
equations, inequalities, progressions, logarithms

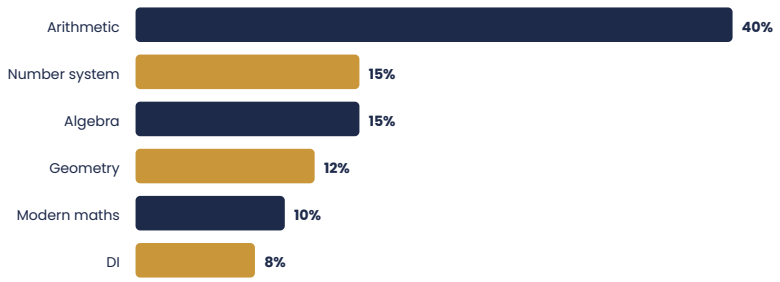
Geometry and mensuration
lines, triangles, circles, areas and volumes, coordinates

Modern maths and DI
permutations, probability, statistics, sets, data interpretation and sufficiency

Speed skills
tables, squares, shortcuts and estimation that make everything faster

1.1.3 Speed and accuracy

Every aptitude test rewards two things at once: **speed** (questions per minute) and **accuracy** (percentage correct). With negative marking, accuracy matters more; with tight sectional time, speed matters more. This course trains both: formulas and tricks for speed, worked solutions and an error log for accuracy.



APPROXIMATE SHARE OF QUESTIONS BY AREA IN TYPICAL PLACEMENT TESTS (%)

1.2 Why do we need it?

1.2.1 The first elimination round

In most campus drives the aptitude test is the **first filter**: only candidates who clear it reach the technical and HR interviews. A strong aptitude score keeps every later opportunity open.

1.2.2 What it tests

Aptitude questions test problem-solving, logical thinking and the ability to work under time pressure, which are skills employers value in every role.

1.2.3 Everyday uses

Situation	Topic used
Comparing discounts in a sale	Percentages, profit and loss
Choosing between loans and EMIs	Simple and compound interest
Planning a trip	Time, speed and distance
Sharing costs or profits	Ratio and partnership
Reading reports and dashboards	Data interpretation

1.3 Where is it tested?

1.3.1 Campus recruitment

Service and product companies such as TCS (NQT), Infosys, Wipro, Accenture, Cognizant, Capgemini, HCLTech and Tech Mahindra include a quantitative section in their written tests.

1.3.2 Assessment platforms

Many companies use third-party platforms: AMCAT, eLitmus, CoCubes, HirePro and Mettl. eLitmus in particular is known for fewer but harder questions with negative marking.

1.3.3 Government exams

Bank PO and clerk exams, SSC and railway exams have large quantitative sections focused on speed: simplification, approximation, number series and arithmetic word problems.

1.3.4 Management entrance exams

CAT, MAT, XAT, GMAT and GRE test the same topics at a higher level, with more emphasis on number system, algebra, geometry and data interpretation.

1.4 Typical test patterns

1.4.1 Questions and time

Patterns change every year, so always read the latest official notice. As a rough guide, placement quantitative sections have 15 to 30 questions in 20 to 40 minutes, about **1 to 1.5 minutes per question**.

1.4.2 Negative marking and cut-offs

Some tests deduct marks for wrong answers (often a quarter or a third of a mark). Many also have **sectional cut-offs**: you must clear a minimum in each section, so you cannot skip quantitative aptitude even if you are strong elsewhere.

1.4.3 Adaptive tests

In adaptive tests the next question gets harder when you answer correctly and easier when you answer wrongly. Accuracy on the early questions matters a lot.

1.4.4 Topic weightage

Percentages, ratios, averages, profit and loss, time and work, and speed and distance together usually make up the largest share of questions. Number system and data interpretation come next.

1.5 How to study aptitude

1.5.1 The learning cycle



THE LEARNING CYCLE USED IN EVERY MODULE

1.5.2 Formula notebook and error log

Keep two notebooks. The **formula notebook** holds every formula and shortcut, one topic per page. The **error log** records every question you got wrong or solved too slowly, with the reason and the correct method. Read both before each mock test.

1.5.3 Building speed

> **TIP**
> Spend 10 minutes every day on mental calculation (Module 2): tables to 30, squares to 50, cubes to 30 and fraction-percentage equivalents. This one habit improves every other topic.

1.6 Diagnostic test

1.6.1 Measuring your starting level

Take the 25 questions below in 30 minutes, without a calculator. Mark each answer, then check with the solutions. Record your score per area (arithmetic, number system, algebra, geometry, modern maths) and set a target for each.

1.6.2 Diagnostic questions

1.6.2.1 Basic level

Q1 What is 25% of 480?

(a) 120 (b) 96 (c) 125 (d) 140

Q2 What is 17 × 13?

(a) 211 (b) 231 (c) 221 (d) 223

Q3 The average of 12, 15, 18, 21 and 24 is

(a) 17 (b) 19 (c) 20 (d) 18

Q4 What is the HCF of 36 and 48?

(a) 6 (b) 4 (c) 12 (d) 24

Q5 Divide 450 in the ratio 2 : 3. What is the larger part?

(a) 180 (b) 225 (c) 270 (d) 300

Q6 A shirt bought for ₹400 is sold for ₹500. What is the profit percentage?

(a) 20% (b) 100% (c) 30% (d) 25%

Q7 What is the simple interest on ₹5,000 at 8% per year for 3 years?

(a) 1,000 (b) 1,500 (c) 400 (d) 1,200

Q8 A car travels 180 km in 3 hours. What is its speed?

(a) 50 km/h (b) 54 km/h (c) 60 km/h (d) 90 km/h

1.6.2.2 Intermediate level

Q9 A can finish a job in 10 days and B in 15 days. Working together, how many days do they take?

(a) 5 days (b) 8 days (c) 6 days (d) 12.5 days

Q10 What is the unit digit of 7^{26}?

(a) 7 (b) 9 (c) 3 (d) 1

Q11 A price rises by 20% and then falls by 20%. What is the net change?

(a) No change **(b)** 4% decrease **(c)** 4% increase **(d)** 2% decrease

Q12 What is the compound interest on ₹10,000 at 10% per year for 2 years?

(a) 2,000 **(b)** 2,200 **(c)** 2,100 **(d)** 1,100

Q13 A father is 3 times as old as his son. In 12 years he will be twice as old. What is the son's present age?

(a) 12 years **(b)** 10 years **(c)** 14 years **(d)** 16 years

Q14 How many numbers between 1 and 100 are divisible by both 4 and 6?

(a) 6 **(b)** 8 **(c)** 9 **(d)** 16

Q15 A train 200 m long passes a pole in 10 seconds. What is its speed in km/h?

(a) 20 km/h **(b)** 72 km/h **(c)** 60 km/h **(d)** 80 km/h

Q16 In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹45/kg?

(a) 1 : 3 **(b)** 3 : 1 **(c)** 2 : 1 **(d)** 1 : 1

1.6.2.3 Advanced level

Q17 In how many ways can the letters of the word APPLE be arranged?

(a) 120 **(b)** 24 **(c)** 60 **(d)** 30

Q18 Two dice are thrown. What is the probability that the sum is 7?

(a) 1/12 **(b)** 5/36 **(c)** 1/6 **(d)** 7/36

Q19 What is the area of a circle of radius 14 cm? ($\pi = 22/7$)

(a) 88 cm² **(b)** 308 cm² **(c)** 1,232 cm² **(d)** 616 cm²

Q20 What is the sum of the first 20 natural numbers?

(a) 210 **(b)** 200 **(c)** 190 **(d)** 420

Q21 What is $\log_2 64$?

(a) 8 **(b)** 6 **(c)** 5 **(d)** 32

Q22 How many trailing zeros does 50! have?

(a) 12 **(b)** 10 **(c)** 11 **(d)** 5

Q23 What is the distance between the points (1, 2) and (4, 6)?

(a) 7 **(b)** 5 **(c)** 4 **(d)** 25

Q24 The 10th term of the arithmetic progression 3, 7, 11, … is

(a) 43 **(b)** 40 **(c)** 35 **(d)** 39

Q25 A sum doubles in 5 years at simple interest. What is the rate of interest per year?

(a) 10% **(b)** 25% **(c)** 20% **(d)** 15%

1.6.3 Solutions

Q1 **Answer: (a) 120**

25% = 1/4, so 480 ÷ 4 = **120**.

Q2 **Answer: (c) 221**

17 × 13 = $15^2 - 2^2$ = 225 - 4 = **221**.

Q3 **Answer: (d) 18**

The numbers are evenly spaced, so the average is the middle one: **18**.

Q4 **Answer: (c) 12**

36 = $2^2 \times 3^2$, 48 = $2^4 \times 3$, HCF = $2^2 \times 3$ = **12**.

Q5 **Answer: (c) 270**

Total parts 5; larger part = 450 × 3/5 = **270**.

Q6 **Answer: (d) 25%**

Profit 100 on cost 400: 100/400 × 100 = **25**%.

Q7 **Answer: (d) 1,200**

SI = 5000 × 8 × 3 / 100 = **₹1,200**.

Q8 **Answer: (c) 60 km/h**

Speed = 180 ÷ 3 = **60 km/h**.

Q9 **Answer: (c) 6 days**

Together they do 1/10 + 1/15 = 1/6 of the job per day: **6 days**.

Q10 **Answer: (b) 9**

Unit digits of powers of 7 repeat 7, 9, 3, 1. 26 = 4 × 6 + 2, so the unit digit is the 2nd in the cycle: **9**.

Q11 **Answer: (b) 4% decrease**

1.2 × 0.8 = 0.96, so the price falls by **4%**.

Q12 **Answer: (c) 2,100**

Amount = 10000 × 1.1^2 = 12100, so CI = **₹2,100**.

Q13 **Answer: (a) 12 years**

Let son = x: 3x + 12 = 2(x + 12) → x = **12** years.

Q14 **Answer: (b) 8**

Divisible by both means divisible by LCM 12: $\lfloor 100/12 \rfloor$ = **8**.

Q15 **Answer: (b) 72 km/h**

Speed = 200/10 = 20 m/s = 20 × 18/5 = **72 km/h**.

Q16 **Answer: (b) 3 : 1**

Alligation: (60 - 45) : (45 - 40) = 15 : 5 = **3 : 1**.

Q17 **Answer: (c) 60**

5 letters with P repeated twice: 5!/2! = **60**.

Q18 **Answer: (c) 1/6**

Favourable pairs: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6 of 36: **1/6**.

Q19 **Answer: (d) 616 cm²**

πr^2 = 22/7 × 196 = **616 cm²**.

Q20 **Answer: (a) 210**

$n(n + 1)/2$ = 20 × 21 / 2 = **210**.

Q21 **Answer: (b) 6**

2^6 = 64, so $\log_2 64$ = **6**.

Q22 **Answer: (a) 12**

$\lfloor 50/5 \rfloor + \lfloor 50/25 \rfloor$ = 10 + 2 = **12**.

Q23 **Answer: (b) 5**

$\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{25}$ = **5**.

Q24 **Answer: (d) 39**

$a + 9d$ = 3 + 36 = **39**.

Q25 **Answer: (c) 20%**

Interest = principal in 5 years: R = 100/5 = **20**% per year.

Quantitative aptitude is fast, accurate school mathematics. Learn each topic in the cycle concept → formula → examples → timed practice → mocks, keep a formula notebook and an error log, and build calculation speed daily.

MODULE 02 · BEGINNER

Speed mathematics and calculation techniques

Minimum duration **3 days** | Level **Beginner**

Fast, accurate calculation is the foundation of every aptitude topic. This module builds mental-maths speed: tables, squares and cubes, fraction-percentage equivalents, multiplication shortcuts (by 5, 9, 11, 25, 99, 125, base method, vertical and crosswise, squares ending in 5 and near 50 and 100), division shortcuts, left-to-right addition and complements, quick square and cube roots, unit digits and the digit-sum check, followed by 25 solved drills.

IN THIS MODULE

2.1 Tables, squares and cubes
2.2 Fraction–percentage equivalents
2.3 Multiplication shortcuts
2.4 Division shortcuts
2.5 Addition and subtraction tricks
2.6 Square roots and cube roots
2.7 Unit digit tricks
2.8 Checking answers
2.9 Practice questions
2.10 Solutions

2.1 Tables, squares and cubes

2.1.1 Multiplication tables up to 30

Tables up to 30 make percentage and ratio questions much faster. Learn 11 to 20 first, then 21 to 30 using (20 + k) × n = 20n + kn: for example 27 × 8 = 160 + 56 = 216.

2.1.2 Squares up to 50

n	n²	n	n²	n	n²	n	n²	n	n²
1	1	11	121	21	441	31	961	41	1681
2	4	12	144	22	484	32	1024	42	1764
3	9	13	169	23	529	33	1089	43	1849
4	16	14	196	24	576	34	1156	44	1936
5	25	15	225	25	625	35	1225	45	2025
6	36	16	256	26	676	36	1296	46	2116
7	49	17	289	27	729	37	1369	47	2209
8	64	18	324	28	784	38	1444	48	2304
9	81	19	361	29	841	39	1521	49	2401

n	n²	n	n²	n	n²	n	n²	n	n²
10	100	20	400	30	900	40	1600	50	2500

2.1.3 Cubes up to 30

n	n³	n	n³	n	n³
1	1	11	1,331	21	9,261
2	8	12	1,728	22	10,648
3	27	13	2,197	23	12,167
4	64	14	2,744	24	13,824
5	125	15	3,375	25	15,625
6	216	16	4,096	26	17,576
7	343	17	4,913	27	19,683
8	512	18	5,832	28	21,952
9	729	19	6,859	29	24,389
10	1,000	20	8,000	30	27,000

2.1.4 Square roots and cube roots of perfect numbers

Reading the tables backwards gives roots: $\sqrt{1764} = 42$ because $42^2 = 1764$; $\sqrt[3]{2197} = 13$ because $13^3 = 2197$. Sections 2.6.1 and 2.6.2 show how to find roots of larger numbers without the table.

2.2 Fraction–percentage equivalents

2.2.1 1/2 to 1/20 as percentages

Fraction	%	Fraction	%	Fraction	%
1/2	50	1/9	11.11	1/16	6.25
1/3	33.33	1/10	10	1/17	5.88
1/4	25	1/11	9.09	1/18	5.56
1/5	20	1/12	8.33	1/19	5.26
1/6	16.67	1/13	7.69	1/20	5
1/7	14.29	1/14	7.14		
1/8	12.5	1/15	6.67		

SHORTCUT TRICK
Multiples follow directly: 3/8 = 3 × 12.5 = 37.5%, 5/6 = 5 × 16.67 = 83.33%, 7/12 = 7 × 8.33 = 58.33%. Converting a percentage to a fraction turns "find 37.5% of 640" into 3/8 × 640 = 240.

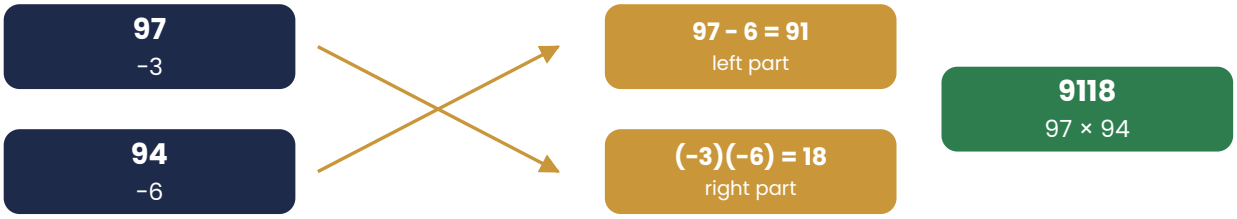
2.3 Multiplication shortcuts

2.3.1 Multiplying by 5, 9, 11, 25, 99, 125

Multiply by	Rule	Example
5	× 10 ÷ 2	86 × 5 = 860 ÷ 2 = 430
9	× 10 − n	47 × 9 = 470 − 47 = 423
11	write the digits with their sums in between	63 × 11 = 6 (6+3) 3 = 693; 87 × 11 = 8 (15) 7 = 957
25	× 100 ÷ 4	68 × 25 = 6800 ÷ 4 = 1700
99	× 100 − n	73 × 99 = 7300 − 73 = 7227
125	× 1000 ÷ 8	48 × 125 = 48000 ÷ 8 = 6000

2.3.2 Base method (numbers near 10, 100, 1000)

For two numbers near a base, write each number's difference from the base. The answer has two parts: (one number + the other's difference) | (product of the differences), with the right part having as many digits as the base has zeros.



BASE METHOD: 97 × 94 = 9118

Product	Differences	Left | right	Answer
97 × 94	−3, −6	91 | 18	9118
104 × 107	+4, +7	111 | 28	11128
996 × 1003	−4, +3	999 | −12 → 998 | 988	998988

2.3.3 Vertical and crosswise method

For two two-digit numbers ab × cd: right digit = b × d, middle = a × d + b × c, left = a × c, carrying as you go from right to left. 43 × 26: right 3 × 6 = 18 (write 8, carry 1); middle 4 × 6 + 3 × 2 + 1 = 31 (write 1, carry 3); left 4 × 2 + 3 = 11. Answer **1118**.

2.3.4 Squaring numbers ending in 5

For a number n5: multiply n by (n + 1) and write 25 after it. $75^2 = 7 \times 8 \mid 25 = 5625$; $115^2 = 11 \times 12 \mid 25 = 13225$.

2.3.5 Squaring numbers near 50 and 100

Near 50: $(50 + k)^2 = (25 + k)$ hundreds + k^2. $53^2 = 28 \mid 09 = 2809$; $47^2 = 22 \mid 09 = 2209$. Near 100: $(100 + k)^2 = (100 + 2k) \mid k^2$ (two digits). $106^2 = 112 \mid 36 = 11236$; $97^2 = 94 \mid 09 = 9409$.

2.3.6 Multiplying two-digit numbers mentally

Use identities: $(a + b)(a - b) = a^2 - b^2$ gives $47 \times 53 = 50^2 - 3^2 = 2491$; and split one number: $36 \times 14 = 36 \times 10 + 36 \times 4 = 360 + 144 = 504$.

2.4 Division shortcuts

2.4.1 Dividing by 5, 25, 125

Divide by	Rule	Example
5	× 2 ÷ 10	365 ÷ 5 = 730 ÷ 10 = 73
25	× 4 ÷ 100	1450 ÷ 25 = 5800 ÷ 100 = 58
125	× 8 ÷ 1000	4250 ÷ 125 = 34000 ÷ 1000 = 34

2.4.2 Quick division by approximation

In approximation questions, round the divisor to a friendly number and adjust: 4987 ÷ 49 ≈ 5000 ÷ 50 = 100 (exact value 101.8). Choose the option closest to your estimate.

2.5 Addition and subtraction tricks

2.5.1 Left-to-right addition

Add the largest place values first, then refine: 487 + 356 = 700 + 130 + 13 = 843. You keep a running total in your head instead of carrying digits.

2.5.2 Complements to 100 and 1000

The complement of a number to 1000 is "all from 9, the last from 10": 1000 - 368 = 632 (9 - 3, 9 - 6, 10 - 8). Use it for change and for subtraction such as 5000 - 2747 = 2253.

2.6 Square roots and cube roots

2.6.1 Square roots of perfect squares

Split off the last two digits. The unit digit of the root comes from the unit digit of the square (1 → 1 or 9, 4 → 2 or 8, 9 → 3 or 7, 6 → 4 or 6, 5 → 5, 0 → 0). The tens digit is the largest number whose square fits in the

remaining part. Choose between the two candidates by comparing with the square of the number ending in 5.

WORKED EXAMPLE

$\sqrt{7056}$

Unit digit 6 → root ends in 4 or 6. The front part 70 lies between $8^2 = 64$ and $9^2 = 81$, so the root is 84 or 86. $85^2 = 7225 > 7056$, so the root is **84**.

2.6.2 Cube roots of perfect cubes

Unit digits of cubes are unique: 1→1, 8→2, 7→3, 4→4, 5→5, 6→6, 3→7, 2→8, 9→9, 0→0. Drop the last three digits; the largest cube not exceeding what remains gives the tens digit.

WORKED EXAMPLE

$\sqrt[3]{157464}$

Unit digit 4 → root ends in 4. Remaining part 157 lies between $5^3 = 125$ and $6^3 = 216$, so the root is **54**.

2.6.3 Estimating non-perfect roots

$\sqrt{n} \approx a + (n - a^2)/(2a)$, where a^2 is the nearest perfect square. $\sqrt{50} \approx 7 + 1/14 \approx 7.07$ (true value 7.071).

2.7 Unit digit tricks

Only the unit digits matter for the unit digit of a sum or product: the unit digit of 347 × 268 is that of 7 × 8 = 56, which is **6**. Powers of 2, 3, 7, 8 repeat in cycles of 4; Module 3 covers this in detail.

2.8 Checking answers

2.8.1 Digit-sum (casting out nines)

Reduce each number to its digit sum (repeat until one digit). The digit sum of a product equals the digit sum of the product of the digit sums. If they do not match, the answer is wrong. (A match does not prove the answer right, but most errors are caught.)

WORKED EXAMPLE

Is 347 × 268 = 92996 correct?

Digit sums: 347 → 14 → 5; 268 → 16 → 7; 5 × 7 = 35 → 8. 92996 → 35 → 8. They match, and indeed 347 × 268 = 92996.

2.9 Practice questions

2.9.1 Basic level

Q1 84 × 5 = ?

(a) 420 **(b)** 410 **(c)** 425 **(d)** 400

Q2 76 × 11 = ?

(a) 826 (b) 846 (c) 836 (d) 776

Q3 65^2 = ?

(a) 4,025 (b) 4,225 (c) 4,125 (d) 4,325

Q4 What is 3/8 as a percentage?

(a) 37.5% (b) 33.33% (c) 38.5% (d) 35%

Q5 √1849 = ?

(a) 43 (b) 41 (c) 47 (d) 37

Q6 64 × 25 = ?

(a) 1,600 (b) 1,500 (c) 1,625 (d) 1,580

Q7 1000 - 458 = ?

(a) 542 (b) 552 (c) 642 (d) 532

Q8 $\sqrt[3]{9261}$ = ?

(a) 19 (b) 21 (c) 23 (d) 31

2.9.2 Intermediate level

Q9 98 × 96 = ?

(a) 9,418 (b) 9,308 (c) 9,508 (d) 9,408

Q10 103 × 108 = ?

(a) 11,124 (b) 11,024 (c) 11,134 (d) 11,224

Q11 48 × 52 = ?

(a) 2,504 (b) 2,486 (c) 2,496 (d) 2,396

Q12 57^2 = ?

(a) 3,149 (b) 3,259 (c) 3,249 (d) 3,349

Q13 86 × 99 = ?

(a) 8,604 (b) 8,504 (c) 8,524 (d) 8,514

Q14 What is 7/12 of 360?

(a) 200 (b) 220 (c) 210 (d) 180

Q15 √6889 = ?

(a) 87 (b) 83 (c) 77 (d) 93

Q16 What is the unit digit of 2347 × 5689?

(a) 7 (b) 3 (c) 9 (d) 1

2.9.3 Advanced level

Q17 994 × 1006 = ?

(a) 999,964 (b) 1,000,036 (c) 999,936 (d) 998,964

Q18 $\sqrt[3]{389017}$ = ?

(a) 77 (b) 63 (c) 83 (d) 73

Q19 Approximately, 7998 ÷ 39.9 ≈ ?

(a) 180 (b) 200 (c) 220 (d) 160

Q20 112^2 = ?

(a) 12,444 (b) 12,554 (c) 12,544 (d) 12,644

Q21 Which product is correct according to the digit-sum check?

(a) 213 × 47 = 10011 (b) 213 × 47 = 10111
(c) 213 × 47 = 10021 (d) 213 × 47 = 10001

Q22 What is 87.5% of 1,280?

(a) 1,100 (b) 1,150 (c) 1,120 (d) 1,080

Q23 64 × 36 = ?

(a) 2,204 (b) 2,314 (c) 2,304 (d) 2,404

Q24 Approximately, $\sqrt{90}$ ≈ ?

(a) 9.3 (b) 9.49 (c) 9.6 (d) 9.8

Q25 4375 ÷ 125 = ?

(a) 33 (b) 37 (c) 35 (d) 45

2.10 Solutions

Q1 **Answer: (a) 420**

84 × 10 ÷ 2 = 840 ÷ 2 = **420**.

Q2 **Answer: (c) 836**

7 (7 + 6) 6 = 7 13 6 → carry: **836**.

Q3 **Answer: (b) 4,225**

6 × 7 = 42, then 25: **4225**.

Q4 **Answer: (a) 37.5%**

3 × 12.5 = **37.5%**.

Q5 **Answer: (a) 43**

Unit digit 9 → 3 or 7; 18 lies between 4^2 and 5^2, so 43 or 47; 45^2 = 2025 > 1849, so **43**.

Q6 **Answer: (a) 1,600**

6400 ÷ 4 = **1600**.

Q7 **Answer: (a) 542**

Complement: 9 - 4, 9 - 5, 10 - 8 = **542**.

Q8 **Answer: (b) 21**

Unit digit 1 → root ends in 1; 9 lies between $2^3 = 8$ and $3^3 = 27$, so **21**.

Q9 **Answer: (d) 9,408**

Base 100: differences -2, -4. Left 98 - 4 = 94, right 2 × 4 = 08: **9408**.

Q10 **Answer: (a) 11,124**

Differences +3, +8: left 103 + 8 = 111, right 24: **11124**.

Q11 **Answer: (c) 2,496**

$50^2 - 2^2 = 2500 - 4 =$ **2496**.

Q12 **Answer: (c) 3,249**

Near 50: (25 + 7) | $7^2 = 32$ | 49 = **3249**.

Q13 **Answer: (d) 8,514**

8600 - 86 = **8514**.

Q14 **Answer: (c) 210**

360 ÷ 12 = 30, × 7 = **210**.

Q15 **Answer: (b) 83**

Unit digit 9 → 3 or 7; 68 lies between 8^2 and 9^2, so 83 or 87; $85^2 = 7225 > 6889$, so **83**.

Q16 **Answer: (b) 3**

Unit digits 7 × 9 = 63: **3**.

Q17 **Answer: (a) 999,964**

Base 1000, differences -6 and +6: (1000 - 6)(1000 + 6) = 1000000 - 36 = **999964**.

Q18 **Answer: (d) 73**

Unit digit 7 → root ends in 3; 389 lies between $7^3 = 343$ and $8^3 = 512$, so **73**.

Q19 **Answer: (b) 200**

8000 ÷ 40 = **200** (exact 200.45).

Q20 **Answer: (c) 12,544**

Near 100: (100 + 24) | $12^2 = 124$ | 144 → 12400 + 144 = **12544**.

Q21 **Answer: (a) 213 × 47 = 10011**

Digit sums: 213 → 6, 47 → 11 → 2; 6 × 2 = 12 → 3. Only 10011 (digit sum 3) passes, and 213 × 47 = 10011.

Q22 **Answer: (c) 1,120**

87.5% = 7/8; 1280 ÷ 8 × 7 = **1120**.

Q23 **Answer: (c) 2,304**

Vertical and crosswise: 4 × 6 = 24 (4, carry 2); 6 × 6 + 4 × 3 + 2 = 50 (0, carry 5); 6 × 3 + 5 = 23 → **2304**.

Q24 **Answer: (b) 9.49**

$\sqrt{90} \approx 9 + (90 - 81)/18 = 9.5$; the closest option is **9.49** (true value 9.487).

Q25 **Answer: (c) 35**

4375 × 8 = 35000; ÷ 1000 = **35**.

Learn tables to 30, squares to 50, cubes to 30 and fractions 1/2 to 1/20 as percentages. Use the base method near 10, 100, 1000; $n5^2 = n(n + 1) \mid 25$; near 50 and 100 use $(25 + k) \mid k^2$ and $(100 + 2k) \mid k^2$. Find roots from unit digits and bounds, estimate to pick options, and check products with digit sums.

MODULE 03 · BEGINNER

Number system

Minimum duration **3 days** | Level **Beginner**

The number system is the most important foundation topic and appears in almost every aptitude test. This module covers the types of numbers, divisibility rules, factors (number, sum, even and odd factors, factor pairs), remainders (including large powers, Fermat, Euler and Wilson), unit and last two digits, trailing zeros and highest powers in n!, standard sums, base conversions and digit problems, followed by 25 solved questions checked by a program.

IN THIS MODULE

3.1	Types of numbers	3.2	Divisibility rules
3.3	Factors	3.4	Remainders
3.5	Unit digit and last two digits	3.6	Trailing zeros and highest power
3.7	Sums of numbers	3.8	Base systems
3.9	Place value and digit problems	3.10	Practice questions
3.11	Solutions		

3.1 Types of numbers

3.1.1 Natural, whole and integers

Set	Members	Note
Natural numbers	1, 2, 3, ...	counting numbers
Whole numbers	0, 1, 2, 3, ...	natural numbers with 0
Integers	..., -2, -1, 0, 1, 2, ...	whole numbers and their negatives

3.1.2 Rational and irrational numbers

A **rational** number can be written as p/q with integers p, q and q ≠ 0; its decimal either ends (0.25) or repeats (0.333...). An **irrational** number cannot: its decimal never ends and never repeats (√2, π, √5).

3.1.3 Real numbers

Rational and irrational numbers together form the real numbers, every point on the number line.



RATIONAL AND IRRATIONAL NUMBERS ON THE NUMBER LINE

3.1.4 Even and odd numbers

Even numbers are divisible by 2; odd numbers leave remainder 1. Even ± even = even, odd ± odd = even, even ± odd = odd; a product is odd only if every factor is odd.

3.1.5 Prime and composite numbers

A **prime** has exactly two factors, 1 and itself (2, 3, 5, 7, 11 …); 2 is the only even prime. A **composite** number has more than two factors. 1 is neither prime nor composite. There are **25 primes below 100**.

Primes below 100
2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

SHORTCUT TRICK

To test whether n is prime, divide only by primes up to √n. For 211: √211 ≈ 14.5, and 211 is not divisible by 2, 3, 5, 7, 11 or 13, so 211 is prime.

3.1.6 Co-primes and twin primes

Two numbers are **co-prime** if their HCF is 1 (8 and 15), even if neither is prime. **Twin primes** are primes that differ by 2: (3, 5), (5, 7), (11, 13), (17, 19), (29, 31), (41, 43), (59, 61), (71, 73).

3.1.7 Perfect numbers

A perfect number equals the sum of its factors other than itself: 6 = 1 + 2 + 3, 28 = 1 + 2 + 4 + 7 + 14. The next are 496 and 8128.

3.2 Divisibility rules

3.2.1 Rules for 2 to 13

Divisor	Rule	Example
2	last digit even	3,574
3	digit sum divisible by 3	4,521 (12)
4	last two digits divisible by 4	7,316 (16)
5	last digit 0 or 5	8,905
6	divisible by 2 and 3	5,412
7	double the last digit and subtract from the rest; repeat	343 → 34 - 6 = 28 ✓
8	last three digits divisible by 8	9,128 (128)
9	digit sum divisible by 9	7,128 (18)
10	last digit 0	450
11	(sum of odd-place digits) - (sum of even-place digits) is 0 or a multiple of 11	918,082: (9+8+8) - (1+0+2) = 22 ✓

Divisor	Rule	Example
12	divisible by 3 and 4	5,412
13	add 4 × last digit to the rest; repeat	351 → 35 + 4 = 39 ✓

3.2.2 Divisibility by composite numbers

To test divisibility by a composite number, split it into **co-prime** factors and test each: 72 = 8 × 9, 88 = 8 × 11. Do not use non-co-prime factors such as 4 × 6 for 24 (use 3 × 8).

WORKED EXAMPLE

For which digit d is 4d76 divisible by 9?

Digit sum 4 + d + 7 + 6 = 17 + d must be a multiple of 9, so d = 1 (sum 18).

3.3 Factors

3.3.1 Prime factorisation

Every integer above 1 is a unique product of primes. Write $n = a^p \times b^q \times c^r \ldots$; all factor questions use these exponents. Example: $360 = 2^3 \times 3^2 \times 5$.

3.3.2 Number of factors

If $n = a^p \times b^q \times c^r$, the number of factors = $(p + 1)(q + 1)(r + 1)$

$360 = 2^3 \times 3^2 \times 5$ has $(3 + 1)(2 + 1)(1 + 1) =$ **24** factors.

3.3.3 Sum of factors

Sum of factors = $(1 + a + \ldots + a^p)(1 + b + \ldots + b^q)(\ldots)$. For 360: $(1 + 2 + 4 + 8)(1 + 3 + 9)(1 + 5) = 15 \times 13 \times 6 =$ **1170**.

3.3.4 Factor pairs

The number of ways to write n as a product of two factors is (number of factors)/2, or (number of factors + 1)/2 if n is a perfect square. 360 → 12 ways; 36 (9 factors) → 5 ways.

3.3.5 Even and odd factors

Odd factors ignore the power of 2: 360 has (2 + 1)(1 + 1) = 6 odd factors and 24 - 6 = 18 even factors.



CHOOSING AN EXPONENT FOR EACH PRIME: 4 × 3 × 2 = 24 FACTORS OF 360

3.4 Remainders

3.4.1 Division identity

Dividend = Divisor × Quotient + Remainder, with $0 \le$ remainder $<$ divisor

3.4.2 Remainders of sums and products

The remainder of a sum (or product) equals the remainder of the sum (or product) of the remainders. $(47 \times 53) \bmod 5 = (2 \times 3) \bmod 5 = 1$.

3.4.3 Negative remainders

Use negative remainders to simplify: $98 \equiv -2 \pmod{100}$, so $98 \times 97 \equiv (-2)(-3) = 6 \pmod{100}$. Similarly $29 \equiv -1 \pmod{10}$.

3.4.4 Remainders of large powers (cyclicity)

WORKED EXAMPLE

Remainder when 2^{100} is divided by 7

$2^3 = 8 \equiv 1 \pmod 7$. $2^{100} = (2^3)^{33} \times 2 \equiv 1 \times 2 =$ **2**.

3.4.5 Fermat's and Euler's theorems

Fermat: if p is prime and a is not a multiple of p, $a^{p-1} \equiv 1 \pmod p$. **Euler**: if a and n are co-prime, $a^{\varphi(n)} \equiv 1 \pmod n$, where $\varphi(n)$ counts numbers below n co-prime to n; $\varphi(n) = n(1 - 1/p_1)(1 - 1/p_2)\ldots$.

WORKED EXAMPLE

Remainder when 3^{2026} is divided by 11

Fermat: $3^{10} \equiv 1 \pmod{11}$. $2026 = 10 \times 202 + 6$, so $3^{2026} \equiv 3^6 = 729 \equiv 729 - 726 =$ **3**.

3.4.6 Wilson's theorem

For a prime p, $(p - 1)! \equiv -1 \pmod p$. So 12! leaves remainder 12 when divided by 13.

3.5 Unit digit and last two digits

3.5.1 Cyclicity of unit digits

Digit	Unit digits of powers	Cycle length
0, 1, 5, 6	always the same	1
4	4, 6	2
9	9, 1	2
2	2, 4, 8, 6	4

Digit	Unit digits of powers	Cycle length
3	3, 9, 7, 1	4
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4

3.5.2 Unit digit of products and powers

Divide the exponent by 4 and use the remainder (a remainder of 0 means the 4th, last, term of the cycle). Unit digit of 3^{87}: 87 = 4 × 21 + 3 → 3rd term of 3, 9, 7, 1 → **7**.

3.5.3 Last two digits

Work mod 100. Numbers ending in 1: last two digits of $(\ldots a1)^n$ are (a × last digit of n)1; e.g. 31^{46} ends in 3 × 6 = 18 → **81**. For others, reduce to such a form or use $2^{10} = 1024 \equiv 24$ and $24^2 \equiv 76 \pmod{100}$, with $76^n \equiv 76$.

3.6 Trailing zeros and highest power

3.6.1 Trailing zeros in n!

Trailing zeros in n! = ⌊n/5⌋ + ⌊n/25⌋ + ⌊n/125⌋ + …

100! has 20 + 4 = **24** trailing zeros (there are always more 2s than 5s).

3.6.2 Highest power of a prime in n!

The same method works for any prime p: highest power of 3 in 50! = 16 + 5 + 1 = **22**. For a composite number, factorise it and take the limiting prime: highest power of 12 = $2^2 \times 3$ in 30! is min(⌊26/2⌋, 14) = 13.

3.7 Sums of numbers

3.7.1 Natural numbers, squares and cubes

Sum of first n	Formula	n = 10
natural numbers	$n(n+1)/2$	55
squares	$n(n+1)(2n+1)/6$	385
cubes	$[n(n+1)/2]^2$	3025

3.7.2 Odd and even numbers

Sum of the first n odd numbers = n^2; sum of the first n even numbers = n(n + 1).

3.8 Base systems

3.8.1 Binary, octal and hexadecimal

To convert from base 10, divide repeatedly by the base and read the remainders from bottom to top: 45 = 101101_2 = 55_8 = $2D_{16}$. To convert to base 10, multiply each digit by the power of the base: 1011_2 = 8 + 0 + 2 + 1 = 11.

3.9 Place value and digit problems

In 4,736 the **face value** of 7 is 7 and its **place value** is 700. A two-digit number with digits a and b is 10a + b; reversing gives 10b + a, and the difference is 9(a - b).

WORKED EXAMPLE

The digits of a two-digit number add up to 9, and reversing it increases it by 27. Find it.

9(b - a) = 27 → b - a = 3, with a + b = 9 → a = 3, b = 6: the number is **36**.

3.10 Practice questions

3.10.1 Basic level

Q1 How many prime numbers are there between 1 and 50?

(a) 14 (b) 16 (c) 15 (d) 13

Q2 Which of these numbers is divisible by 11?

(a) 918,082 (b) 918,081 (c) 918,072 (d) 908,082

Q3 How many factors does 72 have?

(a) 12 (b) 10 (c) 8 (d) 9

Q4 What is the unit digit of 3^{87}?

(a) 7 (b) 3 (c) 9 (d) 1

Q5 How many trailing zeros does 100! have?

(a) 24 (b) 20 (c) 25 (d) 22

Q6 What is the sum of the first 15 odd numbers?

(a) 225 (b) 240 (c) 210 (d) 196

Q7 For which digit d is 4d76 divisible by 9?

(a) 2 (b) 4 (c) 1 (d) 7

Q8 What is 101101_2 in decimal?

(a) 43 (b) 47 (c) 53 (d) 45

3.10.2 Intermediate level

Q9 What is the sum of all factors of 360?

(a) 1,080 **(b)** 1,200 **(c)** 1,050 **(d)** 1,170

Q10 How many odd factors does 360 have?

(a) 8 **(b)** 4 **(c)** 12 **(d)** 6

Q11 What is the remainder when 2^{100} is divided by 7?

(a) 1 **(b)** 4 **(c)** 3 **(d)** 2

Q12 What is the remainder when 47 × 53 × 61 is divided by 5?

(a) 2 **(b)** 3 **(c)** 1 **(d)** 4

Q13 What is the highest power of 3 that divides 50!?

(a) 22 **(b)** 16 **(c)** 21 **(d)** 25

Q14 What is the unit digit of $7^{95} \times 3^{58}$?

(a) 1 **(b)** 3 **(c)** 7 **(d)** 9

Q15 What is the sum of the squares of the first 10 natural numbers?

(a) 355 **(b)** 385 **(c)** 405 **(d)** 3,025

Q16 The digits of a two-digit number add up to 9, and reversing the digits increases the number by 27. What is the number?

(a) 27 **(b)** 36 **(c)** 45 **(d)** 63

3.10.3 Advanced level

Q17 What is the remainder when 3^{2026} is divided by 11?

(a) 3 **(b)** 1 **(c)** 5 **(d)** 9

Q18 What are the last two digits of 31^{46}?

(a) 61 **(b)** 81 **(c)** 91 **(d)** 31

Q19 What is the remainder when 12! is divided by 13?

(a) 1 **(b)** 12 **(c)** 0 **(d)** 11

Q20 In how many ways can 360 be written as a product of two natural numbers (order does not matter)?

(a) 24 **(b)** 11 **(c)** 12 **(d)** 13

Q21 What is the highest power of 12 that divides 30!?

(a) 14 **(b)** 10 **(c)** 13 **(d)** 26

Q22 How many numbers from 1 to 1000 are divisible by 3 or 5?

(a) 466 **(b)** 533 **(c)** 467 **(d)** 400

Q23 What is the remainder when 98 × 97 × 96 is divided by 100?

(a) 76 **(b)** 24 **(c)** 64 **(d)** 36

Q24 $\varphi(36)$, the number of integers from 1 to 36 that are co-prime to 36, is

(a) 18 **(b)** 24 **(c)** 12 **(d)** 9

Q25 What is 2D in hexadecimal written in base 10?

(a) 43 **(b)** 45 **(c)** 29 **(d)** 213

3.11 Solutions

Q1 **Answer: (c) 15**

They are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47: **15**.

Q2 **Answer: (a) 918,082**

For 918082: (9 + 8 + 8) − (1 + 0 + 2) = 22, a multiple of 11. Answer **918082**.

Q3 **Answer: (a) 12**

$72 = 2^3 \times 3^2$: $(3 + 1)(2 + 1) = \mathbf{12}$.

Q4 **Answer: (a) 7**

87 mod 4 = 3 → third term of 3, 9, 7, 1: **7**.

Q5 **Answer: (a) 24**

$\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = \mathbf{24}$.

Q6 **Answer: (a) 225**

Sum of first n odd numbers = $n^2 = \mathbf{225}$.

Q7 **Answer: (c) 1**

Digit sum 17 + d must be a multiple of 9: d = **1**.

Q8 **Answer: (d) 45**

32 + 8 + 4 + 1 = **45**.

Q9 **Answer: (d) 1,170**

$(1 + 2 + 4 + 8)(1 + 3 + 9)(1 + 5) = 15 \times 13 \times 6 = \mathbf{1170}$.

Q10 **Answer: (d) 6**

Ignore 2^3: $3^2 \times 5$ gives $(2 + 1)(1 + 1) = \mathbf{6}$.

Q11 **Answer: (d) 2**

$2^3 \equiv 1 \pmod 7$; $2^{100} = (2^3)^{33} \times 2 \equiv \mathbf{2}$.

Q12 **Answer: (c) 1**

Remainders 2, 3, 1; 2 × 3 × 1 = 6 ≡ **1** (mod 5).

Q13 **Answer: (a) 22**

⌊50/3⌋ + ⌊50/9⌋ + ⌊50/27⌋ = 16 + 5 + 1 = **22**.

Q14 **Answer: (c) 7**

7^{95}: 95 mod 4 = 3 → 3. 3^{58}: 58 mod 4 = 2 → 9. 3 × 9 = 27: unit digit **7**.

Q15 **Answer: (b) 385**

10 × 11 × 21 / 6 = **385**.

Q16 **Answer: (b) 36**

9(b − a) = 27 → b − a = 3 and a + b = 9 → **36**.

Q17 **Answer: (a) 3**

Fermat: $3^{10} \equiv 1 \pmod{11}$; 2026 mod 10 = 6; $3^6 = 729 \equiv$ **3**.

Q18 **Answer: (b) 81**

For …1 bases: tens digit = 3 × 6 = 18 → 8; last two digits **81**.

Q19 **Answer: (b) 12**

Wilson's theorem: $(p-1)! \equiv -1 \equiv 12 \pmod{13}$. Answer **12**.

Q20 **Answer: (c) 12**

360 has 24 factors and is not a perfect square: 24/2 = **12**.

Q21 **Answer: (c) 13**

Powers in 30!: 2 → 26, 3 → 14. $12 = 2^2 \times 3$ needs min(26/2, 14) = **13**.

Q22 **Answer: (c) 467**

333 + 200 − 66 = **467** (inclusion-exclusion).

Q23 **Answer: (a) 76**

Use negative remainders: (−2)(−3)(−4) = −24 ≡ 100 − 24 = **76** (mod 100).

Q24 **Answer: (c) 12**

36 × (1 − 1/2)(1 − 1/3) = **12**.

Q25 **Answer: (b) 45**

2 × 16 + 13 = **45**.

Factorise first: n = a^p b^q c^r gives (p+1)(q+1)(r+1) factors. Use remainders of parts (and negative remainders), cyclicity, Fermat (a^(p-1) ≡ 1) and Euler for large powers. Unit digits follow cycles of 4; trailing zeros in n! come from ⌊n/5⌋ + ⌊n/25⌋ + … . Test primes only up to √n.

MODULE 04 · BEGINNER

HCF and LCM

Minimum duration **1 day** | Level **Beginner**

HCF and LCM problems test number sense and appear in scheduling, tiling and grouping questions. This module covers the meaning of HCF and LCM, the product rule, the prime factorisation and division methods, HCF and LCM of fractions and decimals, and the standard application problems (bells and lights, tiles, given remainders, circular tracks, grouping), followed by 25 solved questions with answers computed by a program.

IN THIS MODULE

4.1 Concepts
4.2 Methods
4.3 Application problems
4.4 Practice questions
4.5 Solutions

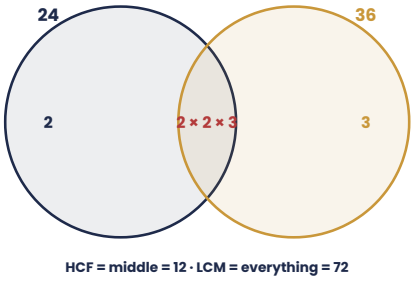
4.1 Concepts

4.1.1 Highest Common Factor (HCF)

The **HCF** (or GCD) of two or more numbers is the largest number that divides all of them exactly. HCF(24, 36) = 12.

4.1.2 Least Common Multiple (LCM)

The **LCM** is the smallest number that is a multiple of all of them. LCM(24, 36) = 72.



PRIME FACTORS OF 24 AND 36: HCF USES THE SHARED PRIMES, LCM USES ALL OF THEM

4.1.3 HCF × LCM = product of two numbers

For two numbers: HCF × LCM = first number × second number

24 × 36 = 864 = 12 × 72. This works for **two** numbers only, not three.

4.2 Methods

4.2.1 Prime factorisation method

Factorise each number. HCF = product of common primes with the **lowest** powers; LCM = product of all primes with the **highest** powers.

Number	Factorisation
60	$2^2 \times 3 \times 5$
84	$2^2 \times 3 \times 7$
126	$2 \times 3^2 \times 7$
HCF	$2 \times 3 = 6$
LCM	$2^2 \times 3^2 \times 5 \times 7 = 1260$

4.2.2 Division method (Euclid)

Divide the larger number by the smaller, then the divisor by the remainder, and repeat until the remainder is 0. The last divisor is the HCF. HCF(391, 667): $667 = 1 \times 391 + 276$; $391 = 1 \times 276 + 115$; $276 = 2 \times 115 + 46$; $115 = 2 \times 46 + 23$; $46 = 2 \times 23 + 0 \rightarrow$ **23**.

4.2.3 Fractions and decimals

HCF of fractions = HCF of numerators ÷ LCM of denominators; LCM of fractions = LCM of numerators ÷ HCF of denominators

HCF(2/3, 4/9, 8/15) = HCF(2, 4, 8)/LCM(3, 9, 15) = 2/45. For decimals, make the decimal places equal, drop the point, find HCF/LCM, then put the point back: HCF(0.6, 1.2, 1.8) = HCF(6, 12, 18)/10 = 0.6.

4.3 Application problems

4.3.1 Bells and traffic lights (LCM)

Events that repeat every a, b, c units happen together again after LCM(a, b, c). Bells ringing every 6, 8 and 12 minutes ring together every 24 minutes.

4.3.2 Largest tile (HCF)

The largest square tile that exactly covers a 6 m 24 cm × 4 m 32 cm floor has side HCF(624, 432) = 48 cm; the number of tiles is (624/48) × (432/48) = 13 × 9 = 117.

4.3.3 Largest number dividing with given remainders (HCF)

The largest number that divides 70 and 125 leaving remainders 5 and 8 divides 70 - 5 = 65 and 125 - 8 = 117: HCF(65, 117) = 13. If the remainder is the same r for all, use HCF of the differences between the numbers.

4.3.4 Smallest number leaving given remainders (LCM)

Condition	Answer
Same remainder r when divided by a, b, c	LCM(a, b, c) + r
Remainders differ, but divisor - remainder = k is the same for all	LCM(a, b, c) - k

WORKED EXAMPLE

Smallest number that leaves remainder 2, 3, 4 when divided by 3, 4, 5

Each divisor - remainder = 1, so the answer is LCM(3, 4, 5) - 1 = 60 - 1 = **59**.

4.3.5 Circular track meetings

Runners who complete a lap in a, b, c minutes are next together at the start after LCM(a, b, c) minutes.

4.3.6 Grouping into equal groups (HCF)

To split 48 apples, 72 oranges and 96 bananas into the largest identical baskets with one kind of fruit per basket, the basket size is HCF(48, 72, 96) = 24, giving 2 + 3 + 4 = 9 baskets.

SHORTCUT TRICK

Ask "bigger or smaller?": the **largest** size that fits into all (tiles, baskets, divisors) is an HCF; the **first time** everything coincides (bells, laps, smallest number) is an LCM.

4.4 Practice questions

4.4.1 Basic level

Q1 HCF of 48 and 180 is

(a) 6 (b) 24 (c) 12 (d) 18

Q2 LCM of 12, 18 and 30 is

(a) 90 (b) 360 (c) 120 (d) 180

Q3 The HCF of two numbers is 12 and their LCM is 144. One number is 36. What is the other?

(a) 24 (b) 36 (c) 48 (d) 72

Q4 Three bells ring every 6, 8 and 12 minutes. They ring together at 9:00 a.m. When do they next ring together?

(a) 9:12 a.m. (b) 9:24 a.m. (c) 9:48 a.m. (d) 9:36 a.m.

Q5 HCF of 391 and 667 is

(a) 17 (b) 23 (c) 29 (d) 13

Q6 What is the largest number that divides 70 and 125 leaving remainders 5 and 8?

(a) 5 (b) 9 (c) 13 (d) 65

Q7 LCM of 2/3, 4/9 and 8/15 is

(a) 8/45 (b) 8/3 (c) 2/45 (d) 16/3

Q8 HCF of 0.6, 1.2 and 1.8 is

(a) 0.3 **(b)** 0.6 **(c)** 1.2 **(d)** 0.06

4.4.2 Intermediate level

Q9 What is the smallest number which, when divided by 12, 15 and 20, leaves remainder 4 in each case?

(a) 64 **(b)** 60 **(c)** 56 **(d)** 124

Q10 What is the smallest number that leaves remainders 2, 3 and 4 when divided by 3, 4 and 5?

(a) 61 **(b)** 119 **(c)** 59 **(d)** 57

Q11 A floor 6 m 24 cm long and 4 m 32 cm wide is to be covered with identical square tiles. What is the least number of tiles needed?

(a) 108 **(b)** 96 **(c)** 117 **(d)** 144

Q12 48 apples, 72 oranges and 96 bananas are packed into the largest possible identical-size baskets, one kind of fruit per basket. How many baskets are needed?

(a) 12 **(b)** 9 **(c)** 6 **(d)** 24

Q13 Three runners take 30, 45 and 60 seconds to complete a lap. After how many seconds will they first be together at the start again?

(a) 180 **(b)** 90 **(c)** 135 **(d)** 360

Q14 Two numbers are in the ratio 3 : 4 and their HCF is 7. What is their LCM?

(a) 42 **(b)** 49 **(c)** 84 **(d)** 168

Q15 The product of two numbers is 2160 and their HCF is 12. How many such pairs of numbers are possible?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q16 What is the greatest number that divides 43, 91 and 183 leaving the same remainder in each case?

(a) 8 **(b)** 4 **(c)** 2 **(d)** 16

4.4.3 Advanced level

Q17 Traffic lights at three crossings change every 48, 72 and 108 seconds. If they change together at 8:20:00 a.m., when do they next change together?

(a) 8:27:12 a.m. **(b)** 8:24:00 a.m. **(c)** 8:26:24 a.m. **(d)** 8:30:00 a.m.

Q18 What is the least 4-digit number that is divisible by 12, 15 and 18?

(a) 1,080 **(b)** 1,020 **(c)** 1,170 **(d)** 1,008

Q19 What is the greatest 4-digit number that leaves remainder 5 when divided by 6, 9 and 12?

(a) 9,972 **(b)** 9,985 **(c)** 9,977 **(d)** 9,941

Q20 The HCF and LCM of two numbers are 11 and 385. If one number lies between 75 and 125, what is it?

(a) 55 **(b)** 99 **(c)** 110 **(d)** 77

Q21 What is the smallest number which, when increased by 3, is divisible by 12, 18, 21 and 30?

(a) 1,263 **(b)** 1,260 **(c)** 1,257 **(d)** 2,517

Q22 Find the least number of square slabs needed to pave a courtyard 15 m 17 cm long and 9 m 2 cm wide.

(a) 407 **(b)** 902 **(c)** 814 **(d)** 1,628

Q23 HCF of 2/3, 4/9 and 8/15 is

(a) 8/3 **(b)** 2/3 **(c)** 2/45 **(d)** 4/45

Q24 Three containers hold 403, 434 and 465 litres of oil. What is the largest measure that can measure each exactly?

(a) 13 **(b)** 41 **(c)** 62 **(d)** 31

Q25 A number leaves remainder 3 when divided by 5, 6 and 8. What is the least such number greater than 3 that is also divisible by 9?

(a) 723 **(b)** 363 **(c)** 243 **(d)** 603

4.5 Solutions

Q1 **Answer: (c) 12**

$48 = 2^4 \times 3$, $180 = 2^2 \times 3^2 \times 5$; common lowest powers $2^2 \times 3 =$ **12**.

Q2 **Answer: (d) 180**

$2^2 \times 3^2 \times 5 =$ **180**.

Q3 **Answer: (c) 48**

Other = HCF × LCM ÷ 36 = 1728 ÷ 36 = **48**.

Q4 **Answer: (b) 9:24 a.m.**

LCM(6, 8, 12) = 24 minutes: **9:24 a.m.**

Q5 **Answer: (b) 23**

By repeated division (4.2.2), the last non-zero divisor is **23**.

Q6 **Answer: (c) 13**

HCF(70 − 5, 125 − 8) = HCF(65, 117) = **13**.

Q7 **Answer: (b) 8/3**

LCM of numerators 8 ÷ HCF of denominators 3 = **8/3**.

Q8 **Answer: (b) 0.6**

HCF(6, 12, 18) = 6 → **0.6**.

Q9 **Answer: (a) 64**

LCM(12, 15, 20) + 4 = 60 + 4 = **64**.

Q10 **Answer: (c) 59**

Divisor - remainder = 1 each time: LCM 60 - 1 = **59**.

Q11 **Answer: (c) 117**

Tile side = HCF(624, 432) = 48 cm; tiles = 13 × 9 = **117**.

Q12 **Answer: (b) 9**

Basket size HCF = 24: 2 + 3 + 4 = **9** baskets.

Q13 **Answer: (a) 180**

LCM(30, 45, 60) = **180** seconds.

Q14 **Answer: (c) 84**

Numbers 21 and 28; LCM = 3 × 4 × 7 = **84**.

Q15 **Answer: (b) 2**

Numbers 12a, 12b with a, b co-prime and ab = 2160/144 = 15: (1, 15), (3, 5). So **2** pairs.

Q16 **Answer: (b) 4**

HCF of the differences 48, 92, 140 = **4**.

Q17 **Answer: (a) 8:27:12 a.m.**

LCM(48, 72, 108) = 432 s = 7 min 12 s: **8:27:12 a.m.**

Q18 **Answer: (a) 1,080**

LCM = 180; the smallest 4-digit multiple is 180 × 6 = **1080**.

Q19 **Answer: (c) 9,977**

LCM = 36; the greatest 4-digit multiple of 36 is 9972; add 5: **9977**.

Q20 **Answer: (d) 77**

Numbers are 11a, 11b with ab = 35 and a, b co-prime: (1, 35) or (5, 7) → 55, 77 or 11, 385. Only **77** lies between 75 and 125.

Q21 **Answer: (c) 1,257**

LCM = 1260; the number is LCM - 3 = **1257**.

Q22 **Answer: (c) 814**

Side = HCF(1517, 902) = 41 cm; slabs = 37 × 22 = **814**.

Q23 **Answer: (c) 2/45**

HCF of numerators 2 ÷ LCM of denominators 45 = **2/45**.

Q24 **Answer: (d) 31**

HCF(403, 434, 465) = **31** litres.

Q25 **Answer: (c) 243**

The numbers are 120k + 3. 120k + 3 is divisible by 9 when 3k + 3 ≡ 0 (mod 9), i.e. k = 2: 240 + 3 = **243**.

HCF = common primes, lowest powers; LCM = all primes, highest powers; HCF × LCM = product of two numbers. Largest size that fits everything → HCF (subtract remainders first); first time everything coincides → LCM (add a common remainder, or subtract a common shortfall). Fractions: HCF = HCF(num)/LCM(den), LCM = LCM(num)/HCF(den).

MODULE 05 · BEGINNER

Simplification, approximation, surds and indices

Minimum duration **2 days** | Level **Beginner**

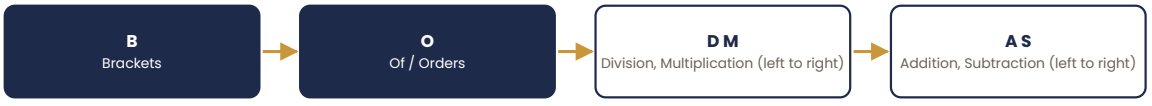
These skills let you calculate complex expressions quickly and are heavily tested in bank and placement exams. This module covers BODMAS and vinculum brackets, simplification with algebraic identities, approximation, the laws of indices (including negative and fractional exponents and comparing powers), surds (simplifying, rationalising, comparing, square roots of surd expressions) and missing or wrong numbers in an equation, followed by 25 solved questions checked by a program.

IN THIS MODULE

5.1	BODMAS	5.2	Simplification
5.3	Approximation	5.4	Indices
5.5	Surds	5.6	Missing and wrong numbers
5.7	Practice questions	5.8	Solutions

5.1 BODMAS

5.1.1 Order of operations



BODMAS

"Of" means multiplication but is done before ordinary division and multiplication: 48 ÷ 4 of 3 = 48 ÷ 12 = 4, while 48 ÷ 4 × 3 = 36.

5.1.2 Vinculum (bar) brackets

A bar over an expression is the innermost bracket and is simplified first. Bracket order: bar, then ( ), then { }, then [ ].

WORKED EXAMPLE

Simplify $18 - [5 - \{6 + 2(7 - \overline{8 - 5})\}]$

Bar: 8 - 5 = 3. Then 7 - 3 = 4; 6 + 8 = 14; 5 - 14 = -9; 18 - (-9) = **27**.

5.2 Simplification

5.2.1 Fractions and mixed expressions

Convert mixed numbers to improper fractions, cancel before multiplying, and use a common denominator only when adding or subtracting.

5.2.2 Algebraic identities

Identity	Expansion
$(a + b)^2$	$a^2 + 2ab + b^2$
$(a - b)^2$	$a^2 - 2ab + b^2$
$a^2 - b^2$	$(a + b)(a - b)$
$(a + b)^3$	$a^3 + b^3 + 3ab(a + b)$
$a^3 + b^3$	$(a + b)(a^2 - ab + b^2)$
$a^3 - b^3$	$(a - b)(a^2 + ab + b^2)$
$a^3 + b^3 + c^3 - 3abc$	$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$; $= 0$ if $a + b + c = 0$

5.2.3 Using identities for fast calculation

WORKED EXAMPLE
$(76 \times 76 \times 76 + 24 \times 24 \times 24) \div (76 \times 76 - 76 \times 24 + 24 \times 24)$
Form $(a^3 + b^3)/(a^2 - ab + b^2) = a + b = 76 + 24 =$ **100**.

WORKED EXAMPLE
$(0.87^2 - 0.13^2) \div 0.74$
$a^2 - b^2 = (a + b)(a - b) = 1 \times 0.74$, so the value is **1**.

5.3 Approximation

5.3.1 Rounding

Bank exams ask for the approximate value of long expressions. Round each number to a convenient value (nearest whole, ten or friendly fraction), keep track of whether you rounded up or down, and pick the closest option.

WORKED EXAMPLE
$\approx 4999 \div 50.02 \times 19.98$
$5000 \div 50 \times 20 =$ **2000** (exact 1996.8).

5.3.2 Square roots and percentages

$\surd(n)$ for n near a perfect square: $\surd 122 \approx 11.05$. Percentages: 24.97% of 801 ≈ 25% of 800 = 200.

5.4 Indices

5.4.1 Laws of exponents

Law	Example
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7$
$a^m \div a^n = a^{m-n}$	$5^6 \div 5^2 = 5^4$
$(a^m)^n = a^{mn}$	$(3^2)^3 = 3^6$
$(ab)^n = a^n b^n$	$6^2 = 2^2 \times 3^2$
$a^0 = 1$	$7^0 = 1$

5.4.2 Negative and fractional exponents

$a^{-n} = 1/a^n$ and $a^{(m/n)} = (\sqrt[n]{a})^m$. So $8^{(2/3)} = (\sqrt[3]{8})^2 = 4$ and $27^{(-1/3)} = 1/3$.

5.4.3 Comparing powers

Make either the bases or the exponents equal. Compare 2^{300} and 3^{200}: write both with exponent 100: $(2^3)^{100} = 8^{100}$ and $(3^2)^{100} = 9^{100}$, so 3^{200} is larger.

5.5 Surds

5.5.1 Simplifying surds

Take out perfect-square factors: $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$; $\sqrt{50} + \sqrt{18} = 5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$.

5.5.2 Rationalising the denominator

Multiply by the conjugate: $1/(\sqrt{5} - \sqrt{3}) = (\sqrt{5} + \sqrt{3})/(5 - 3) = (\sqrt{5} + \sqrt{3})/2$.

5.5.3 Comparing surds

Bring surds to the same root order: compare $\sqrt[3]{4}$ and $\sqrt{3}$ by raising both to the 6th power: $4^2 = 16$ and $3^3 = 27$, so $\sqrt{3}$ is larger. For sums like $\sqrt{7} + \sqrt{3}$ vs $\sqrt{6} + \sqrt{4}$, square both: $10 + 2\sqrt{21}$ vs $10 + 2\sqrt{24}$, so $\sqrt{6} + \sqrt{4}$ is larger.

5.5.4 Square root of a surd expression

$\sqrt{(a + 2\sqrt{b})} = \sqrt{x} + \sqrt{y}$ where $x + y = a$ and $xy = b$. $\sqrt{(8 + 2\sqrt{15})} = \sqrt{5} + \sqrt{3}$, because $5 + 3 = 8$ and $5 \times 3 = 15$.

> **SHORTCUT TRICK**
> For $\sqrt{(a + 2\sqrt{b})}$, look for two numbers that add to a and multiply to b, just as when factorising a quadratic.

5.6 Missing and wrong numbers

For a missing number, simplify the known side and solve backwards. For a "wrong number" in an equation, check each number by recomputing; the digit-sum check of Module 2 quickly finds which side is inconsistent.

WORKED EXAMPLE
? × 12 + 15 = 159
Subtract 15: 144; divide by 12: **12**.

5.7 Practice questions

5.7.1 Basic level

Q1 48 ÷ 4 of 3 + 5 = ?
(a) 41 (b) 21 (c) 17 (d) 9

Q2 Simplify: 18 − [5 − {6 + 2(7 − (8 − 5))}]
(a) 9 (b) 15 (c) 27 (d) 21

Q3 $2^3 \times 2^5 \div 2^4$ = ?
(a) 16 (b) 8 (c) 32 (d) 64

Q4 8^(2/3) = ?
(a) 2 (b) 16 (c) 6 (d) 4

Q5 √72 simplifies to
(a) 3√8 (b) 8√3 (c) 4√2 (d) 6√2

Q6 $(0.87^2 - 0.13^2) \div 0.74$ = ?
(a) 0.74 (b) 2 (c) 1 (d) 0.5

Q7 ? × 12 + 15 = 159
(a) 11 (b) 13 (c) 12 (d) 14

Q8 Approximately, 24.97% of 801 ≈ ?
(a) 180 (b) 200 (c) 220 (d) 250

5.7.2 Intermediate level

Q9 $(76^3 + 24^3) \div (76^2 - 76 \times 24 + 24^2)$ = ?
(a) 52 (b) 76 (c) 152 (d) 100

Q10 √50 + √18 − √8 = ?
(a) 8√2 (b) 6√2 (c) 4√2 (d) 10√2

Q11 27^(−1/3) = ?
(a) 3 (b) 1/3 (c) 1/9 (d) −3

Q12 Which is larger: 2^{300} or 3^{200}?

(a) 3^{200} **(b)** 2^{300}

(c) They are equal **(d)** Cannot be compared

Q13 $1/(\sqrt{5} - \sqrt{3}) = ?$

(a) $\sqrt{5} + \sqrt{3}$ **(b)** $(\sqrt{5} - \sqrt{3})/2$ **(c)** $(\sqrt{5} + \sqrt{3})/8$ **(d)** $(\sqrt{5} + \sqrt{3})/2$

Q14 If $3^x = 81$, what is x?

(a) 3 **(b)** 9 **(c)** 4 **(d)** 27

Q15 Approximately, $4999 \div 50.02 \times 19.98 \approx ?$

(a) 1,500 **(b)** 2,000 **(c)** 2,500 **(d)** 200

Q16 $(1.5)^3 - (0.5)^3 - 3 \times 1.5 \times 0.5 \times 1 = ?$

(a) 3/2 **(b)** 2 **(c)** 0 **(d)** 1

5.7.3 Advanced level

Q17 $\sqrt{(8 + 2\sqrt{15})} = ?$

(a) $\sqrt{5} + \sqrt{3}$ **(b)** $\sqrt{6} + \sqrt{2}$ **(c)** $\sqrt{5} - \sqrt{3}$ **(d)** $2\sqrt{2} + \sqrt{3}$

Q18 Which is larger: $\sqrt[3]{4}$ or $\sqrt{3}$?

(a) $\sqrt[3]{4}$ **(b)** They are equal

(c) $\sqrt{3}$ **(d)** Cannot be compared

Q19 If $2^{x+3} = 32$, then x = ?

(a) 3 **(b)** 5 **(c)** 2 **(d)** 1

Q20 If $a + b + c = 0$, then $(a^3 + b^3 + c^3)/(abc) = ?$

(a) 0 **(b)** 1 **(c)** 9 **(d)** 3

Q21 $(0.2 \times 0.2 + 0.01) \div (0.1 \times 0.1 + 0.02) = ?$

(a) 3/5 **(b)** 2 **(c)** 5/3 **(d)** 1/2

Q22 Which is the largest: $\sqrt{7} + \sqrt{3}$, $\sqrt{6} + \sqrt{4}$, $\sqrt{5} + \sqrt{5}$, $\sqrt{8} + \sqrt{2}$?

(a) $\sqrt{7} + \sqrt{3}$ **(b)** $\sqrt{6} + \sqrt{4}$ **(c)** $\sqrt{8} + \sqrt{2}$ **(d)** $\sqrt{5} + \sqrt{5}$

Q23 $(2^{50} + 2^{50}) \div 2^{48} = ?$

(a) 4 **(b)** 2 **(c)** 16 **(d)** 8

Q24 Find the wrong number in $25 \times 16 + 120 \div 8 = 418$.

(a) 25 **(b)** 418 (should be 415)

(c) 16 **(d)** 120

Q25 If $x = 3 + 2\sqrt{2}$, what is $x + 1/x$?

(a) $4\sqrt{2}$ **(b)** 3 **(c)** 6 **(d)** 2

5.8 Solutions

Q1 **Answer: (d) 9**

"of" first: 4 of 3 = 12; 48 ÷ 12 = 4; 4 + 5 = **9**.

Q2 **Answer: (c) 27**

Innermost first: 8 - 5 = 3, 7 - 3 = 4, 6 + 8 = 14, 5 - 14 = -9, 18 + 9 = **27**.

Q3 **Answer: (a) 16**

$2^{(3 + 5 - 4)} = 2^4 = \mathbf{16}$.

Q4 **Answer: (d) 4**

$(\sqrt[3]{8})^2 = 2^2 = \mathbf{4}$.

Q5 **Answer: (d) $6\sqrt{2}$**

$\sqrt{(36 \times 2)} = \mathbf{6\sqrt{2}}$.

Q6 **Answer: (c) 1**

$(a + b)(a - b) = 1 \times 0.74$; ÷ 0.74 = **1**.

Q7 **Answer: (c) 12**

(159 - 15) ÷ 12 = **12**.

Q8 **Answer: (b) 200**

25% of 800 = **200**.

Q9 **Answer: (d) 100**

$(a^3 + b^3)/(a^2 - ab + b^2) = a + b = \mathbf{100}$.

Q10 **Answer: (b) $6\sqrt{2}$**

$5\sqrt{2} + 3\sqrt{2} - 2\sqrt{2} = \mathbf{6\sqrt{2}}$.

Q11 **Answer: (b) 1/3**

$1/\sqrt[3]{27} = \mathbf{1/3}$.

Q12 **Answer: (a) 3^{200}**

$2^{300} = 8^{100}$ and $3^{200} = 9^{100}$: $\mathbf{3^{200}}$ is larger.

Q13 **Answer: (d) $(\sqrt{5} + \sqrt{3})/2$**

Multiply by the conjugate: $(\sqrt{5} + \sqrt{3})/(5 - 3) = \mathbf{(\sqrt{5} + \sqrt{3})/2}$.

Q14 **Answer: (c) 4**

$81 = 3^4$, so x = **4**.

Q15 **Answer: (b) 2,000**

$5000 \div 50 \times 20 =$ **2000**.

Q16 **Answer: (d) 1**

Since $1.5 - 0.5 = 1$, $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$ gives $a^3 - b^3 - 3ab = 1^3 =$ **1**.

Q17 **Answer: (a) $\sqrt{5} + \sqrt{3}$**

Numbers adding to 8 and multiplying to 15 are 5 and 3: $\mathbf{\sqrt{5} + \sqrt{3}}$.

Q18 **Answer: (c) $\sqrt{3}$**

Sixth powers: $(\sqrt[3]{4})^6 = 16$, $(\sqrt{3})^6 = 27$. $\mathbf{\sqrt{3}}$ is larger.

Q19 **Answer: (c) 2**

$32 = 2^5 \rightarrow x + 3 = 5 \rightarrow x =$ **2**.

Q20 **Answer: (d) 3**

With $a + b + c = 0$, $a^3 + b^3 + c^3 = 3abc$, so the ratio is **3**.

Q21 **Answer: (c) 5/3**

$(0.04 + 0.01)/(0.01 + 0.02) = 0.05/0.03 =$ **5/3**.

Q22 **Answer: (d) $\sqrt{5} + \sqrt{5}$**

Squares are $10 + 2\sqrt{21}$, $10 + 2\sqrt{24}$, $10 + 2\sqrt{25}$, $10 + 2\sqrt{16}$. The largest product is 25: $\mathbf{\sqrt{5} + \sqrt{5}}$.

Q23 **Answer: (d) 8**

$2 \times 2^{50} = 2^{51}$; $2^{51} \div 2^{48} = 2^3 =$ **8**.

Q24 **Answer: (b) 418 (should be 415)**

$25 \times 16 = 400$; $120 \div 8 = 15$; total **415**, so 418 is wrong.

Q25 **Answer: (c) 6**

$1/x = 3 - 2\sqrt{2}$ (rationalise), so $x + 1/x =$ **6**.

BODMAS with "of" before ÷ and ×, and the bar bracket first. Use identities $(a \pm b)^2$, $a^2 - b^2$, $a^3 \pm b^3$ to collapse expressions. Indices: add, subtract, multiply exponents; $a^{-n} = 1/a^n$, $a^{(m/n)} = (\sqrt[n]{a})^m$; compare powers by equalising bases or exponents. Surds: extract squares, rationalise with conjugates, compare by raising to equal powers.

MODULE 06 · BEGINNER

Decimals and fractions

Minimum duration **1 day** | Level **Beginner**

Fractions and decimals appear inside almost every arithmetic problem. This module covers types of fractions, equivalent fractions, comparing and ordering, operations, continued fractions, terminating and recurring decimals, converting recurring decimals to fractions, operations on decimals and fraction word problems, followed by 25 solved questions computed with exact fractions.

IN THIS MODULE

6.1 Fractions
6.2 Decimals
6.3 Word problems with fractions
6.4 Practice questions
6.5 Solutions

6.1 Fractions

6.1.1 Proper, improper and mixed fractions

A **proper** fraction has numerator < denominator (3/5); an **improper** fraction has numerator ≥ denominator (7/4); a **mixed** fraction combines a whole number and a proper fraction (1 3/4 = 7/4).



3/8 of the bar is shaded

BAR MODEL OF A FRACTION

6.1.2 Equivalent fractions

Multiplying or dividing the numerator and denominator by the same number gives an equivalent fraction: 2/3 = 4/6 = 10/15. A fraction is in lowest terms when the HCF of numerator and denominator is 1.

6.1.3 Comparing and ordering

Method	When	Example
Cross-multiplication	two fractions	5/7 vs 7/10: 50 vs 49 → 5/7 larger
Common numerator	same numerators	3/7 > 3/8 (smaller denominator is larger)
Convert to decimals or %	many fractions	3/8 = 37.5%, 2/5 = 40%
Difference rule	numerator and denominator differ by the same amount	4/5 < 5/6 < 6/7 (closer to 1)

6.1.4 Operations

Add or subtract with the LCM of denominators; multiply numerators and denominators (cancel first); divide by multiplying by the reciprocal: 3/4 ÷ 9/8 = 3/4 × 8/9 = 2/3.

6.1.5 Continued fractions

Simplify from the bottom up. 1 + 1/(1 + 1/(1 + 1/2)) = 1 + 1/(1 + 2/3) = 1 + 3/5 = 8/5.

6.2 Decimals

6.2.1 Terminating and recurring decimals

A fraction in lowest terms gives a **terminating** decimal exactly when its denominator has no prime factors other than 2 and 5 (7/40 = 0.175). Otherwise the decimal **recurs**: 1/3 = 0.333..., 5/11 = 0.4545... (written with a bar over the repeating block).

6.2.2 Converting recurring decimals to fractions

Decimal	Rule	Fraction
0.$\bar{a}$ (pure)	repeating block over as many 9s	0.4545... = 45/99 = 5/11
0.a$\bar{b}$ (mixed)	(whole digits - non-repeating part) over 9s followed by 0s	0.1666... = (16 - 1)/90 = 1/6
2.$\dot{3}$	2 + 3/9	7/3

SHORTCUT TRICK

Pure recurring: one 9 for each repeating digit. Mixed recurring: 9s for the repeating digits and 0s for the non-repeating digits after the point.

6.2.3 Operations on decimals

When multiplying, ignore the points, multiply, then place the point so the answer has as many decimal places as the factors together (0.03 × 0.4 = 0.012). When dividing, shift the point in both numbers to make the divisor whole (1.44 ÷ 0.12 = 144 ÷ 12 = 12).

6.3 Word problems with fractions

WORKED EXAMPLE

A man spends 1/3 of his income on rent and 1/4 of the remainder on food, and saves ₹9,000. What is his income?

After rent 2/3 remains; food takes 1/4 of that = 1/6; remaining 2/3 - 1/6 = 1/2 is ₹9,000, so the income is **₹18,000**.



BAR MODEL FOR THE WORKED EXAMPLE

TIP
Work with the fraction of the **original** whole: "1/4 of the remainder" is 1/4 × 2/3 = 1/6 of the income, not 1/4.

6.4 Practice questions

6.4.1 Basic level

Q1 Which is larger: 5/7 or 7/10?

(a) 5/7 (b) 7/10
(c) They are equal (d) Cannot be compared

Q2 3/4 ÷ 9/8 = ?

(a) 27/32 (b) 2/3 (c) 3/2 (d) 1/2

Q3 2/3 + 3/4 - 5/6 = ?

(a) 1/2 (b) 7/12 (c) 5/6 (d) 3/4

Q4 0.4545… (45 recurring) as a fraction is

(a) 5/11 (b) 9/20 (c) 1/2 (d) 4/9

Q5 0.03 × 0.4 = ?

(a) 0.12 (b) 0 (c) 0.01 (d) 1.2

Q6 1.44 ÷ 0.12 = ?

(a) 1.2 (b) 120 (c) 0.12 (d) 12

Q7 Which fraction gives a terminating decimal?

(a) 7/40 (b) 5/12 (c) 2/15 (d) 4/21

Q8 What is 2 3/4 as an improper fraction?

(a) 9/4 (b) 5/2 (c) 11/4 (d) 3/2

6.4.2 Intermediate level

Q9 Arrange in ascending order: 2/3, 3/5, 7/10, 5/8

(a) 2/3, 3/5, 5/8, 7/10 (b) 3/5, 2/3, 5/8, 7/10
(c) 5/8, 3/5, 2/3, 7/10 (d) 3/5, 5/8, 2/3, 7/10

Q10 0.1666… (6 recurring) as a fraction is

(a) 1/9 (b) 1/3 (c) 1/6 (d) 1/5

Q11 1 + 1/(1 + 1/(1 + 1/2)) = ?

(a) 8/5 (b) 5/3 (c) 3/2 (d) 13/8

Q12 A man spends 1/3 of his income on rent and 1/4 of the remainder on food, and saves ₹9,000. What is his income?

(a) 27,000 **(b)** 36,000 **(c)** 15,000 **(d)** 18,000

Q13 What fraction of 3/4 is 2/5?

(a) 3/10 **(b)** 15/8 **(c)** 8/15 **(d)** 5/8

Q14 2.3333… (3 recurring) as a fraction is

(a) 23/10 **(b)** 13/6 **(c)** 7/3 **(d)** 233/100

Q15 If 3/5 of a number is 48, what is 7/8 of the number?

(a) 60 **(b)** 70 **(c)** 80 **(d)** 64

Q16 Which is the smallest: 4/5, 5/6, 6/7, 7/8?

(a) 5/6 **(b)** 6/7 **(c)** 4/5 **(d)** 7/8

6.4.3 Advanced level

Q17 0.2353535… (35 recurring after the 2) as a fraction is

(a) 233/990 **(b)** 233/1000 **(c)** 35/99 **(d)** 47/200

Q18 A tank is 2/5 full. After adding 36 litres it is 7/10 full. What is its capacity?

(a) 100 litres **(b)** 120 litres **(c)** 150 litres **(d)** 90 litres

Q19 In a class, 3/5 of the students are boys and 1/4 of the boys wear glasses. If 18 boys wear glasses, how many students are in the class?

(a) 100 **(b)** 90 **(c)** 72 **(d)** 120

Q20 (0.5555… + 0.4444…) = ?

(a) 1 **(b)** 9/10 **(c)** 99/100 **(d)** 11/10

Q21 Simplify: (1 - 1/2)(1 - 1/3)(1 - 1/4)…(1 - 1/20)

(a) 1/20 **(b)** 1/19 **(c)** 19/20 **(d)** 1/10

Q22 A sum of money is divided among A, B, C so that A gets 1/3, B gets 2/5 of the remainder and C gets ₹600. What is the total?

(a) 1,500 **(b)** 1,200 **(c)** 1,800 **(d)** 2,000

Q23 Which is the largest: 0.3̄ (3 recurring), 0.33, 0.333, 0.3?

(a) 0.33 **(b)** 0.333… (recurring)
(c) 0.333 **(d)** 0.3

Q24 If 1/a + 1/b = 1/6 and a = 10, what is b?

(a) 12 **(b)** 16 **(c)** 4 **(d)** 15

Q25 The numerator of a fraction is 3 less than its denominator. If 2 is added to both, the fraction becomes 4/5. What is the original fraction?

(a) 5/8 **(b)** 7/10 **(c)** 10/13 **(d)** 11/14

6.5 Solutions

Q1 **Answer: (a) 5/7**
Cross-multiply: 5 × 10 = 50 > 7 × 7 = 49, so **5/7** is larger.

Q2 **Answer: (b) 2/3**
3/4 × 8/9 = **2/3**.

Q3 **Answer: (b) 7/12**
LCM 12: 8/12 + 9/12 − 10/12 = **7/12**.

Q4 **Answer: (a) 5/11**
45/99 = **5/11**.

Q5 **Answer: (c) 0.01**
3 × 4 = 12 with 3 decimal places: **0.012**.

Q6 **Answer: (d) 12**
144 ÷ 12 = **12**.

Q7 **Answer: (a) 7/40**
40 = 2^3 × 5 has only 2s and 5s; the others have 3 or 7. Answer **7/40**.

Q8 **Answer: (c) 11/4**
2 × 4 + 3 = 11: **11/4**.

Q9 **Answer: (d) 3/5, 5/8, 2/3, 7/10**
As decimals: 0.667, 0.6, 0.7, 0.625. Ascending: **3/5, 5/8, 2/3, 7/10**.

Q10 **Answer: (c) 1/6**
Mixed recurring: (16 − 1)/90 = 15/90 = **1/6**.

Q11 **Answer: (a) 8/5**
Bottom up: 1 + 1/2 = 3/2; 1 + 2/3 = 5/3; 1 + 3/5 = **8/5**.

Q12 **Answer: (d) 18,000**
Savings = 2/3 × 3/4 = 1/2 of income = 9,000 → income **₹18,000**.

Q13 **Answer: (c) 8/15**
(2/5) ÷ (3/4) = **8/15**.

Q14 **Answer: (c) 7/3**

2 + 3/9 = 2 + 1/3 = **7/3**.

Q15 **Answer: (b) 70**

Number = 48 ÷ 3/5 = 80; 7/8 × 80 = **70**.

Q16 **Answer: (c) 4/5**

Each differs from 1 by 1/denominator; the largest gap is 1/5, so **4/5** is smallest.

Q17 **Answer: (a) 233/990**

Mixed recurring: (235 - 2)/990 = **233/990**.

Q18 **Answer: (b) 120 litres**

36 litres = 7/10 - 2/5 = 3/10 of the tank, so the capacity is **120 litres**.

Q19 **Answer: (d) 120**

Boys = 18 × 4 = 72; students = 72 ÷ 3/5 = **120**.

Q20 **Answer: (a) 1**

5/9 + 4/9 = **1**.

Q21 **Answer: (a) 1/20**

Each term is (k - 1)/k; the product telescopes to **1/20**.

Q22 **Answer: (a) 1,500**

C gets 2/3 × 3/5 = 2/5 of the total = 600 → total **₹1,500**.

Q23 **Answer: (b) 0.333… (recurring)**

0.333… = 1/3 exceeds every terminating decimal 0.3, 0.33, 0.333.

Q24 **Answer: (d) 15**

1/b = 1/6 - 1/10 = 1/15, so b = **15**.

Q25 **Answer: (c) 10/13**

Let the denominator be x: (x - 1)/(x + 2) = 4/5 → 5x - 5 = 4x + 8 → x = 13. The fraction is **10/13**.

Compare fractions by cross-multiplying, cancel before multiplying, divide by multiplying by the reciprocal. A fraction terminates only if its denominator has just 2s and 5s. Pure recurring = block over 9s; mixed recurring = (all digits - non-repeating part) over 9s then 0s. In word problems always convert "of the remainder" into a fraction of the whole.

MODULE 07 · BEGINNER

Averages

Minimum duration **1 day** | Level **Beginner**

Averages are simple but frequently tested, and they are used inside many other topics. This module covers the definition, averages of consecutive numbers, weighted averages, adding, removing and replacing members, wrongly recorded values, average ages, cricket batting averages, average speed, combining groups, and the deviation method, followed by 25 solved questions with answers computed by a program.

IN THIS MODULE

7.1 Concepts
7.2 Problem types
7.3 Shortcut: deviation method
7.4 Practice questions
7.5 Solutions

7.1 Concepts

7.1.1 Average = sum ÷ number of observations

Average = (sum of observations) ÷ (number of observations); Sum = Average × Number

The second form is the one you use most: almost every average problem is solved by working with **totals**, then dividing at the end.

7.1.2 Average of consecutive numbers

For any arithmetic progression (consecutive numbers, consecutive evens, multiples of 7 …) the average is the average of the first and last terms, which is also the middle term. Average of 1 to n = (n + 1)/2; of the first n odd numbers = n; of the first n even numbers = n + 1.

7.1.3 Weighted average

If groups of sizes n_1, n_2 have averages a_1, a_2, the combined average is $(n_1a_1 + n_2a_2)/(n_1 + n_2)$. It always lies between a_1 and a_2, closer to the larger group.

WORKED EXAMPLE

30 boys average 60 marks and 20 girls average 70. Class average?

(30 × 60 + 20 × 70)/50 = 3200/50 = **64**.

7.2 Problem types

7.2.1 Adding or removing a member

New total - old total gives the value of the person added (or removed). Average of 10 is 25; adding one person makes it 26: new person = 11 × 26 - 10 × 25 = 286 - 250 = 36.

> **SHORTCUT TRICK**
> Shortcut: new person = new average + (old count × change in average) = 26 + 10 × 1 = 36.

7.2.2 Replacing a member

If replacing one member changes the average of n members by d, the new member differs from the old one by n × d. A man of 60 kg is replaced in a group of 8 and the average rises by 1.5 kg: the new man weighs 60 + 8 × 1.5 = 72 kg.

7.2.3 Wrongly recorded values

Correct the total, not the average: correct total = recorded total - wrong value + right value.

7.2.4 Average age problems

Every year each person's age rises by 1, so the average age of a fixed group rises by 1 per year. If the average age of 5 members was 30 three years ago, their total now is 5 × 33 = 165.

7.2.5 Cricket batting averages

Batting average = total runs ÷ innings. A batsman averaging 40 in 10 innings who wants an average of 42 after 11 innings must score 11 × 42 - 10 × 40 = 62 in the 11th innings.

7.2.6 Average speed

Average speed = total distance ÷ total time; for equal distances at speeds x and y it is $2xy/(x + y)$

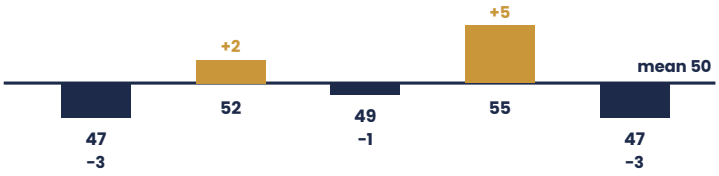
Going at 40 km/h and returning at 60 km/h gives 2 × 40 × 60/100 = 48 km/h, not 50. Module 18 covers this in detail.

7.2.7 Combining groups

Use the weighted average of 7.1.3. If the combined average and both group averages are known, the ratio of group sizes is $(a_2 - a)/(a - a_1)$, which is the alligation rule of Module 11.

7.3 Shortcut: deviation method

Pick an assumed mean close to the numbers, add the deviations, and correct: average = assumed mean + (sum of deviations)/n. For 47, 52, 49, 55, 47 with assumed mean 50: deviations -3, +2, -1, +5, -3 → sum 0 → average **50**.



DEVIATIONS ABOVE AND BELOW THE MEAN CANCEL EXACTLY

7.4 Practice questions

7.4.1 Basic level

Q1 What is the average of 12, 18, 24, 30 and 36?

(a) 22 **(b)** 24 **(c)** 26 **(d)** 30

Q2 What is the average of the first 50 natural numbers?

(a) 25 **(b)** 25.5 **(c)** 26 **(d)** 50

Q3 The average of 10 numbers is 25. What is their sum?

(a) 25 **(b)** 2,500 **(c)** 35 **(d)** 250

Q4 30 boys average 60 marks and 20 girls average 70 marks. What is the class average?

(a) 64 **(b)** 65 **(c)** 63 **(d)** 66

Q5 The average of 10 people is 25. When one more person joins, the average becomes 26. What is the age of the new person?

(a) 26 **(b)** 36 **(c)** 30 **(d)** 35

Q6 What is the average of the first 20 odd numbers?

(a) 21 **(b)** 19 **(c)** 20 **(d)** 40

Q7 A car goes 40 km/h one way and 60 km/h back over the same distance. What is the average speed?

(a) 48 km/h **(b)** 50 km/h **(c)** 45 km/h **(d)** 52 km/h

Q8 Find the average of 47, 52, 49, 55 and 47.

(a) 49 **(b)** 51 **(c)** 52 **(d)** 50

7.4.2 Intermediate level

Q9 In a group of 8, a man weighing 60 kg is replaced by a new man and the average rises by 1.5 kg. What does the new man weigh?

(a) 72 kg **(b)** 66 kg **(c)** 69 kg **(d)** 75 kg

Q10 The average of 25 observations was 40. Later one value was found to be 84 instead of 48. What is the correct average?

(a) 38.56 **(b)** 41.44 **(c)** 40 **(d)** 39

Q11 A batsman has an average of 40 after 10 innings. How many runs must he score in the 11th innings to raise his average to 42?

(a) 62 **(b)** 42 **(c)** 52 **(d)** 84

Q12 The average age of a husband and wife was 25 years when they married 5 years ago. Today the average age of the husband, wife and their child is 21 years. How old is the child?

(a) 2 years **(b)** 4 years **(c)** 5 years **(d)** 3 years

Q13 The average of 5 consecutive even numbers is 36. What is the largest?

(a) 38 **(b)** 40 **(c)** 42 **(d)** 44

Q14 The average of 11 results is 50. The average of the first 6 is 49 and of the last 6 is 52. What is the 6th result?

(a) 50 **(b)** 52 **(c)** 56 **(d)** 48

Q15 The average marks of 40 students is 72. If the top 5 scores (average 92) are removed, what is the new average?

(a) 70 **(b)** 69.14 **(c)** 68 **(d)** 66

Q16 A student's average over 6 subjects is 75. To get an average of 76 over 7 subjects, how much must he score in the 7th?

(a) 82 **(b)** 76 **(c)** 80 **(d)** 85

7.4.3 Advanced level

Q17 The average of three numbers is 30. The first is twice the second, and the second is twice the third. What is the largest number?

(a) 45 **(b)** 51.43 **(c)** 60 **(d)** 48

Q18 A cricketer's average for 20 innings is 32. In his next 5 innings he scores 50, 64, 0, 76 and 40. What is his new average?

(a) 35 **(b)** 36 **(c)** 34.8 **(d)** 33

Q19 Section A has 40 students averaging 50 and section B has 60 students. The combined average is 56. What is section B's average?

(a) 58 **(b)** 62 **(c)** 60 **(d)** 64

Q20 A person travels three equal distances at 20, 30 and 60 km/h. What is the average speed?

(a) 36 km/h **(b)** 40 km/h **(c)** 32 km/h **(d)** 30 km/h

Q21 The average weight of A, B, C is 45 kg. The average of A and B is 40 kg and of B and C is 43 kg. What does B weigh?

(a) 31 kg **(b)** 29 kg **(c)** 33 kg **(d)** 35 kg

Q22 The average of 50 numbers is 38. If two numbers, 45 and 55, are removed, what is the average of the rest?

(a) 36.5 **(b)** 38 **(c)** 37 **(d)** 37.5

Q23 The average salary of 20 workers is ₹15,000. When the manager's salary is added, the average rises by ₹1,000. What is the manager's salary?

(a) 16,000 **(b)** 36,000 **(c)** 35,000 **(d)** 31,000

Q24 The average age of 6 boys is 12 years. If a teacher's age is included, the average rises by 3 years. What is the teacher's age?

(a) 30 years **(b)** 36 years **(c)** 33 years **(d)** 27 years

Q25 The average of 7 consecutive numbers is 20. What is the average of the first 4 of them?

(a) 18.5 **(b)** 18 **(c)** 19 **(d)** 17

7.5 Solutions

Q1 **Answer: (b) 24**

Equally spaced numbers: the middle term **24**.

Q2 **Answer: (b) 25.5**

(1 + 50)/2 = **25.5**.

Q3 **Answer: (d) 250**

Sum = average × count = **250**.

Q4 **Answer: (a) 64**

(1800 + 1400)/50 = **64**.

Q5 **Answer: (b) 36**

11 × 26 − 10 × 25 = **36**.

Q6 **Answer: (c) 20**

Average of first n odd numbers = n = **20**.

Q7 **Answer: (a) 48 km/h**

2 × 40 × 60/(40 + 60) = **48 km/h**.

Q8 **Answer: (d) 50**

Deviations from 50: −3, +2, −1, +5, −3 sum to 0: **50**.

Q9 **Answer: (a) 72 kg**

60 + 8 × 1.5 = **72 kg**.

Q10 **Answer: (a) 38.56**

Correct total = 1000 − 84 + 48 = 964; average = **38.56**.

Q11 **Answer: (a) 62**

11 × 42 − 10 × 40 = **62**.

Q12 **Answer: (d) 3 years**

Husband and wife now total 2 × 30 = 60. Family total = 3 × 21 = 63, so the child is **3** years old.

Q13 **Answer: (b) 40**

The middle number is 36, so the numbers are 32 to 40; largest **40**.

Q14 **Answer: (c) 56**

First 6 + last 6 counts the 6th twice: 294 + 312 − 550 = **56**.

Q15 **Answer: (b) 69.14**

(2880 − 460)/35 = 2420/35 = **69.14** (to two decimals).

Q16 **Answer: (a) 82**

7 × 76 − 6 × 75 = **82**.

Q17 **Answer: (b) 51.43**

Numbers 4x, 2x, x sum to 90 → x = 90/7; largest = 360/7 = **51.43** (to two decimals).

Q18 **Answer: (c) 34.8**

(640 + 230)/25 = 870/25 = **34.8**.

Q19 **Answer: (c) 60**

(5600 − 2000)/60 = **60**.

Q20 **Answer: (d) 30 km/h**

Average speed = 3/(1/20 + 1/30 + 1/60) = 3/(6/60) = **30 km/h**.

Q21 **Answer: (a) 31 kg**

(A + B) + (B + C) − (A + B + C) = 80 + 86 − 135 = **31 kg**.

Q22 **Answer: (d) 37.5**

(1900 − 100)/48 = **37.5**.

Q23 **Answer: (b) 36,000**

Manager = 16,000 + 20 × 1,000 = **₹36,000**.

Q24 **Answer: (c) 33 years**

7 × 15 − 6 × 12 = **33** years.

Q25 **Answer: (a) 18.5**

The numbers are 17 to 23; first four average (17 + 20)/2 = **18.5**.

Work with totals: sum = average × number. Average of an arithmetic progression is the middle term. Combined average is weighted by group size. Adding, removing or replacing a member changes the total by (count × change in average). Correct wrong values in the total. Average speed is total distance ÷ total time, 2xy/(x + y) for two equal distances.

MODULE 08 · BEGINNER

Percentages

Minimum duration **2 days** | Level **Beginner**

Percentages are the backbone of commercial mathematics: profit, interest, data interpretation and more. This module covers the meaning of a percentage, conversions, percentage change, successive changes, percentage points, "more than / less than" comparisons, price and consumption, population and depreciation, elections, exams, income and savings, salary changes, and the multiplying-factor method, followed by 25 solved questions computed exactly.

IN THIS MODULE

8.1 Concepts
8.2 Percentage change
8.3 Comparison problems
8.4 Applications
8.5 Multiplying-factor method
8.6 Practice questions
8.7 Solutions

8.1 Concepts

8.1.1 Percentage as a fraction of 100

"Per cent" means "out of 100": x% = x/100. So 35% = 35/100 = 0.35 = 7/20.

8.1.2 Fraction–percentage conversions

Use the table of Module 2.2: 1/8 = 12.5%, 1/6 = 16.67%, 1/7 = 14.29%, 1/9 = 11.11%, 1/11 = 9.09%, 1/12 = 8.33%. Converting the percentage into a fraction is usually the fastest way to calculate.

8.1.3 Percentage of a quantity

x% of N = x × N/100. Note that x% of y = y% of x: 16% of 25 = 25% of 16 = 4.

8.1.4 One quantity as a percentage of another

A as a percentage of B = (A/B) × 100

8.2 Percentage change

8.2.1 Increase and decrease

Percentage change = (new - old)/old × 100 (always divide by the ORIGINAL value)

8.2.2 Successive percentage changes

Two successive changes of a% and b% give a net change of a + b + ab/100 (use negative signs for decreases)

+20% then -20%: 20 - 20 - 400/100 = -4%, a 4% net decrease. +10% then +10%: 21%.



SUCCESSIVE CHANGES DO NOT CANCEL: +20% THEN -20% GIVES -4%

8.2.3 Percentage points vs percentage change

If an interest rate rises from 8% to 10%, it rises by **2 percentage points**, but by **25 per cent** (2/8 × 100). Data interpretation questions use both; read carefully.

8.3 Comparison problems

8.3.1 If A is x% more than B, B is how much less than A?

If A is x% more than B, then B is x/(100 + x) × 100 % less than A; if A is x% less than B, then B is x/(100 − x) × 100 % more than A

A earns 25% more than B, so B earns 25/125 × 100 = 20% less than A.

> **SHORTCUT TRICK**
> Use 100 as the base: if B = 100, A = 125. Then B is 25 less than 125, i.e. 25/125 = 20% less.

8.4 Applications

8.4.1 Price and consumption

If expenditure stays the same and price rises by r%, consumption must fall by r/(100 + r) × 100 %. Sugar prices rise 25%: consumption must fall by 20% to keep spending unchanged.

8.4.2 Population growth and depreciation

Population after n years = $P(1 + r/100)^n$; value after depreciation = $V(1 - r/100)^n$

8.4.3 Elections and votes

In a two-candidate election where the winner gets 60% of valid votes and wins by 2,400 votes, the margin is 60% - 40% = 20% of valid votes, so valid votes = 2400/0.2 = 12,000.

8.4.4 Exams and pass marks

A student scoring 30% fails by 15 marks; another scoring 40% gets 35 marks more than the pass mark. The 10% gap equals 15 + 35 = 50 marks, so the maximum is 500 and the pass mark is 150 + 15 = 165.

8.4.5 Income, expenditure and savings

Income = expenditure + savings. If income rises 20% and expenditure rises 10%, the change in savings depends on the split; set income = 100 and work with numbers.

8.4.6 Salary hikes and cuts

A 10% cut followed by a 10% raise leaves the salary 1% lower (succ(-10, 10) = -1). To restore a salary after a 20% cut, it needs a 25% raise.

8.5 Multiplying-factor method

Convert every change into a multiplying factor and multiply: +20% → × 1.2, -15% → × 0.85, three years of 10% growth → × 1.1^3 = 1.331. Then the final value is just the original times the product of the factors.

Change	Factor
+5%	1.05
+12.5%	1.125 (9/8)
+25%	1.25 (5/4)
-10%	0.9
-20%	0.8 (4/5)
-33⅓%	2/3

8.6 Practice questions

8.6.1 Basic level

Q1 What is 35% of 640?

(a) 214 **(b)** 220 **(c)** 240 **(d)** 224

Q2 45 is what percentage of 180?

(a) 25% **(b)** 20% **(c)** 30% **(d)** 40%

Q3 A price rises from ₹80 to ₹100. What is the percentage increase?

(a) 20% **(b)** 80% **(c)** 25% **(d)** 15%

Q4 A price rises by 20% and then falls by 20%. What is the net change?

(a) No change **(b)** 4% decrease **(c)** 4% increase **(d)** 2% decrease

Q5 16% of 25 = ?

(a) 4 **(b)** 6.25 **(c)** 2.5 **(d)** 16

Q6 A earns 25% more than B. By what percentage does B earn less than A?

(a) 25% **(b)** 15% **(c)** 20% **(d)** 30%

Q7 The population of a town is 20,000 and grows 10% a year. What will it be after 2 years?

(a) 24,000 **(b)** 22,000 **(c)** 24,200 **(d)** 24,400

Q8 An interest rate rises from 8% to 10%. By what percentage has it risen?

(a) 2% **(b)** 25% **(c)** 20% **(d)** 10%

8.6.2 Intermediate level

Q9 The price of sugar rises by 25%. By what percentage must a family reduce consumption to keep its expenditure the same?

(a) 25% **(b)** 20% **(c)** 15% **(d)** 33.33%

Q10 In an election between two candidates, the winner got 60% of the valid votes and won by 2,400 votes. How many valid votes were cast?

(a) 10,000 **(b)** 12,000 **(c)** 14,000 **(d)** 24,000

Q11 A student scoring 30% fails by 15 marks; another scoring 40% gets 35 marks more than the pass mark. What is the maximum mark?

(a) 400 **(b)** 450 **(c)** 500 **(d)** 600

Q12 In the previous question, what is the pass mark?

(a) 150 **(b)** 175 **(c)** 180 **(d)** 165

Q13 Successive discounts of 10% and 20% are equivalent to a single discount of

(a) 28% **(b)** 30% **(c)** 25% **(d)** 32%

Q14 A machine worth ₹50,000 depreciates by 10% a year. What is its value after 3 years?

(a) 35,000 **(b)** 36,000 **(c)** 37,500 **(d)** 36,450

Q15 If A's salary is 20% less than B's, by what percentage is B's salary more than A's?

(a) 25% **(b)** 20% **(c)** 15% **(d)** 30%

Q16 After a 20% salary cut, by what percentage must the salary be raised to restore it?

(a) 20% **(b)** 22% **(c)** 30% **(d)** 25%

8.6.3 Advanced level

Q17 A number is increased by 10%, then by 20%, then decreased by 25%. What is the net change?

(a) 5% increase **(b)** 1% increase **(c)** No change **(d)** 1% decrease

Q18 A man spends 75% of his income. His income rises by 20% and his expenditure by 10%. By what percentage do his savings rise?

(a) 50% **(b)** 10% **(c)** 30% **(d)** 40%

Q19 In an election, 20% of voters did not vote and 5% of the votes cast were invalid. The winner got 60% of the valid votes and won by 1,520 votes. How many voters were on the list?

(a) 8,000 **(b)** 12,000 **(c)** 10,000 **(d)** 9,500

Q20 The length of a rectangle is increased by 20% and its breadth decreased by 10%. What is the change in area?

(a) 8% increase **(b)** 10% increase **(c)** 2% increase **(d)** 8% decrease

Q21 The population of a village rose by 10% in the first year and fell by 10% in the second, ending at 9,900. What was it at the start?

(a) 9,900 **(b)** 10,000 **(c)** 9,000 **(d)** 10,100

Q22 If x is 25% of y and y is 40% of z, what percentage of z is x?

(a) 15% **(b)** 10% **(c)** 65% **(d)** 20%

Q23 The price of petrol rises by 30% and a car owner increases his spending on petrol by only 17%. By what percentage does his consumption fall?

(a) 13% **(b)** 15% **(c)** 10% **(d)** 17%

Q24 A's income is 40% more than B's, and C's income is 30% less than A's. How does C's income compare with B's?

(a) 10% less **(b)** 2% more **(c)** 2% less **(d)** Equal

Q25 Two numbers are 20% and 50% more than a third number. What percentage is the first of the second?

(a) 70% **(b)** 75% **(c)** 60% **(d)** 80%

8.7 Solutions

Q1 **Answer: (d) 224**

35% = 7/20; 640 × 7/20 = **224**.

Q2 **Answer: (a) 25%**

45/180 × 100 = **25**%.

Q3 **Answer: (c) 25%**

20/80 × 100 = **25**%.

Q4 **Answer: (b) 4% decrease**

20 - 20 - 400/100 = -4: a **4% decrease**.

Q5 **Answer: (a) 4**

16% of 25 = 25% of 16 = **4**.

Q6 **Answer: (c) 20%**

25/125 × 100 = **20**%.

Q7 **Answer: (c) 24,200**

20000 × 1.1² = **24,200**.

Q8 **Answer: (b) 25%**

(10 - 8)/8 × 100 = **25**% (a rise of 2 percentage points).

Q9 **Answer: (b) 20%**

25/125 × 100 = **20**%.

Q10 **Answer: (b) 12,000**

Margin = 20% of valid votes = 2,400 → valid votes **12,000**.

Q11 **Answer: (c) 500**

The 10% gap equals 15 + 35 = 50 marks, so the maximum is **500**.

Q12 **Answer: (d) 165**

30% of 500 = 150; plus 15 = **165**.

Q13 **Answer: (a) 28%**

-10 - 20 + 200/100 = -28: **28**%.

Q14 **Answer: (d) 36,450**

50000 × 0.9³ = **₹36,450**.

Q15 **Answer: (a) 25%**

20/80 × 100 = **25**%.

Q16 **Answer: (d) 25%**

20/80 × 100 = **25**%.

Q17 **Answer: (d) 1% decrease**

1.1 × 1.2 × 0.75 = 0.99: a **1% decrease**.

Q18 **Answer: (a) 50%**

Income 100 → 120; expenditure 75 → 82.5; savings 25 → 37.5: up **50**%.

Q19 **Answer: (c) 10,000**

Margin = 20% of valid votes; valid = 0.95 × 0.8 = 76% of the list. 0.2 × 0.76 × N = 1520 → N = **10,000**.

Q20 **Answer: (a) 8% increase**

20 - 10 - 200/100 = +8: **8% increase**.

Q21 **Answer: (b) 10,000**

1.1 × 0.9 = 0.99, so the start was 9900/0.99 = **10,000**.

Q22 **Answer: (b) 10%**

x = 0.25 × 0.4 z = 0.1 z: **10**%.

Q23 **Answer: (c) 10%**

Consumption factor = 1.17/1.3 = 0.9: a fall of **10**%.

Q24 **Answer: (c) 2% less**

B = 100, A = 140, C = 0.7 × 140 = 98: C earns **2% less** than B.

Q25 **Answer: (d) 80%**

120/150 × 100 = **80**%.

Always divide by the original value. Successive changes: a + b + ab/100, or multiply the factors. If A is x% more than B, B is x/(100 + x) × 100 % less than A. Constant expenditure: a price rise of r% needs a consumption cut of r/(100 + r) × 100 %. Growth and depreciation use $(1 \pm r/100)^n$.

MODULE 09 · BEGINNER

Ratio and proportion

Minimum duration **2 days** | Level **Beginner**

Ratios compare quantities and are used in partnership, mixtures, ages and many other problems. This module covers ratios in simplest form, comparing ratios, compound, duplicate and sub-duplicate ratios, combining ratios, direct and inverse proportion, mean, third and fourth proportionals, componendo and dividendo, variation, and the standard applications (dividing amounts, income and expenditure, coins, salaries), followed by 25 solved questions.

IN THIS MODULE

9.1 Ratio
9.2 Proportion
9.3 Variation
9.4 Applications
9.5 Practice questions
9.6 Solutions

9.1 Ratio

9.1.1 Meaning and simplest form

A ratio a : b compares two quantities of the **same kind and unit**. Divide both terms by their HCF to get the simplest form: 45 : 60 = 3 : 4. Convert units first: 50 cm : 2 m = 50 : 200 = 1 : 4.



₹450 in the ratio 2 : 3 → 5 equal parts of ₹90 → ₹180 and ₹270

A RATIO AS EQUAL PARTS

9.1.2 Comparing ratios

Compare a : b and c : d as fractions a/b and c/d (cross-multiply). 3 : 4 vs 5 : 7: 21 vs 20, so 3 : 4 is greater.

9.1.3 Compound, duplicate, triplicate and sub-duplicate ratios

Name	Of a : b	Example (4 : 9)
Duplicate	$a^2 : b^2$	16 : 81
Triplicate	$a^3 : b^3$	64 : 729
Sub-duplicate	$\sqrt{a} : \sqrt{b}$	2 : 3
Sub-triplicate	$\sqrt[3]{a} : \sqrt[3]{b}$	$\sqrt[3]{4} : \sqrt[3]{9}$
Compound of a : b and c : d	ac : bd	(4 : 9) and (3 : 2) → 12 : 18 = 2 : 3
Inverse	b : a	9 : 4

9.1.4 Combining ratios

To combine A : B = 2 : 3 and B : C = 4 : 5, make B the same: multiply the first by 4 and the second by 3: A : B = 8 : 12, B : C = 12 : 15, so A : B : C = **8 : 12 : 15**.

> **SHORTCUT TRICK**
> Write the ratios one under the other and multiply crosswise: A = 2 × 4, B = 3 × 4, C = 3 × 5.

9.2 Proportion

9.2.1 Direct and inverse proportion

Quantities are in **direct proportion** if their ratio is constant (cost and quantity), and in **inverse proportion** if their product is constant (workers and days for a fixed job).

9.2.2 Mean, third and fourth proportionals

Term	Definition	Formula
Fourth proportional to a, b, c	a : b = c : x	x = bc/a
Third proportional to a, b	a : b = b : x	x = b^2/a
Mean proportional between a, b	a : x = x : b	x = $\sqrt{(ab)}$

9.2.3 Componendo and dividendo

If a/b = c/d, then (a + b)/(a - b) = (c + d)/(c - d)

Useful for equations like (x + 3)/(x - 3) = 5/3: componendo-dividendo gives x/3 = 8/2, so x = 12.

9.3 Variation

9.3.1 Direct, inverse and joint variation

y ∝ x means y = kx; y ∝ 1/x means y = k/x; z ∝ xy (joint) means z = kxy. Find k from the given data, then use it.

> **WORKED EXAMPLE**
> **y varies inversely as x. y = 12 when x = 5. Find y when x = 15.**
> k = xy = 60, so y = 60/15 = **4**.

9.4 Applications

9.4.1 Dividing an amount in a given ratio

Amount × (share of parts)/(total parts). ₹450 in 2 : 3 → 180 and 270.

9.4.2 Income and expenditure

If incomes are in the ratio 5 : 4 and expenditures in 3 : 2 and each saves ₹1,600, write incomes 5x, 4x and expenditures 3y, 2y: 5x - 3y = 1600 and 4x - 2y = 1600 → x = 800, y = 800. Incomes ₹4,000 and ₹3,200.

9.4.3 Coins and notes

A bag has ₹1, 50 p and 25 p coins in the ratio 1 : 2 : 3, worth ₹110 in all. Take the coins in sets of 1 + 2 + 3: each set is worth 1 + 2 × 0.50 + 3 × 0.25 = ₹2.75, so there are 110/2.75 = 40 sets: 40, 80 and 120 coins.

9.4.4 Salary and increment ratios

Salaries in the ratio 2 : 3 each rise by ₹4,000 and the ratio becomes 3 : 4. With salaries 2x and 3x: (2x + 4000)/(3x + 4000) = 3/4 → 8x + 16000 = 9x + 12000 → x = 4000, so the salaries were ₹8,000 and ₹12,000.

9.5 Practice questions

9.5.1 Basic level

Q1 Express 45 : 60 in its simplest form.

(a) 9 : 12 **(b)** 3 : 4 **(c)** 4 : 3 **(d)** 5 : 6

Q2 Divide ₹450 in the ratio 2 : 3. What is the larger share?

(a) 180 **(b)** 225 **(c)** 300 **(d)** 270

Q3 If A : B = 2 : 3 and B : C = 4 : 5, find A : B : C.

(a) 2 : 3 : 5 **(b)** 8 : 12 : 5 **(c)** 2 : 12 : 15 **(d)** 8 : 12 : 15

Q4 What is the fourth proportional to 4, 6 and 10?

(a) 12 **(b)** 20 **(c)** 15 **(d)** 24

Q5 What is the mean proportional between 4 and 16?

(a) 10 **(b)** 12 **(c)** 8 **(d)** 6

Q6 What is the duplicate ratio of 3 : 5?

(a) 6 : 10 **(b)** 9 : 25 **(c)** 27 : 125 **(d)** √3 : √5

Q7 Which ratio is greater: 3 : 4 or 5 : 7?

(a) 5 : 7 **(b)** 3 : 4
(c) They are equal **(d)** Cannot be compared

Q8 What is the third proportional to 9 and 12?

(a) 15 **(b)** 18 **(c)** 16 **(d)** 108

9.5.2 Intermediate level

Q9 The incomes of A and B are in the ratio 5 : 4 and their expenditures in the ratio 3 : 2. If each saves ₹1,600, what is A's income?

(a) 3,200 **(b)** 4,800 **(c)** 4,000 **(d)** 2,400

Q10 If $(x + 3)/(x - 3) = 5/3$, what is x?

(a) 9 **(b)** 15 **(c)** 12 **(d)** 6

Q11 y varies inversely as x, and y = 12 when x = 5. Find y when x = 15.

(a) 36 **(b)** 6 **(c)** 3 **(d)** 4

Q12 A bag has ₹1, 50 p and 25 p coins in the ratio 1 : 2 : 3, worth ₹110 in all. How many 25 p coins are there?

(a) 40 **(b)** 80 **(c)** 60 **(d)** 120

Q13 Two numbers are in the ratio 3 : 5. If 9 is subtracted from each, the ratio becomes 12 : 23. What is the smaller number?

(a) 27 **(b)** 36 **(c)** 33 **(d)** 55

Q14 What number must be added to each term of 7 : 11 to make the ratio 3 : 4?

(a) 5 **(b)** 3 **(c)** 4 **(d)** 6

Q15 The compound ratio of 2 : 3, 6 : 11 and 11 : 2 is

(a) 2 : 1 **(b)** 1 : 2 **(c)** 3 : 2 **(d)** 11 : 6

Q16 A sum is divided among A, B, C in the ratio 1/2 : 1/3 : 1/4. If C gets ₹600, what is the total?

(a) 2,000 **(b)** 2,600 **(c)** 2,400 **(d)** 3,000

9.5.3 Advanced level

Q17 A : B = 3 : 4, B : C = 5 : 6 and C : D = 7 : 8. Find A : D.

(a) 3 : 8 **(b)** 21 : 32 **(c)** 35 : 64 **(d)** 15 : 32

Q18 If a : b = 2 : 3, find (3a + 2b) : (4a - b).

(a) 5 : 12 **(b)** 6 : 5 **(c)** 12 : 7 **(d)** 12 : 5

Q19 The ratio of milk to water in 60 litres of a mixture is 2 : 1. How much water must be added to make the ratio 1 : 2?

(a) 60 litres **(b)** 30 litres **(c)** 40 litres **(d)** 20 litres

Q20 Salaries of A and B are in the ratio 2 : 3. After each gets an increment of ₹4,000, the ratio becomes 3 : 4. What is B's present salary?

(a) 12,000 **(b)** 8,000 **(c)** 16,000 **(d)** 6,000

Q21 The ratio of boys to girls in a class is 4 : 5. If 8 more boys join, the ratio becomes 4 : 3. How many girls are there?

(a) 15 **(b)** 12 **(c)** 20 **(d)** 25

Q22 The cost of an article varies directly as its weight. If 5 kg costs ₹350, what do 12 kg cost?

(a) 800 **(b)** 840 **(c)** 720 **(d)** 900

Q23 The area of a circle varies as the square of its radius. If the radius is increased by 50%, the area increases by

(a) 50% **(b)** 100% **(c)** 125% **(d)** 150%

Q24 If $x : y = 3 : 4$, what is $(x^2 + y^2) : (x^2 - y^2)$?

(a) 25 : 7 **(b)** 7 : 25 **(c)** 25 : -7 **(d)** -7 : 25

Q25 Three numbers are in the ratio 2 : 3 : 5 and the sum of their squares is 1368. What is the largest number?

(a) 25 **(b)** 35 **(c)** 30 **(d)** 18

9.6 Solutions

Q1 **Answer: (b) 3 : 4**

Divide by HCF 15: **3 : 4**.

Q2 **Answer: (d) 270**

450 × 3/5 = **₹270**.

Q3 **Answer: (d) 8 : 12 : 15**

Make B equal (12): **8 : 12 : 15**.

Q4 **Answer: (c) 15**

$x = 6 \times 10/4 =$ **15**.

Q5 **Answer: (c) 8**

$\sqrt{}(4 \times 16) =$ **8**.

Q6 **Answer: (b) 9 : 25**

Square both terms: **9 : 25**.

Q7 **Answer: (b) 3 : 4**

Cross-multiply: $3 \times 7 = 21 > 5 \times 4 = 20$, so **3 : 4**.

Q8 **Answer: (c) 16**

$x = 12^2/9 =$ **16**.

Q9 **Answer: (c) 4,000**

$5x - 3y = 1600$ and $4x - 2y = 1600$ give $x = y = 800$, so A's income = **₹4,000**.

Q10 **Answer: (c) 12**

Componendo-dividendo: $x/3 = (5 + 3)/(5 - 3) = 4 \rightarrow x =$ **12**.

Q11 **Answer: (d) 4**

$xy = 60 \rightarrow y = 60/15 =$ **4**.

Q12 **Answer: (d) 120**

Value per set (1 + 2 + 3 coins) = 1 + 1 + 0.75 = ₹2.75. Sets = 110/2.75 = 40, so 25 p coins = 3 × 40 = **120**.

Q13 **Answer: (c) 33**

$(3x - 9)/(5x - 9) = 12/23 \rightarrow 69x - 207 = 60x - 108 \rightarrow x = 11$. Smaller number = **33**.

Q14 **Answer: (a) 5**

$(7 + x)/(11 + x) = 3/4 \rightarrow 28 + 4x = 33 + 3x \rightarrow x =$ **5**.

Q15 **Answer: (a) 2 : 1**

$(2 \times 6 \times 11) : (3 \times 11 \times 2) = 132 : 66 =$ **2 : 1**.

Q16 **Answer: (b) 2,600**

Multiply by 12: ratio 6 : 4 : 3. C's 3 parts = 600 → 1 part = 200 → total 13 × 200 = **₹2,600**.

Q17 **Answer: (c) 35 : 64**

A/D = (3/4)(5/6)(7/8) = 105/192 = **35 : 64**.

Q18 **Answer: (d) 12 : 5**

Put a = 2, b = 3: (6 + 6) : (8 - 3) = **12 : 5**.

Q19 **Answer: (a) 60 litres**

Milk 40, water 20. For 1 : 2, water must be 80: add **60 litres**.

Q20 **Answer: (a) 12,000**

Let the salaries be 2x and 3x: $(2x + 4000)/(3x + 4000) = 3/4 \rightarrow 8x + 16000 = 9x + 12000 \rightarrow x = 4000$. B's salary = **₹12,000**.

Q21 **Answer: (a) 15**

$(4x + 8)/5x = 4/3 \rightarrow 12x + 24 = 20x \rightarrow x = 3$. Girls = 5 × 3 = **15**.

Q22 **Answer: (b) 840**

Cost per kg 70; 12 × 70 = **₹840**.

Q23 **Answer: (c) 125%**

Factor $(1.5)^2 = 2.25$: an increase of **125**%.

Q24 **Answer: (c) 25 : -7**

x = 3, y = 4: (9 + 16) : (9 - 16) = **25 : -7**.

Q25 **Answer: (c) 30**

$(4 + 9 + 25)k^2 = 38k^2 = 1368 \rightarrow k^2 = 36 \rightarrow k = 6$. Largest = $5k$ = **30**.

Simplify ratios by the HCF and use the same units. Combine A : B and B : C by equalising B. Fourth proportional bc/a, third b^2/a, mean $\sqrt{(ab)}$. Componendo-dividendo: $(a + b)/(a - b) = (c + d)/(c - d)$. For word problems write the quantities as 3x, 5x …, form one equation and solve for x.

MODULE 10 · BEGINNER

Partnership

Minimum duration **1 day** | Level **Beginner**

Partnership problems apply ratios to how profits are shared in a business. This module covers simple and compound partnership, the rule that profit share is proportional to capital × time, working and sleeping partners, salaries for working partners, partners joining or leaving midway, capital changes during the year and finding an investment from a profit share, followed by 25 solved questions computed exactly.

IN THIS MODULE

10.1 Concepts
10.2 Partners
10.3 Problem types
10.4 Practice questions
10.5 Solutions

10.1 Concepts

10.1.1 Simple partnership

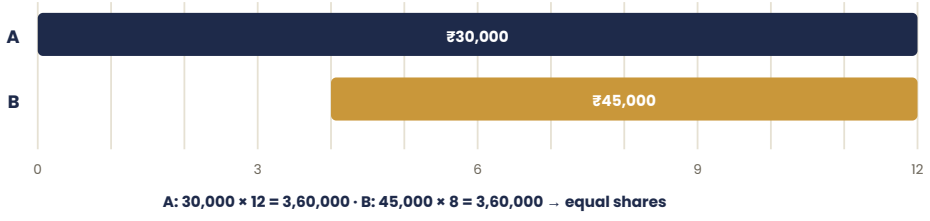
When all partners invest for the **same time**, profit is shared in the ratio of their capitals. A and B invest ₹40,000 and ₹60,000 for a year; a profit of ₹25,000 is shared 2 : 3 → ₹10,000 and ₹15,000.

10.1.2 Compound partnership

When the investment periods differ, each partner's effective investment is capital × time.

10.1.3 Profit share ∝ capital × time

Profit share of a partner ∝ capital × time (in months)



CAPITAL × TIME FOR EACH PARTNER

10.2 Partners

10.2.1 Working and sleeping partners

A **working** partner manages the business; a **sleeping** partner only invests money. The working partner often receives a salary or a fixed percentage of the profit **before** the rest is shared.

10.2.2 Salary for working partners

WORKED EXAMPLE

A (working) and B invest ₹50,000 and ₹30,000. A gets 10% of the profit for managing, and the rest is shared by capital. The profit is ₹40,000. How much does A get?

Management share 4,000; remaining 36,000 shared 5 : 3 → A 22,500. A gets 4,000 + 22,500 = **₹26,500**.

WATCH OUT

Take the salary or management share out of the profit first. Only the remainder is divided in the capital ratio.

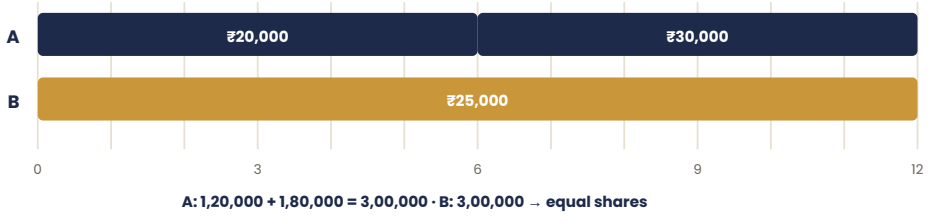
10.3 Problem types

10.3.1 Partners joining or leaving midway

Count each partner's months of investment. A starts with ₹30,000; B joins after 4 months with ₹45,000: their ratio is 30,000 × 12 : 45,000 × 8 = 1 : 1.

10.3.2 Capital changes during the year

Split the year into periods: A invests ₹20,000 for 6 months and then ₹30,000 for 6 months → 20,000 × 6 + 30,000 × 6 = 3,00,000 rupee-months.



A PARTNER ADDING CAPITAL MIDWAY

10.3.3 Finding an investment from a profit share

Work backwards: if the profit ratio and one capital are known, the other capital = (its share/known share) × known capital × (known time/its time).

10.4 Practice questions

10.4.1 Basic level

Q1 A and B invest ₹40,000 and ₹60,000 for a year. Out of a profit of ₹25,000, what is B's share?

(a) 10,000 (b) 12,500 (c) 15,000 (d) 20,000

Q2 A, B and C invest ₹20,000, ₹30,000 and ₹50,000. What is C's share of a profit of ₹12,000?

(a) 3,600 (b) 5,000 (c) 6,000 (d) 2,400

Q3 A invests ₹30,000 for 12 months and B invests ₹45,000 for 8 months. What is the ratio of their profits?

(a) 1 : 1 (b) 2 : 3 (c) 3 : 2 (d) 4 : 3

Q4 A and B share profits in the ratio 3 : 4. If A gets ₹9,000, what is the total profit?

(a) 12,000 (b) 18,000 (c) 24,000 (d) 21,000

Q5 A invests ₹10,000 for 9 months and B invests ₹15,000 for 6 months. What is the ratio of their profits?

(a) 2 : 3 (b) 3 : 2 (c) 9 : 6 (d) 1 : 1

Q6 A, B, C invest in the ratio 2 : 3 : 4 for 6, 4 and 3 months. What is the profit ratio?

(a) 1 : 1 : 1 (b) 2 : 3 : 4 (c) 6 : 4 : 3 (d) 1 : 2 : 2

Q7 A sleeping partner is one who

(a) invests money but does not manage the business
(b) manages but invests nothing
(c) joins in the middle of the year
(d) gets a fixed salary only

Q8 A starts a business with ₹25,000. After 3 months B joins with ₹25,000. What is the profit ratio A : B at the end of the year?

(a) 1 : 1 (b) 3 : 4 (c) 4 : 3 (d) 12 : 3

10.4.2 Intermediate level

Q9 A (working partner) and B invest ₹50,000 and ₹30,000. A gets 10% of the profit for managing and the rest is shared by capital. What does A get from a profit of ₹40,000?

(a) 25,000 (b) 22,500 (c) 28,000 (d) 26,500

Q10 A invests ₹20,000 for 6 months and then ₹30,000 for the next 6 months; B invests ₹25,000 for the whole year. What is the profit ratio?

(a) 1 : 1 (b) 4 : 5 (c) 6 : 5 (d) 5 : 4

Q11 A and B start a business with ₹60,000 and ₹40,000. After 4 months A withdraws ₹20,000. What is A's share of a profit of ₹26,000 at year end?

(a) 13,000 (b) 15,600 (c) 14,000 (d) 12,000

Q12 A invests ₹45,000. B joins later with ₹60,000 and the year-end profit is shared equally. After how many months did B join?

(a) 3 months (b) 4 months (c) 6 months (d) 2 months

Q13 A, B and C invest ₹1,20,000, ₹1,35,000 and ₹1,50,000. Out of a profit of ₹56,700, what is B's share?

(a) 18,900 (b) 16,800 (c) 21,000 (d) 18,000

Q14 A's capital is twice B's, and B's is three times C's. What is C's share of a profit of ₹50,000?

(a) 10,000 (b) 5,000 (c) 15,000 (d) 3,000

Q15 A and B invest in the ratio 3 : 5 and their profit ratio is 9 : 10. If B invested for 6 months, for how long did A invest?

(a) 9 months (b) 6 months (c) 8 months (d) 10 months

Q16 A, B, C rent a pasture. A grazes 10 oxen for 7 months, B 12 oxen for 5 months, C 15 oxen for 3 months. The rent is ₹175. What does C pay?

(a) 35 (b) 60 (c) 45 (d) 70

10.4.3 Advanced level

Q17 A starts with ₹40,000. After 3 months B joins with ₹50,000, and after another 3 months C joins with ₹60,000. What is C's share of a profit of ₹43,000?

(a) 16,000 (b) 12,000 (c) 15,000 (d) 10,000

Q18 A and B invest ₹30,000 and ₹40,000. After 6 months A adds ₹10,000 and B withdraws ₹10,000. What is A's share of a profit of ₹35,000?

(a) 17,500 (b) 15,000 (c) 20,000 (d) 14,000

Q19 A, B, C invest ₹5,000, ₹6,000 and ₹8,000. A is a working partner and gets 20% of the profit; the rest is shared by capital. If A gets ₹3,900 in all, what is the total profit?

(a) 9,500 (b) 9,000 (c) 12,000 (d) 7,800

Q20 Two partners invest ₹1,25,000 and ₹85,000. 60% of the profit is divided equally and the rest by capital. If the first partner gets ₹320 more than the second, what is the total profit?

(a) 3,000 (b) 4,000 (c) 4,200 (d) 4,800

Q21 A, B and C invest money in the ratio 5 : 6 : 8. At the end of the year profits are shared 5 : 4 : 2. What is the ratio of their investment periods?

(a) 12 : 8 : 3 (b) 5 : 4 : 2 (c) 3 : 2 : 1 (d) 6 : 4 : 3

Q22 A invests ₹64,000 and B ₹48,000. After 3 months A withdraws half his capital, and after another 3 months B withdraws a third of his. What is the profit ratio at year end?

(a) 4 : 3 (b) 1 : 1 (c) 5 : 4 (d) 3 : 2

Q23 A and B invest ₹10,000 and ₹20,000. A gets a salary of ₹500 a month for managing. If the year's profit is ₹18,000, what is B's share?

(a) 12,000 (b) 9,000 (c) 8,000 (d) 6,000

Q24 A, B, C share a profit in the ratio of their capitals 1/2 : 1/3 : 1/4. If the profit is ₹39,000, what is A's share?

(a) 18,000 **(b)** 13,000 **(c)** 12,000 **(d)** 9,000

Q25 A invests ₹3,000 for 4 months, ₹4,000 for the next 4 months and ₹5,000 for the last 4 months. B invests a constant amount for the year and they share profits equally. What did B invest?

(a) 3,500 **(b)** 4,000 **(c)** 4,500 **(d)** 3,000

10.5 Solutions

Q1 **Answer: (c) 15,000**

Ratio 2 : 3; B = 25000 × 3/5 = **₹15,000**.

Q2 **Answer: (c) 6,000**

Ratio 2 : 3 : 5; C = 12000 × 5/10 = **₹6,000**.

Q3 **Answer: (a) 1 : 1**

3,60,000 : 3,60,000 = **1 : 1**.

Q4 **Answer: (d) 21,000**

A's 3 parts = 9000 → 1 part = 3000; total 7 parts = **₹21,000**.

Q5 **Answer: (d) 1 : 1**

90,000 : 90,000 = **1 : 1**.

Q6 **Answer: (a) 1 : 1 : 1**

12 : 12 : 12 = **1 : 1 : 1**.

Q7 **Answer: (a) invests money but does not manage the business**

A sleeping partner only invests money.

Q8 **Answer: (c) 4 : 3**

25,000 × 12 : 25,000 × 9 = **4 : 3**.

Q9 **Answer: (d) 26,500**

4,000 + 36,000 × 5/8 = 4,000 + 22,500 = **₹26,500**.

Q10 **Answer: (a) 1 : 1**

3,00,000 : 3,00,000 = **1 : 1**.

Q11 **Answer: (c) 14,000**

A: 60,000 × 4 + 40,000 × 8 = 5,60,000; B: 4,80,000. Ratio 7 : 6; A = 26000 × 7/13 = **₹14,000**.

Q12 **Answer: (a) 3 months**

B invested for 45,000 × 12/60,000 = 9 months, so B joined after **3** months.

Q13 **Answer: (a) 18,900**

Ratio 8 : 9 : 10; B = 56700 × 9/27 = **₹18,900**.

Q14 **Answer: (b) 5,000**

A : B : C = 6 : 3 : 1; C = 50000/10 = **₹5,000**.

Q15 **Answer: (a) 9 months**

3 × tA : 5 × 6 = 9 : 10 → 30 tA = 270 → tA = **9** months.

Q16 **Answer: (c) 45**

Ox-months 70 : 60 : 45 = 14 : 12 : 9; C = 175 × 9/35 = **₹45**.

Q17 **Answer: (b) 12,000**

Ratio 4,80,000 : 4,50,000 : 3,60,000 = 16 : 15 : 12; C = 43000 × 12/43 = **₹12,000**.

Q18 **Answer: (a) 17,500**

A: 1,80,000 + 2,40,000 = 4,20,000; B: 2,40,000 + 1,80,000 = 4,20,000. Equal: A = **₹17,500**.

Q19 **Answer: (a) 9,500**

A gets 0.2P + 0.8P × 5/19 = P(19/95 + 20/95) = 39P/95. So P = 3900 × 95/39 = **₹9,500**.

Q20 **Answer: (c) 4,200**

The equal part gives no difference; the 40% part is split 125 : 85, a difference of 40/210 of it. 0.4P × 40/210 = 320 → P = **₹4,200**.

Q21 **Answer: (a) 12 : 8 : 3**

Time ∝ profit/capital: 5/5 : 4/6 : 2/8 = 1 : 2/3 : 1/4 = **12 : 8 : 3**.

Q22 **Answer: (b) 1 : 1**

A: 64,000 × 3 + 32,000 × 9 = 4,80,000; B: 48,000 × 6 + 32,000 × 6 = 4,80,000. Ratio **1 : 1**.

Q23 **Answer: (c) 8,000**

Salary 12 × 500 = 6,000; remaining 12,000 shared 1 : 2; B = **₹8,000**.

Q24 **Answer: (a) 18,000**

Multiply by 12: 6 : 4 : 3; A = 39000 × 6/13 = **₹18,000**.

Q25 **Answer: (b) 4,000**

A's rupee-months = 48,000; B × 12 = 48,000 → B = **₹4,000**.

Profit share ∝ capital × time. Split the year into periods when capital changes or partners join or leave. Take a working partner's salary or management share out first, then divide the rest in the capital-time ratio. Work backwards from a profit ratio to find an unknown capital or time.

MODULE 11 · BEGINNER

Mixtures and alligation

Minimum duration **1 day** | Level **Beginner**

Alligation is a powerful shortcut for mixing two or more items at different prices or concentrations. This module covers mean price, the rule of alligation (cross method), mixing two varieties, milk-and-water ratios, adding a component to change a ratio, repeated removal and replacement, mixing two mixtures, and alligation in profit, speed and interest problems, followed by 25 solved questions computed exactly.

IN THIS MODULE

11.1 Concepts
11.2 Problem types
11.3 Practice questions
11.4 Solutions

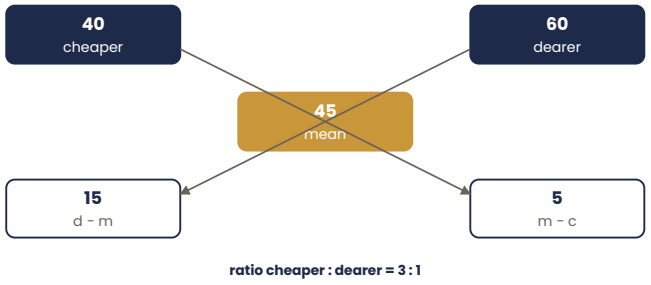
11.1 Concepts

11.1.1 Mixtures and mean price

When two items are mixed, the **mean price** (or mean concentration) of the mixture is the weighted average of the two: mean = $(q_1c + q_2d)/(q_1 + q_2)$. It always lies between the two prices.

11.1.2 The rule of alligation

Quantity of cheaper : quantity of dearer = (dearer - mean) : (mean - cheaper)



ALLIGATION CROSS: RICE AT ₹40 AND ₹60 MIXED TO GET ₹45 PER KG

SHORTCUT TRICK

The quantities go to the opposite corners: the amount of the cheaper item is the distance of the mean from the dearer price, and vice versa.

11.2 Problem types

11.2.1 Mixing two varieties

Rice at ₹40/kg and ₹60/kg mixed to sell at a mean price of ₹45/kg: ratio (60 - 45) : (45 - 40) = 3 : 1.

11.2.2 Milk and water ratios

Treat water as costing ₹0 (or as 0% milk). A 40-litre mixture with milk : water = 3 : 1 has 30 litres of milk and 10 of water.

11.2.3 Adding a component to change the ratio

The quantity that is **not** added stays fixed; use it as the anchor. To change 30 milk : 10 water into 2 : 1, milk stays 30, so water must become 15: add 5 litres of water.

11.2.4 Repeated removal and replacement

After n operations of removing x litres from a vessel of V litres and replacing with water, milk left = $V(1 - x/V)^n$

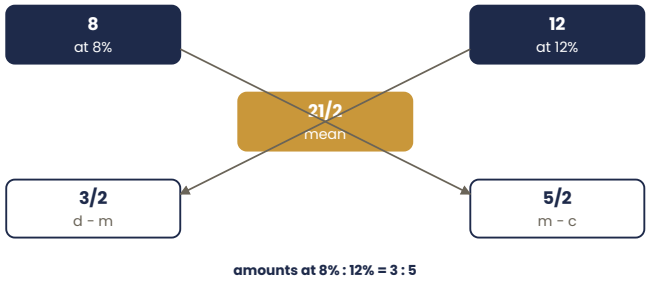
A 40-litre can of milk: 4 litres are removed and replaced by water, three times. Milk left = $40 \times (36/40)^3$ = 29.16 litres.

11.2.5 Mixing two mixtures

Convert each mixture to the fraction of one component, then apply alligation to those fractions. Mixtures with milk fractions 2/3 and 3/5 mixed to give 5/8 milk: ratio (5/8 - 3/5) : (2/3 - 5/8) = 1/40 : 1/24 = 3 : 5.

11.2.6 Alligation in other topics

Any weighted average can be solved with alligation: two parts of a sum lent at 8% and 12% earning 10.5% overall are in the ratio (12 - 10.5) : (10.5 - 8) = 3 : 5; a journey at 40 and 60 km/h averaging 50 km/h for time split equally; profit rates on two parts of a stock.



ALLIGATION WITH INTEREST RATES

11.3 Practice questions

11.3.1 Basic level

Q1 In what ratio must rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹45/kg?

(a) 1 : 3 **(b)** 2 : 1 **(c)** 1 : 2 **(d)** 3 : 1

Q2 Tea at ₹120/kg and ₹180/kg is mixed in the ratio 2 : 1. What is the price of the mixture per kg?

(a) 150 **(b)** 160 **(c)** 135 **(d)** 140

Q3 A 40-litre mixture has milk and water in the ratio 3 : 1. How much water is in it?

(a) 30 litres **(b)** 12 litres **(c)** 8 litres **(d)** 10 litres

Q4 In what ratio must water be mixed with milk costing ₹60/litre to get a mixture worth ₹50/litre?

(a) 1 : 5 **(b)** 5 : 1 **(c)** 1 : 6 **(d)** 1 : 4

Q5 How much water must be added to 30 litres of milk to make a mixture with 75% milk?

(a) 10 litres **(b)** 7.5 litres **(c)** 12 litres **(d)** 15 litres

Q6 Two varieties of sugar at ₹36/kg and ₹48/kg are mixed in equal quantities. What is the mean price?

(a) 40 **(b)** 42 **(c)** 44 **(d)** 45

Q7 A 60-litre mixture contains 20% alcohol. How many litres of alcohol does it contain?

(a) 12 litres **(b)** 10 litres **(c)** 15 litres **(d)** 20 litres

Q8 In what ratio should wheat at ₹22/kg and ₹28/kg be mixed to get wheat worth ₹24/kg?

(a) 1 : 2 **(b)** 2 : 1 **(c)** 3 : 1 **(d)** 3 : 2

11.3.2 Intermediate level

Q9 A 40-litre mixture of milk and water is 3 : 1. How much water must be added to make it 2 : 1?

(a) 10 litres **(b)** 4 litres **(c)** 5 litres **(d)** 8 litres

Q10 From a 40-litre can of milk, 4 litres are removed and replaced with water. This is done three times. How much milk is left?

(a) 28 litres **(b)** 30 litres **(c)** 32.4 litres **(d)** 29.16 litres

Q11 Two mixtures have milk fractions 2/3 and 3/5. In what ratio must they be mixed to get a mixture that is 5/8 milk?

(a) 5 : 3 **(b)** 2 : 3 **(c)** 1 : 1 **(d)** 3 : 5

Q12 A sum of ₹8,000 is lent partly at 8% and partly at 12% simple interest. The total interest for a year is ₹840. How much was lent at 8%?

(a) 3,000 **(b)** 5,000 **(c)** 4,000 **(d)** 2,500

Q13 A shopkeeper mixes 20 kg of rice at ₹30/kg with 30 kg at ₹40/kg. At what price per kg must he sell the mixture to gain 20%?

(a) 42 **(b)** 45 **(c)** 48 **(d)** 43.2

Q14 How many kilograms of tea at ₹180/kg must be mixed with 15 kg at ₹240/kg to get a mixture worth ₹200/kg?

(a) 30 kg **(b)** 20 kg **(c)** 15 kg **(d)** 25 kg

Q15 A vessel holds 60 litres of milk. 6 litres are removed and replaced with water, twice. What is the ratio of milk to water now?

(a) 81 : 19 **(b)** 81 : 100 **(c)** 9 : 1 **(d)** 4 : 1

Q16 A man travels 120 km, partly at 40 km/h and partly at 60 km/h, in 2.5 hours. How far did he travel at 40 km/h?

(a) 40 km **(b)** 80 km **(c)** 50 km **(d)** 60 km

11.3.3 Advanced level

Q17 Vessel A has milk : water = 5 : 3 and vessel B has 1 : 3. In what ratio must they be mixed to get a 1 : 1 mixture?

(a) 1 : 2 **(b)** 3 : 1 **(c)** 1 : 1 **(d)** 2 : 1

Q18 A 75-litre mixture contains milk and water in the ratio 2 : 1. How much milk must be added to make the ratio 4 : 1?

(a) 25 litres **(b)** 50 litres **(c)** 75 litres **(d)** 40 litres

Q19 A container of wine has some quantity removed and replaced with water. After this is done twice, the wine left is 49/64 of the original. What fraction was removed each time?

(a) 1/7 **(b)** 1/8 **(c)** 1/4 **(d)** 15/64

Q20 A merchant sells part of 50 kg of sugar at 8% profit and the rest at 18% profit, gaining 14% overall. How much did he sell at 18% profit?

(a) 30 kg **(b)** 20 kg **(c)** 25 kg **(d)** 35 kg

Q21 Three types of rice at ₹30, ₹40 and ₹50/kg are mixed in the ratio 2 : 3 : 5. What is the price of the mixture?

(a) 40 **(b)** 42 **(c)** 43 **(d)** 45

Q22 A jar holds 80 litres of a 40% acid solution. How much pure acid must be added to make it a 50% solution?

(a) 8 litres **(b)** 16 litres **(c)** 20 litres **(d)** 10 litres

Q23 A milkman mixes water with milk costing ₹40/litre and sells the mixture at ₹40/litre, gaining 25%. What is the ratio of water to milk?

(a) 1 : 5 **(b)** 4 : 1 **(c)** 1 : 3 **(d)** 1 : 4

Q24 Two alloys contain gold and copper in the ratios 7 : 2 and 7 : 11. Equal weights are melted together. What is the ratio of gold to copper in the new alloy?

(a) 7 : 5 **(b)** 7 : 13 **(c)** 14 : 13 **(d)** 1 : 1

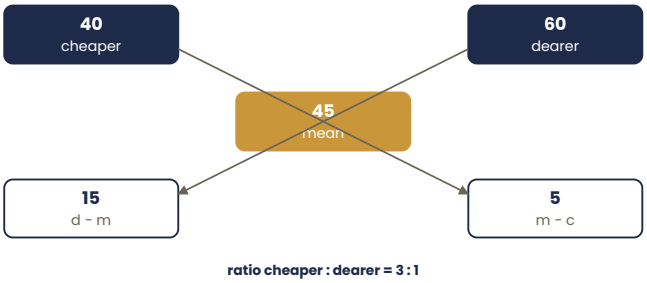
Q25 A 100-litre tank has milk and water in the ratio 3 : 2. 20 litres of the mixture are removed and replaced with water. What is the new ratio of milk to water?

(a) 12 : 13 **(b)** 3 : 2 **(c)** 1 : 1 **(d)** 2 : 3

11.4 Solutions

Q1 **Answer: (d) 3 : 1**

(60 − 45) : (45 − 40) = 15 : 5 = **3 : 1**.



ratio cheaper : dearer = 3 : 1

Q2 **Answer: (d) 140**

(240 + 180)/3 = **₹140**.

Q3 **Answer: (d) 10 litres**

Water = 40 × 1/4 = **10 litres**.

Q4 **Answer: (a) 1 : 5**

Water costs 0: water : milk = (60 − 50) : (50 − 0) = 10 : 50 = **1 : 5**.

Q5 **Answer: (a) 10 litres**

Milk 30 = 75% of the new total, so the total is 40: add **10 litres**.

Q6 **Answer: (b) 42**

Equal quantities: (36 + 48)/2 = **₹42**.

Q7 **Answer: (a) 12 litres**

20% of 60 = **12 litres**.

Q8 **Answer: (b) 2 : 1**

(28 − 24) : (24 − 22) = 4 : 2 = **2 : 1**.

Q9 **Answer: (c) 5 litres**

Milk stays 30; for 2 : 1 water must be 15: add **5 litres**.

Q10 **Answer: (d) 29.16 litres**

40 × $(9/10)^3$ = **29.16 litres**.

Q11 **Answer: (d) 3 : 5**

(5/8 - 3/5) : (2/3 - 5/8) = 1/40 : 1/24 = **3 : 5**.

Q12 **Answer: (a) 3,000**

Average rate 840/8000 = 10.5%. Alligation: (12 - 10.5) : (10.5 - 8) = 3 : 5; at 8% = 8000 × 3/8 = **₹3,000**.

Q13 **Answer: (d) 43.2**

Cost price = (600 + 1200)/50 = 36; with 20% gain: 36 × 1.2 = **₹43.20**.

Q14 **Answer: (a) 30 kg**

Ratio cheap : dear = 40 : 20 = 2 : 1, so cheap tea = 2 × 15 = **30 kg**.

Q15 **Answer: (a) 81 : 19**

Milk = 60 × 0.81 = 48.6, water = 11.4: ratio 48.6 : 11.4 = **81 : 19**.

Q16 **Answer: (d) 60 km**

40t + 60(2.5 - t) = 120 → 150 - 20t = 120 → t = 3/2 h. Distance at 40 km/h = **60 km**.

Q17 **Answer: (d) 2 : 1**

Milk fractions 5/8 and 1/4, target 1/2: (1/2 - 1/4) : (5/8 - 1/2) = 1/4 : 1/8 = **2 : 1**.

Q18 **Answer: (b) 50 litres**

Water stays 25; for 4 : 1 milk must be 100, so add **50 litres**.

Q19 **Answer: (b) 1/8**

$(1 - f)^2 = 49/64 \rightarrow 1 - f = 7/8 \rightarrow f =$ **1/8**.

Q20 **Answer: (a) 30 kg**

Alligation: (18 - 14) : (14 - 8) = 4 : 6 = 2 : 3 (8% : 18%). At 18% = 50 × 3/5 = **30 kg**.

Q21 **Answer: (c) 43**

(60 + 120 + 250)/10 = **₹43**.

Q22 **Answer: (b) 16 litres**

Alligation of 40% with 100%: added : original = (50 - 40) : (100 - 50) = 1 : 5, so add 80/5 = **16 litres**.

Q23 **Answer: (d) 1 : 4**

Cost price of mixture = 40/1.25 = 32. Alligation: water : milk = (40 - 32) : (32 - 0) = 8 : 32 = **1 : 4**.

Q24 **Answer: (a) 7 : 5**

Take 18 kg of each: the first has 14 kg gold and 4 kg copper, the second 7 kg gold and 11 kg copper. Total gold 21, copper 15: ratio **7 : 5**.

Q25 **Answer: (a) 12 : 13**

Removing 20 litres removes 12 milk and 8 water: milk 48, water 32 + 20 = 52. Ratio 48 : 52 = **12 : 13**.

Cheaper : dearer = (dearer - mean) : (mean - cheaper). Treat water as price 0 or 0% concentration. When adding one component, the other stays fixed. Repeated replacement leaves $V(1 - x/V)^n$. For two mixtures, apply alligation to the fractions of one component. Any weighted average (interest, profit, speed) can be solved the same way.

MODULE 12 · BEGINNER

Problems on ages

Minimum duration **1 day** | Level **Beginner**

Age problems test forming and solving simple equations from word statements. This module covers present, past and future ages, the constant age difference, ratio of ages, sums and differences, father-son and family problems, average age of a family, and both the algebraic and option-checking methods, followed by 25 solved questions whose answers were checked by a program.

IN THIS MODULE

12.1 Concepts
12.2 Problem types
12.3 Methods
12.4 Practice questions
12.5 Solutions

12.1 Concepts

12.1.1 Present, past and future ages

If a person's present age is x, then n years ago it was x - n and after n years it will be x + n. Write every age in terms of the present age of one person.

12.1.2 The age difference never changes



AGES SHIFT TOGETHER; THE DIFFERENCE IS CONSTANT

The difference between two people's ages is the same at every point in time; the ratio of their ages changes

12.2 Problem types

12.2.1 Ratio of ages

WORKED EXAMPLE

The ratio of A's and B's ages is 4 : 3. After 6 years it will be 5 : 4. Find A's present age.

Let ages be 4x and 3x: (4x + 6)/(3x + 6) = 5/4 → 16x + 24 = 15x + 30 → x = 6. A is **24**.

SHORTCUT TRICK

Because the difference is fixed, make the difference equal in both ratios. 4 : 3 and 5 : 4 both have difference 1 part, so 1 part now = 1 part later, and the 6 years change 4 parts into 5 parts: 1 part = 6, A = 24.

12.2.2 Sum and difference of ages

Sum S and difference D give the ages (S + D)/2 and (S - D)/2. The sum of the ages of n people rises by n every year.

12.2.3 Father-son and family problems

WORKED EXAMPLE

A father is 3 times as old as his son. In 12 years he will be twice as old. Find their ages.

Son x, father 3x: 3x + 12 = 2(x + 12) → x = 12. Son 12, father 36.



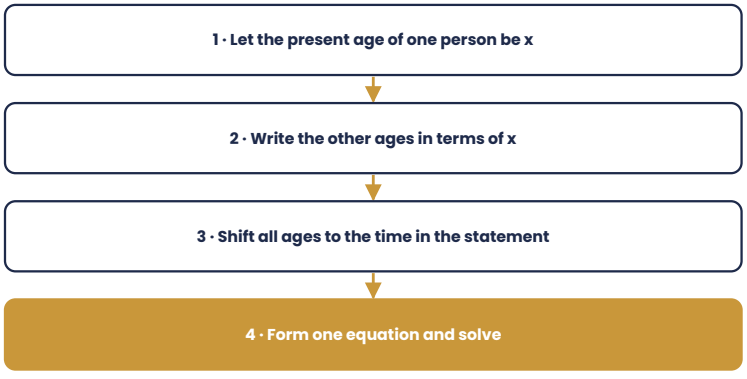
THE WORKED EXAMPLE ON A TIMELINE

12.2.4 Average age of a family

Use totals: total age = average × number of members. If the average age of a family of 4 is 25 now, their total is 100; in 5 years it will be 120.

12.3 Methods

12.3.1 Algebraic method



THE ALGEBRAIC METHOD

12.3.2 Option-checking method

With four options, it is often faster to test each option against the statements. Start with the middle values; one check usually eliminates two options at once.

12.4 Practice questions

12.4.1 Basic level

Q1 A father is 3 times as old as his son. In 12 years he will be twice as old. What is the son's present age?

(a) 12 years (b) 10 years (c) 14 years (d) 16 years

Q2 The sum of the ages of A and B is 50 and the difference is 10. What is the elder one's age?

(a) 25 years (b) 20 years (c) 35 years (d) 30 years

Q3 The ratio of A's and B's ages is 4 : 3. After 6 years it will be 5 : 4. What is A's present age?

(a) 18 years (b) 20 years (c) 30 years (d) 24 years

Q4 Five years ago Ravi was 20. How old will he be in 7 years?

(a) 32 years (b) 27 years (c) 30 years (d) 25 years

Q5 The average age of a family of 4 is 25 years. What will be the total of their ages after 5 years?

(a) 105 (b) 125 (c) 120 (d) 100

Q6 A is 6 years older than B. In 4 years, what will the difference in their ages be?

(a) 6 years (b) 10 years (c) 2 years (d) 4 years

Q7 The present ages of A and B are in the ratio 5 : 7 and their sum is 48. What is B's age?

(a) 28 years (b) 20 years (c) 24 years (d) 32 years

Q8 A mother is twice as old as her daughter. If the mother is 46, how old was the daughter 5 years ago?

(a) 23 years (b) 18 years (c) 20 years (d) 16 years

12.4.2 Intermediate level

Q9 Ten years ago, A was half as old as B. If their present ages are in the ratio 3 : 4, what is the total of their present ages?

(a) 28 years (b) 42 years (c) 35 years (d) 21 years

Q10 The sum of the ages of a father and son is 45. Five years ago the product of their ages was 34. What is the father's age?

(a) 36 years (b) 40 years (c) 30 years (d) 39 years

Q11 A man is 24 years older than his son. In two years he will be twice as old as his son. What is the son's age?

(a) 24 years (b) 20 years (c) 18 years (d) 22 years

Q12 The ages of A and B are in the ratio 6 : 5 and their sum is 44. What will the ratio be after 8 years?

(a) 7 : 6 (b) 6 : 5 (c) 8 : 7 (d) 9 : 8

Q13 Q is as much younger than R as he is older than T. If R + T = 50, how old is Q?

(a) 20 years (b) 30 years (c) 50 years (d) 25 years

Q14 The average age of 3 children is 12. If the father's age is included, the average becomes 20. How old is the father?

(a) 40 years **(b)** 44 years **(c)** 48 years **(d)** 36 years

Q15 Six years ago the ratio of the ages of Kunal and Sagar was 6 : 5. Four years from now it will be 11 : 10. What is Sagar's present age?

(a) 16 years **(b)** 24 years **(c)** 20 years **(d)** 26 years

Q16 A is 2 years older than B, who is twice as old as C. The sum of their ages is 27. How old is B?

(a) 10 years **(b)** 8 years **(c)** 12 years **(d)** 7 years

12.4.3 Advanced level

Q17 The present age of a father is 3 years more than three times his son's age. Three years hence the father's age will be 10 years more than twice the son's. What is the father's present age?

(a) 30 years **(b)** 33 years **(c)** 36 years **(d)** 39 years

Q18 The ratio of the present ages of a mother and daughter is 7 : 2. Four years from now it will be 8 : 3. What is the mother's present age?

(a) 35 years **(b)** 21 years **(c)** 28 years **(d)** 42 years

Q19 A man's age is 125% of his wife's. When they married 10 years ago, his age was 150% of hers. How old is the wife now?

(a) 30 years **(b)** 20 years **(c)** 25 years **(d)** 35 years

Q20 The total age of A and B is 12 years more than the total age of B and C. C is how many years younger than A?

(a) 6 years **(b)** 12 years **(c)** 24 years **(d)** 10 years

Q21 The average age of a husband, wife and child 3 years ago was 27, and that of the wife and child 5 years ago was 20. What is the husband's present age?

(a) 40 years **(b)** 35 years **(c)** 37 years **(d)** 45 years

Q22 A father said to his son, "I was as old as you are now at the time of your birth." If the father is 38 now, how old was the son 5 years ago?

(a) 14 years **(b)** 19 years **(c)** 9 years **(d)** 12 years

Q23 The sum of the ages of 5 children born at intervals of 3 years is 50. What is the age of the youngest?

(a) 5 years **(b)** 4 years **(c)** 8 years **(d)** 10 years

Q24 Eight years ago, the father was 4 times as old as his son. Eight years from now he will be twice as old. What is the present age of the son?

(a) 12 years **(b)** 20 years **(c)** 24 years **(d)** 16 years

Q25 The ages of two persons differ by 16 years. Six years ago the elder was 3 times as old as the younger. What is the elder's present age?

(a) 24 years **(b)** 30 years **(c)** 32 years **(d)** 36 years

12.5 Solutions

Q1 **Answer: (a) 12 years**

$3x + 12 = 2(x + 12) \rightarrow x =$ **12**.

Q2 **Answer: (d) 30 years**

$(50 + 10)/2 =$ **30**.

Q3 **Answer: (d) 24 years**

$(4x + 6)/(3x + 6) = 5/4 \rightarrow x = 6 \rightarrow A =$ **24**.

Q4 **Answer: (a) 32 years**

Now 25; in 7 years **32**.

Q5 **Answer: (c) 120**

Now 100; each gains 5: $100 + 20 =$ **120**.

Q6 **Answer: (a) 6 years**

The difference never changes: **6** years.

Q7 **Answer: (a) 28 years**

$48 \times 7/12 =$ **28**.

Q8 **Answer: (b) 18 years**

Daughter is 23 now; 5 years ago **18**.

Q9 **Answer: (c) 35 years**

$2(3x - 10) = 4x - 10 \rightarrow x = 5$. Ages 15 and 20; total **35**.

Q10 **Answer: (d) 39 years**

If the father is f: $(f - 5)(40 - f) = 34 \rightarrow f =$ **39** (son 6).

Q11 **Answer: (d) 22 years**

$x + 26 = 2x + 4 \rightarrow x =$ **22**.

Q12 **Answer: (c) 8 : 7**

Ages 24 and 20; after 8 years 32 : 28 = **8 : 7**.

Q13 **Answer: (d) 25 years**

$R - Q = Q - T \rightarrow 2Q = R + T = 50 \rightarrow Q =$ **25**.

Q14 **Answer: (b) 44 years**

4 × 20 − 3 × 12 = **44**.

Q15 **Answer: (a) 16 years**

Six years ago 6x and 5x; four years from now 6x + 10 and 5x + 10. (6x + 10)/(5x + 10) = 11/10 → 60x + 100 = 55x + 110 → x = 2. Sagar now = 5x + 6 = **16**.

Q16 **Answer: (a) 10 years**

2c + 2 + 2c + c = 27 → c = 5 → B = **10**.

Q17 **Answer: (b) 33 years**

f = 3s + 3 and 3s + 6 = 2s + 16 → s = 10 → f = **33**.

Q18 **Answer: (c) 28 years**

(7x + 4)/(2x + 4) = 8/3 → 21x + 12 = 16x + 32 → x = 4 → mother = **28**.

Q19 **Answer: (b) 20 years**

Husband = 1.25w. 1.25w − 10 = 1.5(w − 10) → 1.25w − 10 = 1.5w − 15 → w = **20**.

Q20 **Answer: (b) 12 years**

A + B − (B + C) = 12 → A − C = **12**.

Q21 **Answer: (a) 40 years**

Now: husband + wife + child = 3 × 30 = 90; wife + child = 2 × 25 = 50. Husband = **40**.

Q22 **Answer: (a) 14 years**

Father's age at the son's birth = son's present age = s, and the difference is s: 2s = 38 → s = 19. Five years ago **14**.

Q23 **Answer: (b) 4 years**

5y + 30 = 50 → y = **4**.

Q24 **Answer: (d) 16 years**

Father = 4(s − 8) + 8; in 8 years: 4(s − 8) + 16 = 2(s + 8) → 4s − 16 = 2s + 16 → s = **16**.

Q25 **Answer: (b) 30 years**

y + 10 = 3y − 18 → y = 14 → elder **30**.

Write every age in terms of one unknown present age, shift all ages to the time mentioned, and form one equation. Age differences never change, so equalise the difference parts in two ratios for a quick solution. Sums of ages rise by the number of people each year. Check options when an equation looks messy.

LEVEL 2

Intermediate level

Solve profit and loss, interest, time and work, pipes, speed and distance, boats, trains, races, algebra, progressions and logarithms with shortcuts

12 MODULES · 18 DAYS MINIMUM

Module 13 · Profit, loss and discount	2 days
Module 14 · Simple interest	1 day
Module 15 · Compound interest	2 days
Module 16 · Time and work	2 days
Module 17 · Pipes and cisterns	1 day
Module 18 · Time, speed and distance	2 days
Module 19 · Boats and streams	1 day
Module 20 · Problems on trains	1 day
Module 21 · Races and games	1 day
Module 22 · Algebra: equations and inequalities	2 days
Module 23 · Progressions	2 days
Module 24 · Logarithms	1 day

MODULE 13 · INTERMEDIATE

Profit, loss and discount

Minimum duration **2 days** | Level **Intermediate**

Commercial mathematics problems about buying and selling are among the most frequently asked. This module covers cost, selling and marked price, profit and loss percentages, the SP-CP formulas, the same-SP rule, discounts and successive discounts, marking up, "buy x get y free" offers, dishonest dealers and false weights, profit on part of the goods, articles bought and sold in different quantities, and GST-inclusive prices, followed by 25 solved questions computed exactly.

IN THIS MODULE

13.1 Basic terms
13.2 Formulas and shortcuts
13.3 Discount
13.4 Special problem types
13.5 Practice questions
13.6 Solutions

13.1 Basic terms

13.1.1 Cost price, selling price, marked price

Cost price (CP) is what the seller pays; **selling price (SP)** is what the buyer pays; **marked price (MP)** or list price is the price printed on the label. Discount is given on MP; profit and loss are measured on CP.



CP ₹400, MP ₹600 (marked up 50%), 15% discount → SP ₹510, profit ₹110

CP, SP AND MP ON ONE LINE

13.1.2 Profit and loss

Profit = SP − CP when SP > CP; loss = CP − SP when SP < CP.

13.1.3 Profit and loss percentage

Profit % = profit/CP × 100; Loss % = loss/CP × 100 (always on CP unless stated otherwise)

13.2 Formulas and shortcuts

13.2.1 SP from CP

SP = CP × (100 + profit%)/100, or CP × (100 − loss%)/100. CP ₹800 with 15% profit: SP = 800 × 1.15 = ₹920.

13.2.2 CP from SP

CP = SP × 100/(100 + profit%). SP ₹920 with 15% profit: CP = 920/1.15 = ₹800.

13.2.3 Same SP, one profit and one loss of x%

Two articles sold at the same SP, one at x% profit and one at x% loss → overall loss of $x^2/100$ %

Two phones sold at ₹9,900 each, one at 10% profit and one at 10% loss: overall loss = 1%. CPs are 9000 and 11000, total 20000, SP total 19800.

13.3 Discount

13.3.1 Discount on marked price

Discount = MP − SP; discount % = discount/MP × 100.

13.3.2 Successive discounts

Discounts of a% and b% equal a single discount of a + b − ab/100 %. 20% and 10% → 28%.

13.3.3 Marking up and offering a discount

If goods are marked up m% and sold at d% discount, profit % = m − d − md/100. Marked up 50%, 15% discount: 50 − 15 − 7.5 = 27.5%.

> **SHORTCUT TRICK**
> Use multiplying factors: CP × (1 + m/100) × (1 − d/100) = SP. Marked up 50% and 15% off: 1.5 × 0.85 = 1.275 → 27.5% profit.

13.3.4 "Buy x, get y free" offers

The customer pays for x items and receives x + y, so the effective discount is y/(x + y) × 100 %. "Buy 3 get 1 free" = 25% discount.

13.4 Special problem types

13.4.1 Dishonest dealer and false weights

Profit % of a dealer using false weights (selling at cost price) = (true weight - false weight)/false weight × 100

A grocer uses a 900 g weight for 1 kg and sells at cost price: profit = 100/900 × 100 = 11.11%. If he also adds a markup, combine the factors: 1000/900 × (1 + markup).

13.4.2 Profit on part of the goods

Find the total SP required for the overall target, subtract what has already been earned, and see what the rest must fetch.

13.4.3 Articles bought and sold in different quantities

Make the number of articles equal using the LCM. Buying 6 for ₹5 and selling 5 for ₹6: take 30 articles: CP ₹25, SP ₹36, profit 11/25 = 44%.

13.4.4 GST and tax-included prices

A tax-inclusive price = base price × (1 + GST%/100). An item costing ₹1,180 including 18% GST has a base price of 1180/1.18 = ₹1,000.

13.5 Practice questions

13.5.1 Basic level

Q1 A shirt bought for ₹400 is sold for ₹500. What is the profit percentage?

(a) 20% **(b)** 100% **(c)** 30% **(d)** 25%

Q2 An article costing ₹800 is sold at 15% profit. What is the selling price?

(a) 920 **(b)** 880 **(c)** 950 **(d)** 680

Q3 An article sold for ₹920 gives 15% profit. What is its cost price?

(a) 782 **(b)** 820 **(c)** 800 **(d)** 760

Q4 A watch marked at ₹2,000 is sold at a 15% discount. What is the selling price?

(a) 1,800 **(b)** 1,750 **(c)** 1,650 **(d)** 1,700

Q5 Successive discounts of 20% and 10% are equal to a single discount of

(a) 30% **(b)** 25% **(c)** 28% **(d)** 32%

Q6 "Buy 3, get 1 free" is equivalent to a discount of

(a) 33.33% **(b)** 20% **(c)** 30% **(d)** 25%

Q7 By selling an article for ₹450 a man loses 10%. What is the cost price?

(a) 500 **(b)** 495 **(c)** 405 **(d)** 550

Q8 An item costs ₹1,180 including 18% GST. What is its price before GST?

(a) 967.6 **(b)** 1,018 **(c)** 980 **(d)** 1,000

13.5.2 Intermediate level

Q9 Two phones are sold for ₹9,900 each, one at 10% profit and one at 10% loss. What is the overall result?

(a) 1% loss **(b)** No profit, no loss
(c) 1% profit **(d)** 2% loss

Q10 A trader marks goods 50% above cost and gives a 15% discount. What is his profit percentage?

(a) 35% **(b)** 30% **(c)** 27.5% **(d)** 25%

Q11 A grocer sells at cost price but uses a 900 g weight instead of 1 kg. What is his profit percentage?

(a) 10% **(b)** 9.09% **(c)** 11.11% **(d)** 12.5%

Q12 A man buys 6 pens for ₹5 and sells 5 pens for ₹6. What is his profit percentage?

(a) 20% **(b)** 36% **(c)** 50% **(d)** 44%

Q13 The cost price of 20 articles equals the selling price of 16 articles. What is the profit percentage?

(a) 20% **(b)** 16% **(c)** 30% **(d)** 25%

Q14 At what price must an article costing ₹600 be marked so that after a 20% discount there is a 20% profit?

(a) 840 **(b)** 960 **(c)** 900 **(d)** 720

Q15 By selling a fan for ₹1,140 a man loses 5%. At what price should he sell it to gain 5%?

(a) 1,260 **(b)** 1,200 **(c)** 1,197 **(d)** 1,300

Q16 A shopkeeper allows a 10% discount and still makes a 26% profit. If the cost price is ₹500, what is the marked price?

(a) 630 **(b)** 650 **(c)** 700 **(d)** 720

13.5.3 Advanced level

Q17 A dealer marks his goods 20% above cost, gives a 10% discount and uses a 900 g weight for 1 kg. What is his profit percentage?

(a) 20% **(b)** 8% **(c)** 18% **(d)** 12.5%

Q18 A man sold 1/3 of his goods at 15% loss. At what profit percentage must he sell the rest to make an overall profit of 10%?

(a) 20% **(b)** 22.5% **(c)** 25% **(d)** 15%

Q19 Successive discounts of 10%, 20% and 25% are equivalent to a single discount of

(a) 46% **(b)** 55% **(c)** 50% **(d)** 45%

Q20 If selling an article at ₹1,920 gives the same profit as the loss when it is sold at ₹1,280, what is the cost price?

(a) 1,500 **(b)** 1,650 **(c)** 1,700 **(d)** 1,600

Q21 A sold an article to B at 20% profit, and B sold it to C at 10% loss. If C paid ₹2,160, what did A pay for it?

(a) 2,000 **(b)** 1,800 **(c)** 2,160 **(d)** 1,944

Q22 A shopkeeper offers "buy 4, get 1 free" and still makes a 20% profit. If the marked price of each item is ₹150, what is the cost price of each item?

(a) 125 **(b)** 120 **(c)** 100 **(d)** 96

Q23 By selling 45 lemons for ₹40 a man loses 20%. How many lemons should he sell for ₹24 to gain 20%?

(a) 16 **(b)** 20 **(c)** 24 **(d)** 18

Q24 A trader gains 25% by selling an article. If he had bought it for 20% less and sold it for ₹63 less, he would still have gained 25%. What is the cost price?

(a) 300 **(b)** 360 **(c)** 210 **(d)** 252

Q25 An article is marked 40% above cost. After two successive discounts of 10% and x%, the profit is 13.4%. Find x.

(a) 5 **(b)** 10 **(c)** 15 **(d)** 12

13.6 Solutions

Q1 **Answer: (d) 25%**

100/400 × 100 = **25**%.

Q2 **Answer: (a) 920**

800 × 1.15 = **₹920**.

Q3 **Answer: (c) 800**

920/1.15 = **₹800**.

Q4 **Answer: (d) 1,700**

2000 × 0.85 = **₹1,700**.

Q5 **Answer: (c) 28%**

20 + 10 − 2 = **28**%.

Q6 **Answer: (d) 25%**

1/(3 + 1) × 100 = **25**%.

Q7 **Answer: (a) 500**

CP = 450/0.9 = **₹500**.

Q8 **Answer: (d) 1,000**

1180/1.18 = **₹1,000**.

Q9 **Answer: (a) 1% loss**

Same SP, x% profit and x% loss → loss of $x^2/100$ = **1%**.

Q10 **Answer: (c) 27.5%**

1.5 × 0.85 = 1.275 → **27.5**%.

Q11 **Answer: (c) 11.11%**

100/900 × 100 = **11.11**%.

Q12 **Answer: (d) 44%**

For 30 pens: CP 25, SP 36, profit 11/25 = **44**%.

Q13 **Answer: (d) 25%**

Profit on 16 articles = 4 articles' cost: 4/16 × 100 = **25**%.

Q14 **Answer: (c) 900**

SP = 720 = 0.8 × MP → MP = **₹900**.

Q15 **Answer: (a) 1,260**

CP = 1140/0.95 = 1200; SP for 5% gain = **₹1,260**.

Q16 **Answer: (c) 700**

SP = 500 × 1.26 = 630 = 0.9 × MP → MP = **₹700**.

Q17 **Answer: (a) 20%**

Factor = 1.2 × 0.9 × 1000/900 = 1.2 → profit **20**%.

Q18 **Answer: (b) 22.5%**

Take CP = 300: 100 sold for 85; needs total 330, so the remaining 200 must fetch 245: profit 45/200 = **22.5**%.

Q19 **Answer: (a) 46%**

0.9 × 0.8 × 0.75 = 0.54 → single discount **46**%.

Q20 **Answer: (d) 1,600**

CP is midway: (1920 + 1280)/2 = **₹1,600**.

Q21 **Answer: (a) 2,000**

2160/(1.2 × 0.9) = 2160/1.08 = **₹2,000**.

Q22 **Answer: (c) 100**

Selling 5 items for 4 × 150 = 600 → SP per item 120 = 1.2 × CP → CP = **₹100**.

Q23 **Answer: (d) 18**

CP of 45 lemons = 40/0.8 = 50. For a 20% gain, SP of 45 lemons = 60. For ₹24: 45 × 24/60 = **18** lemons.

Q24 **Answer: (d) 252**

Original SP = 1.25C. New CP 0.8C, new SP 1.25 × 0.8C = C. So 1.25C - 63 = C → C = **₹252**.

Q25 **Answer: (b) 10**

1.4 × 0.9 × (1 - x/100) = 1.134 → 1 - x/100 = 0.9 → x = **10**.

Profit and loss are percentages of CP; discount is a percentage of MP. SP = CP × (1 ± r/100); successive changes multiply. Same SP with x% profit and x% loss gives an $x^2/100$ % loss. False weights: profit = (true - false)/false. Equalise quantities with the LCM when buying and selling rates differ.

MODULE 14 · INTERMEDIATE

Simple interest

Minimum duration **1 day** | Level **Intermediate**

Simple interest is calculated only on the original principal. This module covers principal, rate, time, interest and amount, the formula SI = PRT/100, finding any one of P, R or T, money doubling and tripling, different rates for different periods, instalments, and lending and borrowing at different rates, followed by 25 solved questions computed exactly.

IN THIS MODULE

14.1 Concepts
14.2 Problem types
14.3 Practice questions
14.4 Solutions

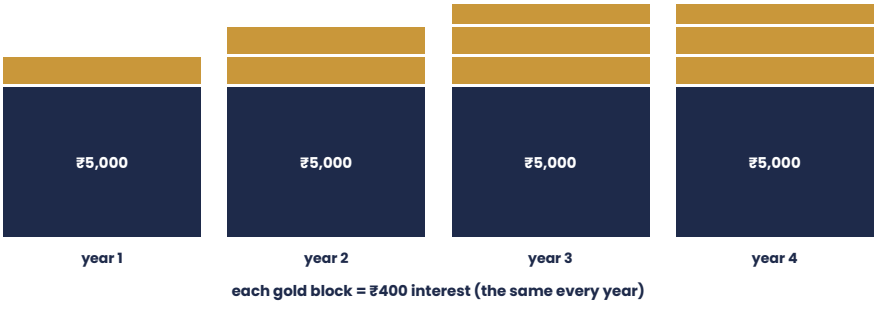
14.1 Concepts

14.1.1 Principal, rate, time, interest and amount

The **principal** P is the sum lent or borrowed; the **rate** R is the interest per ₹100 per year; the **time** T is in years. The **amount** A is principal plus interest.

14.1.2 The formula

$$SI = P \times R \times T / 100;\ A = P + SI = P(1 + RT/100)$$



SIMPLE INTEREST GROWS BY THE SAME AMOUNT EVERY YEAR

WATCH OUT

Convert months to years (9 months = 3/4 year) and days to years (73 days = 1/5 year) before using the formula.

14.2 Problem types

14.2.1 Finding principal, rate or time

Unknown	Formula
P	P = 100 × SI/(R × T)
R	R = 100 × SI/(P × T)
T	T = 100 × SI/(P × R)

14.2.2 Money doubling or tripling

A sum becomes n times when the interest equals (n - 1)P: R × T = 100(n - 1). A sum doubles in 8 years → R = 12.5%; at that rate it becomes 4 times in 24 years.

> SHORTCUT TRICK
> If a sum becomes n_1 times in T_1 years, it becomes n_2 times in $T_1 \times (n_2 - 1)/(n_1 - 1)$ years.

14.2.3 Different rates for different periods

Add the interest for each period: ₹10,000 at 6% for 2 years and then 8% for 3 years: 1200 + 2400 = ₹3,600.

14.2.4 Instalments with simple interest

For a debt repaid in equal annual instalments x at simple interest R%, each instalment earns interest from its payment date to the end of the term. For a debt A due after n years: A = nx + x × R/100 × (0 + 1 + … + (n - 1)).

> WORKED EXAMPLE
> **What annual instalment discharges a debt of ₹1,092 due in 3 years at 12% simple interest?**
> 3x + x × 0.12 × (2 + 1 + 0) = 3.36x = 1092 → x = **₹325**.

14.2.5 Lending and borrowing at different rates

Gain per year = P × (lending rate - borrowing rate)/100. Borrowing ₹50,000 at 6% and lending at 8% gains ₹1,000 a year.

14.3 Practice questions

14.3.1 Basic level

Q1 What is the simple interest on ₹5,000 at 8% per year for 3 years?

(a) 1,000 **(b)** 1,500 **(c)** 400 **(d)** 1,200

Q2 What is the amount on ₹12,000 at 10% simple interest for 2 years?

(a) 14,000 **(b)** 14,400 **(c)** 14,520 **(d)** 13,200

Q3 At what rate will ₹4,000 earn ₹960 simple interest in 3 years?

(a) 6% **(b)** 8% **(c)** 9% **(d)** 10%

Q4 In how many years will ₹2,500 earn ₹750 at 6% simple interest?

(a) 5 years **(b)** 4 years **(c)** 6 years **(d)** 3 years

Q5 What principal earns ₹1,350 simple interest in 3 years at 9%?

(a) 4,500 **(b)** 5,000 **(c)** 6,000 **(d)** 4,000

Q6 A sum doubles in 8 years at simple interest. What is the rate?

(a) 10% **(b)** 8% **(c)** 12.5% **(d)** 16%

Q7 What is the simple interest on ₹7,300 at 10% for 73 days?

(a) 73 **(b)** 146 **(c)** 160 **(d)** 120

Q8 What is the simple interest on ₹6,000 at 5% for 9 months?

(a) 225 **(b)** 300 **(c)** 200 **(d)** 450

14.3.2 Intermediate level

Q9 A sum becomes ₹9,600 in 2 years and ₹12,000 in 5 years at simple interest. What is the rate?

(a) 8% **(b)** 15% **(c)** 12% **(d)** 10%

Q10 A sum doubles in 8 years at simple interest. In how many years will it become 4 times?

(a) 16 years **(b)** 24 years **(c)** 32 years **(d)** 20 years

Q11 ₹10,000 is invested at 6% for 2 years and then at 8% for 3 more years (simple interest). What is the total interest?

(a) 3,000 **(b)** 4,200 **(c)** 3,400 **(d)** 3,600

Q12 A man borrows ₹50,000 at 6% and lends it at 8% simple interest. What is his gain in 2 years?

(a) 1,000 **(b)** 1,500 **(c)** 2,000 **(d)** 4,000

Q13 The simple interest on a sum for 5 years is 2/5 of the sum. What is the rate?

(a) 5% **(b)** 10% **(c)** 4% **(d)** 8%

Q14 A sum of ₹8,000 is lent partly at 5% and partly at 8% simple interest. The total interest for a year is ₹520. How much was lent at 8%?

(a) 3,000 **(b)** 4,000 **(c)** 5,000 **(d)** 2,500

Q15 What annual instalment will discharge a debt of ₹1,092 due in 3 years at 12% simple interest?

(a) 300 **(b)** 364 **(c)** 325 **(d)** 350

Q16 Simple interest on a certain sum for 3 years at 8% is ₹480 more than the simple interest for 2 years at 9%. What is the sum?

(a) 8,000 **(b)** 6,000 **(c)** 10,000 **(d)** 9,600

14.3.3 Advanced level

Q17 A sum lent at simple interest amounts to ₹1,120 in 2 years and to ₹1,240 in 5 years. What is the principal?

(a) 1,040 (b) 1,000 (c) 1,060 (d) 960

Q18 At what rate of simple interest will a sum become 7/4 of itself in 5 years?

(a) 15% (b) 12% (c) 20% (d) 14%

Q19 A man lends ₹10,000 in two parts, one at 10% and the other at 12% simple interest, and gets ₹1,140 interest in a year. What is the ratio of the parts (at 10% : at 12%)?

(a) 3 : 7 (b) 7 : 3 (c) 1 : 1 (d) 2 : 3

Q20 The rate of simple interest falls from 8% to 6.5%, and a man's yearly income falls by ₹90. What is his capital?

(a) 5,000 (b) 9,000 (c) 4,500 (d) 6,000

Q21 Divide ₹2,600 between A and B so that A's amount after 5 years equals B's amount after 7 years at 5% simple interest. What is A's share?

(a) 1,250 (b) 1,350 (c) 1,300 (d) 1,400

Q22 A sum of money at simple interest doubles in 6 years. In how many years will it triple?

(a) 9 years (b) 18 years (c) 15 years (d) 12 years

Q23 What sum will give ₹1,200 simple interest in 4 years at 5% for the first 2 years and 7% for the next 2 years?

(a) 4,000 (b) 4,800 (c) 6,000 (d) 5,000

Q24 Ravi took a loan at 6% simple interest for as many years as the rate. If he paid ₹864 as interest, what was the loan?

(a) 2,000 (b) 2,400 (c) 2,600 (d) 3,600

Q25 A sum of ₹1,500 is lent partly at 5% and partly at 6% simple interest; the total yearly interest is ₹85. How much is lent at 5%?

(a) 750 (b) 1,000 (c) 1,200 (d) 500

14.4 Solutions

Q1 **Answer: (d) 1,200**

5000 × 8 × 3/100 = **₹1,200**.

Q2 **Answer: (b) 14,400**

SI = 2,400; amount = **₹14,400**.

Q3 **Answer: (b) 8%**

R = 100 × 960/(4000 × 3) = **8**%.

Q4 **Answer: (a) 5 years**

T = 100 × 750/(2500 × 6) = **5** years.

Q5 **Answer: (b) 5,000**

P = 100 × 1350/27 = **₹5,000**.

Q6 **Answer: (c) 12.5%**

R × 8 = 100 → R = **12.5**%.

Q7 **Answer: (b) 146**

73 days = 1/5 year: 7300 × 10 × 1/5 ÷ 100 = **₹146**.

Q8 **Answer: (a) 225**

9 months = 3/4 year: 6000 × 5 × 3/4 ÷ 100 = **₹225**.

Q9 **Answer: (d) 10%**

Interest for 3 years = 2,400 → per year 800. Principal = 9600 - 2 × 800 = 8,000. Rate = 800/8,000 × 100 = **10%**.

Q10 **Answer: (b) 24 years**

Interest of 3P takes 3 × 8 = **24** years.

Q11 **Answer: (d) 3,600**

1,200 + 2,400 = **₹3,600**.

Q12 **Answer: (c) 2,000**

Gain = 50000 × 2% × 2 = **₹2,000**.

Q13 **Answer: (d) 8%**

R × 5 = 40 → R = **8**%.

Q14 **Answer: (b) 4,000**

If x is at 8%: 0.08x + 0.05(8000 - x) = 520 → 0.03x = 120 → x = **₹4,000**.

Q15 **Answer: (c) 325**

3x + 0.12x(2 + 1 + 0) = 3.36x = 1092 → x = **₹325**.

Q16 **Answer: (a) 8,000**

Interest rates × time: 24% - 18% = 6% of the sum = 480 → sum = **₹8,000**.

Q17 **Answer: (a) 1,040**

Interest for 3 years = 120 → ₹40 a year. Principal = 1120 − 80 = **₹1,040**.

Q18 **Answer: (a) 15%**

Interest = 3/4 of P in 5 years: R = 75/5 = **15**%.

Q19 **Answer: (a) 3 : 7**

Average 11.4%. Alligation: (12 − 11.4) : (11.4 − 10) = 0.6 : 1.4 = **3 : 7**.

Q20 **Answer: (d) 6,000**

1.5% of the capital = 90 → capital = **₹6,000**.

Q21 **Answer: (b) 1,350**

A × 1.25 = B × 1.35 → A : B = 27 : 25 → A = 2600 × 27/52 = **₹1,350**.

Q22 **Answer: (d) 12 years**

Doubling: interest P in 6 years; tripling needs 2P: **12** years.

Q23 **Answer: (d) 5,000**

Total rate × time = 10 + 14 = 24% → P = 1200/0.24 = **₹5,000**.

Q24 **Answer: (b) 2,400**

R = T = 6: P × 36/100 = 864 → P = **₹2,400**.

Q25 **Answer: (d) 500**

0.05x + 0.06(1500 − x) = 85 → 90 − 0.01x = 85 → x = **₹500**.

SI = PRT/100 and A = P + SI; convert months and days to years. Rearrange for P, R or T. A sum becomes n times when RT = 100(n − 1). Add interest period by period when rates change. For two parts at different rates, use alligation on the rates. For instalments, each payment earns interest until the due date.

MODULE 15 · INTERMEDIATE

Compound interest

Minimum duration **2 days** | Level **Intermediate**

Compound interest charges interest on interest and is used in loans, investments and growth problems. This module covers the amount formula, annual, half-yearly and quarterly compounding, fractional periods, the difference between CI and SI for 2 and 3 years, finding the rate or time, different rates in different years, amounts becoming n times, equal annual instalments, growth and depreciation, fixed deposits and EMIs, and the rule of 72, followed by 25 solved questions computed exactly.

IN THIS MODULE

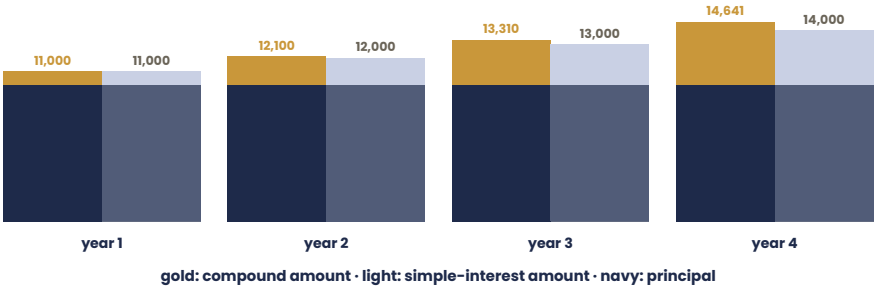
15.1 Concepts
15.2 CI vs SI
15.3 Problem types
15.4 Real-life applications
15.5 Practice questions
15.6 Solutions

15.1 Concepts

15.1.1 The amount formula

$$A = P(1 + R/100)^n;\ CI = A - P$$

Each year the interest is added to the principal, so the next year's interest is calculated on a larger sum.



COMPOUND INTEREST PULLS AHEAD OF SIMPLE INTEREST EVERY YEAR

15.1.2 Half-yearly and quarterly compounding

Compounded half-yearly: use R/2 per period and 2n periods. Quarterly: R/4 and 4n periods. ₹8,000 at 10% p.a. compounded half-yearly for 1 year: $8000 \times 1.05^2 = ₹8{,}820$.

15.1.3 Fractional periods

For 2½ years at R%: $A = P(1 + R/100)^2 \times (1 + (R/2)/100)$. Simple interest is used for the fractional part.

15.2 CI vs SI

15.2.1 Difference for 2 and 3 years

CI − SI for 2 years = $P(R/100)^2$; for 3 years = $P(R/100)^2(3 + R/100)$

On ₹5,000 at 10%: 2 years → 5000 × 0.01 = ₹50; 3 years → 5000 × 0.01 × 3.1 = ₹155.

SHORTCUT TRICK
For 2 years, CI = SI + interest on the first year's interest. At 10% on ₹5,000: SI 1,000 + 10% of 500 = 1,050.

15.3 Problem types

15.3.1 Finding rate or time

Divide A by P and recognise the power: 12100/10000 = 1.21 = 1.1^2 → 10% for 2 years. Learn 1.1^2 = 1.21, 1.1^3 = 1.331, 1.2^2 = 1.44, 1.05^2 = 1.1025, 1.25^2 = 1.5625.

15.3.2 Different rates in different years

$A = P(1 + R_1/100)(1 + R_2/100)\ldots$: ₹10,000 at 10% then 20%: 10000 × 1.1 × 1.2 = ₹13,200.

15.3.3 Amount becoming n times

If a sum doubles in t years at compound interest, it becomes 4 times in 2t years and 8 times in 3t years (powers of 2).

15.3.4 Equal annual instalments

A loan P repaid in n equal annual instalments x at R%: $P = x/(1 + r) + x/(1 + r)^2 + \ldots + x/(1 + r)^n$, with r = R/100

WORKED EXAMPLE
What equal annual instalment repays ₹21,000 in 2 years at 10% compound interest?
$x/1.1 + x/1.21 = 21000 \rightarrow x(1.1 + 1)/1.21 = 21000 \rightarrow x = 21000 \times 1.21/2.1$ = **₹12,100**.

15.3.5 Growth and depreciation

Population $P(1 + r/100)^n$; depreciation $V(1 - r/100)^n$. A machine worth ₹1,00,000 depreciating 10% a year is worth ₹81,000 after 2 years.

15.4 Real-life applications

15.4.1 Fixed deposits and EMIs

Banks compound fixed deposits quarterly. A loan EMI is an equal monthly instalment computed with the instalment formula of 15.3.4 using the monthly rate. Exams ask only simple versions; understanding the idea is enough.

15.4.2 Rule of 72

At R% compound interest a sum roughly doubles in 72/R years: at 8%, about 9 years (exact 9.006).

15.5 Practice questions

15.5.1 Basic level

Q1 What is the compound interest on ₹10,000 at 10% for 2 years?

(a) 2,000 (b) 2,200 (c) 2,100 (d) 1,100

Q2 What is the amount on ₹8,000 at 10% p.a. compounded half-yearly for 1 year?

(a) 8,800 (b) 8,840 (c) 8,400 (d) 8,820

Q3 What is the difference between CI and SI on ₹5,000 at 10% for 2 years?

(a) 100 (b) 25 (c) 500 (d) 50

Q4 What is the amount on ₹15,625 at 4% compound interest for 3 years?

(a) 17,576 (b) 17,500 (c) 17,575 (d) 17,600

Q5 A sum becomes ₹12,100 in 2 years at 10% compound interest. What is the sum?

(a) 11,000 (b) 9,900 (c) 10,100 (d) 10,000

Q6 At what rate does ₹10,000 become ₹14,400 in 2 years at compound interest?

(a) 22% (b) 18% (c) 44% (d) 20%

Q7 Using the rule of 72, about how long does a sum take to double at 9% compound interest?

(a) 9 years (b) 7 years (c) 11 years (d) 8 years

Q8 A machine worth ₹1,00,000 depreciates by 10% a year. What is its value after 2 years?

(a) 80,000 (b) 90,000 (c) 81,000 (d) 82,000

15.5.2 Intermediate level

Q9 What is the difference between CI and SI on ₹5,000 at 10% for 3 years?

(a) 155 (b) 150 (c) 50 (d) 165

Q10 ₹10,000 is invested at 10% in the first year and 20% in the second, compounded annually. What is the amount?

(a) 13,000 (b) 13,310 (c) 13,200 (d) 12,100

Q11 The difference between CI and SI on a sum for 2 years at 8% is ₹64. What is the sum?

(a) 8,000 **(b)** 10,000 **(c)** 12,800 **(d)** 6,400

Q12 A sum doubles in 5 years at compound interest. In how many years will it become 8 times?

(a) 10 years **(b)** 15 years **(c)** 20 years **(d)** 40 years

Q13 What is the CI on ₹16,000 at 20% p.a. for 9 months compounded quarterly?

(a) 2,400 **(b)** 2,600 **(c)** 2,522 **(d)** 2,500

Q14 What equal annual instalment repays a loan of ₹21,000 in 2 years at 10% compound interest?

(a) 12,100 **(b)** 11,000 **(c)** 11,550 **(d)** 12,600

Q15 A sum amounts to ₹6,050 in 2 years and ₹6,655 in 3 years at compound interest. What is the rate?

(a) 5% **(b)** 12% **(c)** 15% **(d)** 10%

Q16 The population of a town is 1,60,000. It grows 5% a year. What will it be after 2 years?

(a) 176,000 **(b)** 175,000 **(c)** 176,500 **(d)** 176,400

15.5.3 Advanced level

Q17 In the previous question, what was the principal? (A sum amounts to ₹6,050 in 2 years and ₹6,655 in 3 years at compound interest.)

(a) 5,500 **(b)** 4,800 **(c)** 5,000 **(d)** 5,200

Q18 What is the amount on ₹10,000 for 2½ years at 10% compound interest (compounded annually)?

(a) 12,500 **(b)** 12,650 **(c)** 12,705 **(d)** 13,310

Q19 The CI on a sum for 2 years is ₹832 and the SI is ₹800. What is the rate?

(a) 4% **(b)** 6% **(c)** 8% **(d)** 10%

Q20 ₹1,000 becomes ₹1,331 in 3 years at compound interest. What is the rate?

(a) 11% **(b)** 10% **(c)** 33.1% **(d)** 9%

Q21 What equal annual instalment repays a loan of ₹3,310 in 3 years at 10% compound interest?

(a) 1,210 **(b)** 1,331 **(c)** 1,100 **(d)** 1,250

Q22 A sum invested at compound interest becomes ₹2,420 in 2 years and ₹2,662 in 3 years. What is the sum?

(a) 1,800 **(b)** 2,000 **(c)** 2,200 **(d)** 2,100

Q23 The difference between CI compounded half-yearly and CI compounded annually on ₹20,000 for 1 year at 10% is

(a) 25 **(b)** 100 **(c)** 40 **(d)** 50

Q24 A man invests ₹50,000 in a fixed deposit at 8% compounded annually. After how many whole years will the amount first exceed ₹62,000?

(a) 3 years **(b)** 2 years **(c)** 4 years **(d)** 5 years

Q25 At compound interest, a sum triples in 4 years. In how many years will it become 9 times?

(a) 12 years **(b)** 16 years **(c)** 8 years **(d)** 6 years

15.6 Solutions

Q1 **Answer: (c) 2,100**

10000 × 1.21 − 10000 = **₹2,100**.

Q2 **Answer: (d) 8,820**

8000 × 1.05^2 = **₹8,820**.

Q3 **Answer: (d) 50**

$P(R/100)^2$ = 5000 × 0.01 = **₹50**.

Q4 **Answer: (a) 17,576**

15625 × 1.04^3 = **₹17,576**.

Q5 **Answer: (d) 10,000**

12100/1.21 = **₹10,000**.

Q6 **Answer: (d) 20%**

14400/10000 = 1.44 = 1.2^2 → **20**%.

Q7 **Answer: (d) 8 years**

72/9 = **8** years.

Q8 **Answer: (c) 81,000**

100000 × 0.9^2 = **₹81,000**.

Q9 **Answer: (a) 155**

5000 × 0.01 × 3.1 = **₹155**.

Q10 **Answer: (c) 13,200**

10000 × 1.1 × 1.2 = **₹13,200**.

Q11 **Answer: (b) 10,000**

P × 0.0064 = 64 → P = **₹10,000**.

Q12 **Answer: (b) 15 years**

8 = 2^3, so 3 × 5 = **15** years.

Q13 **Answer: (c) 2,522**

5% per quarter for 3 quarters: 16000 × 1.05^3 − 16000 = **₹2,522**.

Q14 **Answer: (a) 12,100**

$x/1.1 + x/1.21 = 21000 \rightarrow x =$ **₹12,100**.

Q15 **Answer: (d) 10%**

Interest in the third year = 605 on 6050 = **10**%.

Q16 **Answer: (d) 176,400**

$160000 \times 1.1025 =$ **1,76,400**.

Q17 **Answer: (c) 5,000**

Rate 10%, so P = 6050/1.21 = **₹5,000**.

Q18 **Answer: (c) 12,705**

$10000 \times 1.21 \times 1.05 =$ **₹12,705**.

Q19 **Answer: (c) 8%**

Difference 32 = interest on one year's SI (400): rate = $32/400 \times 100 =$ **8**%.

Q20 **Answer: (b) 10%**

$1331/1000 = 1.331 = 1.1^3 \rightarrow$ **10**%.

Q21 **Answer: (b) 1,331**

$x(1/1.1 + 1/1.21 + 1/1.331) = 3{,}310$. Multiply by 1.331: $x(1.21 + 1.1 + 1) = 3.31x = 4{,}405.61 \rightarrow x =$ **₹1,331**.

Q22 **Answer: (b) 2,000**

Rate = 242/2420 = 10%; P = 2420/1.21 = **₹2,000**.

Q23 **Answer: (d) 50**

Half-yearly: $20000 \times 1.1025 = 22{,}050$; annually 22,000. Difference **₹50**.

Q24 **Answer: (a) 3 years**

$50000 \times 1.08^2 = 58{,}320$; $\times 1.08^3 = 62{,}985.6 > 62{,}000$: **3** years.

Q25 **Answer: (c) 8 years**

$9 = 3^2$, so $2 \times 4 =$ **8** years.

A = $P(1 + R/100)^n$; halve the rate and double the periods for half-yearly compounding. CI − SI = $P(R/100)^2$ for 2 years and $P(R/100)^2(3 + R/100)$ for 3. Recognise powers (1.21, 1.331, 1.44) to find rates and times. Instalments: loan = sum of present values $x/(1 + r)^k$. Rule of 72 estimates doubling time.

MODULE 16 · INTERMEDIATE

Time and work

Minimum duration **2 days** | Level **Intermediate**

Time and work problems are about rates of working, and they are asked in almost every placement test. This module covers work = rate × time, one-day work, the LCM (total units) method, people working together, joining or leaving midway, alternate days, efficiency comparisons, men, women and children, wages, and the man-days chain rule, followed by 25 solved questions computed exactly.

IN THIS MODULE

16.1 Concepts
16.2 Problem types
16.3 Practice questions
16.4 Solutions

16.1 Concepts

16.1.1 Work = rate × time

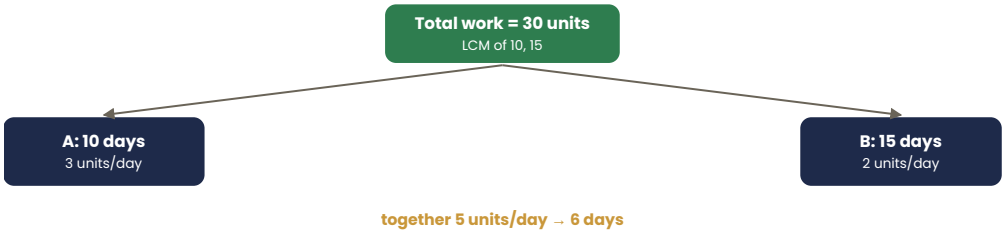
If a person finishes a job in n days, their rate is 1/n of the job per day. Rates of people working together **add**; time is the reciprocal of the total rate.

16.1.2 One-day work and total work

If A takes a days and B takes b days, together they take ab/(a + b) days

16.1.3 The LCM (total units) method

Let the total work be the LCM of the given days. Each person's daily work is then a whole number of units, and all calculations use whole numbers.



LCM METHOD: A AND B TOGETHER FINISH IN 6 DAYS

SHORTCUT TRICK

The LCM method avoids fractions completely: A 10 days, B 15 days → total 30 units, A does 3 and B does 2 units a day, together 5 → 30/5 = 6 days.

16.2 Problem types

16.2.1 Two or three people together

Three people taking 10, 12 and 15 days: LCM 60, rates 6 + 5 + 4 = 15 units/day → 4 days.

16.2.2 People joining or leaving midway

Count the units done in each phase. A (10 days) and B (15 days) work together for 2 days (10 units of 30), then A leaves: B finishes the remaining 20 units at 2 units/day in 10 days. Total 12 days.

16.2.3 Working on alternate days

Group the days into cycles. A (10 days) and B (15 days) work on alternate days starting with A: each 2-day cycle does 5 units; 5 cycles (10 days) do 25 units; day 11 (A) does 3 → 28; day 12 (B) does 2 → 30. Total **12 days**.

16.2.4 Efficiency comparisons

If A is twice as efficient as B, A takes half the time. If together they take 10 days and A is twice as efficient, rates 2 : 1 → total 3 units/day for 30 units: A alone takes 15 days, B 30 days.

16.2.5 Men, women and children

Convert everyone to one unit: if 2 men = 3 women, express all workers in women and use the chain rule.

16.2.6 Work and wages

Wages are shared in the ratio of work done, i.e. rate × days worked. If A and B (rates 3 : 2) work together for the whole job, a payment of ₹1,500 is split ₹900 : ₹600.

16.2.7 Man-days and the chain rule

$$M_1 \times D_1 \times H_1 / W_1 = M_2 \times D_2 \times H_2 / W_2 \text{ (men, days, hours per day, work)}$$

If 12 men working 8 hours a day build a wall in 10 days, then 16 men working 6 hours a day need $12 \times 8 \times 10/(16 \times 6) = 10$ days for the same wall.

16.3 Practice questions

16.3.1 Basic level

Q1 A can finish a job in 10 days and B in 15 days. In how many days can they finish it together?

(a) 6 days **(b)** 5 days **(c)** 8 days **(d)** 12.5 days

Q2 A, B and C can do a job in 10, 12 and 15 days. How long do they take together?

(a) 5 days **(b)** 3 days **(c)** 4 days **(d)** 6 days

Q3 A and B together finish a job in 12 days, and A alone in 20 days. How long does B alone take?

(a) 25 days **(b)** 30 days **(c)** 32 days **(d)** 15 days

Q4 A does 1/4 of a job in 3 days. How long will he take to do the whole job?

(a) 9 days **(b)** 16 days **(c)** 12 days **(d)** 7 days

Q5 If 12 men can build a wall in 10 days, how many days will 15 men take?

(a) 8 days **(b)** 12 days **(c)** 6 days **(d)** 9 days

Q6 A is twice as efficient as B. B alone takes 18 days. How long does A take?

(a) 36 days **(b)** 9 days **(c)** 12 days **(d)** 6 days

Q7 A takes 6 days and B takes 12 days. What fraction of the work do they do together in 2 days?

(a) 1/4 **(b)** 1/2 **(c)** 1/3 **(d)** 2/3

Q8 If 8 men can do a job in 15 days, how many men are needed to do it in 10 days?

(a) 10 **(b)** 12 **(c)** 16 **(d)** 9

16.3.2 Intermediate level

Q9 A (10 days) and B (15 days) work together for 2 days, then A leaves. How many days in total does the job take?

(a) 12 days **(b)** 10 days **(c)** 11 days **(d)** 14 days

Q10 A (10 days) and B (15 days) work on alternate days, starting with A. How many days does the job take?

(a) 12 days **(b)** 11 days **(c)** 12.5 days **(d)** 13 days

Q11 A and B together take 10 days, and A is twice as efficient as B. How long does B alone take?

(a) 15 days **(b)** 20 days **(c)** 30 days **(d)** 25 days

Q12 A and B can do a job in 12 days, B and C in 15 days, and C and A in 20 days. How long do A, B and C take together?

(a) 12 days **(b)** 8 days **(c)** 10 days **(d)** 15 days

Q13 A can do a job in 20 days and B in 30 days. They are paid ₹1,500 for the job working together. What is A's share?

(a) 750 **(b)** 600 **(c)** 900 **(d)** 1,000

Q14 12 men working 8 hours a day build a wall in 10 days. How many days will 16 men working 6 hours a day take?

(a) 8 days **(b)** 12 days **(c)** 10 days **(d)** 15 days

Q15 2 men or 3 women can do a job in 12 days. How long will 1 man and 1 woman take together?

(a) 15 days **(b)** 14.4 days **(c)** 18 days **(d)** 20 days

Q16 A can do a job in 15 days. After working for 5 days he is joined by B, and together they finish the rest in 4 days. How long would B alone take?

(a) 10 days **(b)** 8 days **(c)** 12 days **(d)** 15 days

16.3.3 Advanced level

Q17 A, B and C together can do a job in 6 days. A alone takes 12 days and B alone 18 days. How long does C alone take?

(a) 24 days (b) 30 days (c) 48 days (d) 36 days

Q18 A is 50% more efficient than B. Together they finish a job in 18 days. How long would A take alone?

(a) 27 days (b) 36 days (c) 45 days (d) 30 days

Q19 20 men can finish a job in 30 days. After 10 days, 5 men leave. How many more days are needed to finish?

(a) 20 days (b) 26.67 days (c) 25 days (d) 30 days

Q20 A can do a job in 18 days and B in 24 days. They start together, but A leaves 4 days before the job is finished. How long does the job take in total?

(a) 10 days (b) 12 days (c) 12.57 days (d) 16 days

Q21 A and B can do a job in 8 days, B and C in 12 days, and A, B and C together in 6 days. How long do A and C take together?

(a) 10 days (b) 9 days (c) 8 days (d) 12 days

Q22 10 men or 15 women can do a job in 30 days. In how many days can 6 men and 6 women do it?

(a) 25 days (b) 30 days (c) 40 days (d) 36 days

Q23 A completes 40% of a job in 12 days, and B completes 30% in 6 days. How long will they take together to finish the whole job?

(a) 12 days (b) 10 days (c) 15 days (d) 18 days

Q24 A, B and C are paid ₹2,400 for a job. A and B together do 7/10 of it, and B and C together do 8/15. What does B get?

(a) 400 (b) 720 (c) 560 (d) 600

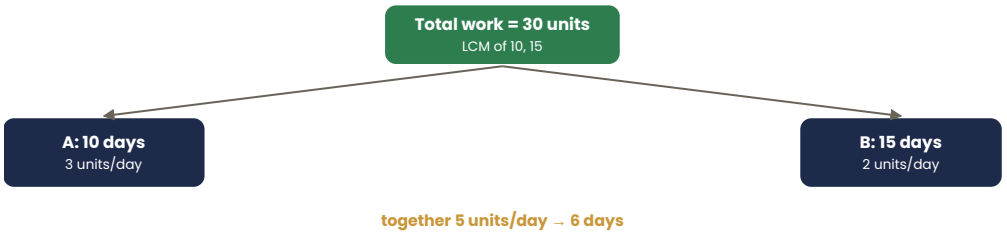
Q25 Working 5 hours a day, 9 men finish a job in 12 days. How many hours a day must 6 men work to finish it in 15 days?

(a) 6 hours (b) 5 hours (c) 7 hours (d) 8 hours

16.4 Solutions

Q1 **Answer: (a) 6 days**

LCM 30: rates 3 + 2 = 5 units/day → **6 days**.



Q2 **Answer: (c) 4 days**

LCM 60: 6 + 5 + 4 = 15 units/day → **4 days**.

Q3 **Answer: (b) 30 days**

1/12 - 1/20 = 1/30 → **30 days**.

Q4 **Answer: (c) 12 days**

3 × 4 = **12 days**.

Q5 **Answer: (a) 8 days**

Man-days = 120 → 120/15 = **8 days**.

Q6 **Answer: (b) 9 days**

Twice as efficient → half the time: **9 days**.

Q7 **Answer: (b) 1/2**

Daily 1/6 + 1/12 = 1/4; in 2 days **1/2**.

Q8 **Answer: (b) 12**

Man-days 120 → 120/10 = **12** men.

Q9 **Answer: (a) 12 days**

2 days × 5 units = 10 of 30; B does the remaining 20 at 2/day in 10 days. Total **12 days**.

Q10 **Answer: (a) 12 days**

2-day cycles of 5 units: 5 cycles = 25 units in 10 days; day 11 (A) +3 → 28; day 12 (B) +2 → 30. **12 days**.

Q11 **Answer: (c) 30 days**

Rates 2 : 1 → 3 units/day for 30 units. B alone at 1 unit/day: **30 days**.

Q12 **Answer: (c) 10 days**

2(A + B + C) = 1/12 + 1/15 + 1/20 = 1/5 → A + B + C = 1/10 → **10 days**.

Q13 **Answer: (c) 900**

Rates 3 : 2 → A = 1500 × 3/5 = **₹900**.

Q14 **Answer: (c) 10 days**

12 × 8 × 10/(16 × 6) = **10 days**.

Q15 **Answer: (b) 14.4 days**

One man takes 24 days, one woman 36 days: 1/24 + 1/36 = 5/72 → **14.4 days**.

Q16 **Answer: (a) 10 days**

A does 9/15 in 9 days; B does the remaining 6/15 in 4 days → B alone 10 days.

Q17 **Answer: (d) 36 days**

1/6 - 1/12 - 1/18 = 1/36 → **36 days**.

Q18 **Answer: (d) 30 days**

Rates 3 : 2, total 5 units/day × 18 = 90 units; A alone 90/3 = **30 days**.

Q19 **Answer: (b) 26.67 days**

Total 600 man-days; 200 done; remaining 400 by 15 men → 400/15 = **26.67 days** (26 2/3).

Q20 **Answer: (c) 12.57 days**

If the job takes T days: A works T - 4 days. (T - 4)/18 + T/24 = 1 → 4(T - 4) + 3T = 72 → 7T = 88 → T = **12.57** days (88/7).

Q21 **Answer: (c) 8 days**

C = 1/6 - 1/8 = 1/24; A = 1/6 - 1/12 = 1/12. A + C = 1/12 + 1/24 = 1/8 → **8 days**.

Q22 **Answer: (b) 30 days**

One man: 300 days, one woman: 450 days. 6/300 + 6/450 = 1/50 + 1/75 = 1/30 → **30 days**.

Q23 **Answer: (a) 12 days**

A alone 30 days, B alone 20 days: together 12 days.

Q24 **Answer: (c) 560**

A + B + C = 1, so B = 7/10 + 8/15 - 1 = 7/30 of the work: 2400 × 7/30 = **₹560**.

Q25 **Answer: (a) 6 hours**

9 × 5 × 12 = 6 × h × 15 → h = **6** hours.

Rates add: together = $1/(1/a + 1/b) = ab/(a + b)$. Take total work as the LCM of the days so every rate is a whole number of units. Count units phase by phase when people join, leave or alternate. Efficiency ratio = inverse of time ratio. Wages follow work done. Chain rule: $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$.

MODULE 17 · INTERMEDIATE

Pipes and cisterns

Minimum duration **1 day** | Level **Intermediate**

Pipes and cisterns problems apply time and work ideas, with inlets adding and outlets removing work. This module covers inlet and outlet pipes, pipes working together, leaks, pipes opened or closed at different times and pipes opened alternately, followed by 25 solved questions computed exactly with the LCM (total units) method.

IN THIS MODULE

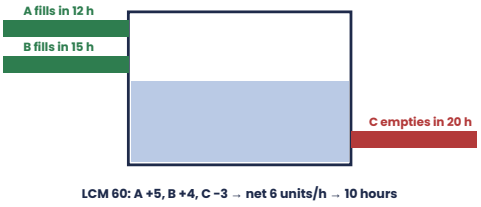
17.1 Concepts
17.2 Problem types
17.3 Practice questions
17.4 Solutions

17.1 Concepts

17.1.1 Inlet pipes and outlet pipes

An **inlet** fills the tank: its work is positive. An **outlet** (or leak) empties it: its work is negative. If a pipe fills a tank in a hours, it fills 1/a of the tank per hour; an outlet that empties it in b hours removes 1/b per hour.

Net rate = Σ(1/inlet times) - Σ(1/outlet times); time = 1/net rate



TWO INLETS AND ONE OUTLET

17.2 Problem types

17.2.1 Pipes working together

Add the rates. A (12 h) and B (15 h) together: LCM 60, 5 + 4 = 9 units/h → 60/9 = 6⅔ hours.

17.2.2 Leaks

A pipe fills a tank in 6 hours, but because of a leak it takes 8 hours. The leak alone empties the full tank in 1/(1/6 - 1/8) = 24 hours.

17.2.3 Pipes opened or closed at different times

Split the time into phases and add the units filled in each phase. If A (12 h) and B (15 h) are opened together and B is closed after 4 hours: 4 × 9 = 36 of 60 units; A fills the remaining 24 in 24/5 = 4.8 h; total 8.8 h.

17.2.4 Alternate opening of pipes

Group the hours into cycles. A (12 h) and B (15 h) opened alternately for an hour each, starting with A: each 2-hour cycle fills 9 units; 6 cycles (12 h) fill 54; hour 13 (A) fills 5 → 59; B needs 1/4 hour for the last unit. Total 13¼ hours.

SHORTCUT TRICK
For alternate opening, run full cycles until less than one cycle of work remains, then add pipes one at a time in their order.

WATCH OUT
In alternate problems with an outlet, check whether the tank overflows (or empties) **during** a cycle, before the cycle ends.

17.3 Practice questions

17.3.1 Basic level

Q1 Pipe A fills a tank in 12 hours and pipe B in 15 hours. How long do they take together?

(a) 6 hours (b) 7.5 hours (c) 27 hours (d) 6.67 hours

Q2 A pipe fills a tank in 10 hours and an outlet empties it in 15 hours. With both open, how long does it take to fill the tank?

(a) 25 hours (b) 12 hours (c) 30 hours (d) 6 hours

Q3 A pipe fills a tank in 6 hours; because of a leak it takes 8 hours. How long would the leak take to empty the full tank?

(a) 14 hours (b) 12 hours (c) 48 hours (d) 24 hours

Q4 Pipes A, B and C fill a tank in 10, 15 and 30 hours. How long do they take together?

(a) 5 hours (b) 6 hours (c) 4 hours (d) 8 hours

Q5 Two pipes fill a tank in 20 and 30 minutes. How long do they take together?

(a) 12 minutes (b) 10 minutes (c) 15 minutes (d) 25 minutes

Q6 A tap fills 1/5 of a tank in 3 hours. How long will it take to fill the whole tank?

(a) 12 hours (b) 15 hours (c) 18 hours (d) 8 hours

Q7 Pipe A fills a tank in 12 hours, pipe B in 15 hours, and pipe C empties it in 20 hours. With all three open, how long does it take to fill?

(a) 10 hours (b) 8 hours (c) 12 hours (d) 15 hours

Q8 An outlet empties a full tank in 8 hours. What fraction of the tank does it empty in 3 hours?

(a) 1/8 (b) 5/8 (c) 3/8 (d) 3/5

17.3.2 Intermediate level

Q9 A (12 h) and B (15 h) are opened together, and B is closed after 4 hours. How long does it take in total to fill the tank?

(a) 8.8 hours **(b)** 8 hours **(c)** 9 hours **(d)** 9.6 hours

Q10 A (12 h) and B (15 h) are opened alternately for an hour each, starting with A. How long does it take to fill the tank?

(a) 13.25 hours **(b)** 13 hours **(c)** 13.5 hours **(d)** 14 hours

Q11 Two pipes fill a tank in 15 and 20 minutes, and a third empties it in 30 minutes. If all are opened together, how long does it take to fill the tank?

(a) 12 minutes **(b)** 10 minutes **(c)** 15 minutes **(d)** 18 minutes

Q12 A pipe can fill a tank in 20 minutes. After it has been open for 5 minutes, a leak develops, and the tank fills 10 minutes later than expected. How long would the leak take to empty a full tank?

(a) 40 minutes **(b)** 60 minutes **(c)** 25 minutes **(d)** 50 minutes

Q13 Pipe A fills a tank twice as fast as pipe B. Together they fill it in 12 minutes. How long does B alone take?

(a) 24 minutes **(b)** 18 minutes **(c)** 48 minutes **(d)** 36 minutes

Q14 Three taps A, B, C fill a tank in 12, 15 and 20 hours. A is open all the time, B and C are open for 1 hour each alternately, starting with B. How long does it take to fill the tank?

(a) 6 hours **(b)** 7 hours **(c)** 8 hours **(d)** 7.5 hours

Q15 A cistern has a leak that would empty it in 8 hours. A tap filling 6 litres a minute is opened, and now the cistern empties in 12 hours. What is the capacity of the cistern?

(a) 5,760 litres **(b)** 6,480 litres **(c)** 7,200 litres **(d)** 8,640 litres

Q16 Two pipes A and B fill a tank in 6 and 8 hours. Both are opened, and A is closed after 2 hours. How much longer does B take to fill the tank?

(a) 3 hours **(b)** 4 hours **(c)** 5 hours **(d)** 3.33 hours

17.3.3 Advanced level

Q17 Pipes A and B fill a tank in 10 and 15 hours, and pipe C empties it in 12 hours. They are opened at 6 a.m., 7 a.m. and 8 a.m. respectively. When will the tank be full?

(a) 4:00 p.m. **(b)** 5:12 p.m. **(c)** 3:48 p.m. **(d)** 4:48 p.m.

Q18 A tank has two inlets that fill it in 24 and 36 minutes and an outlet that empties it in 30 minutes. All are opened when the tank is empty. After how long will the tank be full?

(a) 20 minutes **(b)** 36 minutes **(c)** 27.69 minutes **(d)** 24 minutes

Q19 Two pipes can fill a tank in 10 and 12 hours, and an outlet empties it in 20 hours. All three are opened and, after 2 hours, the outlet is closed. How long does it take in total to fill the tank?

(a) 6.5 hours **(b)** 6 hours **(c)** 7 hours **(d)** 5 hours

Q20 A tank is half full. Pipe A fills the whole tank in 12 minutes and pipe B empties it in 8 minutes. If both are opened, after how long will the tank be empty?

(a) 24 minutes **(b)** 12 minutes **(c)** 6 minutes **(d)** 48 minutes

Q21 One pipe fills a tank 3 times as fast as another. Together they fill it in 36 minutes. How long does the slower pipe alone take?

(a) 108 minutes **(b)** 144 minutes **(c)** 72 minutes **(d)** 48 minutes

Q22 Pipes A and B fill a tank in 20 and 30 minutes. Both are opened together, but after some time A is closed and the tank is full 18 minutes after the start. After how many minutes was A closed?

(a) 6 minutes **(b)** 8 minutes **(c)** 10 minutes **(d)** 12 minutes

Q23 A pipe fills a tank in 3 hours. Due to a leak, it takes 3½ hours. If the tank is full, how long will the leak take to empty it?

(a) 21 hours **(b)** 14 hours **(c)** 7 hours **(d)** 24 hours

Q24 A tap fills a tank in 6 hours. After half the tank is filled, three more identical taps are opened. What is the total time to fill the tank?

(a) 3.75 hours **(b)** 4 hours **(c)** 3.5 hours **(d)** 4.5 hours

Q25 Two pipes fill a cistern in 14 and 16 hours. Opened together, they take 32 minutes longer because of a leak. How long would the leak take to empty the full cistern?

(a) 96 hours **(b)** 120 hours **(c)** 144 hours **(d)** 112 hours

17.4 Solutions

Q1 **Answer: (d) 6.67 hours**

LCM 60: 5 + 4 = 9 units/h → 60/9 = **6.67 hours** (6 h 40 min).

Q2 **Answer: (c) 30 hours**

1/10 − 1/15 = 1/30 → **30 hours**.

Q3 **Answer: (d) 24 hours**

1/6 − 1/8 = 1/24 → **24 hours**.

Q4 **Answer: (a) 5 hours**

LCM 30: 3 + 2 + 1 = 6 units/h → **5 hours**.

Q5 **Answer: (a) 12 minutes**

20 × 30/50 = **12 minutes**.

Q6 **Answer: (b) 15 hours**

3 × 5 = **15 hours**.

Q7 **Answer: (a) 10 hours**

LCM 60: 5 + 4 − 3 = 6 units/h → **10 hours**.

Q8 **Answer: (c) 3/8**

3 × 1/8 = **3/8**.

Q9 **Answer: (a) 8.8 hours**

4 × 9 = 36 of 60 units; A fills 24 more in 24/5 = 4.8 h. Total **8.8 hours**.

Q10 **Answer: (a) 13.25 hours**

6 cycles fill 54 units in 12 h; A fills 5 in hour 13 (59); B takes 1/4 h for the last unit: **13.25 hours**.

Q11 **Answer: (a) 12 minutes**

LCM 60: 4 + 3 − 2 = 5 units/min → **12 minutes**.

Q12 **Answer: (d) 50 minutes**

After 5 minutes, 3/4 remains, which should take 15 minutes but takes 25. Net rate = (3/4)/25 = 3/100; leak = 1/20 − 3/100 = 2/100 = 1/50 → **50 minutes**.

Q13 **Answer: (d) 36 minutes**

Rates 2 : 1 → 3 units/min × 12 = 36 units; B alone 36/1 = **36 minutes**.

Q14 **Answer: (b) 7 hours**

LCM 60: A 5/h, B 4/h, C 3/h. Every 2 hours: 10 + 4 + 3 = 17 units; 3 cycles (6 h) = 51; hour 7 (A + B) adds 9 → 60. **7 hours**.

Q15 **Answer: (d) 8,640 litres**

Tap rate = 1/8 − 1/12 = 1/24 of the cistern per hour = 360 litres/h → capacity = 360 × 24 = **8,640 litres**.

Q16 **Answer: (d) 3.33 hours**

LCM 24: A 4/h, B 3/h. 2 hours → 14 units; B fills the remaining 10 in 10/3 = **3.33 hours** (3 h 20 min).

Q17 **Answer: (d) 4:48 p.m.**

LCM 60: A +6, B +4, C −5 units/h. 6–7 a.m.: 6 units; 7–8 a.m.: 10 units (16 so far). From 8 a.m. the net rate is 5 units/h, so the remaining 44 units take 8.8 h = 8 h 48 min: **4:48 p.m.**

Q18 **Answer: (c) 27.69 minutes**

LCM 360: 15 + 10 − 12 = 13 units/min → 360/13 = **27.69 minutes**.

Q19 **Answer: (b) 6 hours**

LCM 60: +6, +5, −3. In 2 hours: 16 units. The remaining 44 units at 11/h take 4 h. Total **6 hours**.

Q20 **Answer: (b) 12 minutes**

Net emptying = 1/8 − 1/12 = 1/24 per minute. Half a tank takes **12 minutes**.

Q21 **Answer: (b) 144 minutes**

Rates 3 : 1 → 4 units/min × 36 = 144 units; slower pipe: **144 minutes**.

Q22 **Answer: (b) 8 minutes**

B works all 18 minutes: 18/30 = 3/5. A fills the remaining 2/5, which takes 2/5 × 20 = **8 minutes**.

Q23 **Answer: (a) 21 hours**

1/3 - 2/7 = 1/21 → **21 hours**.

Q24 **Answer: (a) 3.75 hours**

Half the tank takes 3 h. Then 4 taps fill at 4/6 per hour: half a tank in 3/4 h. Total **3.75 hours** (3 h 45 min).

Q25 **Answer: (d) 112 hours**

Together without the leak: 14 × 16/30 = 112/15 h = 7 h 28 min; with the leak 8 h. Leak rate = 15/112 - 1/8 = 1/112 → **112 hours**.

Inlets add, outlets subtract: net rate = Σ1/inlet - Σ1/outlet. Use the LCM of the times as total units. Work phase by phase when pipes are opened or closed at different times; run full cycles for alternate opening and finish pipe by pipe. A leak's rate = normal rate - observed rate.

MODULE 18 · INTERMEDIATE

Time, speed and distance

Minimum duration **2 days** | Level **Intermediate**

This is one of the largest aptitude topics, with many variations built on one simple formula. This module covers distance = speed × time, unit conversion, average speed for equal distances and equal times, relative speed, late and early arrival, meeting points, speed and time ratios, stoppage time, circular tracks, chase problems and clocks as a speed problem, followed by 25 solved questions computed exactly.

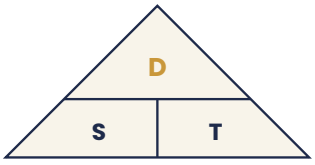
IN THIS MODULE

18.1 Concepts
18.2 Average speed
18.3 Relative speed
18.4 Problem types
18.5 Practice questions
18.6 Solutions

18.1 Concepts

18.1.1 Distance = speed × time

Distance = Speed × Time; Speed = Distance/Time; Time = Distance/Speed



COVER THE QUANTITY YOU WANT: D = S × T, S = D/T, T = D/S

18.1.2 Unit conversion

1 km/h = 5/18 m/s and 1 m/s = 18/5 km/h. So 72 km/h = 20 m/s and 15 m/s = 54 km/h.

18.2 Average speed

18.2.1 Equal distances

Average speed over equal distances at speeds x and y = 2xy/(x + y)

It is the harmonic mean, always less than the simple average (x + y)/2. For three equal distances: 3xyz/(xy + yz + zx).

18.2.2 Equal times

If equal **times** are spent at speeds x and y, the average speed is the simple average (x + y)/2.

> **WATCH OUT**
> Average speed is total distance ÷ total time. Never average the speeds unless the times are equal.

18.3 Relative speed

18.3.1 Same and opposite directions

Directions	Relative speed	Use
Opposite (towards each other)	u + v	time to meet = gap/(u + v)
Same direction	|u - v|	time to catch up = gap/(u - v)



MEETING FROM OPPOSITE ENDS

18.4 Problem types

18.4.1 Late and early arrival

If a person is late by t_1 at speed s_1 and early by t_2 at speed s_2, the distance D satisfies $D/s_1 - D/s_2 = t_1 + t_2$. Walking at 4 km/h a boy is 10 minutes late; at 5 km/h he is 5 minutes early: $D(1/4 - 1/5) = 15/60 \rightarrow D = 5$ km.

18.4.2 Meeting point

Two people starting together from opposite ends cover the distance in the ratio of their speeds. If they meet and then take t_1 and t_2 more to reach the other ends, their speeds are in the ratio $\sqrt{t_2} : \sqrt{t_1}$.

18.4.3 Speed and time ratios

For a fixed distance, time ∝ 1/speed. If speed becomes 3/4 of the usual, time becomes 4/3 of the usual; if this makes a person 20 minutes late, the usual time is 60 minutes.

18.4.4 Stoppage time per hour

Stoppage time per hour = (speed without stops - speed with stops)/speed without stops × 60 minutes

A bus averages 60 km/h without stops and 45 km/h with stops: it stops 15/60 × 60 = 15 minutes per hour.

18.4.5 Circular tracks

On a track of length L, two runners meet for the first time after L/(u + v) if running in opposite directions and L/(u - v) if in the same direction. They are together at the start again after the LCM of their lap times.

18.4.6 Chase problems

A thief has a head start of d; the police chase at a higher speed. Time to catch = d/(police speed - thief speed). Remember to convert the head start (often given as time) into a distance.

18.4.7 Clocks as a speed problem

The minute hand gains 5.5° per minute on the hour hand, so the hands coincide every 360/5.5 = 65 5/11 minutes. This is a relative-speed problem; see the clocks module of Logical Reasoning for details.

18.5 Practice questions

18.5.1 Basic level

Q1 A car travels 180 km in 3 hours. What is its speed?

(a) 50 km/h **(b)** 54 km/h **(c)** 60 km/h **(d)** 90 km/h

Q2 Convert 72 km/h into m/s.

(a) 20 m/s **(b)** 18 m/s **(c)** 25 m/s **(d)** 72 m/s

Q3 Convert 15 m/s into km/h.

(a) 45 km/h **(b)** 54 km/h **(c)** 50 km/h **(d)** 60 km/h

Q4 A man goes to a place at 40 km/h and returns at 60 km/h. What is his average speed?

(a) 50 km/h **(b)** 48 km/h **(c)** 45 km/h **(d)** 52 km/h

Q5 Two cars start 300 km apart and drive towards each other at 40 km/h and 60 km/h. After how long do they meet?

(a) 2 hours **(b)** 3 hours **(c)** 5 hours **(d)** 7.5 hours

Q6 How long does a car at 54 km/h take to cover 270 km?

(a) 5 hours **(b)** 4 hours **(c)** 6 hours **(d)** 4.5 hours

Q7 A man walks 6 km in 1 hour 30 minutes. What is his speed?

(a) 3 km/h **(b)** 4 km/h **(c)** 4.5 km/h **(d)** 5 km/h

Q8 A bus averages 60 km/h without stops and 45 km/h with stops. How many minutes per hour does it stop?

(a) 10 minutes **(b)** 20 minutes **(c)** 25 minutes **(d)** 15 minutes

18.5.2 Intermediate level

Q9 Walking at 4 km/h a boy reaches school 10 minutes late; at 5 km/h he is 5 minutes early. How far is the school?

(a) 5 km **(b)** 4 km **(c)** 6 km **(d)** 7.5 km

Q10 A thief runs at 8 km/h. A policeman starts chasing 10 minutes later at 10 km/h. How long after the policeman starts does he catch the thief?

(a) 40 minutes **(b)** 30 minutes **(c)** 50 minutes **(d)** 60 minutes

Q11 Walking at 3/4 of his usual speed, a man is 20 minutes late. What is his usual time?

(a) 60 minutes **(b)** 45 minutes **(c)** 80 minutes **(d)** 30 minutes

Q12 Two runners on a 400 m circular track run in opposite directions at 5 m/s and 3 m/s. After how many seconds do they first meet?

(a) 80 seconds **(b)** 100 seconds **(c)** 50 seconds **(d)** 40 seconds

Q13 A car covers the first 60 km at 30 km/h and the next 60 km at 60 km/h. What is its average speed?

(a) 45 km/h **(b)** 42 km/h **(c)** 35 km/h **(d)** 40 km/h

Q14 A man travels for 2 hours at 40 km/h and 3 hours at 60 km/h. What is his average speed?

(a) 50 km/h **(b)** 48 km/h **(c)** 52 km/h **(d)** 55 km/h

Q15 Two trains start at the same time from A and B, 450 km apart, towards each other. After meeting, they take 4 hours and 9 hours to reach B and A. What is the ratio of their speeds (train from A : train from B)?

(a) 3 : 2 **(b)** 2 : 3 **(c)** 4 : 9 **(d)** 9 : 4

Q16 Ravi covers a distance at 12 km/h and returns at 8 km/h, taking 5 hours in all. What is the distance one way?

(a) 20 km **(b)** 24 km **(c)** 30 km **(d)** 25 km

18.5.3 Advanced level

Q17 A man covers three equal distances at 10, 20 and 30 km/h. What is his average speed?

(a) 20 km/h **(b)** 15 km/h **(c)** 18 km/h **(d)** 16.36 km/h

Q18 If a man increases his speed by 25%, he takes 30 minutes less. What is his usual time?

(a) 120 minutes **(b)** 125 minutes **(c)** 180 minutes **(d)** 150 minutes

Q19 Two people A and B start from P and Q, 60 km apart, towards each other at 4 km/h and 6 km/h. A dog starts with A at 10 km/h, runs to B, back to A, and so on until they meet. How far does the dog run?

(a) 60 km **(b)** 50 km **(c)** 40 km **(d)** 80 km

Q20 A and B run on a 600 m circular track in the same direction at 6 m/s and 4 m/s. After how long will A first overtake B (lapping him)?

(a) 300 seconds **(b)** 150 seconds **(c)** 60 seconds **(d)** 120 seconds

Q21 A car travels from P to Q at 50 km/h and returns at 75 km/h. If the round trip takes 5 hours, how far apart are P and Q?

(a) 150 km **(b)** 120 km **(c)** 125 km **(d)** 180 km

Q22 A train leaves at 6 a.m. at 60 km/h. Another leaves the same station at 8 a.m. in the same direction at 80 km/h. At what time does the second catch the first?

(a) 12 noon **(b)** 2 p.m. **(c)** 1 p.m. **(d)** 4 p.m.

Q23 Excluding stoppages a train runs at 54 km/h; including stoppages, at 45 km/h. For how many minutes does it stop per hour?

(a) 9 minutes **(b)** 12 minutes **(c)** 15 minutes **(d)** 10 minutes

Q24 A and B start at the same time from two points 68 km apart, towards each other. A walks at 4 km/h; B walks 1 km in the first hour, 2 km in the second, 3 km in the third, and so on. After how many hours do they meet?

(a) 7 hours **(b)** 9 hours **(c)** 10 hours **(d)** 8 hours

Q25 A person cycles from home to office at 15 km/h and is 6 minutes late; at 20 km/h he is 4 minutes early. What is the distance?

(a) 8 km **(b)** 12 km **(c)** 10 km **(d)** 15 km

18.6 Solutions

Q1 **Answer: (c) 60 km/h**

180/3 = **60 km/h**.

Q2 **Answer: (a) 20 m/s**

72 × 5/18 = **20 m/s**.

Q3 **Answer: (b) 54 km/h**

15 × 18/5 = **54 km/h**.

Q4 **Answer: (b) 48 km/h**

2 × 40 × 60/100 = **48 km/h**.

Q5 **Answer: (b) 3 hours**

300/(40 + 60) = **3 hours**.



Q6 **Answer: (a) 5 hours**

270/54 = **5 hours**.

Q7 **Answer: (b) 4 km/h**

6/1.5 = **4 km/h**.

Q8 **Answer: (d) 15 minutes**

(60 - 45)/60 × 60 = **15 minutes**.

Q9 **Answer: (a) 5 km**

D(1/4 - 1/5) = 15/60 → D/20 = 1/4 → D = **5 km**.

Q10 **Answer: (a) 40 minutes**

Head start = 8 × 10/60 = 4/3 km. Relative speed 2 km/h → (4/3)/2 = 2/3 h = **40 minutes**.

Q11 **Answer: (a) 60 minutes**

Time becomes 4/3 of usual: extra 1/3 of usual = 20 min → usual = **60 minutes**.

Q12 **Answer: (c) 50 seconds**

400/(5 + 3) = **50 seconds**.

Q13 **Answer: (d) 40 km/h**

Times 2 h and 1 h; 120/3 = **40 km/h**.

Q14 **Answer: (c) 52 km/h**

Distance 260 km in 5 h = **52 km/h**.

Q15 **Answer: (a) 3 : 2**

Speeds are in the ratio $\sqrt{t_2} : \sqrt{t_1} = \sqrt{9} : \sqrt{4}$ = **3 : 2**.

Q16 **Answer: (b) 24 km**

D/12 + D/8 = 5 → 5D/24 = 5 → D = **24 km**.

Q17 **Answer: (d) 16.36 km/h**

3xyz/(xy + yz + zx) = 18000/1100 = **16.36 km/h**.

Q18 **Answer: (d) 150 minutes**

New time = 4/5 of usual; the saving 1/5 of usual = 30 min → usual = **150 minutes**.

Q19 **Answer: (a) 60 km**

They meet after 60/10 = 6 hours; the dog runs 10 × 6 = **60 km**.

Q20 **Answer: (a) 300 seconds**

Relative speed 2 m/s; to gain a full lap: 600/2 = **300 seconds**.

Q21 **Answer: (a) 150 km**

D/50 + D/75 = 5 → D(5/150) = 5 → D = **150 km**.

Q22 **Answer: (b) 2 p.m.**

Head start 120 km; relative speed 20 km/h → 6 hours after 8 a.m.: **2 p.m.**

Q23 **Answer: (d) 10 minutes**

9/54 × 60 = **10 minutes**.

Q24 **Answer: (d) 8 hours**

After n hours: $4n + n(n + 1)/2 = 68 \rightarrow n^2 + 9n - 136 = 0 \rightarrow (n - 8)(n + 17) = 0 \rightarrow n =$ **8**.

Q25 **Answer: (c) 10 km**

$D(1/15 - 1/20) = 10/60 \rightarrow D/60 = 1/6 \rightarrow D =$ **10 km**.

D = S × T; km/h × 5/18 = m/s. Average speed = total distance/total time: 2xy/(x + y) for equal distances, (x + y)/2 for equal times. Relative speed is u + v towards each other and u - v in the same direction. Late/early: $D(1/s_1 - 1/s_2)$ = total time difference. At constant distance, time is inversely proportional to speed.

MODULE 19 · INTERMEDIATE

Boats and streams

Minimum duration **1 day** | Level **Intermediate**

Boats and streams problems apply relative speed to movement with or against a current. This module covers speed in still water and stream speed, downstream and upstream speeds, finding both from given speeds, round trips, time ratios, and swimming and escalator variants, followed by 25 solved questions computed exactly.

IN THIS MODULE

19.1 Concepts
19.2 Problem types
19.3 Practice questions
19.4 Solutions

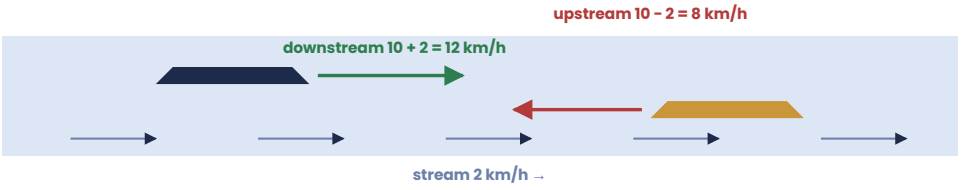
19.1 Concepts

19.1.1 Speed in still water and speed of the stream

Let u be the speed of the boat in still water and v the speed of the stream (current).

19.1.2 Downstream and upstream

Downstream speed D = u + v; upstream speed U = u − v



THE STREAM HELPS DOWNSTREAM AND HINDERS UPSTREAM

19.1.3 Finding u and v

u = (D + U)/2; v = (D − U)/2

A boat goes 24 km downstream in 2 hours and 12 km upstream in 2 hours: D = 12, U = 6, so u = 9 km/h and v = 3 km/h.

19.2 Problem types

19.2.1 Round trips

Time for a round trip of distance d each way = $d/(u + v) + d/(u - v) = 2ud/(u^2 - v^2)$. A boat at 10 km/h in still water with a 2 km/h current takes 24/12 + 24/8 = 5 hours to go 24 km and back.

19.2.2 Time ratios

If going upstream takes k times as long as going downstream over the same distance, then $(u + v)/(u - v) = k$, so $u : v = (k + 1) : (k - 1)$. Upstream twice as long → $u : v = 3 : 1$.

> **SHORTCUT TRICK**
> Time ratio upstream : downstream = speed ratio downstream : upstream. Turn it into u and v with componendo-dividendo.

19.2.3 Swimming and escalators

A swimmer in a river is a boat. An escalator is a stream: walking up a moving-up escalator adds speeds (downstream), walking up a moving-down escalator subtracts them (upstream). Count steps instead of kilometres.

19.3 Practice questions

19.3.1 Basic level

Q1 A boat's speed in still water is 10 km/h and the stream flows at 2 km/h. What is its downstream speed?

(a) 8 km/h **(b)** 10 km/h **(c)** 20 km/h **(d)** 12 km/h

Q2 With the same boat and stream (10 and 2 km/h), what is its upstream speed?

(a) 12 km/h **(b)** 6 km/h **(c)** 5 km/h **(d)** 8 km/h

Q3 A boat goes downstream at 15 km/h and upstream at 9 km/h. What is its speed in still water?

(a) 12 km/h **(b)** 3 km/h **(c)** 6 km/h **(d)** 24 km/h

Q4 In the previous question, what is the speed of the stream?

(a) 3 km/h **(b)** 6 km/h **(c)** 12 km/h **(d)** 2 km/h

Q5 A boat covers 24 km downstream in 2 hours. What is its downstream speed?

(a) 24 km/h **(b)** 6 km/h **(c)** 48 km/h **(d)** 12 km/h

Q6 A man rows 18 km upstream in 3 hours and the stream is 2 km/h. What is his speed in still water?

(a) 6 km/h **(b)** 4 km/h **(c)** 8 km/h **(d)** 10 km/h

Q7 A boat at 10 km/h in still water goes 24 km downstream and back in a 2 km/h stream. How long does the round trip take?

(a) 4.8 hours **(b)** 4 hours **(c)** 5 hours **(d)** 6 hours

Q8 A swimmer swims at 5 km/h in still water. The river flows at 1 km/h. How long will he take to swim 12 km downstream?

(a) 3 hours **(b)** 2 hours **(c)** 2.4 hours **(d)** 1.5 hours

19.3.2 Intermediate level

Q9 A man takes twice as long to row upstream as downstream over the same distance. What is the ratio of his speed in still water to the stream speed?

(a) 2 : 1 **(b)** 3 : 1 **(c)** 1 : 3 **(d)** 4 : 1

Q10 A boat goes 30 km downstream in 2 hours and returns in 3 hours. What is the speed of the stream?

(a) 2.5 km/h **(b)** 5 km/h **(c)** 1.5 km/h **(d)** 3 km/h

Q11 A man rows at 6 km/h in still water. The river flows at 2 km/h. He rows to a place and back in 3 hours. How far is the place?

(a) 8 km **(b)** 12 km **(c)** 6 km **(d)** 9 km

Q12 A boat travels 36 km downstream in the same time it takes to go 24 km upstream. If the stream flows at 2 km/h, what is the boat's speed in still water?

(a) 8 km/h **(b)** 10 km/h **(c)** 12 km/h **(d)** 6 km/h

Q13 A man can row 8 km/h in still water. If the river flows at 2 km/h, what is his average speed for a round trip?

(a) 8 km/h **(b)** 7.5 km/h **(c)** 7 km/h **(d)** 7.2 km/h

Q14 Speed downstream is 20 km/h and upstream 12 km/h. How long does the boat take to go 48 km upstream?

(a) 4 hours **(b)** 2.4 hours **(c)** 3 hours **(d)** 6 hours

Q15 A man walks up a moving-up escalator at 1 step per second and reaches the top in 20 seconds after taking 20 steps. When he stands still, the escalator alone takes 60 seconds. How many steps are visible?

(a) 40 **(b)** 20 **(c)** 60 **(d)** 30

Q16 A boat goes 40 km upstream in 8 hours and 36 km downstream in 6 hours. What is its speed in still water?

(a) 5.5 km/h **(b)** 6 km/h **(c)** 5 km/h **(d)** 1 km/h

19.3.3 Advanced level

Q17 A boat's speed in still water is 14 km/h. It goes 48 km downstream and comes back in 7 hours. What is the speed of the stream?

(a) 2 km/h **(b)** 1 km/h **(c)** 3 km/h **(d)** 4 km/h

Q18 A man rows downstream 32 km and upstream 14 km, taking 6 hours each time. What is the speed of the current?

(a) 2 km/h **(b)** 3 km/h **(c)** 1 km/h **(d)** 1.5 km/h

Q19 A motorboat whose speed in still water is 15 km/h goes 30 km downstream and comes back in a total of 4 hours 30 minutes. What is the speed of the stream?

(a) 4 km/h **(b)** 5 km/h **(c)** 6 km/h **(d)** 3 km/h

Q20 A man can row 3 times as fast downstream as upstream. If the stream flows at 4 km/h, what is his speed in still water?

(a) 12 km/h **(b)** 6 km/h **(c)** 8 km/h **(d)** 16 km/h

Q21 A boat goes 6 km upstream and 6 km back downstream in 2 hours 15 minutes. If the stream flows at 2 km/h, what is the boat's speed in still water?

(a) 5 km/h **(b)** 6 km/h **(c)** 8 km/h **(d)** 7 km/h

Q22 Two boats start towards each other from points 90 km apart on a river: one goes downstream at 12 km/h (still water) and the other upstream at 18 km/h (still water). The stream flows at 3 km/h. When do they meet?

(a) 2.5 hours **(b)** 3 hours **(c)** 3.5 hours **(d)** 5 hours

Q23 A swimmer swims 2 km upstream in 1 hour and back in 20 minutes. What is the speed of the current?

(a) 1 km/h **(b)** 3 km/h **(c)** 4 km/h **(d)** 2 km/h

Q24 A boat's speed in still water is 8 km/h. It takes 1 hour more to go 15 km upstream than to go 15 km downstream. What is the speed of the stream?

(a) 3 km/h **(b)** 1 km/h **(c)** 4 km/h **(d)** 2 km/h

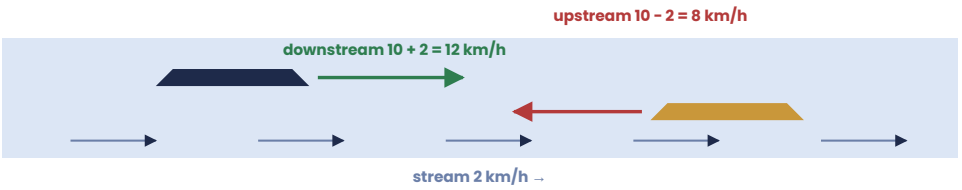
Q25 A man rows to a place 48 km away and back in 14 hours. He can row 4 km downstream in the time he rows 3 km upstream. What is the speed of the stream?

(a) 2 km/h **(b)** 1 km/h **(c)** 1.5 km/h **(d)** 3 km/h

19.4 Solutions

Q1 **Answer: (d) 12 km/h**

10 + 2 = **12 km/h**.



Q2 **Answer: (d) 8 km/h**

10 − 2 = **8 km/h**.

Q3 **Answer: (a) 12 km/h**

(15 + 9)/2 = **12 km/h**.

Q4 **Answer: (a) 3 km/h**

(15 - 9)/2 = **3 km/h**.

Q5 **Answer: (d) 12 km/h**

24/2 = **12 km/h**.

Q6 **Answer: (c) 8 km/h**

Upstream 6 = u - 2 → u = **8 km/h**.

Q7 **Answer: (c) 5 hours**

24/12 + 24/8 = 2 + 3 = **5 hours**.

Q8 **Answer: (b) 2 hours**

12/(5 + 1) = **2 hours**.

Q9 **Answer: (b) 3 : 1**

(u + v)/(u - v) = 2 → u = 3v → **3 : 1**.

Q10 **Answer: (a) 2.5 km/h**

D = 15, U = 10 → v = **2.5 km/h**.

Q11 **Answer: (a) 8 km**

d/8 + d/4 = 3 → 3d/8 = 3 → d = **8 km**.

Q12 **Answer: (b) 10 km/h**

(u + 2)/(u - 2) = 36/24 = 3/2 → 2u + 4 = 3u - 6 → u = **10 km/h**.

Q13 **Answer: (b) 7.5 km/h**

Average of 10 and 6 over equal distances: 2 × 10 × 6/16 = **7.5 km/h**.

Q14 **Answer: (a) 4 hours**

48/12 = **4 hours**.

Q15 **Answer: (d) 30**

Escalator speed = N/60. In 20 s: 20 + 20N/60 = N → 20 = 2N/3 → N = **30**.

Q16 **Answer: (a) 5.5 km/h**

U = 5, D = 6 → u = **5.5 km/h**.

Q17 **Answer: (a) 2 km/h**

48/(14 + v) + 48/(14 - v) = 7 → 1344 = 7(196 - v^2) → v^2 = 4 → v = **2 km/h**.

Q18 **Answer: (d) 1.5 km/h**

D = 16/3, U = 7/3 → v = (16/3 - 7/3)/2 = **1.5 km/h**.

Q19 **Answer: (b) 5 km/h**

30/(15 + v) + 30/(15 - v) = 9/2 → 900 = 4.5(225 - v^2) → v^2 = 25 → v = **5 km/h**.

Q20 **Answer: (c) 8 km/h**

(u + 4) = 3(u - 4) → u = **8 km/h**.

Q21 **Answer: (b) 6 km/h**

6/(u - 2) + 6/(u + 2) = 9/4 → 12u × 4 = 9(u^2 - 4) → $9u^2$ - 48u - 36 = 0 → u = **6 km/h**.

Q22 **Answer: (b) 3 hours**

The stream cancels out: relative speed = 12 + 18 = 30 km/h → **3 hours**.

Q23 **Answer: (d) 2 km/h**

U = 2, D = 6 km/h → v = **2 km/h**.

Q24 **Answer: (d) 2 km/h**

15/(8 - v) - 15/(8 + v) = 1 → 30v = 64 - v^2 → v^2 + 30v - 64 = 0 → v = **2 km/h**.

Q25 **Answer: (b) 1 km/h**

D : U = 4 : 3; write D = 4k, U = 3k. 48/4k + 48/3k = 14 → 28/k = 14 → k = 2. D = 8, U = 6, so v = (8 - 6)/2 = **1 km/h**.

Downstream u + v, upstream u - v; u = (D + U)/2, v = (D - U)/2. Round trip time = d/(u + v) + d/(u - v). Upstream : downstream time = (u + v) : (u - v). Boats moving towards each other on the same river: the stream cancels out. Escalators work exactly like streams.

MODULE 20 · INTERMEDIATE

Problems on trains

Minimum duration **1 day** | Level **Intermediate**

Train problems use relative speed with the length of the train included in the distance. This module covers crossing a pole or a person, crossing a platform or bridge, two trains in the same and opposite directions, trains crossing each other, trains passing moving people or cars, and trains starting from different stations, followed by 25 solved questions computed exactly.

IN THIS MODULE

20.1 Concepts
20.2 Problem types
20.3 Practice questions
20.4 Solutions

20.1 Concepts

20.1.1 Crossing a pole or a person

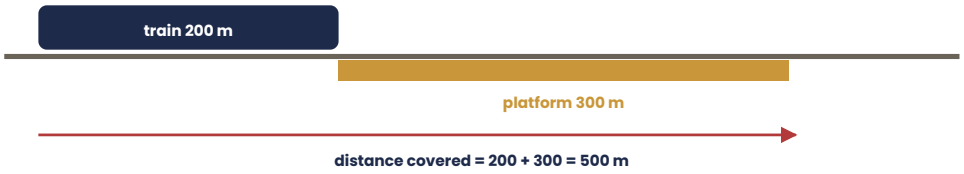
To pass a pole, a standing person or a signal post, the front of the train must travel its own **length**: distance = length of the train.



PASSING A POLE

20.1.2 Crossing a platform or bridge

Distance to cross a platform or bridge = length of train + length of platform



CROSSING A PLATFORM

WATCH OUT
Convert speeds to m/s (× 5/18) when lengths are in metres and times in seconds.

20.2 Problem types

20.2.1 Two trains: same and opposite directions

Time for two trains to cross = $(L_1 + L_2)$/relative speed; relative speed = u + v (opposite) or |u - v| (same direction)

20.2.2 Trains crossing each other

Trains 150 m and 100 m long at 60 km/h and 30 km/h in opposite directions cross in 250/(90 × 5/18) = 250/25 = 10 seconds; in the same direction 250/(30 × 5/18) = 30 seconds.

20.2.3 Passing a moving person or car

A person has (almost) zero length: time = train length/relative speed. A 180 m train at 60 km/h passes a man walking at 6 km/h in the same direction in 180/(54 × 5/18) = 12 s.

20.2.4 Trains starting from different stations

Use relative speed with the starting gap, or write each train's distance from one station as a function of time. If two trains start towards each other from stations 300 km apart at 50 and 70 km/h, they meet after 2.5 hours, 125 km from the first station.

SHORTCUT TRICK
If two trains, after meeting, take t_1 and t_2 hours to reach their destinations, their speeds are in the ratio $\sqrt{t_2} : \sqrt{t_1}$.

20.3 Practice questions

20.3.1 Basic level

Q1 A train 200 m long passes a pole in 10 seconds. What is its speed in km/h?

(a) 20 km/h **(b)** 72 km/h **(c)** 60 km/h **(d)** 80 km/h

Q2 A train 150 m long runs at 54 km/h. How long does it take to pass a pole?

(a) 15 seconds **(b)** 8 seconds **(c)** 10 seconds **(d)** 12 seconds

Q3 A 200 m train at 72 km/h crosses a 300 m platform. How long does it take?

(a) 15 seconds **(b)** 10 seconds **(c)** 30 seconds **(d)** 25 seconds

Q4 A train crosses a 250 m bridge in 30 seconds at 36 km/h. What is its length?

(a) 50 m **(b)** 100 m **(c)** 300 m **(d)** 60 m

Q5 Two trains 150 m and 100 m long run in opposite directions at 60 and 30 km/h. How long do they take to cross each other?

(a) 30 seconds **(b)** 15 seconds **(c)** 12 seconds **(d)** 10 seconds

Q6 The same two trains run in the same direction. How long does the faster take to cross the slower?

(a) 10 seconds **(b)** 30 seconds **(c)** 20 seconds **(d)** 25 seconds

Q7 A 180 m train at 60 km/h passes a man walking at 6 km/h in the same direction. How long does it take?

(a) 10 seconds **(b)** 12 seconds **(c)** 9.82 seconds **(d)** 15 seconds

Q8 A 120 m train passes a man standing on a platform in 6 seconds. What is its speed?

(a) 72 km/h **(b)** 60 km/h **(c)** 20 km/h **(d)** 80 km/h

20.3.2 Intermediate level

Q9 A train passes a pole in 15 seconds and a 100 m platform in 25 seconds. What is its length?

(a) 100 m **(b)** 125 m **(c)** 150 m **(d)** 200 m

Q10 A 240 m train at 54 km/h passes a man running at 6 km/h in the opposite direction. How long does it take?

(a) 16 seconds **(b)** 12 seconds **(c)** 14.4 seconds **(d)** 18 seconds

Q11 Two trains start at the same time from stations 300 km apart and travel towards each other at 50 and 70 km/h. How far from the first station do they meet?

(a) 150 km **(b)** 175 km **(c)** 125 km **(d)** 100 km

Q12 A train 300 m long crosses a 450 m tunnel in 50 seconds? What is its speed in km/h?

(a) 45 km/h **(b)** 60 km/h **(c)** 54 km/h **(d)** 50 km/h

Q13 Two trains of equal length take 10 s and 15 s to pass a pole. How long do they take to cross each other in opposite directions?

(a) 12.5 seconds **(b)** 12 seconds **(c)** 15 seconds **(d)** 25 seconds

Q14 A 100 m train crosses a 150 m train coming from the opposite direction in 9 seconds. If the first runs at 50 km/h, what is the speed of the second?

(a) 50 km/h **(b)** 40 km/h **(c)** 54 km/h **(d)** 60 km/h

Q15 A train passes a station platform in 36 seconds and a man on the platform in 20 seconds. If the train runs at 54 km/h, how long is the platform?

(a) 240 m **(b)** 200 m **(c)** 300 m **(d)** 225 m

Q16 Two trains, after meeting, take 9 hours and 16 hours to reach their destinations. What is the ratio of their speeds?

(a) 3 : 4 **(b)** 4 : 3 **(c)** 16 : 9 **(d)** 9 : 16

20.3.3 Advanced level

Q17 A train overtakes two people walking in the same direction at 2 km/h and 4 km/h in 9 and 10 seconds respectively. What is the length of the train?

(a) 50 m (b) 45 m (c) 60 m (d) 72 m

Q18 In the previous question, what is the speed of the train?

(a) 20 km/h (b) 24 km/h (c) 22 km/h (d) 26 km/h

Q19 A goods train runs at 72 km/h and crosses a 250 m platform in 26 seconds. What is the length of the goods train?

(a) 520 m (b) 270 m (c) 250 m (d) 290 m

Q20 Two trains leave stations A and B, 245 km apart, at 7 a.m. and 8 a.m., travelling towards each other at 25 km/h and 30 km/h. At what time do they meet?

(a) 11 a.m. (b) 12 noon (c) 1 p.m. (d) 10 a.m.

Q21 A train 125 m long passes a man running at 5 km/h in the same direction in 10 seconds. What is the speed of the train?

(a) 45 km/h (b) 55 km/h (c) 40 km/h (d) 50 km/h

Q22 A 150 m train crosses a 250 m train running in the same direction at 60 km/h in 60 seconds. What is the speed of the faster train?

(a) 80 km/h (b) 84 km/h (c) 90 km/h (d) 96 km/h

Q23 Two trains start together from A and B towards each other at 50 and 60 km/h. When they meet, the second has travelled 120 km more than the first. How far apart are A and B?

(a) 1,100 km (b) 1,200 km (c) 1,440 km (d) 1,320 km

Q24 A train at 48 km/h crosses another train, half its length, travelling in the opposite direction at 42 km/h, in 12 seconds. It also passes a platform in 45 seconds. What is the length of the platform?

(a) 500 m (b) 400 m (c) 300 m (d) 600 m

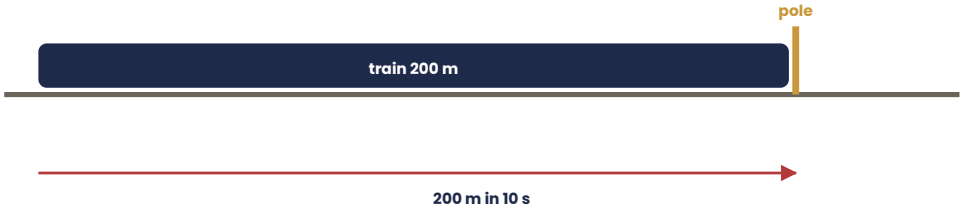
Q25 A train 110 m long passes a man running at 6 km/h in the opposite direction in 6 seconds. What is the speed of the train?

(a) 60 km/h (b) 66 km/h (c) 54 km/h (d) 72 km/h

20.4 Solutions

Q1 **Answer: (b) 72 km/h**

200/10 = 20 m/s = **72 km/h**.



Q2 **Answer: (c) 10 seconds**

54 km/h = 15 m/s; 150/15 = **10 seconds**.

Q3 **Answer: (d) 25 seconds**

500 m at 20 m/s = **25 seconds**.

Q4 **Answer: (a) 50 m**

10 m/s × 30 s = 300 m = train + 250 → train = **50 m**.

Q5 **Answer: (d) 10 seconds**

Relative speed 90 km/h = 25 m/s; 250/25 = **10 seconds**.

Q6 **Answer: (b) 30 seconds**

Relative speed 30 km/h = 25/3 m/s; 250 × 3/25 = **30 seconds**.

Q7 **Answer: (b) 12 seconds**

Relative 54 km/h = 15 m/s; 180/15 = **12 seconds**.

Q8 **Answer: (a) 72 km/h**

120/6 = 20 m/s = **72 km/h**.

Q9 **Answer: (c) 150 m**

In 10 extra seconds it covers 100 m → 10 m/s. Length = 15 × 10 = **150 m**.

Q10 **Answer: (c) 14.4 seconds**

Relative 60 km/h = 50/3 m/s; 240 × 3/50 = **14.4 seconds**.

Q11 **Answer: (c) 125 km**

They meet after 300/120 = 2.5 h; the first has gone 50 × 2.5 = **125 km**.

Q12 **Answer: (c) 54 km/h**

750 m in 50 s = 15 m/s = **54 km/h**.

Q13 **Answer: (b) 12 seconds**

Speeds L/10 and L/15: 2L/(L/10 + L/15) = 2/(1/6) = **12 seconds**.

Q14 **Answer: (a) 50 km/h**

250/9 m/s = 100 km/h relative; second = 100 − 50 = **50 km/h**.

Q15 **Answer: (a) 240 m**

15 m/s; the platform takes the extra 16 s: 15 × 16 = **240 m**.

Q16 **Answer: (b) 4 : 3**

Speeds are in the ratio $\sqrt{16} : \sqrt{9} =$ **4 : 3**.

Q17 **Answer: (a) 50 m**

$L = 9(v - 2) \times 5/18 = 10(v - 4) \times 5/18 \rightarrow 9v - 18 = 10v - 40 \rightarrow v = 22$ km/h. $L = 9 \times 20 \times 5/18 =$ **50 m**.

Q18 **Answer: (c) 22 km/h**

From $9(v - 2) = 10(v - 4)$: $v =$ **22 km/h**.

Q19 **Answer: (b) 270 m**

$20 \times 26 = 520$ m = train + 250 → **270 m**.

Q20 **Answer: (b) 12 noon**

By 8 a.m. the first has gone 25 km; 220 km remain at 25 + 30 = 55 km/h → 4 hours after 8 a.m.: **12 noon**.

Q21 **Answer: (d) 50 km/h**

Relative speed 12.5 m/s = 45 km/h; train = 45 + 5 = **50 km/h**.

Q22 **Answer: (b) 84 km/h**

Relative speed = 400/60 = 20/3 m/s = 24 km/h; faster = 60 + 24 = **84 km/h**.

Q23 **Answer: (d) 1,320 km**

Gap in distance grows at 10 km/h → they meet after 12 h. Total = 110 × 12 = **1,320 km**.

Q24 **Answer: (b) 400 m**

Relative speed 90 km/h = 25 m/s; 25 × 12 = 300 m = L + L/2 → L = 200 m. At 48 km/h = 40/3 m/s, in 45 s it covers 600 m = 200 + platform → platform = **400 m**.

Q25 **Answer: (a) 60 km/h**

110/6 m/s = 66 km/h relative; train = 66 - 6 = **60 km/h**.

Distance = train length (pole, person) or train + platform (platform, bridge, tunnel). For two trains, distance = sum of lengths and speed = sum (opposite) or difference (same direction). Convert km/h to m/s with × 5/18. For trains from different stations, use the head start then relative speed.

MODULE 21 · INTERMEDIATE

Races and games

Minimum duration **1 day** | Level **Intermediate**

Race problems compare the speeds of competitors using head starts and winning margins. This module covers head starts, beating by distance or by time, dead heats, three-runner races, circular races and first meetings, and games of points such as billiards, followed by 25 solved questions computed exactly.

IN THIS MODULE

21.1 Concepts
21.2 Problem types
21.3 Practice questions
21.4 Solutions

21.1 Concepts

21.1.1 Head start

If A gives B a **start** of x metres in a race of L metres, B starts x metres ahead and runs only L − x. If A gives B a start of t seconds, B starts t seconds earlier.

21.1.2 Beating by distance or by time

"A beats B by x metres" means that when A finishes L metres, B has run L − x. "A beats B by t seconds" means B finishes t seconds after A. Both together give B's speed: B runs x metres in t seconds.



A beats B by 50 m: when A finishes 1000 m, B has run 950 m

BEATING BY DISTANCE

In the time A runs L, B runs L − x; so speed ratio A : B = L : (L − x)

21.1.3 Dead heat

A **dead heat** means both reach the finish at the same moment. With a start of x metres for B, a dead heat means A covers L while B covers L − x in the same time.

21.2 Problem types

21.2.1 Three runners

Chain the speed ratios. In a 100 m race A beats B by 10 m and B beats C by 10 m: A : B = 100 : 90 and B : C = 100 : 90, so A : C = 100 : 81: A beats C by 19 m.

> **SHORTCUT TRICK**
> Never add the margins (10 + 10 = 20 is wrong). Multiply the ratios: C runs 90% of 90% = 81% of A's distance.

21.2.2 Circular races and first meeting

On a circular track of length L, runners at speeds a and b meet first after L/(a + b) (opposite directions) or L/(a - b) (same direction), and are together at the start after the LCM of their lap times.

21.2.3 Games of points

"A can give B 20 points in a game of 100" means that when A scores 100, B scores 80, so their scoring rates are 100 : 80. Chain the ratios exactly as in races.

21.3 Practice questions

21.3.1 Basic level

Q1 In a 1000 m race A beats B by 50 m. What is the ratio of their speeds?

(a) 19 : 20 **(b)** 10 : 9 **(c)** 20 : 19 **(d)** 21 : 20

Q2 In a 100 m race A beats B by 10 m and B beats C by 10 m. By how many metres does A beat C?

(a) 20 m **(b)** 18 m **(c)** 19 m **(d)** 21 m

Q3 A runs 100 m in 10 s and B in 12.5 s. By how many metres does A beat B in a 100 m race?

(a) 25 m **(b)** 20 m **(c)** 15 m **(d)** 12.5 m

Q4 In a 200 m race A beats B by 20 m or 4 seconds. What is B's time for 200 m?

(a) 40 seconds **(b)** 36 seconds **(c)** 44 seconds **(d)** 50 seconds

Q5 In the previous race, what is A's time?

(a) 40 seconds **(b)** 36 seconds **(c)** 32 seconds **(d)** 34 seconds

Q6 A can give B a start of 20 m in a 100 m race. What is the ratio of their speeds?

(a) 4 : 5 **(b)** 6 : 5 **(c)** 5 : 4 **(d)** 5 : 3

Q7 In a game of 100 points, A can give B 20 points. How many points does B score when A scores 50?

(a) 30 **(b)** 40 **(c)** 45 **(d)** 35

Q8 Two runners on a 400 m circular track run in opposite directions at 6 m/s and 4 m/s. When do they first meet?

(a) 100 seconds **(b)** 200 seconds **(c)** 40 seconds **(d)** 66.67 seconds

21.3.2 Intermediate level

Q9 A runs 1¾ times as fast as B. If A gives B a start of 60 m, how long should the course be for a dead heat?

(a) 140 m **(b)** 100 m **(c)** 120 m **(d)** 160 m

Q10 In a 1 km race A beats B by 40 m and B beats C by 50 m. By how much does A beat C?

(a) 88 m **(b)** 90 m **(c)** 86 m **(d)** 92 m

Q11 A and B run a 1500 m race on a 300 m circular track at 25 m/s and 15 m/s, starting together. How many times does A overtake B before A finishes?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q12 In a 500 m race the speeds of A and B are in the ratio 3 : 4. A has a start of 140 m. By how much does A win?

(a) 10 m **(b)** 40 m **(c)** 30 m **(d)** 20 m

Q13 In a 100 m race A beats B by 25 m and B beats C by 4 m. By how much does A beat C?

(a) 29 m **(b)** 27 m **(c)** 30 m **(d)** 28 m

Q14 At a game of billiards A can give B 15 points in 60 and A can give C 20 points in 60. How many points can B give C in a game of 90?

(a) 12 **(b)** 15 **(c)** 10 **(d)** 20

Q15 A can run 22.5 m while B runs 25 m. In a kilometre race B beats A by

(a) 50 m **(b)** 100 m **(c)** 75 m **(d)** 125 m

Q16 A and B run around a 600 m circular track starting together in the same direction at 8 m/s and 5 m/s. After how long are they first together again at the starting point?

(a) 120 seconds **(b)** 75 seconds **(c)** 300 seconds **(d)** 600 seconds

21.3.3 Advanced level

Q17 In a 100 m race A gives B a start of 10 m and still beats him by 10 m. What is the ratio of their speeds?

(a) 10 : 9 **(b)** 9 : 8 **(c)** 11 : 9 **(d)** 5 : 4

Q18 A beats B by 30 m in a 300 m race and C by 60 m in the same race. By how much does B beat C in a 270 m race?

(a) 20 m **(b)** 30 m **(c)** 24 m **(d)** 36 m

Q19 In a 1 km race A beats B by 100 m or 20 seconds. What is A's speed?

(a) 5 m/s **(b)** 6 m/s **(c)** 5.56 m/s **(d)** 4.5 m/s

Q20 A, B, C run a 3 km circular race at 6, 5 and 4 km/h, starting together in the same direction. After how many hours are all three together again at the start?

(a) 3 hours **(b)** 1.5 hours **(c)** 6 hours **(d)** 12 hours

Q21 In a 400 m race A gives B a start of 5 seconds and beats him by 15 m. In another 400 m race A beats B by 7 1/7 seconds. What is A's speed?

(a) 8 m/s **(b)** 7 m/s **(c)** 7.5 m/s **(d)** 9 m/s

Q22 A runs twice as fast as B and gives him a start of 50 m. How long must the race be for a dead heat?

(a) 150 m **(b)** 100 m **(c)** 200 m **(d)** 75 m

Q23 In a 1 km race A beats B by 28 m or 7 seconds. What is A's time?

(a) 250 seconds **(b)** 236 seconds **(c)** 243 seconds **(d)** 244 seconds

Q24 Two runners A and B start from the same point on a 500 m circular track and run in opposite directions at 3 m/s and 2 m/s. How many times do they meet in 10 minutes (excluding the start)?

(a) 5 **(b)** 6 **(c)** 12 **(d)** 3

Q25 In a game of 100 points, A can give B 20 points and C 28 points. How many points can B give C in a game of 100?

(a) 10 **(b)** 8 **(c)** 12 **(d)** 9

21.4 Solutions

Q1 **Answer: (c) 20 : 19**

A runs 1000 while B runs 950: 1000 : 950 = **20 : 19**.



B is 50 m short when A finishes

Q2 **Answer: (c) 19 m**

C runs 0.9 × 0.9 = 81% of A's distance: A beats C by **19 m**.

Q3 **Answer: (b) 20 m**

B's speed = 8 m/s; in 10 s B runs 80 m: A wins by **20 m**.

Q4 **Answer: (a) 40 seconds**

B runs 20 m in 4 s → 5 m/s → 200/5 = **40 seconds**.

Q5 **Answer: (b) 36 seconds**

A finishes 4 s before B: 40 − 4 = **36 seconds**.

Q6 **Answer: (c) 5 : 4**

A runs 100 while B runs 80: **5 : 4**.

Q7 **Answer: (b) 40**

Rates 100 : 80; when A scores 50, B scores **40**.

Q8 **Answer: (c) 40 seconds**

400/(6 + 4) = **40 seconds**.

Q9 **Answer: (a) 140 m**

In the same time B runs 4/7 of A's distance. A runs L, B runs L - 60 = 4L/7 → 3L/7 = 60 → L = **140 m**.

Q10 **Answer: (a) 88 m**

C runs 0.96 × 0.95 = 0.912 of A's distance: A beats C by **88 m**.

Q11 **Answer: (b) 2**

A finishes in 1500/25 = 60 s. In that time he gains 60 × 10 = 600 m = 2 laps on B: he overtakes B **2** times.

Q12 **Answer: (d) 20 m**

A runs 360 m; in that time B runs 360 × 4/3 = 480 m. A wins by **20 m**.

Q13 **Answer: (d) 28 m**

C runs 0.75 × 0.96 = 0.72 of A's distance: A beats C by **28 m**.

Q14 **Answer: (c) 10**

A : B = 60 : 45, A : C = 60 : 40, so B : C = 45 : 40. In 90, C scores 80: B gives C **10** points.

Q15 **Answer: (b) 100 m**

Speed ratio A : B = 9 : 10; when B runs 1000 m, A runs 900 m: B wins by **100 m**.

Q16 **Answer: (d) 600 seconds**

Lap times 75 s and 120 s; LCM = **600 seconds**.

Q17 **Answer: (d) 5 : 4**

B runs 100 - 10 - 10 = 80 m while A runs 100: **5 : 4**.

Q18 **Answer: (b) 30 m**

B : C = 270 : 240. When B runs 270, C runs 240: B wins by **30 m**.

Q19 **Answer: (c) 5.56 m/s**

B runs 100 m in 20 s → 5 m/s → B takes 200 s, A 180 s. A's speed = 1000/180 = **5.56 m/s**.

Q20 **Answer: (a) 3 hours**

Lap times 1/2 h, 3/5 h and 3/4 h; the LCM of 1/2, 3/5, 3/4 = LCM(1, 3, 3)/HCF(2, 5, 4) = 3/1 = **3 hours**.

Q21 **Answer: (a) 8 m/s**

With the 5 s start A wins by 50/7 - 5 = 15/7 s less, which equals the 15 m B is short: B's speed = 15/(15/7) = 7 m/s. B takes 400/7 s, A takes 400/7 - 50/7 = 50 s → A's speed = **8 m/s**.

Q22 **Answer: (b) 100 m**

B runs half of A's distance: L - 50 = L/2 → L = **100 m**.

Q23 **Answer: (c) 243 seconds**

B: 28 m in 7 s = 4 m/s → 250 s. A = 250 - 7 = **243 seconds**.

Q24 **Answer: (b) 6**

They meet every 500/5 = 100 s; in 600 s they meet **6** times.

Q25 **Answer: (a) 10**

B : C = 80 : 72. When B scores 100, C scores 90: B gives C **10** points.

"A beats B by x m in L m" means speed ratio L : (L − x). Beating by x metres and t seconds together give B's speed x/t. Chain ratios for three runners (multiply, never add margins). Circular tracks: first meeting L/(a ± b); together at the start after the LCM of lap times. Points games use the same ratios.

MODULE 22 · INTERMEDIATE

Algebra: equations and inequalities

Minimum duration **2 days** | Level **Intermediate**

Algebra skills turn word problems into equations and appear directly in many tests. This module covers linear equations in one and two variables, consistency of systems, word problems, quadratic equations (factorisation, formula, discriminant, sum and product of roots, forming equations), the bank-exam "compare x and y" questions, remainder and factor theorems, linear, quadratic and modulus inequalities, and functions, followed by 25 solved questions checked by a program.

IN THIS MODULE

22.1 Linear equations
22.2 Quadratic equations
22.3 Polynomials
22.4 Inequalities
22.5 Functions
22.6 Practice questions
22.7 Solutions

22.1 Linear equations

22.1.1 Equations in one variable

Collect the variable terms on one side and the constants on the other. $3(x - 4) + 5 = 2x + 7 \rightarrow 3x - 7 = 2x + 7 \rightarrow x = 14$.

22.1.2 Two variables: substitution and elimination

Elimination: multiply the equations so one variable has equal coefficients, then subtract. $2x + 3y = 13$ and $3x - y = 3$: multiply the second by 3 and add: $11x = 22 \rightarrow x = 2, y = 3$.

22.1.3 Consistency of systems

For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$	Solutions
$a_1/a_2 \neq b_1/b_2$	exactly one (lines cross)
$a_1/a_2 = b_1/b_2 = c_1/c_2$	infinitely many (same line)
$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	none (parallel lines)

22.1.4 Word problems

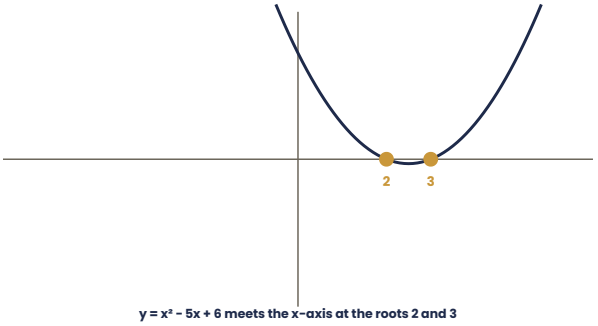
Name the unknowns, translate each sentence into an equation, solve, and check against the story. Common types: digits, coins, ages, tickets, and "the sum of two numbers is … and their difference is …".

22.2 Quadratic equations

22.2.1 Factorisation and the formula

$ax^2 + bx + c = 0 \rightarrow x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$

Factorise by splitting the middle term: $x^2 - 5x + 6 = (x - 2)(x - 3) \rightarrow x = 2$ or 3.



ROOTS OF A QUADRATIC ARE WHERE ITS GRAPH MEETS THE X-AXIS

22.2.2 Nature of roots

Discriminant $D = b^2 - 4ac$	Roots
$D > 0$, perfect square	two distinct rational roots
$D > 0$, not a perfect square	two distinct irrational roots
$D = 0$	two equal roots
$D < 0$	no real roots

22.2.3 Sum and product of roots

Sum of roots = $-b/a$; product of roots = c/a

22.2.4 Forming equations from roots

With roots α and β: $x^2 - (\alpha + \beta)x + \alpha\beta = 0$. Roots 3 and $-5 \rightarrow x^2 + 2x - 15 = 0$.

22.2.5 Comparing two quantities (bank-exam style)

Solve both equations, then compare every root of x with every root of y. The five standard answers are:

Option	Meaning
(a)	$x > y$
(b)	$x < y$
(c)	$x \geq y$

Option	Meaning
(d)	x ≤ y
(e)	x = y or the relation cannot be established

WORKED EXAMPLE

I. $x^2 - 7x + 12 = 0$ II. $y^2 - 5y + 6 = 0$

x = 3 or 4; y = 2 or 3. Every x ≥ every y (3 = 3 is possible), so **x ≥ y**.

SHORTCUT TRICK

Sign shortcut: if both equations have the pattern + + (b and c positive), the roots are negative; - + gives positive roots. A quick sign check often decides the comparison before solving.

22.3 Polynomials

22.3.1 Remainder and factor theorems

The remainder when p(x) is divided by (x - a) is p(a). If p(a) = 0, then (x - a) is a factor. Remainder of $x^3 - 2x^2 + 5 \div (x - 2) = 8 - 8 + 5 = 5$.

22.4 Inequalities

22.4.1 Linear inequalities

Solve like equations, but **reverse the sign** when multiplying or dividing by a negative number: $-2x > 6 \rightarrow x < -3$.

22.4.2 Quadratic inequalities

Find the roots and test the intervals: $(x - 2)(x - 5) < 0$ holds between the roots, $2 < x < 5$; > 0 holds outside them.

22.4.3 Modulus equations and inequalities

$|x - 3| = 5 \rightarrow x = 8$ or -2. $|x - 3| < 5 \rightarrow -2 < x < 8$. $|x| > a \rightarrow x > a$ or $x < -a$.

22.5 Functions

22.5.1 Domain, range, maximum and minimum

The minimum of $ax^2 + bx + c$ $(a > 0)$ is at $x = -b/(2a)$ and equals $c - b^2/(4a)$. $x^2 - 6x + 13$ has minimum 4 at x = 3. The domain of $\sqrt{(x - 2)}$ is $x \geq 2$.

22.6 Practice questions

22.6.1 Basic level

Q1 Solve $3(x - 4) + 5 = 2x + 7$.

(a) 12 **(b)** 10 **(c)** 14 **(d)** 16

Q2 If $2x + 3y = 13$ and $3x - y = 3$, what is $x + y$?

(a) 4 **(b)** 5 **(c)** 6 **(d)** 7

Q3 What are the roots of $x^2 - 5x + 6 = 0$?

(a) -2 and -3 **(b)** 1 and 6 **(c)** -1 and 6 **(d)** 2 and 3

Q4 What is the sum of the roots of $2x^2 - 8x + 5 = 0$?

(a) -4 **(b)** 5/2 **(c)** 4 **(d)** 8

Q5 What is the product of the roots of $3x^2 + 7x - 6 = 0$?

(a) 2 **(b)** -7/3 **(c)** 6 **(d)** -2

Q6 Solve $-2x > 6$.

(a) $x > -3$ **(b)** $x > 3$ **(c)** $x < 3$ **(d)** $x < -3$

Q7 Which quadratic has roots 3 and -5?

(a) $x^2 + 2x - 15 = 0$ **(b)** $x^2 - 2x - 15 = 0$
(c) $x^2 + 2x + 15 = 0$ **(d)** $x^2 - 8x + 15 = 0$

Q8 Solve $|x - 3| = 5$.

(a) $x = 8$ only **(b)** $x = -8$ or 2
(c) $x = 8$ or $x = -2$ **(d)** $x = 2$ or 8

22.6.2 Intermediate level

Q9 I. $x^2 - 7x + 12 = 0$ II. $y^2 - 5y + 6 = 0$

(a) $x > y$
(b) $x < y$
(c) $x \geq y$
(d) $x \leq y$
(e) $x = y$ or the relation cannot be established

Q10 I. $x^2 + 7x + 12 = 0$ II. $y^2 + 9y + 20 = 0$

(a) $x > y$
(b) $x < y$
(c) $x \geq y$
(d) $x \leq y$
(e) $x = y$ or the relation cannot be established

Q11 I. $2x^2 - 11x + 15 = 0$ II. $y^2 - 6y + 9 = 0$

(a) $x > y$
(b) $x < y$
(c) $x \geq y$
(d) $x \leq y$
(e) $x = y$ or the relation cannot be established

Q12 I. $x^2 - 1x - 12 = 0$ II. $y^2 - 8y + 15 = 0$

(a) $x > y$
(b) $x < y$
(c) $x \geq y$
(d) $x \leq y$
(e) $x = y$ or the relation cannot be established

Q13 For what value of k does $x^2 - kx + 9 = 0$ have equal roots ($k > 0$)?

(a) 3 **(b)** 9 **(c)** 18 **(d)** 6

Q14 What is the remainder when $x^3 - 2x^2 + 5$ is divided by $(x - 2)$?

(a) 1 **(b)** 13 **(c)** 0 **(d)** 5

Q15 Solve $(x - 2)(x - 5) < 0$.

(a) $2 < x < 5$ **(b)** $x < 2$ or $x > 5$ **(c)** $x > 5$ **(d)** $-5 < x < -2$

Q16 The sum of two numbers is 25 and their difference is 7. What is the larger number?

(a) 9 **(b)** 16 **(c)** 18 **(d)** 14

22.6.3 Advanced level

Q17 I. $x^2 + 5x + 6 = 0$ II. $y^2 + 3y + 2 = 0$

(a) $x > y$
(b) $x < y$
(c) $x \geq y$
(d) $x \leq y$
(e) $x = y$ or the relation cannot be established

Q18 I. $x^2 - 9x + 20 = 0$ II. $y^2 - 11y + 30 = 0$

(a) $x > y$
(b) $x < y$
(c) $x \geq y$
(d) $x \leq y$
(e) $x = y$ or the relation cannot be established

Q19 For what value of k do the equations $2x + 3y = 5$ and $4x + ky = 10$ have infinitely many solutions?

(a) 3 **(b)** 9 **(c)** 12 **(d)** 6

Q20 For what value of k do $3x - y = 4$ and $6x - 2y = k$ have no solution?

(a) $k = 8$ **(b)** any $k \neq 8$ **(c)** $k = 4$ **(d)** no such k

Q21 If α and β are the roots of $x^2 - 5x + 6 = 0$, what is $\alpha^2 + \beta^2$?

(a) 25 **(b)** 12 **(c)** 13 **(d)** 1

Q22 What is the minimum value of $x^2 - 6x + 13$?

(a) 3 **(b)** 13 **(c)** 0 **(d)** 4

Q23 Solve $|2x - 1| < 5$.

(a) $x < 3$ **(b)** $-2 < x < 3$ **(c)** $-3 < x < 2$ **(d)** $x > -2$

Q24 A two-digit number is 4 times the sum of its digits, and its units digit is 2 more than its tens digit. Find the number.

(a) 24 **(b)** 36 **(c)** 48 **(d)** 12

Q25 If $(x - 1)$ is a factor of $x^3 + kx^2 - 4x + 2$, what is k?

(a) 1 **(b)** -1 **(c)** 2 **(d)** 3

22.7 Solutions

Q1 **Answer: (c) 14**

$3x - 7 = 2x + 7 \rightarrow x =$ **14**.

Q2 **Answer: (b) 5**

$x = 2, y = 3 \rightarrow$ **5**.

Q3 **Answer: (d) 2 and 3**

$(x - 2)(x - 3) = 0 \rightarrow$ **2 and 3**.

Q4 **Answer: (c) 4**

$-b/a = 8/2 =$ **4**.

Q5 **Answer: (d) -2**

$c/a = -6/3 =$ **-2**.

Q6 **Answer: (d) $x < -3$**

Dividing by -2 reverses the sign: **$x < -3$**.

Q7 **Answer: (a) $x^2 + 2x - 15 = 0$**

Sum -2, product -15: $x^2 - (-2)x + (-15) =$ **$x^2 + 2x - 15 = 0$**.

Q8 **Answer: (c) $x = 8$ or $x = -2$**

$x - 3 = \pm 5 \rightarrow$ **$x = 8$ or $x = -2$**.

Q9 **Answer: (c) x ≥ y**

x = 3, 4; y = 2, 3. Comparing every pair gives **x ≥ y**.

Q10 **Answer: (c) x ≥ y**

x = -4, -3; y = -5, -4. Comparing every pair gives **x ≥ y**.

Q11 **Answer: (d) x ≤ y**

x = 5/2, 3; y = 3. Comparing every pair gives **x ≤ y**.

Q12 **Answer: (e) x = y or the relation cannot be established**

x = -3, 4; y = 3, 5. Comparing every pair gives **x = y or the relation cannot be established**.

Q13 **Answer: (d) 6**

$D = k^2 - 36 = 0 \rightarrow k = \mathbf{6}$.

Q14 **Answer: (d) 5**

$p(2) = 8 - 8 + 5 = \mathbf{5}$.

Q15 **Answer: (a) 2 < x < 5**

The product is negative between the roots: **2 < x < 5**.

Q16 **Answer: (b) 16**

$(25 + 7)/2 = \mathbf{16}$.

Q17 **Answer: (d) x ≤ y**

x = -3, -2; y = -2, -1. Comparing every pair gives **x ≤ y**.

Q18 **Answer: (d) x ≤ y**

x = 4, 5; y = 5, 6. Comparing every pair gives **x ≤ y**.

Q19 **Answer: (d) 6**

Need $2/4 = 3/k = 5/10 \rightarrow k = \mathbf{6}$.

Q20 **Answer: (b) any k ≠ 8**

3/6 = -1/-2, so the lines are parallel; they coincide only if k = 8. No solution for **any k ≠ 8**.

Q21 **Answer: (c) 13**

$(\alpha + \beta)^2 - 2\alpha\beta = 25 - 12 = \mathbf{13}$.

Q22 **Answer: (d) 4**

At x = 3: 9 - 18 + 13 = **4**.

Q23 **Answer: (b) -2 < x < 3**

$-5 < 2x - 1 < 5 \rightarrow -4 < 2x < 6 \rightarrow$ **$-2 < x < 3$**.

Q24 **Answer: (a) 24**

$10t + u = 4(t + u) \rightarrow 6t = 3u \rightarrow u = 2t$. With $u = t + 2$: $2t = t + 2 \rightarrow t = 2$, $u = 4$. The number is **24**.

Q25 **Answer: (a) 1**

$p(1) = 1 + k - 4 + 2 = 0 \rightarrow k =$ **1**.

Linear systems: eliminate one variable; compare coefficient ratios for consistency. Quadratics: roots by factorising or the formula; $D = b^2 - 4ac$ decides their nature; sum $-b/a$, product c/a. In "compare x and y", check every pair of roots. Reverse an inequality when multiplying by a negative; quadratic < 0 lies between the roots.

MODULE 23 · INTERMEDIATE

Progressions

Minimum duration **2 days** | Level **Intermediate**

Sequences and series appear in direct questions and inside number-series reasoning. This module covers arithmetic progressions (nth term, sum, arithmetic mean, choosing terms), geometric progressions (nth term, sum, infinite sum, geometric mean), harmonic progressions and the AM ≥ GM ≥ HM relation, sums of squares and cubes, telescoping series and word problems, followed by 25 solved questions computed exactly.

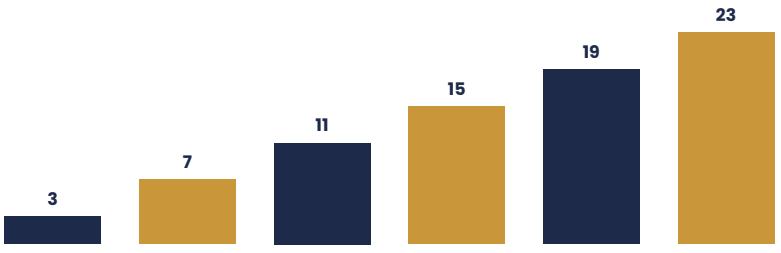
IN THIS MODULE

23.1 Arithmetic progression
23.2 Geometric progression
23.3 Harmonic progression
23.4 Special series
23.5 Word problems
23.6 Practice questions
23.7 Solutions

23.1 Arithmetic progression (AP)

23.1.1 nth term and sum

$t_n = a + (n - 1)d$; $S_n = n/2\,[2a + (n - 1)d] = n/2$ (first + last)



AP 3, 7, 11, …: each term is 4 more than the previous one

AN AP GROWS BY A CONSTANT DIFFERENCE

For 3, 7, 11, …: $t_{20} = 3 + 19 \times 4 = 79$ and $S_{20} = 10 \times (3 + 79) = 820$.

23.1.2 Arithmetic mean

The arithmetic mean of a and b is (a + b)/2. In an AP, every term is the mean of the terms equidistant from it.

23.1.3 Selecting terms in an AP

For three terms in AP use a - d, a, a + d; for four terms use a - 3d, a - d, a + d, a + 3d. The sum then gives a immediately.

WORKED EXAMPLE

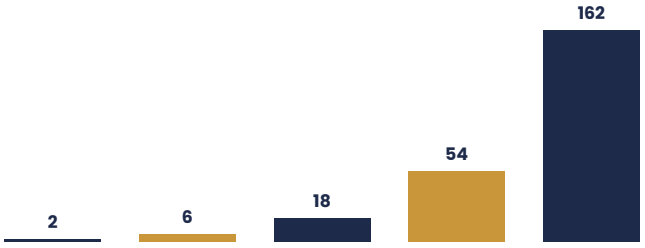
Three numbers in AP have sum 27 and product 504. Find them.

a - d + a + a + d = 27 → a = 9. (9 - d)(9)(9 + d) = 504 → 81 - d^2 = 56 → d = 5. Numbers 4, 9, 14.

23.2 Geometric progression (GP)

23.2.1 nth term and sum

$t_n = ar^{n-1}$; $S_n = a(r^n - 1)/(r - 1)$ for $r \neq 1$



GP 2, 6, 18, ...: each term is 3 times the previous one

A GP GROWS BY A CONSTANT RATIO

23.2.2 Sum of an infinite GP

For $|r| < 1$: $S_\infty = a/(1 - r)$

1 + 1/2 + 1/4 + ... = 1/(1 - 1/2) = 2. A recurring decimal is an infinite GP: 0.333... = 0.3/(1 - 0.1) = 1/3.

23.2.3 Geometric mean

The geometric mean of a and b is $\sqrt{(ab)}$. Three terms in GP: a/r, a, ar.

23.3 Harmonic progression (HP)

23.3.1 Harmonic mean

Numbers are in HP if their reciprocals are in AP. The harmonic mean of a and b is 2ab/(a + b), the same expression as the average speed over equal distances.

23.3.2 AM ≥ GM ≥ HM

For positive a and b: AM ≥ GM ≥ HM, with equality only when a = b; also $GM^2 = AM \times HM$

For 4 and 16: AM 10, GM 8, HM 6.4, and $8^2 = 10 \times 6.4$.

23.4 Special series

23.4.1 Sums of squares and cubes

$\Sigma k^2 = n(n+1)(2n+1)/6$; $\Sigma k^3 = [n(n+1)/2]^2$. Sum of squares 1^2 to $20^2 = 2870$.

23.4.2 Telescoping series

Split each term so that consecutive terms cancel: $1/(k(k+1)) = 1/k - 1/(k+1)$, so $1/(1\cdot2) + 1/(2\cdot3) + \ldots + 1/(n(n+1)) = 1 - 1/(n+1) = n/(n+1)$.

> **SHORTCUT TRICK**
> If the terms are fractions with products of consecutive numbers in the denominator, try partial fractions: most of the sum cancels.

23.5 Word problems with progressions

Savings rising by a fixed amount each month form an AP; populations or bacteria multiplying by a fixed factor form a GP. A ball dropped from height h that rebounds to a fraction r each time travels $h + 2hr/(1-r)$.

23.6 Practice questions

23.6.1 Basic level

Q1 What is the 20th term of the AP 3, 7, 11, …?

(a) 83 **(b)** 76 **(c)** 80 **(d)** 79

Q2 What is the sum of the first 20 terms of 3, 7, 11, …?

(a) 800 **(b)** 840 **(c)** 790 **(d)** 820

Q3 What is the 6th term of the GP 2, 6, 18, …?

(a) 162 **(b)** 1,458 **(c)** 486 **(d)** 243

Q4 What is the sum to infinity of 1 + 1/2 + 1/4 + …?

(a) 3/2 **(b)** 2 **(c)** 7/4 **(d)** 3

Q5 What is the arithmetic mean of 12 and 20?

(a) 16 **(b)** 15 **(c)** 18 **(d)** 14

Q6 What is the geometric mean of 4 and 16?

(a) 10 **(b)** 6.4 **(c)** 12 **(d)** 8

Q7 How many terms are there in the AP 7, 11, 15, ..., 99?

(a) 23 **(b)** 24 **(c)** 25 **(d)** 22

Q8 What is the harmonic mean of 4 and 16?

(a) 8 **(b)** 6.4 **(c)** 10 **(d)** 6

23.6.2 Intermediate level

Q9 Three numbers in AP have sum 27 and product 504. What is the largest?

(a) 13 **(b)** 15 **(c)** 12 **(d)** 14

Q10 The 5th term of an AP is 17 and the 10th term is 32. What is the 1st term?

(a) 2 **(b)** 4 **(c)** 3 **(d)** 5

Q11 What is the sum of the first 8 terms of the GP 3, 6, 12, ...?

(a) 768 **(b)** 765 **(c)** 381 **(d)** 1,533

Q12 What is the sum of all two-digit multiples of 7?

(a) 742 **(b)** 735 **(c)** 728 **(d)** 700

Q13 What is $1/(1\cdot2) + 1/(2\cdot3) + \ldots + 1/(9\cdot10)$?

(a) 9/10 **(b)** 1/10 **(c)** 1 **(d)** 10/11

Q14 Express 0.272727... as a fraction using an infinite GP.

(a) 3/11 **(b)** 27/100 **(c)** 2/7 **(d)** 3/10

Q15 What is the sum $1^2 + 2^2 + \ldots + 20^2$?

(a) 2,780 **(b)** 2,900 **(c)** 4,410 **(d)** 2,870

Q16 The first term of a GP is 5 and its 4th term is 135. What is the common ratio?

(a) 9 **(b)** 3 **(c)** 4 **(d)** 27

23.6.3 Advanced level

Q17 A man saves ₹500 in the first month and increases his savings by ₹100 each month. How much does he save in the first year?

(a) 12,000 **(b)** 13,200 **(c)** 12,600 **(d)** 11,400

Q18 A ball is dropped from 10 m and rebounds to half its height each time. What total distance does it travel before coming to rest?

(a) 20 m **(b)** 25 m **(c)** 40 m **(d)** 30 m

Q19 The sum of the first n terms of an AP is $3n^2 + 2n$. What is its 10th term?

(a) 61 **(b)** 63 **(c)** 59 **(d)** 60

Q20 If a, b, c are in GP with a = 4 and c = 9, what is b (positive)?

(a) 13/2 **(b)** 5 **(c)** 6 **(d)** 36

Q21 The AM of two numbers is 25 and their GM is 15. What are the numbers?

(a) 45 and 5 **(b)** 30 and 20 **(c)** 40 and 10 **(d)** 35 and 15

Q22 Bacteria double every hour. Starting with 50, how many are there after 6 hours?

(a) 3,000 **(b)** 3,200 **(c)** 1,600 **(d)** 6,400

Q23 What is the sum $1^3 + 2^3 + \ldots + 10^3$?

(a) 3,000 **(b)** 3,025 **(c)** 385 **(d)** 2,025

Q24 How many terms of the AP 9, 17, 25, … must be taken to give a sum of 636?

(a) 10 **(b)** 12 **(c)** 14 **(d)** 11

Q25 Insert two numbers between 3 and 81 so that the four numbers are in GP. What is the sum of the inserted numbers?

(a) 30 **(b)** 36 **(c)** 45 **(d)** 24

23.7 Solutions

Q1 **Answer: (d) 79**

$3 + 19 \times 4 = \mathbf{79}$.

Q2 **Answer: (d) 820**

$10 \times (3 + 79) = \mathbf{820}$.

Q3 **Answer: (c) 486**

$2 \times 3^5 = \mathbf{486}$.

Q4 **Answer: (b) 2**

$1/(1 - 1/2) = \mathbf{2}$.

Q5 **Answer: (a) 16**

$(12 + 20)/2 = \mathbf{16}$.

Q6 **Answer: (d) 8**

$\sqrt{64} = \mathbf{8}$.

Q7 **Answer: (b) 24**

$(99 - 7)/4 + 1 = \mathbf{24}$.

Q8 **Answer: (b) 6.4**

$2 \times 4 \times 16/20 = \mathbf{6.4}$.

Q9 **Answer: (d) 14**

$a = 9$; $81 - d^2 = 56 \rightarrow d = 5 \rightarrow 4, 9, 14$: largest **14**.

Q10 **Answer: (d) 5**

$d = (32 - 17)/5 = 3$; $a = 17 - 12 = \mathbf{5}$.

Q11 **Answer: (b) 765**

$3(2^8 - 1) = 3 \times 255 = \mathbf{765}$.

Q12 **Answer: (c) 728**

Terms 14 to 98: 13 terms, sum $13 \times (14 + 98)/2 = \mathbf{728}$.

Q13 **Answer: (a) 9/10**

Telescoping: $1 - 1/10 = \mathbf{9/10}$.

Q14 **Answer: (a) 3/11**

$0.27/(1 - 0.01) = 27/99 = \mathbf{3/11}$.

Q15 **Answer: (d) 2,870**

$20 \times 21 \times 41/6 = \mathbf{2870}$.

Q16 **Answer: (b) 3**

$5r^3 = 135 \rightarrow r^3 = 27 \rightarrow r = \mathbf{3}$.

Q17 **Answer: (c) 12,600**

$S_{12} = 6 \times (2 \times 500 + 11 \times 100) = 6 \times 2100 = \mathbf{₹12,600}$.

Q18 **Answer: (d) 30 m**

$10 + 2 \times 5/(1 - 1/2) = 10 + 20 = \mathbf{30\ m}$.

Q19 **Answer: (c) 59**

$t_{10} = S_{10} - S_9 = 320 - 261 = \mathbf{59}$.

Q20 **Answer: (c) 6**

$b = \sqrt{(ac)} = \mathbf{6}$.

Q21 **Answer: (a) 45 and 5**

$a + b = 50$, $ab = 225 \rightarrow$ roots of $x^2 - 50x + 225 = 0$: **45 and 5**.

Q22 **Answer: (b) 3,200**

$50 \times 2^6 = \mathbf{3,200}$.

Q23 **Answer: (b) 3,025**

$(10 \times 11/2)^2 = 55^2 =$ **3,025**.

Q24 **Answer: (b) 12**

$n/2[18 + 8(n - 1)] = 636 \rightarrow n(4n + 5) = 636 \rightarrow 4n^2 + 5n - 636 = 0 \rightarrow n =$ **12**.

Q25 **Answer: (b) 36**

$r^3 = 81/3 = 27 \rightarrow r = 3$: 3, 9, 27, 81. Sum of inserted = **36**.

AP: $t_n = a + (n - 1)d$, $S_n = n/2$ (first + last). GP: $t_n = ar^{n-1}$, $S_n = a(r^n - 1)/(r - 1)$, $S_\infty = a/(1 - r)$ for $|r| < 1$. Means: AM $(a + b)/2$, GM $\sqrt{(ab)}$, HM $2ab/(a + b)$; AM ≥ GM ≥ HM and $GM^2 = AM \times HM$. Choose symmetric terms $(a - d, a, a + d)$ and telescope fractions.

MODULE 24 · INTERMEDIATE

Logarithms

Minimum duration **1 day** | Level **Intermediate**

Logarithms simplify calculations with powers and appear in placement and management exams. This module covers the definition, common and natural logarithms, the product, quotient and power rules, change of base, the special values log_a a = 1 and log_a 1 = 0, simplifying expressions, solving logarithmic equations and finding the number of digits in large powers, followed by 25 solved questions checked by a program.

IN THIS MODULE

24.1 Concepts
24.2 Laws of logarithms
24.3 Problem types
24.4 Practice questions
24.5 Solutions

24.1 Concepts

24.1.1 Definition

$$\log_a x = y \Leftrightarrow a^y = x \ (a > 0, a \neq 1, x > 0)$$

A logarithm answers "to what power must the base be raised?". $\log_2 32 = 5$ because $2^5 = 32$; $\log_{10} 0.01 = -2$ because $10^{-2} = 0.01$.



THE SAME FACT IN TWO FORMS

24.1.2 Common and natural logarithms

log x usually means base 10 (common log); ln x means base e ≈ 2.718 (natural log). Useful values: log 2 ≈ 0.3010, log 3 ≈ 0.4771, log 5 = 1 - log 2 ≈ 0.6990, log 7 ≈ 0.8451.

24.2 Laws of logarithms

24.2.1 Product, quotient and power rules

Rule	Formula	Example
Product	log(mn) = log m + log n	log 6 = log 2 + log 3

Rule	Formula	Example
Quotient	log(m/n) = log m - log n	log 5 = log 10 - log 2
Power	log m^k = k log m	log 8 = 3 log 2

24.2.2 Change of base

log_a x = log_b x / log_b a; in particular log_a b × log_b a = 1

24.2.3 Special values

log_a a = 1 and log_a 1 = 0 for every valid base. Also a^(log_a x) = x.

> **WATCH OUT**
> There is no rule for log(m + n). log(2 + 3) is log 5, not log 2 + log 3 (which is log 6).

24.3 Problem types

24.3.1 Simplifying expressions

Combine with the laws: log 8 + log 125 = log 1000 = 3; $\log_2 48 - \log_2 3 = \log_2 16 = 4$.

24.3.2 Solving equations

Convert to exponential form or combine logs first: $\log_3(x - 1) = 2 \rightarrow x - 1 = 9 \rightarrow x = 10$. log x + log(x - 3) = 1 → x(x - 3) = 10 → x = 5 (x = -2 is rejected since logs need positive arguments).

24.3.3 Number of digits in large powers

Number of digits of $N = \lfloor \log_{10} N \rfloor + 1$

2^{100}: log = 100 × 0.30103 = 30.103 → 31 digits. 3^{50}: 50 × 0.4771 = 23.86 → 24 digits.

24.4 Practice questions

24.4.1 Basic level

Q1 $\log_2 64$ = ?

(a) 8 **(b)** 5 **(c)** 6 **(d)** 32

Q2 $\log_{10} 0.001$ = ?

(a) 3 **(b)** -2 **(c)** -3 **(d)** 1/1000

Q3 log 8 + log 125 = ?

(a) 2 **(b)** 1,000 **(c)** 133 **(d)** 3

Q4 $\log_5 1 = ?$

(a) 0 **(b)** 1 **(c)** 5 **(d)** -1

Q5 $\log_7 7 = ?$

(a) 1 **(b)** 0 **(c)** 7 **(d)** 49

Q6 If $\log_3 x = 4$, then $x = ?$

(a) 81 **(b)** 12 **(c)** 64 **(d)** 27

Q7 $\log_2 48 - \log_2 3 = ?$

(a) 16 **(b)** 4 **(c)** 5 **(d)** 3

Q8 Given log 2 = 0.3010, find log 8.

(a) 0.6020 **(b)** 0.9030 **(c)** 2.4080 **(d)** 0.0903

24.4.2 Intermediate level

Q9 Given log 2 = 0.3010 and log 3 = 0.4771, find log 12.

(a) 0.7781 **(b)** 1.1761 **(c)** 1.0791 **(d)** 0.9542

Q10 $\log_3(x - 1) = 2$. Find x.

(a) 9 **(b)** 10 **(c)** 8 **(d)** 7

Q11 log x + log(x - 3) = 1 (base 10). Find x.

(a) -2 **(b)** 5 **(c)** 2 **(d)** 10

Q12 $\log_2 3 \times \log_3 4 = ?$

(a) 1 **(b)** 12 **(c)** 4/3 **(d)** 2

Q13 How many digits does 2^{100} have? (log 2 = 0.30103)

(a) 30 **(b)** 32 **(c)** 100 **(d)** 31

Q14 Given log 2 = 0.3010, find log 5.

(a) 0.3010 **(b)** 0.6505 **(c)** 0.6990 **(d)** 1.3010

Q15 $\log_4 8 = ?$

(a) 2 **(b)** 2/3 **(c)** 3/2 **(d)** 32

Q16 If $\log_{10} x = -1$, what is x?

(a) -10 **(b)** 0.1 **(c)** 0.01 **(d)** 10

24.4.3 Advanced level

Q17 How many digits does 3^{50} have? (log 3 = 0.4771)

(a) 23 **(b)** 24 **(c)** 25 **(d)** 50

Q18 Simplify $\log(a^2/bc) + \log(b^2/ac) + \log(c^2/ab)$.

(a) 1 **(b)** log abc **(c)** 3 **(d)** 0

Q19 If log 2 = x and log 3 = y, what is log 45?

(a) 2y - x + 1 **(b)** 2y + x + 1 **(c)** y - 2x + 1 **(d)** 2x + y

Q20 Solve $2^x = 5$ using log 2 = 0.3010 and log 5 = 0.6990.

(a) 2.32 **(b)** 2.5 **(c)** 2.12 **(d)** 2.42

Q21 If $\log_8 x + \log_4 x + \log_2 x = 11$, what is x?

(a) 16 **(b)** 32 **(c)** 128 **(d)** 64

Q22 What is the value of $\log_{10} 25 + \log_{10} 4$?

(a) 1 **(b)** 29 **(c)** 2 **(d)** 5/2

Q23 If $\log(x + 1) - \log(x - 1) = \log 3$, find x.

(a) 3 **(b)** 1 **(c)** 4 **(d)** 2

Q24 Which is larger: $\log_2 10$ or $\log_3 25$?

(a) $\log_3 25$ **(b)** They are equal

(c) Cannot be determined **(d)** $\log_2 10$

Q25 If $\log_5 2 = a$, express $\log_{10} 2$ in terms of a? (Use $\log_5 10 = 1 + \log_5 2$.)

(a) 1/(1 + a) **(b)** a/(1 - a) **(c)** a/(1 + a) **(d)** (1 + a)/a

24.5 Solutions

Q1 **Answer: (c) 6**

$2^6 = 64 \rightarrow$ **6**.

Q2 **Answer: (c) -3**

$10^{-3} = 0.001 \rightarrow$ **-3**.

Q3 **Answer: (d) 3**

$\log(8 \times 125) = \log 1000 =$ **3**.

Q4 **Answer: (a) 0**

Any log of 1 is **0**.

Q5 **Answer: (a) 1**

log_a a = **1**.

Q6 **Answer: (a) 81**

$x = 3^4 =$ **81**.

Q7 **Answer: (b) 4**

$\log_2(48/3) = \log_2 16 =$ **4**.

Q8 **Answer: (b) 0.9030**

$3 \times 0.3010 =$ **0.9030**.

Q9 **Answer: (c) 1.0791**

log 12 = 2 log 2 + log 3 = 0.6020 + 0.4771 = **1.0791**.

Q10 **Answer: (b) 10**

x − 1 = 9 → x = **10**.

Q11 **Answer: (b) 5**

x(x − 3) = 10 → $x^2 - 3x - 10 = 0$ → x = 5 or −2; logs need x > 3, so x = **5**.

Q12 **Answer: (d) 2**

Change of base: (log 3/log 2)(log 4/log 3) = log 4/log 2 = **2**.

Q13 **Answer: (d) 31**

100 × 0.30103 = 30.103 → 30 + 1 = **31**.

Q14 **Answer: (c) 0.6990**

log 5 = log 10 − log 2 = 1 − 0.3010 = **0.6990**.

Q15 **Answer: (c) 3/2**

4^(3/2) = 8 → **3/2**.

Q16 **Answer: (b) 0.1**

$x = 10^{-1}$ = **0.1**.

Q17 **Answer: (b) 24**

50 × 0.4771 = 23.855 → **24**.

Q18 **Answer: (d) 0**

Product inside = $a^2b^2c^2/(a^2b^2c^2)$ = 1 → log 1 = **0**.

Q19 **Answer: (a) 2y − x + 1**

45 = 9 × 10/2 → 2y + 1 − x = **2y − x + 1**.

Q20 **Answer: (a) 2.32**

x = log 5/log 2 = 0.6990/0.3010 ≈ **2.32**.

Q21 **Answer: (d) 64**

In base 2: x/3 + x/2 + x terms → $(1/3 + 1/2 + 1) \log_2 x = 11$ → $(11/6) \log_2 x = 11$ → $\log_2 x = 6$ → x = **64**.

Q22 **Answer: (c) 2**

log(25 × 4) = log 100 = **2**.

Q23 **Answer: (d) 2**

$(x + 1)/(x - 1) = 3 \rightarrow x + 1 = 3x - 3 \rightarrow x = \mathbf{2}$.

Q24 **Answer: (d) $\log_2 10$**

$\log_2 10 \approx 3.32$ and $\log_3 25 \approx 2.93$, so $\mathbf{\log_2 10}$ is larger.

Q25 **Answer: (c) a/(1 + a)**

$\log_{10} 2 = \log_5 2/\log_5 10 = \mathbf{a/(1 + a)}$.

log_a x = y means $a^y = x$. Product, quotient and power rules turn products into sums; change of base gives log_a x = log x/log a and log_a b × log_b a = 1. Check that every log argument is positive. Number of digits of N = $\lfloor \log_{10} N \rfloor + 1$.

LEVEL 3

Advanced level

Master permutations, probability, geometry, mensuration, coordinate geometry, data interpretation and data sufficiency, and perform in timed company-specific tests

9 MODULES · 19 DAYS MINIMUM

Module 25 · Permutations and combinations	**2 days**
Module 26 · Probability	**2 days**
Module 27 · Geometry	**3 days**
Module 28 · Mensuration	**2 days**
Module 29 · Coordinate geometry and trigonometry	**2 days**
Module 30 · Data interpretation	**3 days**
Module 31 · Data sufficiency	**1 day**
Module 32 · Statistics, set theory and miscellaneous topics	**1 day**
Module 33 · Exam strategy, mock tests and company-specific preparation	**3 days**

MODULE 25 · ADVANCED

Permutations and combinations

Minimum duration **2 days** | Level **Advanced**

Counting principles are essential for probability and are tested in most placement exams. This module covers the addition and multiplication rules, factorials, permutations (with repetition, identical objects, letters together or apart, circular arrangements, rank of a word), combinations (properties, at least / at most, committees, selecting from groups), distributions, diagonals and triangles from points, forming numbers from digits and derangements, followed by 25 solved questions verified by brute-force counting.

IN THIS MODULE

25.1 Fundamental principles
25.2 Permutations
25.3 Combinations
25.4 Advanced counting
25.5 Practice questions
25.6 Solutions

25.1 Fundamental principles

25.1.1 Addition and multiplication rules

If a task can be done in m ways **or** in n ways (but not both), there are m + n ways (addition rule). If it is done in steps with m ways **and then** n ways, there are m × n ways (multiplication rule).



4 choices × 3 choices × 2 choices = 24

FILLING 3 POSITIONS FROM 4 PEOPLE

25.1.2 Factorials

n! = n × (n - 1) × … × 1, with 0! = 1. 5! = 120, 6! = 720, 7! = 5040, 8! = 40320.

25.2 Permutations

25.2.1 The formula

nPr = n!/(n - r)! (ordered arrangements of r out of n distinct objects)

25.2.2 Arrangements with repetition

If each of r positions can take any of n objects, there are n^r arrangements: 4-digit PINs from 10 digits = 10^4.

25.2.3 Identical objects

Arrangements of n objects with groups of p, q, r identical = n!/(p! q! r!)

BANANA: 6!/(3! 2!) = 60.

25.2.4 Letters together or apart

Together: tie them into one block, arrange the blocks, then arrange inside the block. **Never together**: total - together, or place the other letters first and insert into the gaps.

WORKED EXAMPLE

Arrangements of LEADER with the vowels together

LEADER has vowels E, A, E and consonants L, D, R. Block (EAE) + L, D, R = 4 units: 4! = 24; inside the block 3!/2! = 3. Total 72.

25.2.5 Circular permutations

n distinct people around a round table: (n - 1)!; necklaces or garlands (flip allowed): (n - 1)!/2

25.2.6 Rank of a word

Count the words that come before it in dictionary order, letter by letter. For RACE: letters A, C, E, R. Words starting with A, C or E: 3 × 3! = 18. Then R A ...: next letters C, E — RACE is the first: rank 19.

25.3 Combinations

25.3.1 Formula and properties

nCr = n!/(r!(n - r)!); nCr = nC(n - r); nC0 = nCn = 1; nCr + nC(r - 1) = (n + 1)Cr

25.3.2 At least and at most

Add the cases, or use total - complement. Committees of 4 from 6 men and 4 women with at least one woman: C(10, 4) - C(6, 4) = 210 - 15 = 195.

25.3.3 Committees and teams

Multiply the choices from each group: 3 men from 7 and 2 women from 5 = C(7, 3) × C(5, 2) = 35 × 10 = 350.

25.3.4 Selecting from groups

Selecting at least one item from n distinct items: $2^n - 1$. From identical items (p of one kind, q of another): $(p + 1)(q + 1) - 1$.

> **SHORTCUT TRICK**
> Arrangement or selection? If swapping two chosen items gives a **different** outcome (president/secretary, digits of a number), use permutations; if not (a committee, a handful of fruit), use combinations.

25.4 Advanced counting

25.4.1 Distributions

n identical objects into r distinct boxes (empty allowed): $C(n + r - 1, r - 1)$. Each box non-empty: $C(n - 1, r - 1)$. n distinct objects into r distinct boxes: r^n.

25.4.2 Lines, triangles and diagonals

From n points with no three collinear: lines $C(n, 2)$, triangles $C(n, 3)$. A polygon with n sides has $n(n - 3)/2$ diagonals.

25.4.3 Forming numbers from digits

Fill the most restricted place first (the first digit cannot be 0; an even number needs an even last digit).

25.4.4 Derangements

A derangement is an arrangement where no object is in its original place. $D_1 = 0$, $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$.

25.5 Practice questions

25.5.1 Basic level

Q1 In how many ways can 3 people be chosen as president, secretary and treasurer from 4 candidates?

(a) 12 **(b)** 4 **(c)** 24 **(d)** 64

Q2 In how many ways can the letters of APPLE be arranged?

(a) 120 **(b)** 60 **(c)** 24 **(d)** 30

Q3 How many ways can 5 people sit around a round table?

(a) 24 **(b)** 120 **(c)** 60 **(d)** 12

Q4 $C(10, 3) = ?$

(a) 120 **(b)** 720 **(c)** 210 **(d)** 100

Q5 How many 4-digit PINs can be formed from the digits 0–9 if digits may repeat?

(a) 10,000 **(b)** 5,040 **(c)** 9,000 **(d)** 1,000

Q6 How many diagonals does a hexagon have?

(a) 6 **(b)** 15 **(c)** 9 **(d)** 12

Q7 In how many ways can a committee of 3 be chosen from 8 people?

(a) 336 **(b)** 24 **(c)** 112 **(d)** 56

Q8 How many triangles can be formed from 8 points, no three of which are collinear?

(a) 56 **(b)** 28 **(c)** 336 **(d)** 64

25.5.2 Intermediate level

Q9 In how many ways can the letters of BANANA be arranged?

(a) 60 **(b)** 720 **(c)** 120 **(d)** 36

Q10 In how many ways can the letters of LEADER be arranged so that the vowels are together?

(a) 36 **(b)** 144 **(c)** 72 **(d)** 360

Q11 A committee of 5 (3 men and 2 women) is chosen from 7 men and 5 women. How many ways?

(a) 792 **(b)** 175 **(c)** 350 **(d)** 455

Q12 How many committees of 4 from 6 men and 4 women include at least one woman?

(a) 210 **(b)** 180 **(c)** 195 **(d)** 200

Q13 How many 3-digit even numbers can be formed from 0, 1, 2, 3, 4, 5 without repetition?

(a) 60 **(b)** 52 **(c)** 48 **(d)** 40

Q14 What is the rank of RACE in the dictionary order of its letters?

(a) 18 **(b)** 20 **(c)** 19 **(d)** 24

Q15 In how many ways can 8 beads of different colours be strung into a necklace?

(a) 2,520 **(b)** 5,040 **(c)** 40,320 **(d)** 20,160

Q16 In how many ways can 6 people be seated in a row if two particular people must sit together?

(a) 120 **(b)** 720 **(c)** 480 **(d)** 240

25.5.3 Advanced level

Q17 In how many ways can 6 people be seated in a row if two particular people must NOT sit together?

(a) 480 **(b)** 240 **(c)** 360 **(d)** 600

Q18 In how many ways can 10 identical balls be put into 4 distinct boxes (boxes may be empty)?

(a) 210 **(b)** 84 **(c)** 1,048 **(d)** 286

Q19 In how many ways can 10 identical balls be put into 4 distinct boxes so that no box is empty?

(a) 84 **(b)** 120 **(c)** 286 **(d)** 70

Q20 How many words can be formed from MATHEMATICS?

(a) 9,979,200 **(b)** 19,958,400 **(c)** 6,652,800 **(d)** 4,989,600

Q21 From 5 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

(a) 7,200 **(b)** 720 **(c)** 60 **(d)** 3,600

Q22 How many ways are there to select at least one fruit from 3 apples, 4 mangoes and 2 oranges (fruits of a kind are identical)?

(a) 59 **(b)** 60 **(c)** 9 **(d)** 511

Q23 In how many ways can 4 letters be put into 4 addressed envelopes so that no letter goes into the right envelope?

(a) 8 **(b)** 9 **(c)** 11 **(d)** 23

Q24 12 points lie in a plane, 5 of them on one line and no other three collinear. How many straight lines can be drawn through pairs of them?

(a) 66 **(b)** 57 **(c)** 56 **(d)** 55

Q25 How many numbers greater than 4000 can be formed with the digits 2, 3, 4, 5, 6 without repetition (4 or 5 digits)?

(a) 192 **(b)** 120 **(c)** 168 **(d)** 240

25.6 Solutions

Q1 **Answer: (c) 24**

$4 \times 3 \times 2 = \mathbf{24}$.

Q2 **Answer: (b) 60**

$5!/2! = \mathbf{60}$.

Q3 **Answer: (a) 24**

$(5-1)! = \mathbf{24}$.

Q4 **Answer: (a) 120**

$10 \times 9 \times 8/6 = \mathbf{120}$.

Q5 **Answer: (a) 10,000**

$10^4 = \mathbf{10{,}000}$.

Q6 **Answer: (c) 9**

$n(n-3)/2 = 6 \times 3/2 = \mathbf{9}$.

Q7 **Answer: (d) 56**

$C(8, 3) = \mathbf{56}$.

Q8 **Answer: (a) 56**

$C(8, 3) = 56$.

Q9 **Answer: (a) 60**

$6!/(3!\,2!) = 60$.

Q10 **Answer: (c) 72**

Block EAE + L, D, R: $4! \times 3!/2! = 24 \times 3 = 72$.

Q11 **Answer: (c) 350**

$35 \times 10 = 350$.

Q12 **Answer: (c) 195**

$210 - 15 = 195$.

Q13 **Answer: (b) 52**

Last digit 0: $5 \times 4 = 20$. Last digit 2 or 4: first digit 4 choices (not 0, not the last), middle 4 → $2 \times 16 = 32$. Total **52**.

Q14 **Answer: (c) 19**

Words beginning with A, C, E: $3 \times 3! = 18$; RACE is the first word starting with R: **19**.

Q15 **Answer: (a) 2,520**

$(8 - 1)!/2 = 5040/2 = 2{,}520$.

Q16 **Answer: (d) 240**

Tie the pair: $5! \times 2 = 240$.

Q17 **Answer: (a) 480**

$720 - 240 = 480$.

Q18 **Answer: (d) 286**

$C(10 + 4 - 1, 3) = C(13, 3) = 286$.

Q19 **Answer: (a) 84**

$C(10 - 1, 4 - 1) = C(9, 3) = 84$.

Q20 **Answer: (d) 4,989,600**

M, A, T each appear twice: $11!/(2!\,2!\,2!) = 4{,}989{,}600$.

Q21 **Answer: (a) 7,200**

Choose: 10 × 6 = 60 sets; arrange each in 5! = 120 ways: **7,200**.

Q22 **Answer: (a) 59**

$(3 + 1)(4 + 1)(2 + 1) - 1 =$ **59**.

Q23 **Answer: (b) 9**

Derangements of 4: $D_4 =$ **9**.

Q24 **Answer: (b) 57**

$C(12, 2) - C(5, 2) + 1 = 66 - 10 + 1 =$ **57**.

Q25 **Answer: (a) 192**

Four-digit: first digit 4, 5 or 6 → 3 × 4 × 3 × 2 = 72. Five-digit: 5! = 120. Total **192**.

Multiply for "and then", add for "or". nPr = n!/(n - r)! for arrangements, nCr = n!/(r!(n - r)!) for selections. Divide by p!q! for identical objects; (n - 1)! around a table, half that for necklaces. Together: treat as a block; apart: total - together. At least one: total - none. Identical objects into boxes: C(n + r - 1, r - 1).

MODULE 26 · ADVANCED

Probability

Minimum duration **2 days** | Level **Advanced**

Probability measures how likely an event is; it builds directly on permutations and combinations. This module covers sample spaces and events, the classical formula, complementary events, the addition and multiplication rules, conditional probability, Bayes' theorem, and the classic problem types (coins, dice, cards, bags with and without replacement, selections of people), odds and expected value, followed by 25 solved questions verified by enumerating every outcome.

IN THIS MODULE

26.1 Basics
26.2 Rules
26.3 Classic problem types
26.4 Practice questions
26.5 Solutions

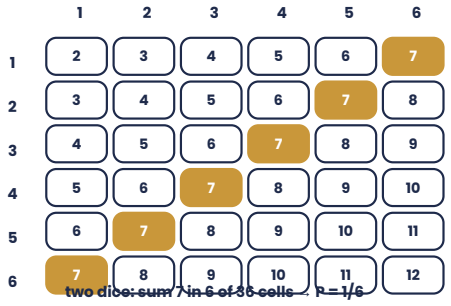
26.1 Basics

26.1.1 Experiment, sample space and events

An **experiment** has uncertain outcomes; the **sample space** S is the set of all outcomes; an **event** is a subset of S. Tossing two coins: S = {HH, HT, TH, TT}.

26.1.2 Classical probability

P(E) = number of favourable outcomes / total number of equally likely outcomes; 0 ≤ P(E) ≤ 1



THE 36 OUTCOMES OF TWO DICE

26.1.3 Complementary events

P(not E) = 1 − P(E)

"At least one" questions are easiest by complement: P(at least one six in 4 throws) = 1 - $(5/6)^4$ = 671/1296.

26.2 Rules

26.2.1 Addition rule

P(A or B) = P(A) + P(B) - P(A and B); = P(A) + P(B) if A and B are mutually exclusive

26.2.2 Multiplication rule

For independent events, P(A and B) = P(A) × P(B). Two coins both heads: 1/2 × 1/2 = 1/4.

26.2.3 Conditional probability

P(A | B) = P(A and B)/P(B)

Two dice show a sum of 8. Probability that one die shows 3: outcomes with sum 8 are (2,6), (3,5), (4,4), (5,3), (6,2); two contain a 3 → 2/5.

26.2.4 Bayes' theorem

Bayes' theorem reverses a conditional probability: P(A | B) = P(B | A)P(A)/P(B). Exams use it in simple two-bag forms: pick a bag at random, draw a ball, and ask which bag it probably came from.

26.3 Classic problem types

26.3.1 Coins

n coins give 2^n equally likely outcomes; the number with exactly k heads is C(n, k).

26.3.2 Dice

One die: 6 outcomes; two dice: 36; three dice: 216. Use the 6 × 6 grid for two dice.

26.3.3 Playing cards

Cards in a deck	Count
Total	52
Suits (spades, clubs black; hearts, diamonds red)	4 × 13
Face cards (J, Q, K)	12
Aces	4
Red cards	26

26.3.4 Balls from bags

Without replacement, use combinations: 2 balls from a bag of 5 red and 3 blue, both red: C(5, 2)/C(8, 2) = 10/28 = 5/14. With replacement, multiply: $(5/8)^2 = 25/64$.

26.3.5 Selecting people

Committees chosen at random: favourable selections/total selections, using combinations.

26.3.6 Odds

Odds in favour of E = P(E) : P(not E); odds against = P(not E) : P(E). If odds in favour are 3 : 5, P(E) = 3/8.

26.3.7 Expected value

Expected value = Σ (value × probability). A game paying ₹6 on a six and nothing otherwise has expected value 6 × 1/6 = ₹1 per throw.

> **SHORTCUT TRICK**
> For "at least one", always try the complement first. For "without replacement", combinations on top and bottom avoid order mistakes.

26.4 Practice questions

26.4.1 Basic level

Q1 A die is thrown. What is the probability of getting an even number?

(a) 1/3 **(b)** 1/2 **(c)** 1/6 **(d)** 2/3

Q2 Two dice are thrown. What is the probability that the sum is 7?

(a) 1/12 **(b)** 5/36 **(c)** 1/6 **(d)** 7/36

Q3 Three coins are tossed. What is the probability of exactly two heads?

(a) 1/2 **(b)** 1/4 **(c)** 1/8 **(d)** 3/8

Q4 A card is drawn from a deck. What is the probability that it is a king?

(a) 1/4 **(b)** 1/26 **(c)** 1/13 **(d)** 1/52

Q5 A bag has 5 red and 3 blue balls. One ball is drawn. What is the probability that it is blue?

(a) 5/8 **(b)** 3/5 **(c)** 1/3 **(d)** 3/8

Q6 The probability that it will rain tomorrow is 0.3. What is the probability that it will not rain?

(a) 0.7 **(b)** 0.3 **(c)** 0.6 **(d)** 1

Q7 Two coins are tossed. What is the probability of at least one head?

(a) 1/2 **(b)** 1/4 **(c)** 3/4 **(d)** 1

Q8 If the odds in favour of an event are 3 : 5, what is its probability?

(a) 3/8 **(b)** 3/5 **(c)** 5/8 **(d)** 2/5

26.4.2 Intermediate level

Q9 Two balls are drawn without replacement from a bag of 5 red and 3 blue. What is the probability that both are red?

(a) 25/64 (b) 5/8 (c) 5/14 (d) 3/14

Q10 A card is drawn from a deck. What is the probability that it is a heart or a king?

(a) 4/13 (b) 17/52 (c) 1/4 (d) 1/13

Q11 Two dice are thrown. What is the probability that the sum is at least 10?

(a) 1/12 (b) 5/36 (c) 1/6 (d) 1/4

Q12 What is the probability of getting at least one six in 4 throws of a die?

(a) 2/3 (b) 1/2 (c) 671/1296 (d) 625/1296

Q13 Two dice show a sum of 8. What is the probability that one of them shows 3?

(a) 1/6 (b) 1/3 (c) 1/18 (d) 2/5

Q14 A and B hit a target with probabilities 1/2 and 1/3 independently. What is the probability that the target is hit?

(a) 2/3 (b) 5/6 (c) 1/6 (d) 1/2

Q15 From 4 men and 3 women, a committee of 3 is chosen at random. What is the probability that it has exactly 2 women?

(a) 3/35 (b) 12/35 (c) 18/35 (d) 4/35

Q16 Two cards are drawn without replacement. What is the probability that both are aces?

(a) 1/221 (b) 1/169 (c) 1/13 (d) 2/221

26.4.3 Advanced level

Q17 A bag has 4 red and 6 black balls. Two are drawn with replacement. What is the probability that they are of different colours?

(a) 8/15 (b) 6/25 (c) 13/25 (d) 12/25

Q18 Three dice are thrown. What is the probability that all show the same number?

(a) 1/216 (b) 1/36 (c) 1/6 (d) 1/72

Q19 Bag I has 3 red and 2 blue balls; bag II has 2 red and 3 blue. A bag is chosen at random and a ball drawn is red. What is the probability that it came from bag I?

(a) 1/2 (b) 3/5 (c) 2/5 (d) 3/10

Q20 The letters of the word LEVEL are arranged at random. What is the probability that the arrangement starts and ends with L?

(a) 1/5 (b) 1/30 (c) 2/5 (d) 1/10

Q21 A number is chosen from 1 to 100. What is the probability that it is divisible by 3 or 5?

(a) 53/100 (b) 1/2 (c) 2/5 (d) 47/100

Q22 Two dice are thrown. What is the expected value of the sum?

(a) 6 (b) 7 (c) 6.5 (d) 12

Q23 Four coins are tossed. What is the probability of getting at least 3 heads?

(a) 1/4 **(b)** 3/8 **(c)** 5/16 **(d)** 1/16

Q24 A box has 10 bulbs, of which 3 are defective. Two are chosen at random. What is the probability that at least one is defective?

(a) 7/15 **(b)** 3/10 **(c)** 8/15 **(d)** 1/15

Q25 The probability that A speaks the truth is 3/4 and that B does is 4/5. In what percentage of cases are they likely to contradict each other on the same fact?

(a) 30% **(b)** 40% **(c)** 25% **(d)** 35%

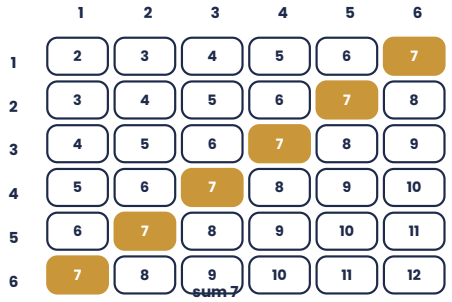
26.5 Solutions

Q1 **Answer: (b) 1/2**

3 of 6 outcomes: **1/2**.

Q2 **Answer: (c) 1/6**

6 of 36: **1/6**.



Q3 **Answer: (d) 3/8**

HHT, HTH, THH: 3 of 8 → **3/8**.

Q4 **Answer: (c) 1/13**

4/52 = **1/13**.

Q5 **Answer: (d) 3/8**

3 of 8: **3/8**.

Q6 **Answer: (a) 0.7**

1 − 0.3 = **0.7**.

Q7 **Answer: (c) 3/4**

1 − P(TT) = 1 − 1/4 = **3/4**.

Q8 **Answer: (a) 3/8**

3/(3 + 5) = **3/8**.

Q9 **Answer: (c) 5/14**

C(5, 2)/C(8, 2) = 10/28 = **5/14**.

Q10 **Answer: (a) 4/13**

13 + 4 - 1 = 16 cards: 16/52 = **4/13**.

Q11 **Answer: (c) 1/6**

Sums 10, 11, 12: 3 + 2 + 1 = 6 of 36 → **1/6**.

Q12 **Answer: (c) 671/1296**

$1 - (5/6)^4 = 1 - 625/1296 =$ **671/1296**.

Q13 **Answer: (d) 2/5**

Sum-8 outcomes: 5; with a 3: (3,5), (5,3) → **2/5**.

Q14 **Answer: (a) 2/3**

1 - P(both miss) = 1 - 1/2 × 2/3 = **2/3**.

Q15 **Answer: (b) 12/35**

C(3, 2) × C(4, 1)/C(7, 3) = 12/35 → **12/35**.

Q16 **Answer: (a) 1/221**

C(4, 2)/C(52, 2) = 6/1326 = **1/221**.

Q17 **Answer: (d) 12/25**

2 × 0.4 × 0.6 = 0.48 = **12/25**.

Q18 **Answer: (b) 1/36**

6 outcomes of 216: **1/36**.

Q19 **Answer: (b) 3/5**

Bayes: (1/2 × 3/5)/(1/2 × 3/5 + 1/2 × 2/5) = 3/5 → **3/5**.

Q20 **Answer: (d) 1/10**

Arrangements 5!/(2! 2!) = 30; with L at both ends: E, V, E in the middle → 3 ways. 3/30 = **1/10**.

Q21 **Answer: (d) 47/100**

33 + 20 - 6 = 47 numbers: **47/100**.

Q22 **Answer: (b) 7**

Each die has expectation 3.5; the sum has **7**.

Q23 **Answer: (c) 5/16**

$(4 + 1)/16 = \mathbf{5/16}$.

Q24 **Answer: (c) 8/15**

$1 - C(7, 2)/C(10, 2) = 1 - 21/45 = \mathbf{8/15}$.

Q25 **Answer: (d) 35%**

$P(\text{contradict}) = 3/4 \times 1/5 + 1/4 \times 4/5 = 3/20 + 4/20 = 7/20 = \mathbf{35}\%$.

P(E) = favourable/total for equally likely outcomes; P(not E) = 1 − P(E). P(A or B) = P(A) + P(B) − P(A and B); for independent events P(A and B) = P(A)P(B). Use combinations for draws without replacement, and the complement for "at least one". Conditional probability restricts the sample space to the given event.

MODULE 27 · ADVANCED

Geometry

Minimum duration **3 days** | Level **Advanced**

Geometry questions test properties of shapes and are common in management exams and some placement tests. This module covers angles and parallel lines, triangles (types, angle properties, Pythagoras and triplets, congruence and similarity, the four centres, area formulas including Heron's, and the 30-60-90 and 45-45-90 triangles), quadrilaterals and polygons, and circles (chords, arcs, tangents, angle properties, cyclic quadrilaterals), with a diagram for each idea and 25 solved questions checked by a program.

IN THIS MODULE

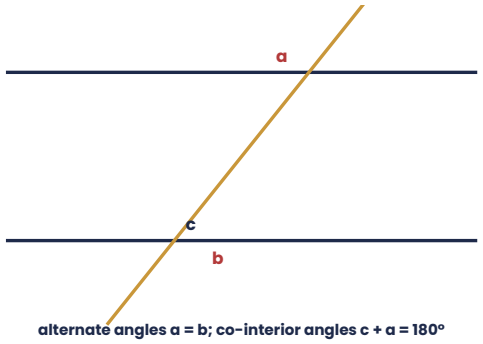
27.1 Lines and angles
27.2 Triangles
27.3 Quadrilaterals and polygons
27.4 Circles
27.5 Practice questions
27.6 Solutions

27.1 Lines and angles

27.1.1 Types of angles

Angle	Measure
Acute	between 0° and 90°
Right	90°
Obtuse	between 90° and 180°
Straight	180°
Reflex	between 180° and 360°
Complementary pair	sum 90°
Supplementary pair	sum 180°

27.1.2 Parallel lines and a transversal



PARALLEL LINES CUT BY A TRANSVERSAL

Corresponding angles are equal, alternate interior angles are equal, and co-interior (same-side interior) angles add to 180°.

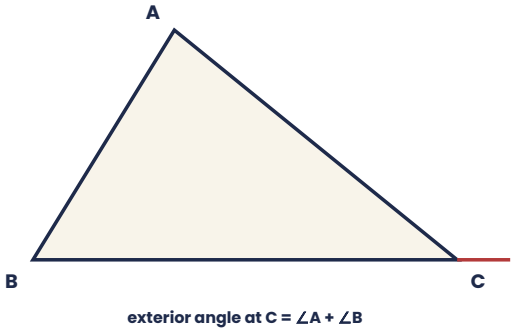
27.2 Triangles

27.2.1 Types of triangles

By sides: equilateral (all equal), isosceles (two equal), scalene (none equal). By angles: acute, right, obtuse. The sum of any two sides is greater than the third side.

27.2.2 Angle sum and exterior angle

Angles of a triangle add to 180°; an exterior angle equals the sum of the two opposite interior angles



THE EXTERIOR ANGLE PROPERTY

27.2.3 Pythagoras and triplets

In a right triangle: hypotenuse² = sum of squares of the other two sides

Triplets: 3-4-5, 5-12-13, 8-15-17, 7-24-25, 20-21-29, 9-40-41 and their multiples.

27.2.4 Congruence and similarity

Congruent triangles are identical (tests SSS, SAS, ASA, RHS). Similar triangles have equal angles and proportional sides (AA, SSS ratio, SAS ratio); the ratio of their areas is the square of the ratio of their sides.

27.2.5 Centres of a triangle

Centre	Meeting point of	Property
Centroid	medians	divides each median 2 : 1 from the vertex
Incentre	angle bisectors	centre of the inscribed circle
Circumcentre	perpendicular bisectors of sides	centre of the circumscribed circle; midpoint of the hypotenuse in a right triangle
Orthocentre	altitudes	at the right-angle vertex in a right triangle

27.2.6 Area formulas

Given	Area
base b, height h	½ b h
two sides and included angle	½ ab sin C
three sides (Heron)	$\sqrt{[s(s-a)(s-b)(s-c)]}$, $s = (a+b+c)/2$
equilateral side a	$(\sqrt{3}/4)\,a^2$
inradius r	r × s

WORKED EXAMPLE

Area of a triangle with sides 13, 14, 15

s = 21; area = $\sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7056} =$ **84**.

27.2.7 Special triangles

30-60-90: sides in the ratio 1 : $\sqrt{3}$: 2. 45-45-90: sides in the ratio 1 : 1 : $\sqrt{2}$.

27.3 Quadrilaterals and polygons

27.3.1 Quadrilaterals

Shape	Key properties
Parallelogram	opposite sides parallel and equal; diagonals bisect each other
Rectangle	parallelogram with right angles; diagonals equal
Rhombus	all sides equal; diagonals bisect at right angles; area = ½ d_1 d_2
Square	rectangle and rhombus together

Shape	Key properties
Trapezium	one pair of parallel sides; area = ½ (sum of parallel sides) × height

27.3.2 Angles of polygons

Sum of interior angles of an n-gon = (n - 2) × 180°; sum of exterior angles = 360°

27.3.3 Regular polygons and diagonals

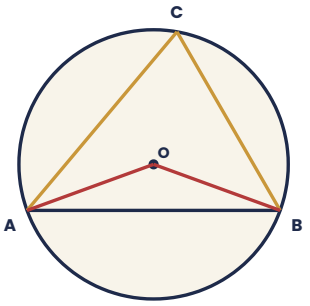
Each interior angle of a regular n-gon = (n - 2) × 180°/n; each exterior angle = 360°/n. Number of diagonals = n(n - 3)/2.

27.4 Circles

27.4.1 Chords, arcs, tangents and secants

The perpendicular from the centre to a chord bisects it. Equal chords are equidistant from the centre. A tangent touches the circle at one point and is perpendicular to the radius there.

27.4.2 Angle properties



angle at the centre ∠AOB = 2 × angle at the circumference ∠ACB

ANGLE AT THE CENTRE IS TWICE THE ANGLE AT THE CIRCUMFERENCE

The angle in a semicircle is 90°. Angles in the same segment are equal.

27.4.3 Cyclic quadrilaterals

Opposite angles of a cyclic quadrilateral add to 180°; an exterior angle equals the interior opposite angle.

27.4.4 Tangent properties

Tangents from an external point are equal in length. If PT is a tangent and a secant from P meets the circle at A and B, then $PT^2 = PA \times PB$. The angle between a tangent and a chord equals the angle in the alternate segment.

27.5 Practice questions

27.5.1 Basic level

Q1 Two angles of a triangle are 50° and 60°. What is the third?

(a) 80° **(b)** 60° **(c)** 90° **(d)** 70°

Q2 What is the hypotenuse of a right triangle with legs 9 and 12?

(a) 15 **(b)** 21 **(c)** 13 **(d)** 16

Q3 What is the sum of the interior angles of a hexagon?

(a) 540° **(b)** 720° **(c)** 900° **(d)** 360°

Q4 Each exterior angle of a regular polygon is 40°. How many sides does it have?

(a) 8 **(b)** 10 **(c)** 9 **(d)** 12

Q5 What is the area of a triangle with base 10 cm and height 6 cm?

(a) 60 cm² **(b)** 16 cm² **(c)** 15 cm² **(d)** 30 cm²

Q6 What is the complement of 35°?

(a) 145° **(b)** 55° **(c)** 65° **(d)** 45°

Q7 An exterior angle of a triangle is 110° and one opposite interior angle is 45°. What is the other opposite interior angle?

(a) 65° **(b)** 70° **(c)** 55° **(d)** 135°

Q8 What is the area of a rhombus with diagonals 10 cm and 24 cm?

(a) 240 cm² **(b)** 60 cm² **(c)** 130 cm² **(d)** 120 cm²

27.5.2 Intermediate level

Q9 What is the area of a triangle with sides 13, 14 and 15?

(a) 91 **(b)** 72 **(c)** 105 **(d)** 84

Q10 What is the area of an equilateral triangle with side 6 cm?

(a) $18\sqrt{3}$ cm² **(b)** 36 cm² **(c)** $6\sqrt{3}$ cm² **(d)** $9\sqrt{3}$ cm²

Q11 In a 30-60-90 triangle the shortest side is 5. What is the hypotenuse?

(a) $5\sqrt{3}$ **(b)** $5\sqrt{2}$ **(c)** 15 **(d)** 10

Q12 Each interior angle of a regular octagon is

(a) 120° **(b)** 140° **(c)** 150° **(d)** 135°

Q13 How many diagonals does a decagon have?

(a) 45 **(b)** 35 **(c)** 40 **(d)** 30

Q14 The angle at the centre of a circle subtended by an arc is 100°. What is the angle subtended by the same arc at the circumference?

(a) 100° **(b)** 200° **(c)** 50° **(d)** 80°

Q15 In a cyclic quadrilateral ABCD, ∠A = 75°. What is ∠C?

(a) 75° **(b)** 115° **(c)** 105° **(d)** 95°

Q16 Two similar triangles have corresponding sides in the ratio 3 : 5. What is the ratio of their areas?

(a) 9 : 25 **(b)** 3 : 5 **(c)** 27 : 125 **(d)** 6 : 10

27.5.3 Advanced level

Q17 A chord of length 16 cm is 6 cm from the centre of a circle. What is the radius?

(a) 8 cm **(b)** 10 cm **(c)** 12 cm **(d)** 11 cm

Q18 From a point 13 cm from the centre of a circle of radius 5 cm, what is the length of the tangent?

(a) 12 cm **(b)** 8 cm **(c)** 10 cm **(d)** 14 cm

Q19 In triangle ABC, the median from A is 9 cm long. How far is the centroid from A?

(a) 3 cm **(b)** 4.5 cm **(c)** 6 cm **(d)** 7 cm

Q20 PT is a tangent and PAB a secant of a circle with PA = 4 and PB = 9. What is PT?

(a) 5 **(b)** 6 **(c)** 6.5 **(d)** 13

Q21 In a right triangle with legs 6 and 8, what is the circumradius?

(a) 10 **(b)** 4 **(c)** 7 **(d)** 5

Q22 What is the inradius of a right triangle with sides 6, 8, 10?

(a) 2 **(b)** 3 **(c)** 2.4 **(d)** 4

Q23 The sum of the interior angles of a polygon is 1440°. How many sides does it have?

(a) 8 **(b)** 10 **(c)** 9 **(d)** 12

Q24 Two parallel lines are cut by a transversal. One co-interior angle is (2x + 10)° and the other (3x - 30)°. What is x?

(a) 40 **(b)** 30 **(c)** 32 **(d)** 44

Q25 ABCD is a trapezium with parallel sides 12 cm and 20 cm and height 8 cm. What is its area?

(a) 160 cm² **(b)** 96 cm² **(c)** 256 cm² **(d)** 128 cm²

27.6 Solutions

Q1 **Answer: (d) 70°**

180 - 110 = **70°**.

Q2 **Answer: (a) 15**

√(81 + 144) = √225 = **15**.

Q3 **Answer: (b) 720°**

(6 - 2) × 180 = **720°**.

Q4 **Answer: (c) 9**

360/40 = **9**.

Q5 **Answer: (d) 30 cm²**

½ × 10 × 6 = **30 cm²**.

Q6 **Answer: (b) 55°**

90 − 35 = **55°**.

Q7 **Answer: (a) 65°**

110 − 45 = **65°**.

Q8 **Answer: (d) 120 cm²**

½ × 10 × 24 = **120 cm²**.

Q9 **Answer: (d) 84**

s = 21; √(21 × 8 × 7 × 6) = **84**.

Q10 **Answer: (d) 9√3 cm²**

(√3/4) × 36 = **9√3 cm²**.

Q11 **Answer: (d) 10**

Sides 1 : √3 : 2, so the hypotenuse is **10**.

Q12 **Answer: (d) 135°**

(8 − 2) × 180/8 = **135°**.

Q13 **Answer: (b) 35**

10 × 7/2 = **35**.

Q14 **Answer: (c) 50°**

Half the central angle: **50°**.

Q15 **Answer: (c) 105°**

Opposite angles add to 180°: **105°**.

Q16 **Answer: (a) 9 : 25**

Area ratio = square of side ratio: **9 : 25**.

Q17 **Answer: (b) 10 cm**

Half chord 8, distance 6: radius = √(64 + 36) = **10 cm**.

Q18 **Answer: (a) 12 cm**

$\sqrt{(13^2 - 5^2)} =$ **12 cm**.

Q19 **Answer: (c) 6 cm**

The centroid divides a median 2 : 1 from the vertex: **6 cm**.

Q20 **Answer: (b) 6**

$PT^2 = PA \times PB = 36 \rightarrow$ **6**.

Q21 **Answer: (d) 5**

Circumcentre = midpoint of the hypotenuse (10): radius **5**.

Q22 **Answer: (a) 2**

Area 24, $s = 12 \rightarrow r = 24/12 =$ **2**.

Q23 **Answer: (b) 10**

$(n - 2) \times 180 = 1440 \rightarrow n =$ **10**.

Q24 **Answer: (a) 40**

$(2x + 10) + (3x - 30) = 180 \rightarrow 5x = 200 \rightarrow x =$ **40**.

Q25 **Answer: (d) 128 cm²**

$\frac{1}{2} \times 32 \times 8 =$ **128 cm²**.

Triangle angles add to 180° and an exterior angle equals the two opposite interior angles. Pythagoras and its triplets solve most right triangles; 30-60-90 is $1 : \sqrt{3} : 2$ and 45-45-90 is $1 : 1 : \sqrt{2}$. Polygon angle sum $(n - 2)180°$, exterior sum 360°. In circles, the central angle is twice the inscribed angle, opposite angles of a cyclic quadrilateral add to 180°, and $PT^2 = PA \times PB$.

MODULE 28 · ADVANCED

Mensuration

Minimum duration **2 days** | Level **Advanced**

Mensuration calculates areas, perimeters, surface areas and volumes of shapes. This module covers plane figures (triangle, rectangle, square, parallelogram, rhombus, trapezium, circle, semicircle, sector, ring), paths around and inside fields, the effect of percentage changes in dimensions, solids (cube, cuboid, cylinder, cone, frustum, sphere, hemisphere, prism, pyramid), melting and recasting, painted-cube problems, water flowing through pipes and costing problems, followed by 25 solved questions computed exactly (π = 22/7 unless stated).

IN THIS MODULE

28.1 2D mensuration
28.2 3D mensuration
28.3 Problem types
28.4 Practice questions
28.5 Solutions

28.1 2D mensuration

28.1.1 Rectilinear figures

Figure	Perimeter	Area
Rectangle l × b	2(l + b)	lb
Square side a	4a	a^2 (= $d^2/2$)
Triangle	a + b + c	½ bh or Heron
Parallelogram	2(a + b)	base × height
Rhombus	4a	½ d_1d_2
Trapezium	sum of sides	½ (a + b) h

28.1.2 Circle, semicircle, sector, ring

Figure	Perimeter	Area
Circle	2πr	πr^2
Semicircle	πr + 2r	½ πr^2
Sector of angle θ	2r + (θ/360) 2πr	(θ/360) πr^2
Ring (radii R > r)	—	$\pi(R^2 - r^2)$

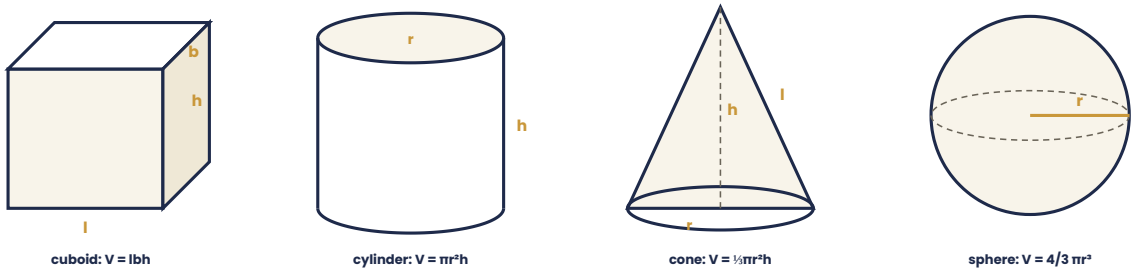
28.1.3 Paths around and inside fields

A path of width w **outside** an l × b field: area = (l + 2w)(b + 2w) - lb. A path **inside**: lb - (l - 2w)(b - 2w). Two crossroads of width w through the middle: w(l + b) - w².

28.1.4 Effect of percentage changes

If every length changes by x%, area changes by (2x + x²/100)% and volume by the cube factor. Length +20% and breadth -10%: area changes by 20 - 10 - 2 = +8%.

28.2 3D mensuration



THE MAIN SOLIDS AND THEIR VOLUMES

28.2.1 Cube and cuboid

Cuboid: V = lbh, total surface area 2(lb + bh + hl), diagonal √(l² + b² + h²). Cube: V = a³, TSA 6a², diagonal √3 a.

28.2.2 Cylinder

V = πr²h; curved surface area 2πrh; total surface area 2πr(r + h).

28.2.3 Cone and frustum

Cone: V = ⅓πr²h, slant height l = √(r² + h²), CSA πrl, TSA πr(l + r). Frustum with radii R, r and height h: V = ⅓πh(R² + Rr + r²).

28.2.4 Sphere and hemisphere

Sphere: V = 4/3 πr³, surface 4πr². Hemisphere: V = 2/3 πr³, curved surface 2πr², total 3πr².

28.2.5 Prism and pyramid

Prism: V = base area × height. Pyramid: V = ⅓ × base area × height.

28.3 Problem types

28.3.1 Melting and recasting

Volume is conserved: number of new solids = volume of original / volume of one new solid

28.3.2 Painted cubes

A painted cube of side n cut into n^3 unit cubes: 3 faces painted = 8 (corners); 2 faces = 12(n - 2) (edges); 1 face = $6(n - 2)^2$; no face = $(n - 2)^3$. For n = 4: 8, 24, 24, 8.

28.3.3 Water flowing through pipes

Volume per unit time = cross-section area × speed of flow. A pipe of radius 1 cm with water at 7 m/s delivers $\pi \times 1^2 \times 700 = 2200$ cm^3 per second.

28.3.4 Costing

Cost of fencing = perimeter × rate; flooring or painting = area × rate. Convert units first (1 m^2 = 10,000 cm^2; 1 m^3 = 1000 litres).

28.4 Practice questions

28.4.1 Basic level

Q1 What is the area of a circle of radius 14 cm?

(a) 88 cm^2 **(b)** 308 cm^2 **(c)** 1,232 cm^2 **(d)** 616 cm^2

Q2 What is the volume of a cube of side 5 cm?

(a) 125 cm^3 **(b)** 25 cm^3 **(c)** 150 cm^3 **(d)** 75 cm^3

Q3 What is the perimeter of a rectangle 12 m by 8 m?

(a) 40 m **(b)** 96 m **(c)** 20 m **(d)** 48 m

Q4 What is the volume of a cylinder with radius 7 cm and height 10 cm?

(a) 1,540 cm^3 **(b)** 1,100 cm^3 **(c)** 2,200 cm^3 **(d)** 770 cm^3

Q5 What is the total surface area of a cube of side 4 cm?

(a) 64 cm^2 **(b)** 96 cm^2 **(c)** 48 cm^2 **(d)** 16 cm^2

Q6 What is the area of a square whose diagonal is 10 cm?

(a) 100 cm^2 **(b)** 25 cm^2 **(c)** 50 cm^2 **(d)** 40 cm^2

Q7 What is the circumference of a circle of radius 21 cm?

(a) 66 cm **(b)** 1,386 cm **(c)** 121 cm **(d)** 132 cm

Q8 What is the volume of a cuboid 10 cm × 8 cm × 5 cm?

(a) 400 cm^3 **(b)** 230 cm^3 **(c)** 340 cm^3 **(d)** 450 cm^3

28.4.2 Intermediate level

Q9 What is the volume of a sphere of radius 21 cm?

(a) 9,702 cm^3 **(b)** 38,808 cm^3 **(c)** 5,544 cm^3 **(d)** 19,404 cm^3

Q10 A cone has radius 6 cm and height 8 cm. What is its curved surface area? (π = 3.14)

(a) 188.4 cm^2 **(b)** 150.72 cm^2 **(c)** 301.44 cm^2 **(d)** 113.04 cm^2

Q11 A path 2 m wide runs around the outside of a 20 m × 15 m garden. What is its area?

(a) 140 m² **(b)** 128 m² **(c)** 160 m² **(d)** 156 m²

Q12 The length of a rectangle increases by 20% and its breadth decreases by 10%. What is the change in area?

(a) 10% increase **(b)** 2% increase **(c)** 8% increase **(d)** 8% decrease

Q13 What is the area of a sector of radius 14 cm and angle 90°?

(a) 154 cm² **(b)** 308 cm² **(c)** 77 cm² **(d)** 616 cm²

Q14 A 4 cm cube is painted on all faces and cut into 1 cm cubes. How many small cubes have exactly two faces painted?

(a) 8 **(b)** 16 **(c)** 32 **(d)** 24

Q15 What is the curved surface area of a cylinder of radius 7 cm and height 12 cm?

(a) 264 cm² **(b)** 616 cm² **(c)** 528 cm² **(d)** 1,848 cm²

Q16 What is the length of the longest rod that fits in a box 12 m × 4 m × 3 m?

(a) 13 m **(b)** 12 m **(c)** 19 m **(d)** 14 m

28.4.3 Advanced level

Q17 A metal sphere of radius 6 cm is melted and recast into small spheres of radius 1 cm. How many are made?

(a) 216 **(b)** 36 **(c)** 72 **(d)** 108

Q18 Water flows through a pipe of radius 1 cm at 7 m/s. How many litres flow in one minute?

(a) 66 litres **(b)** 132 litres **(c)** 220 litres **(d)** 2,200 litres

Q19 A cone and a cylinder have the same base and height. What is the ratio of their volumes?

(a) 3 : 1 **(b)** 1 : 2 **(c)** 2 : 3 **(d)** 1 : 3

Q20 A cylinder of radius 3 cm and height 8 cm is melted and recast into a cone of radius 6 cm. What is the height of the cone?

(a) 6 cm **(b)** 8 cm **(c)** 4 cm **(d)** 12 cm

Q21 By what percentage does the volume of a sphere increase if its radius increases by 10%?

(a) 33.1% **(b)** 30% **(c)** 21% **(d)** 33%

Q22 A 5 cm cube is painted and cut into 1 cm cubes. How many small cubes have no face painted?

(a) 64 **(b)** 9 **(c)** 27 **(d)** 54

Q23 What is the cost of fencing a circular field of radius 35 m at ₹10 per metre?

(a) 1,100 **(b)** 3,850 **(c)** 2,100 **(d)** 2,200

Q24 A hemispherical bowl of radius 21 cm is full of water. How many litres does it hold?

(a) 19.4 litres **(b)** 9.7 litres **(c)** 38.81 litres **(d)** 13.86 litres

Q25 Two cubes of side 6 cm are joined end to end. What is the surface area of the resulting cuboid?

(a) 432 cm² **(b)** 396 cm² **(c)** 361 cm² **(d)** 360 cm²

28.5 Solutions

Q1 **Answer: (d) 616 cm²**
22/7 × 196 = **616 cm²**.

Q2 **Answer: (a) 125 cm³**
5^3 = **125 cm³**.

Q3 **Answer: (a) 40 m**
2(12 + 8) = **40 m**.

Q4 **Answer: (a) 1,540 cm³**
22/7 × 49 × 10 = **1,540 cm³**.

Q5 **Answer: (b) 96 cm²**
6 × 16 = **96 cm²**.

Q6 **Answer: (c) 50 cm²**
$d^2/2$ = **50 cm²**.

Q7 **Answer: (d) 132 cm**
2 × 22/7 × 21 = **132 cm**.

Q8 **Answer: (a) 400 cm³**
10 × 8 × 5 = **400 cm³**.

Q9 **Answer: (b) 38,808 cm³**
4/3 × 22/7 × 9261 = **38,808 cm³**.

Q10 **Answer: (a) 188.4 cm²**
Slant height √(36 + 64) = 10; CSA = πrl = 3.14 × 60 = **188.4 cm²**.

Q11 **Answer: (d) 156 m²**
(24 × 19) − 300 = 456 − 300 = **156 m²**.

Q12 **Answer: (c) 8% increase**
20 − 10 − 2 = **+8%**.

Q13 **Answer: (a) 154 cm²**
¼ × 616 = **154 cm²**.

Q14 **Answer: (d) 24**

Edge cubes: 12(n - 2) = **24**.

Q15 **Answer: (c) 528 cm²**

2 × 22/7 × 7 × 12 = **528 cm²**.

Q16 **Answer: (a) 13 m**

√(144 + 16 + 9) = √169 = **13 m**.

Q17 **Answer: (a) 216**

Volume ratio $(6/1)^3$ = **216**.

Q18 **Answer: (b) 132 litres**

Per second π × 1^2 × 700 = 2200 cm^3; per minute 1,32,000 cm^3 = **132 litres**.

Q19 **Answer: (d) 1 : 3**

Cone = ⅓ of the cylinder: **1 : 3**.

Q20 **Answer: (a) 6 cm**

$\pi r^2 h$ is conserved: π × 9 × 8 = ⅓ π × 36 × h → h = 216/36 = **6 cm**.

Q21 **Answer: (a) 33.1%**

1.1^3 = 1.331 → **33.1%** increase.

Q22 **Answer: (c) 27**

$(n - 2)^3$ = **27**.

Q23 **Answer: (d) 2,200**

Circumference = 2 × 22/7 × 35 = 220 m → **₹2,200**.

Q24 **Answer: (a) 19.4 litres**

2/3 × 22/7 × 9261 = 19,404 cm^3 = **19.404 litres**.

Q25 **Answer: (d) 360 cm²**

Cuboid 12 × 6 × 6: 2(72 + 36 + 72) = **360 cm²**.

Learn the area formulas for plane figures and the volume and surface formulas for solids (cuboid lbh, cylinder $\pi r^2 h$, cone ⅓$\pi r^2 h$, sphere 4/3 πr^3). Volume is conserved in melting and recasting. Percentage changes multiply. Painted cubes: 8 corners, 12(n - 2) edges, $6(n - 2)^2$ faces, $(n - 2)^3$ inside. Convert units before costing.

MODULE 29 · ADVANCED

Coordinate geometry and trigonometry

Minimum duration **2 days** | Level **Advanced**

These topics appear in management exams and some technical aptitude tests. This module covers the distance, section and midpoint formulas, the area of a triangle from coordinates, slopes and equations of lines, parallel and perpendicular lines, the equation of a circle, trigonometric ratios, standard-angle values, basic identities, and heights and distances with one and two observation points, followed by 25 solved questions checked by a program.

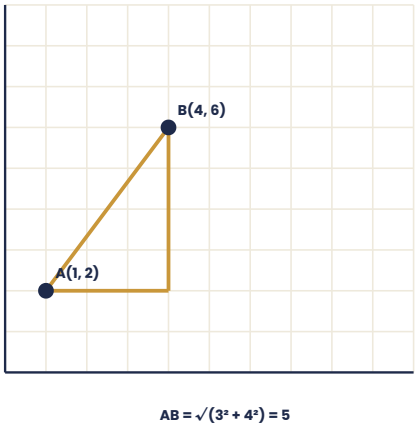
IN THIS MODULE

29.1 Coordinate geometry
29.2 Trigonometry
29.3 Heights and distances
29.4 Practice questions
29.5 Solutions

29.1 Coordinate geometry

29.1.1 Distance formula

Distance between (x_1, y_1) and $(x_2, y_2) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$



DISTANCE AS THE HYPOTENUSE OF A RIGHT TRIANGLE

29.1.2 Section formula and midpoint

The point dividing (x_1, y_1)–(x_2, y_2) internally in the ratio m : n is $((mx_2 + nx_1)/(m + n), (my_2 + ny_1)/(m + n))$. The midpoint is $((x_1 + x_2)/2, (y_1 + y_2)/2)$. The centroid of a triangle is the average of the three vertices.

29.1.3 Area of a triangle

Area = ½ $|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$; area 0 means the points are collinear

29.1.4 Slope and equation of a line

Slope m = $(y_2 - y_1)/(x_2 - x_1)$. Forms: $y = mx + c$; $y - y_1 = m(x - x_1)$; $ax + by + c = 0$ has slope $-a/b$, x-intercept $-c/a$, y-intercept $-c/b$.

29.1.5 Parallel and perpendicular lines

Parallel lines have equal slopes; perpendicular lines have slopes whose product is -1. The distance from (x_0, y_0) to $ax + by + c = 0$ is $|ax_0 + by_0 + c|/\sqrt{(a^2 + b^2)}$.

29.1.6 Equation of a circle

Centre (h, k), radius r: $(x - h)^2 + (y - k)^2 = r^2$. $x^2 + y^2 + 2gx + 2fy + c = 0$ has centre (-g, -f) and radius $\sqrt{(g^2 + f^2 - c)}$.

29.2 Trigonometry

29.2.1 Trigonometric ratios

In a right triangle with angle θ: $\sin\theta$ = opposite/hypotenuse, $\cos\theta$ = adjacent/hypotenuse, $\tan\theta$ = opposite/adjacent; cosec, sec and cot are their reciprocals.

29.2.2 Standard angles

θ	0°	30°	45°	60°	90°
$\sin\theta$	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0
$\tan\theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	not defined

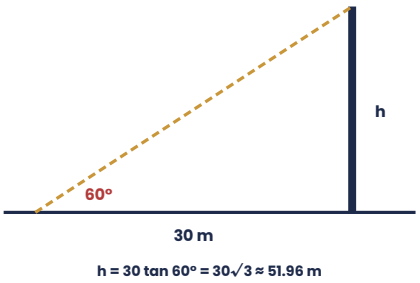
29.2.3 Basic identities

$\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \text{cosec}^2\theta$

29.3 Heights and distances

29.3.1 Angles of elevation and depression

Looking up at an object, the angle above the horizontal is the **angle of elevation**; looking down, the angle below the horizontal is the **angle of depression**. Both lead to a right triangle: height = distance × tan(angle).



A TOWER SEEN AT AN ANGLE OF ELEVATION OF 60°

29.3.2 Two observation points

If the elevation changes from α to β as an observer walks d towards a tower, then $h(\cot\alpha - \cot\beta) = d$. From 30° to 60° over 20 m: $h(\sqrt{3} - 1/\sqrt{3}) = 20 \rightarrow h = 10\sqrt{3} \approx 17.32$ m.

> **SHORTCUT TRICK**
> Learn tan 30° = 1/√3, tan 45° = 1, tan 60° = √3. With 30° and 60°, the two distances are in the ratio 3 : 1 for the same height.

29.4 Practice questions

29.4.1 Basic level

Q1 What is the distance between (1, 2) and (4, 6)?

(a) 7 **(b)** 4 **(c)** 25 **(d)** 5

Q2 What is the midpoint of (2, 8) and (6, 4)?

(a) (8, 12) **(b)** (2, 2) **(c)** (4, 6) **(d)** (4, 4)

Q3 What is the slope of the line through (1, 3) and (4, 9)?

(a) 2 **(b)** 1/2 **(c)** 3 **(d)** 6

Q4 sin 30° + cos 60° = ?

(a) 1/2 **(b)** √3 **(c)** 0 **(d)** 1

Q5 tan 45° = ?

(a) 1 **(b)** 0 **(c)** √3 **(d)** 1/√3

Q6 What is the slope of the line 2x + 3y - 6 = 0?

(a) 2/3 **(b)** -3/2 **(c)** 2 **(d)** -2/3

Q7 $\sin^2 40° + \cos^2 40°$ = ?

(a) 0 **(b)** 1/2 **(c)** 1 **(d)** 2

Q8 A tower casts a shadow 30 m long when the sun's elevation is 45°. How tall is the tower?

(a) 30 m **(b)** 30√3 m **(c)** 15 m **(d)** 10√3 m

29.4.2 Intermediate level

Q9 What is the area of the triangle with vertices (0, 0), (4, 0) and (0, 3)?

(a) 12 **(b)** 7 **(c)** 5 **(d)** 6

Q10 Find the point dividing the join of (1, 2) and (7, 8) internally in the ratio 1 : 2.

(a) (5, 6) **(b)** (4, 5) **(c)** (2, 3) **(d)** (3, 4)

Q11 Are the points (1, 1), (2, 3) and (3, 5) collinear?

(a) No **(b)** Only if extended
(c) Cannot be determined **(d)** Yes

Q12 What is the slope of a line perpendicular to $y = 3x + 2$?

(a) $-1/3$ **(b)** 3 **(c)** $1/3$ **(d)** -3

Q13 A tower stands 30 m from a point; the angle of elevation of its top is 60°. What is its height?

(a) $10\sqrt{3}$ m **(b)** $30\sqrt{3}$ m **(c)** 15 m **(d)** 60 m

Q14 If $\sin\theta = 3/5$ (θ acute), what is $\tan\theta$?

(a) 4/3 **(b)** 4/5 **(c)** 3/4 **(d)** 3/5

Q15 What are the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$?

(a) centre (−3, 2), radius 5 **(b)** centre (3, −2), radius 12
(c) centre (6, −4), radius 5 **(d)** centre (3, −2), radius 5

Q16 From the top of a 50 m cliff, the angle of depression of a boat is 30°. How far is the boat from the foot of the cliff?

(a) 50 m **(b)** $50/\sqrt{3}$ m **(c)** $50\sqrt{3}$ m **(d)** 100 m

29.4.3 Advanced level

Q17 The angle of elevation of a tower changes from 30° to 60° as an observer walks 20 m towards it. What is its height?

(a) $20\sqrt{3}$ m **(b)** 10 m **(c)** 20 m **(d)** $10\sqrt{3}$ m

Q18 What is the distance from the point (3, 4) to the line $3x + 4y - 5 = 0$?

(a) 5 **(b)** 4 **(c)** 3 **(d)** 2

Q19 What is the centroid of the triangle with vertices (2, 3), (4, 7) and (6, 2)?

(a) (4, 6) **(b)** (6, 4) **(c)** (3, 4) **(d)** (4, 4)

Q20 What is the equation of the line through (2, 3) with slope 2?

(a) $y = 2x - 1$ **(b)** $y = 2x + 3$ **(c)** $y = 2x + 1$ **(d)** $y = x + 1$

Q21 $(1 - \cos^2\theta)\sec^2\theta = ?$

(a) $\tan^2\theta$ **(b)** $\sin^2\theta$ **(c)** $\cos^2\theta$ **(d)** 1

Q22 What is the value of $\sin 60° \cos 30° + \cos 60° \sin 30°$?

(a) 1/2 **(b)** 1 **(c)** 3/4 **(d)** $\sqrt{3}/2$

Q23 A ladder 10 m long leans against a wall, making 60° with the ground. How high up the wall does it reach?

(a) 5 m **(b)** 10√3 m **(c)** 20 m **(d)** 5√3 m

Q24 For what value of k are the lines 2x + ky = 5 and 3x − 6y = 7 perpendicular?

(a) −1 **(b)** 4 **(c)** −4 **(d)** 1

Q25 Find the area of the quadrilateral with vertices (0, 0), (4, 0), (4, 3), (0, 5).

(a) 20 **(b)** 12 **(c)** 14 **(d)** 16

29.5 Solutions

Q1 **Answer: (d) 5**

√(9 + 16) = **5**.

Q2 **Answer: (c) (4, 6)**

((2 + 6)/2, (8 + 4)/2) = **(4, 6)**.

Q3 **Answer: (a) 2**

(9 − 3)/(4 − 1) = **2**.

Q4 **Answer: (d) 1**

1/2 + 1/2 = **1**.

Q5 **Answer: (a) 1**

tan 45° = **1**.

Q6 **Answer: (d) −2/3**

Slope = −a/b = **−2/3**.

Q7 **Answer: (c) 1**

$\sin^2\theta + \cos^2\theta$ = **1** for every θ.

Q8 **Answer: (a) 30 m**

h = 30 tan 45° = **30 m**.

Q9 **Answer: (d) 6**

½ × 4 × 3 = **6**.

Q10 **Answer: (d) (3, 4)**

((1 × 7 + 2 × 1)/3, (1 × 8 + 2 × 2)/3) = **(3, 4)**.

Q11 **Answer: (d) Yes**

Area = ½|1(3 − 5) + 2(5 − 1) + 3(1 − 3)| = ½|−2 + 8 − 6| = 0 → **Yes**.

Q12 **Answer: (a) −1/3**

Product of slopes −1 → **−1/3**.

Q13 **Answer: (b) 30√3 m**

h = 30 tan 60° = **30√3 m** (≈ 51.96 m).

Q14 **Answer: (c) 3/4**

cos θ = 4/5, so tan θ = 3/4 → **3/4**.

Q15 **Answer: (d) centre (3, −2), radius 5**

g = −3, f = 2: centre (3, −2), radius √(9 + 4 + 12) = 5.

Q16 **Answer: (c) 50√3 m**

Distance = 50/tan 30° = **50√3 m**.

Q17 **Answer: (d) 10√3 m**

h(cot 30° − cot 60°) = 20 → h(√3 − 1/√3) = 20 → h(2/√3) = 20 → h = **10√3 m**.

Q18 **Answer: (b) 4**

|9 + 16 − 5|/√(9 + 16) = 20/5 = **4**.

Q19 **Answer: (d) (4, 4)**

((2 + 4 + 6)/3, (3 + 7 + 2)/3) = **(4, 4)**.

Q20 **Answer: (a) y = 2x − 1**

y − 3 = 2(x − 2) → **y = 2x − 1**.

Q21 **Answer: (a) tan²θ**

sin²θ/cos²θ = **tan²θ**.

Q22 **Answer: (b) 1**

(√3/2)(√3/2) + (1/2)(1/2) = 3/4 + 1/4 = **1** (= sin 90°).

Q23 **Answer: (d) 5√3 m**

Height = 10 sin 60° = **5√3 m**.

Q24 **Answer: (d) 1**

Slopes −2/k and 1/2: (−2/k)(1/2) = −1 → k = **1**.

Q25 **Answer: (d) 16**

Split into two triangles: 6 + 10 = **16**.

Distance $\sqrt{(\Delta x^2 + \Delta y^2)}$; section $((mx_2 + nx_1)/(m + n), \ldots)$; area $\frac{1}{2}|x_1(y_2 - y_3) + \ldots|$ (0 means collinear). Slope $(y_2 - y_1)/(x_2 - x_1)$; parallel lines share slopes, perpendicular slopes multiply to -1. Learn the standard trig values and $\sin^2 + \cos^2 = 1$. Heights: h = d tan(angle); with two points, $h(\cot \alpha - \cot \beta) = d$.

MODULE 30 · ADVANCED

Data interpretation

Minimum duration **3 days** | Level **Advanced**

Data interpretation (DI) tests reading charts and calculating quickly; it is a major section in placement and bank exams. This module covers the DI approach (read the questions first, approximate, compute percentage changes, ratios and averages), every common chart type (tables, simple and multiple bar graphs, line graphs, pie charts, mixed graphs, caselets, missing-data sets, radar and scatter charts) and set-selection strategy, followed by five complete DI sets with 25 solved questions computed from the data.

IN THIS MODULE

30.1 DI fundamentals
30.2 Types of DI sets
30.3 Strategy
30.4 Practice sets
30.5 Solutions

30.1 DI fundamentals

30.1.1 Read the questions before the data

Skim the five questions first. They tell you which rows, years or categories matter, so you can ignore the rest of the data and avoid reading everything twice.

30.1.2 Approximation for speed

Most DI options are far apart. Round to two significant figures: 4,872/6,135 ≈ 4,900/6,100 ≈ 0.80. Calculate exactly only when two options are close.

30.1.3 Percentage change, ratios and averages

Percentage change = (new − old)/old × 100; share % = part/total × 100; pie angle = share % × 3.6°

SHORTCUT TRICK

Know 1/7 = 14.29%, 1/9 = 11.11%, 1/11 = 9.09%, 1/12 = 8.33%, 1/16 = 6.25%. Converting a ratio like 13/16 to 81.25% takes seconds with these.

30.2 Types of DI sets

30.2.1 Tables

Tables give exact values; questions ask for totals, averages, growth rates and ratios. Add along rows or down columns only for the items you need.

30.2.2 Bar graphs

Simple bars show one series; multiple (grouped) bars compare series side by side; stacked bars show parts of a total. Read values from the labels when given; estimate from the scale only if necessary.

30.2.3 Line graphs

Line graphs show trends over time. The steepest segment shows the largest absolute change, not necessarily the largest percentage change.

30.2.4 Pie charts

Pie charts show shares of a total, in percentages or degrees (360° = 100%). Convert a sector to a value with value = total × share.

30.2.5 Mixed graphs

Two charts share a common variable (for example a pie of totals and a bar of shares). Combine them step by step: find the total from one, the share from the other.

30.2.6 Caselets

Data is given in a paragraph. Convert it into a small table before answering; most caselet errors come from misreading a sentence.

30.2.7 Missing-data DI

Some cells are blank and must be found from totals, averages or ratios given elsewhere. Fill them in first, then answer.

30.2.8 Radar and scatter charts

Radar charts plot several variables on spokes from a centre; scatter charts plot pairs of values as points. Both are read exactly like a table once you identify the axes.

30.3 Strategy

30.3.1 Choosing which sets to attempt

Spend 30 seconds scanning each set. Attempt first the sets with simple charts, round numbers and direct questions; leave caselets with heavy conditions for the end.

30.3.2 Avoiding calculation traps

- Percentage change uses the **earlier** value as the base.

- Percentage points (difference of two percentages) are not percentage change.
- Check units (thousands, lakhs, crores) in the chart title.
- “Approximately” allows rounding; “exactly” does not.

30.4 Practice sets

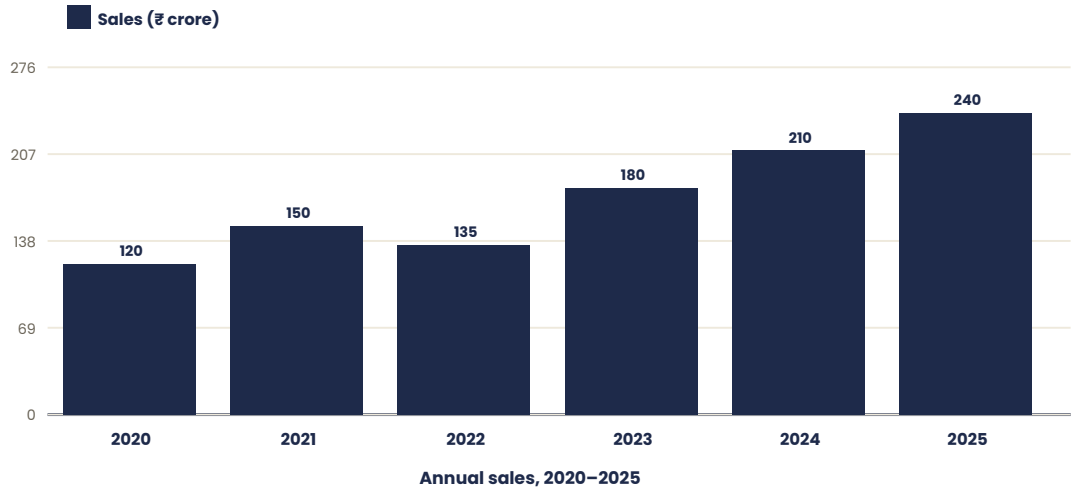
SET A · TABLE

Set A (questions 1 to 5): cars produced by four companies P, Q, R and S (in thousands).

Company	2021	2022	2023	2024	2025
P	40	45	50	60	66
Q	55	50	60	66	72
R	30	36	45	48	54
S	25	30	35	42	48

SET B · BAR GRAPH

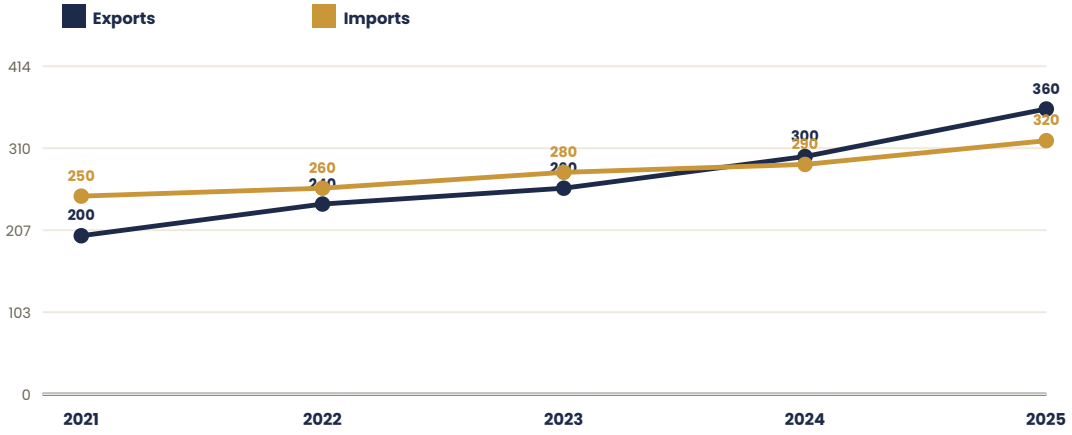
Set B (questions 6 to 10): annual sales of a company (₹ crore).



SET B DATA

SET C · LINE GRAPH

Set C (questions 11 to 15): exports and imports of a country (₹ thousand crore).

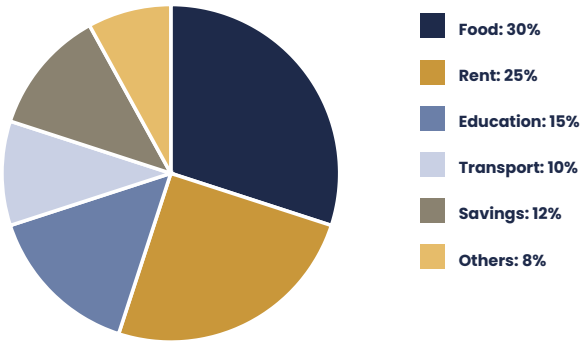


Exports and imports, 2021–2025

SET C DATA

SET D · PIE CHART

Set D (questions 16 to 20): monthly expenditure of a family with income ₹60,000.



Distribution of monthly income

SET D DATA

SET E · CASELET

Set E (questions 21 to 25): A college has 1,200 students. 45% study commerce, 35% science and the rest arts. Among commerce students, 60% are girls; among science students, 40% are girls; among arts students, 75% are girls.

30.4.1 Questions on Sets A to E

30.4.1.1 Basic level

Q1 Set A: what was the total production of the four companies in 2023 (thousands)?

(a) 180 **(b)** 185 **(c)** 190 **(d)** 200

Q2 Set A: what was the percentage increase in P's production from 2021 to 2025?

(a) 60% **(b)** 66% **(c)** 26% **(d)** 65%

Q3 Set A: what was the average production of R over the five years (thousands)?

(a) 42.6 **(b)** 43 **(c)** 45 **(d)** 40

Q4 Set A: in which year was the total production the highest?

(a) 2024 (b) 2025 (c) 2023 (d) 2022

Q5 Set A: what is the ratio of Q's production in 2022 to S's production in 2024?

(a) 25 : 21 (b) 21 : 25 (c) 5 : 4 (d) 6 : 5

Q6 Set B: what was the percentage increase in sales from 2020 to 2021?

(a) 30% (b) 20% (c) 25% (d) 15%

Q7 Set B: in which year did sales fall compared with the previous year?

(a) 2022 (b) 2021 (c) 2023 (d) 2024

Q8 Set B: what were the average annual sales over the six years (₹ crore)?

(a) 170 (b) 180 (c) 165 (d) 172.5

30.4.1.2 Intermediate level

Q9 Set B: total sales in 2023–2025 are what percentage of total sales in 2020–2022?

(a) 155.56% (b) 140% (c) 160% (d) 150%

Q10 Set B: in which year was the percentage increase over the previous year the highest?

(a) 2021 (b) 2024 (c) 2023 (d) 2025

Q11 Set C: in which year was the trade deficit (imports - exports) the largest?

(a) 2021 (b) 2022 (c) 2023 (d) 2024

Q12 Set C: in which year did exports first exceed imports?

(a) 2023 (b) 2024 (c) 2025 (d) 2022

Q13 Set C: what was the percentage increase in exports from 2021 to 2025?

(a) 60% (b) 80% (c) 70% (d) 90%

Q14 Set C: what is the ratio of total exports to total imports over the five years?

(a) 1 : 1 (b) 27 : 28 (c) 34 : 35 (d) 28 : 27

Q15 Set C: what were the average imports per year?

(a) 270 (b) 280 (c) 285 (d) 290

Q16 Set D: how much does the family spend on food each month?

(a) 18,000 (b) 15,000 (c) 20,000 (d) 12,000

30.4.1.3 Advanced level

Q17 Set D: what is the central angle of the education sector?

(a) 54° (b) 45° (c) 60° (d) 15°

Q18 Set D: by how much does spending on rent exceed spending on transport each month?

(a) 6,000 (b) 12,000 (c) 15,000 (d) 9,000

Q19 Set D: if the income rises by 10% and savings stay at the same amount, what percentage of the new income are savings?

(a) 12% (b) 11% (c) 10.91% (d) 13.2%

Q20 Set D: what is the ratio of spending on food to spending on education and transport together?

(a) 5 : 6 **(b)** 6 : 5 **(c)** 3 : 2 **(d)** 2 : 1

Q21 Set E: how many girls are there in the college?

(a) 600 **(b)** 672 **(c)** 660 **(d)** 720

Q22 Set E: how many boys study science?

(a) 168 **(b)** 252 **(c)** 210 **(d)** 240

Q23 Set E: what percentage of all girls study commerce?

(a) 45% **(b)** 50% **(c)** 60% **(d)** 48.21%

Q24 Set E: what is the ratio of boys in commerce to boys in arts?

(a) 9 : 4 **(b)** 4 : 1 **(c)** 3 : 1 **(d)** 18 : 5

Q25 Set E: by how many does the number of girls exceed the number of boys?

(a) 60 **(b)** 144 **(c)** 120 **(d)** 48

30.5 Solutions

Q1 **Answer: (c) 190**

50 + 60 + 45 + 35 = **190** thousand.

Q2 **Answer: (d) 65%**

(66 − 40)/40 × 100 = **65**%.

Q3 **Answer: (a) 42.6**

(30 + 36 + 45 + 48 + 54)/5 = 213/5 = **42.6** thousand.

Q4 **Answer: (b) 2025**

Totals: 2021: 150, 2022: 161, 2023: 190, 2024: 216, 2025: 240. Highest in **2025**.

Q5 **Answer: (a) 25 : 21**

50 : 42 = **25 : 21**.

Q6 **Answer: (c) 25%**

(150 − 120)/120 × 100 = **25**%.

Q7 **Answer: (a) 2022**

Sales fell from 150 to 135 in **2022**.

Q8 **Answer: (d) 172.5**

1035/6 = **172.5**.

Q9 **Answer: (a) 155.56%**

630/405 × 100 ≈ **155.56**%.

Q10 **Answer: (c) 2023**

Increases: 2021 25%, 2023 33.3%, 2024 16.7%, 2025 14.3%. Highest in **2023**.

Q11 **Answer: (a) 2021**

Deficits: 50, 20, 20, -10, -40. Largest in **2021**.

Q12 **Answer: (b) 2024**

In 2024 exports 300 > imports 290: **2024**.

Q13 **Answer: (b) 80%**

(360 - 200)/200 × 100 = **80**%.

Q14 **Answer: (c) 34 : 35**

1360 : 1400 = **34 : 35**.

Q15 **Answer: (b) 280**

1400/5 = **280**.

Q16 **Answer: (a) 18,000**

30% of 60,000 = **₹18,000**.

Q17 **Answer: (a) 54°**

15% × 3.6 = **54°**.

Q18 **Answer: (d) 9,000**

(25 - 10)% of 60,000 = **₹9,000**.

Q19 **Answer: (c) 10.91%**

Savings ₹7,200; new income ₹66,000; 7200/66000 × 100 ≈ **10.91**%.

Q20 **Answer: (b) 6 : 5**

30 : (15 + 10) = 30 : 25 = **6 : 5**.

Q21 **Answer: (b) 672**

Commerce 324, science 168, arts 180: total **672**.

Q22 **Answer: (b) 252**

Science students 420; girls 168; boys **252**.

Q23 **Answer: (d) 48.21%**

324/672 × 100 ≈ **48.21**%.

Q24 **Answer: (d) 18 : 5**

Commerce boys 216, arts boys 60: ratio **18 : 5**.

Q25 **Answer: (b) 144**

Girls 672, boys 528: difference **144**.

Read the questions first, approximate whenever options are far apart, and always use the earlier value as the base for percentage change. Pie chart: value = total × share, angle = share × 3.6°. Convert caselets into a table before answering. Check units and the difference between percentage points and percentage change.

MODULE 31 · ADVANCED

Data sufficiency

Minimum duration **1 day** | Level **Advanced**

Data sufficiency asks whether the given information is enough to answer a question, without fully solving it. This module covers the standard answer options, the test-each-statement-then-combine method, how to stop before solving, the common traps, quantitative DS across arithmetic, algebra, geometry and number system, and three-statement DS, followed by 25 solved questions with a written reason for each answer.

IN THIS MODULE

31.1 Format
31.2 Approach
31.3 Problem types
31.4 Practice questions
31.5 Solutions

31.1 Format

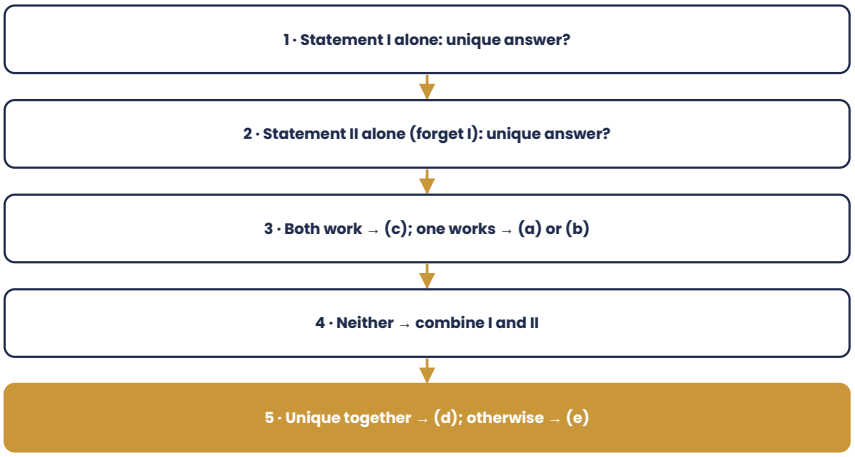
31.1.1 Standard answer options

Option	Meaning
(a)	I alone is sufficient, II alone is not
(b)	II alone is sufficient, I alone is not
(c)	Either I alone or II alone is sufficient
(d)	I and II together are needed (neither alone is enough)
(e)	Even I and II together are not sufficient

Sufficient = the data fix a single value (or a definite yes or no). You never need the value itself.

31.2 Approach

31.2.1 Test each statement, then combine



THE DATA-SUFFICIENCY PROCEDURE

31.2.2 Do not solve fully

Count equations and unknowns: two independent linear equations in two unknowns are sufficient; you do not need to find the numbers. For geometry, ask whether the figure is fixed by the data.

31.2.3 Common traps

- Two equations that are multiples of each other ($2x + 4y = 10$ and $x + 2y = 5$) are really one equation.
- A quadratic often gives two values: not sufficient unless one is rejected (for example, a length cannot be negative).
- A definite "no" is sufficient.
- Do not assume integers, positive numbers or a figure drawn to scale unless the question says so.

31.3 Problem types

31.3.1 DS across topics

Topic	Usually sufficient when you know …
Percentages and profit	any two of cost, selling price and profit %
Interest	principal, rate and time (or the amount and two of them)
Speed	any two of distance, speed and time
Algebra	as many independent equations as unknowns
Geometry	enough lengths and angles to fix the figure
Number system	a condition that leaves one number

31.3.2 Three-statement DS

With three statements, the options ask which statements are needed (for example "only I and II", "any two of the three", "all three"). Test each statement alone, then pairs, then all three; choose the smallest set that is sufficient.

> WORKED EXAMPLE
> **What is the value of x + y? I. x − y = 4. II. 2x + 2y = 18. III. x = y + 4.**
> II alone gives x + y = 9, so II is sufficient on its own. I and III say the same thing (x − y = 4), so even together they cannot give x + y. Answer: only II is needed.

31.4 Practice questions

31.4.1 Basic level

Q1 What is the cost price of the article?
I. It was sold for ₹1,200.
II. The profit was 20%.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q2 What is the speed of the train?
I. It covers 300 km in 5 hours.
II. It is 200 m long.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q3 What is x?
I. $3x + 5 = 20$.
II. $x^2 = 25$.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q4 What is the area of the rectangle?
I. Its perimeter is 30 cm.
II. Its length is 10 cm.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q5 Is n even?
I. n^2 is even.
II. n + 3 is odd.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q6 What is the simple interest on the sum?
I. The sum is ₹5,000 and the rate is 8% per year.
II. The time is 3 years.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q7 How old is Ravi?
I. Ravi is 4 years older than Sita.
II. Sita is half as old as Ravi.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q8 What is the average of five numbers?
I. Their sum is 75.
II. The largest is 25.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

31.4.2 Intermediate level

Q9 What are x and y?
I. $2x + 4y = 10$.
II. $x + 2y = 5$.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q10 What is the radius of the circle?
I. Its area is 154 cm^2.
II. Its circumference is 44 cm.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q11 How many days will A and B take together to finish the job?
I. A alone takes 10 days.
II. B alone takes 15 days.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q12 What is the two-digit number?
I. The sum of its digits is 9.
II. The number is 9 more than the number with its digits reversed.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q13 Is $x > y$?
I. $x - y > 0$.
II. $x^2 > y^2$.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q14 What is the compound interest for 2 years?
I. The simple interest for 2 years at the same rate is ₹400.
II. The rate is 10% per year.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q15 What is the value of x?
I. $x^2 - 5x + 6 = 0$.
II. x is an odd number.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q16 By what percentage did the price rise?
I. The old price was ₹80.
II. The new price is ₹100.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

31.4.3 Advanced level

Q17 What is the length of the hypotenuse of the right triangle?
I. Its area is 24 cm^2.
II. One leg is 6 cm.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q18 What is the value of a + b?
I. $a^2 - b^2 = 21$.
II. $a - b = 3$.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q19 How many students are there in the class?
I. The ratio of boys to girls is 3 : 2.
II. There are 12 more boys than girls.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q20 Is the integer n divisible by 6?
I. n is divisible by 3.
II. n is divisible by 12.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q21 What is the speed of the boat in still water?
I. It goes 24 km downstream in 2 hours.
II. It goes 24 km upstream in 3 hours.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q22 What is the value of x?
I. x is a prime number between 20 and 30.
II. x + 2 is also prime.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q23 What is the area of triangle ABC?
I. AB = 5, BC = 12.
II. Angle B = 90°.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q24 What is the profit percentage?
I. The cost price of 20 articles equals the selling price of 16.
II. The selling price of each article is ₹50.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q25 What is the value of k?
I. The equation $x^2 - kx + 9 = 0$ has equal roots.
II. k is positive.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

31.5 Solutions

Q1 **Answer: (d) I and II together are needed (neither alone is enough)**
I alone gives only the selling price; II alone only the profit percentage. Together CP = 1200/1.2 = ₹1,000. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q2 **Answer: (a) I alone is sufficient, II alone is not**
I gives 300/5 = 60 km/h. II gives only the length. Answer: **(a) I alone is sufficient, II alone is not**.

Q3 **Answer: (a) I alone is sufficient, II alone is not**
I gives x = 5. II gives x = 5 or -5: not unique. Answer: **(a) I alone is sufficient, II alone is not**.

Q4 **Answer: (d) I and II together are needed (neither alone is enough)**
Together breadth = 15 - 10 = 5, area 50 cm². Neither alone is enough. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q5 **Answer: (c) Either I alone or II alone is sufficient**
I: if n^2 is even, n is even. II: n + 3 odd means n is even. Either alone suffices. Answer: **(c) Either I alone or II alone is sufficient**.

Q6 **Answer: (d) I and II together are needed (neither alone is enough)**
SI needs P, R and T: only together, 5000 × 8 × 3/100 = ₹1,200. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q7 **Answer: (d) I and II together are needed (neither alone is enough)**

Together R = S + 4 and S = R/2 → R = 8. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q8 **Answer: (a) I alone is sufficient, II alone is not**

I gives 75/5 = 15. II gives one number only. Answer: **(a) I alone is sufficient, II alone is not**.

Q9 **Answer: (e) Even I and II together are not sufficient**

II is I divided by 2: really one equation in two unknowns, so even together not sufficient. Answer: **(e) Even I and II together are not sufficient**.

Q10 **Answer: (c) Either I alone or II alone is sufficient**

I: $\pi r^2 = 154 \rightarrow r = 7$. II: $2\pi r = 44 \rightarrow r = 7$. Either alone. Answer: **(c) Either I alone or II alone is sufficient**.

Q11 **Answer: (d) I and II together are needed (neither alone is enough)**

Together 1/10 + 1/15 = 1/6 → 6 days; neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q12 **Answer: (d) I and II together are needed (neither alone is enough)**

II: 9(t - u) = 9 → t - u = 1; with I: t = 5, u = 4 → 54. Neither alone is unique. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q13 **Answer: (a) I alone is sufficient, II alone is not**

I directly says x > y. II holds for x = -3, y = 1 (x < y) and for x = 3, y = 1: not sufficient. Answer: **(a) I alone is sufficient, II alone is not**.

Q14 **Answer: (d) I and II together are needed (neither alone is enough)**

Together P = 400 × 100/(10 × 2) = 2000, CI = 2000 × 1.21 - 2000 = ₹420. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q15 **Answer: (d) I and II together are needed (neither alone is enough)**

I gives x = 2 or 3; with II, x = 3. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q16 **Answer: (d) I and II together are needed (neither alone is enough)**

Both prices are needed: (100 - 80)/80 = 25%. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q17 **Answer: (d) I and II together are needed (neither alone is enough)**

Together the other leg = 2 × 24/6 = 8 → hypotenuse 10 cm. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q18 **Answer: (d) I and II together are needed (neither alone is enough)**

Together $(a + b)(a - b) = 21 \rightarrow a + b = 7$. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q19 **Answer: (d) I and II together are needed (neither alone is enough)**

Together the difference of 1 part = 12 → total 5 parts = 60. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q20 **Answer: (b) II alone is sufficient, I alone is not**

II: divisible by 12 implies divisible by 6. I: 9 is divisible by 3 but not 6. Answer: **(b) II alone is sufficient, I alone is not**.

Q21 **Answer: (d) I and II together are needed (neither alone is enough)**

Together D = 12, U = 8 → u = 10 km/h. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q22 **Answer: (d) I and II together are needed (neither alone is enough)**

I gives 23 or 29; II alone allows many values. Together, 23 + 2 = 25 is not prime while 29 + 2 = 31 is, so x = 29. Both statements are needed. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q23 **Answer: (d) I and II together are needed (neither alone is enough)**

Together it is a right triangle with legs 5 and 12: area 30. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q24 **Answer: (a) I alone is sufficient, II alone is not**

I gives profit = 4/16 = 25%. II gives no cost information. Answer: **(a) I alone is sufficient, II alone is not**.

Q25 **Answer: (d) I and II together are needed (neither alone is enough)**

I: $k^2 = 36 \rightarrow k = \pm 6$; II picks k = 6. Neither alone. Answer: **(d) I and II together are needed (neither alone is enough)**.

Test I alone, II alone (forgetting I), then together; a single value or a definite yes/no is sufficient. Count independent equations against unknowns and stop before solving. Watch for disguised duplicate equations, two-valued quadratics and hidden assumptions about integers or signs.

MODULE 32 · ADVANCED

Statistics, set theory and miscellaneous topics

Minimum duration **1 day** | Level **Advanced**

This module covers smaller topics that appear regularly in tests: mean, median and mode, range, variance and standard deviation, the effect of adding or multiplying all values, sets and Venn diagrams (two and three sets, maximum and minimum overlap), clock angles and odd days from the quantitative side, maxima and minima using AM-GM and algebra, and an overview of stocks, shares and unit conversions, followed by 25 solved questions computed by a program.

IN THIS MODULE

32.1 Statistics
32.2 Set theory and Venn diagrams
32.3 Clocks and calendars
32.4 Maxima and minima
32.5 Miscellaneous
32.6 Practice questions
32.7 Solutions

32.1 Statistics

32.1.1 Mean, median and mode

The **mean** is the average; the **median** is the middle value after sorting (the average of the two middle values for an even count); the **mode** is the most frequent value. For 3, 7, 7, 8, 10: mean 7, median 7, mode 7.

Empirical relation for moderately skewed data: mode ≈ 3 median − 2 mean

32.1.2 Range, variance and standard deviation

Range = maximum − minimum. Variance = average of squared deviations from the mean; standard deviation (SD) = √variance. For 2, 4, 4, 4, 5, 5, 7, 9: mean 5, variance 4, SD 2.

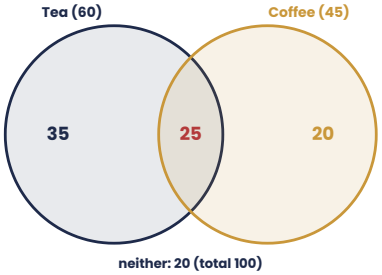
32.1.3 Effect of changing all values

Operation on every value	Mean, median, mode	Range, SD
add k	increase by k	unchanged
multiply by k	multiplied by k	multiplied by |k|
(variance)		multiplied by k^2

32.2 Set theory and Venn diagrams

32.2.1 Union, intersection, complement

$$n(A \cup B) = n(A) + n(B) - n(A \cap B);\ n(A \cup B \cup C) = \Sigma A - \Sigma(\text{pairs}) + n(A \cap B \cap C)$$



OF 100 PEOPLE, 60 LIKE TEA, 45 COFFEE AND 25 BOTH: 20 LIKE NEITHER

32.2.2 Two- and three-set problems

Fill a Venn diagram from the inside out: the centre (all three) first, then the regions for exactly two, then exactly one. "Exactly two" = $\Sigma(\text{pairs}) - 3 \times (\text{all three})$.

32.2.3 Maximum and minimum overlap

For two sets in a universe of N: maximum overlap = $\min(n(A), n(B))$; minimum overlap = $\max(0, n(A) + n(B) - N)$. If 70% read paper X and 60% read Y, at least 30% read both.

32.3 Clocks and calendars (quantitative approach)

32.3.1 Angle between the hands

Angle = $|30H - 5.5M|$; the hands coincide every $720/11$ minutes. See the clocks module of the Logical Reasoning book for full coverage.

32.3.2 Odd days

Days beyond complete weeks: an ordinary year has 1 odd day, a leap year 2. Full coverage is in the calendars module of the Logical Reasoning book.

32.4 Maxima and minima

32.4.1 Using AM-GM and algebra

For positive numbers with a fixed sum, the product is largest when they are equal; with a fixed product, the sum is smallest when they are equal. $x + 1/x \geq 2$ for $x > 0$. A quadratic $ax^2 + bx + c$ ($a > 0$) has its minimum at $x = -b/2a$.

WORKED EXAMPLE
x + y = 20 (x, y > 0). Maximum of xy?
Largest when x = y = 10: xy = **100**.

32.5 Miscellaneous

32.5.1 Stocks, shares and unit conversions

Income from shares = number of shares × face value × dividend %; the market price decides the cost. Rate of return = income/investment × 100. Common conversions: 1 km = 1000 m; 1 hectare = 10,000 m^2; 1 m^3 = 1000 litres; 1 lakh = 10^5; 1 crore = 10^7.

32.6 Practice questions

32.6.1 Basic level

Q1 What is the median of 12, 15, 11, 18, 14, 20?

(a) 14 **(b)** 15 **(c)** 14.5 **(d)** 15.5

Q2 What is the mode of 4, 6, 6, 7, 8, 8, 8, 9?

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

Q3 What is the range of 23, 45, 12, 67, 34?

(a) 45 **(b)** 44 **(c)** 67 **(d)** 55

Q4 In a class of 100, 60 like tea, 45 like coffee and 25 like both. How many like neither?

(a) 25 **(b)** 15 **(c)** 0 **(d)** 20

Q5 If 5 is added to every value of a data set, what happens to the standard deviation?

(a) It increases by 5 **(b)** It increases by 25
(c) It is multiplied by 5 **(d)** It is unchanged

Q6 n(A) = 30, n(B) = 25 and n(A ∩ B) = 10. What is n(A ∪ B)?

(a) 55 **(b)** 65 **(c)** 45 **(d)** 35

Q7 What is the mean of 3, 7, 7, 8, 10?

(a) 7 **(b)** 6 **(c)** 8 **(d)** 7.5

Q8 If x + y = 20 with x, y > 0, what is the maximum value of xy?

(a) 96 **(b)** 99 **(c)** 100 **(d)** 400

32.6.2 Intermediate level

Q9 What is the variance of 2, 4, 4, 4, 5, 5, 7, 9?

(a) 2 **(b)** 16 **(c)** 4 **(d)** 5

Q10 Each value of a data set is multiplied by 3. The original SD was 4. What is the new SD?

(a) 4 **(b)** 36 **(c)** 12 **(d)** 7

Q11 In a survey, 70% read paper X and 60% read paper Y. What is the minimum percentage that read both?

(a) 10% **(b)** 40% **(c)** 60% **(d)** 30%

Q12 For a distribution the mean is 30 and the median is 32. Estimate the mode.

(a) 34 **(b)** 31 **(c)** 36 **(d)** 28

Q13 In a group of 50, 30 play cricket and 25 play football, and every person plays at least one. How many play both?

(a) 10 **(b)** 5 **(c)** 15 **(d)** 20

Q14 What is the minimum value of $x + 9/x$ for $x > 0$?

(a) 9 **(b)** 10 **(c)** 6 **(d)** 3

Q15 The mean of 10 numbers is 20. If each number is increased by 2 and then doubled, what is the new mean?

(a) 44 **(b)** 42 **(c)** 40 **(d)** 22

Q16 What is the angle between the hands of a clock at 2:30?

(a) 90° **(b)** 100° **(c)** 105° **(d)** 120°

32.6.3 Advanced level

Q17 In a group of 100 students, 50 study maths, 40 physics and 30 chemistry; 20 study maths and physics, 15 physics and chemistry, 10 maths and chemistry, and 5 all three. How many study none?

(a) 15 **(b)** 20 **(c)** 25 **(d)** 30

Q18 In the previous question, how many study exactly two subjects?

(a) 45 **(b)** 35 **(c)** 25 **(d)** 30

Q19 What is the median of the first 10 prime numbers?

(a) 11 **(b)** 13 **(c)** 12 **(d)** 10

Q20 The average of 5 numbers is 18. If one number is excluded, the average becomes 16. What is the excluded number?

(a) 24 **(b)** 18 **(c)** 30 **(d)** 26

Q21 Of 60 employees, 35 speak Hindi and 40 speak English. What is the maximum number who speak both?

(a) 40 **(b)** 15 **(c)** 60 **(d)** 35

Q22 The product of two positive numbers is 64. What is the minimum value of their sum?

(a) 8 **(b)** 32 **(c)** 20 **(d)** 16

Q23 What is the SD of 4, 4, 4, 4, 4?

(a) 0 **(b)** 4 **(c)** 1 **(d)** 2

Q24 A man invests ₹9,600 in shares of face value ₹100 at ₹120 each, paying an 8% dividend. What is his annual income?

(a) 768 **(b)** 640 **(c)** 960 **(d)** 600

Q25 The sum of the first n natural numbers is 210. What is their mean?

(a) 10 **(b)** 11 **(c)** 10.5 **(d)** 21

32.7 Solutions

Q1 **Answer: (c) 14.5**

Sorted: 11, 12, 14, 15, 18, 20; median = (14 + 15)/2 = **14.5**.

Q2 **Answer: (c) 8**

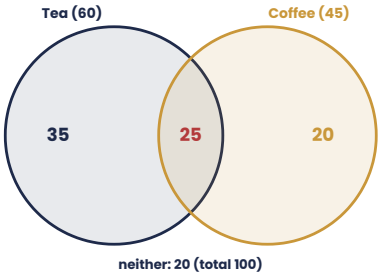
8 appears three times: **8**.

Q3 **Answer: (d) 55**

67 − 12 = **55**.

Q4 **Answer: (d) 20**

100 − (60 + 45 − 25) = **20**.



Q5 **Answer: (d) It is unchanged**

Adding a constant shifts all values equally: the SD is **unchanged**.

Q6 **Answer: (c) 45**

30 + 25 − 10 = **45**.

Q7 **Answer: (a) 7**

35/5 = **7**.

Q8 **Answer: (c) 100**

Equal parts: 10 × 10 = **100**.

Q9 **Answer: (c) 4**

Mean 5; squared deviations 9, 1, 1, 1, 0, 0, 4, 16 sum to 32; 32/8 = **4**.

Q10 **Answer: (c) 12**

SD is multiplied by 3: **12**.

Q11 **Answer: (d) 30%**

70 + 60 − 100 = **30**%.

Q12 **Answer: (c) 36**

Mode ≈ 3 × 32 − 2 × 30 = **36**.

Q13 **Answer: (b) 5**

30 + 25 − 50 = **5**.

Q14 **Answer: (c) 6**

AM-GM: $x + 9/x \geq 2\sqrt{9}$ = **6** (at $x = 3$).

Q15 **Answer: (a) 44**

(20 + 2) × 2 = **44**.

Q16 **Answer: (c) 105°**

|60 − 165| = **105°**.

Q17 **Answer: (b) 20**

Union = 120 − 45 + 5 = 80; none = **20**.

Q18 **Answer: (d) 30**

Σ pairs − 3 × all three = 45 − 15 = **30**.

Q19 **Answer: (c) 12**

Primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29; median = (11 + 13)/2 = **12**.

Q20 **Answer: (d) 26**

90 − 64 = **26**.

Q21 **Answer: (d) 35**

Maximum overlap = min(35, 40) = **35**.

Q22 **Answer: (d) 16**

Equal numbers 8 and 8: sum **16**.

Q23 **Answer: (a) 0**

All values are equal: no spread, SD = **0**.

Q24 **Answer: (b) 640**

80 shares × ₹8 = **₹640**.

Q25 **Answer: (c) 10.5**

210 = n(n + 1)/2 → n = 20; mean = (1 + 20)/2 = **10.5**.

Mean = sum/count, median = middle value, mode = most frequent; adding k shifts the centre but not the spread; multiplying by k scales both. n(A ∪ B) = n(A) + n(B) − n(A ∩ B); minimum overlap = n(A) + n(B) − N. With a fixed sum, the product is maximum for equal parts; x + a/x ≥ 2√a.

MODULE 33 · ADVANCED

Exam strategy, mock tests and company-specific preparation

Minimum duration **3 days** | Level **Advanced**

The final module turns knowledge into exam performance through strategy and full-length practice. It covers classifying questions in seconds, attempt order, time per question, negative marking, option elimination and back-substitution, checking with approximation and unit digits, company-specific patterns (TCS NQT, Infosys, Wipro, Accenture, Cognizant, Capgemini, AMCAT, eLitmus, bank exams), mock analysis, the error log, a revision plan and exam-day preparation, followed by a 25-question mixed mock test, fully solved.

IN THIS MODULE

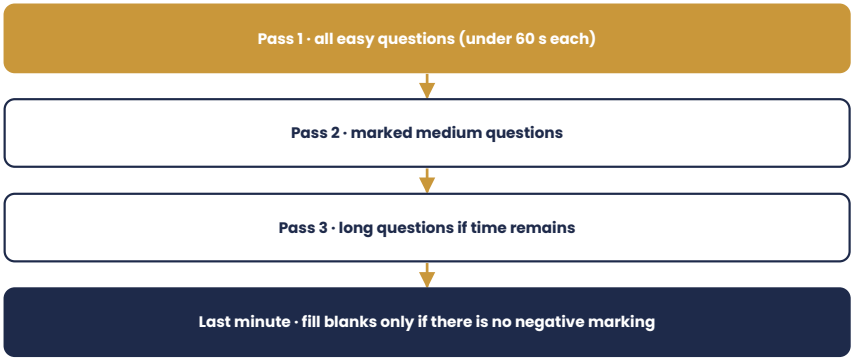
33.1 Exam strategy
33.2 Company-specific patterns
33.3 Mock tests
33.4 Revision plan
33.5 Exam-day preparation
33.6 Mixed mock test
33.7 Solutions

33.1 Exam strategy

33.1.1 Classify in seconds

Read the last line of the question first (what is asked), then glance at the numbers. In 5 to 10 seconds decide: **easy** (do now), **medium** (mark and return), **long** (leave for the end).

33.1.2 Attempt order



A THREE-PASS ATTEMPT ORDER

33.1.3 Time per question

Divide the section time by the number of questions to get your average budget (usually 60 to 90 seconds). If you pass twice the budget without a clear path, skip and return later.

33.1.4 Negative marking

With -1/4 per wrong answer and four options, a random guess has expected value 1/4 × 1 - 3/4 × 1/4 = 1/16 > 0, a slight gain. With -1/3, the expected value is 0. Eliminating even one option makes guessing worthwhile in both cases.

33.1.5 Option elimination and back-substitution

Many questions are faster backwards: put each option into the question. Start with a middle option; if it is too big, the answer is smaller, which eliminates two options at once.

33.1.6 Checking with approximation and unit digits

Before marking, check the size (does 23% of 4,800 ≈ 1,100 make sense?) and, for products, the unit digit (the unit digit of 347 × 268 must be 6).

33.2 Company-specific patterns

Patterns change every year; always read the latest official notice. The guide below describes the broad shape of each test and what to prioritise.

Test	Quantitative focus	Priority topics
TCS NQT (numerical ability)	moderate difficulty, sectional timing	percentages, ratios, time and work, speed, number system, DI
Infosys	harder, more reasoning-heavy maths	number system, P&C, probability, DI, algebra
Wipro, Accenture, Cognizant, Capgemini	short arithmetic questions	percentages, profit, interest, averages, ratios, speed
AMCAT	adaptive, topic-wise	number system, arithmetic, basic algebra
eLitmus	few, deep questions; negative marking	number system, P&C, probability, geometry, algebra
Bank and government exams	speed: simplification and DI	approximation, series, quadratic comparison, DI, arithmetic

33.3 Mock tests

33.3.1 Weekly full-length mocks

From the second half of the course, take one full-length timed mock each week, in one sitting, in the same time slot as your real exam.

33.3.2 Analysing mocks

After each mock, record three numbers per topic: attempted, correct and average time. Topics with low accuracy need concept revision; topics with high accuracy but slow speed need timed drills.

33.3.3 Error log

Topic	Question	Error type	Correct method / rule
Percentages	mock 3, Q7	used new value as base	percentage change uses the old value
Time and work	mock 4, Q12	slow (fractions)	use the LCM units method
Probability	mock 4, Q20	counted order twice	without replacement: combinations above and below

33.4 Revision plan

33.4.1 Formula sheet

Keep a one-page formula sheet per level and read it every day in the last two weeks. Every key-idea box in these notes is a line on that sheet.

33.4.2 Spaced repetition of mistakes

Re-solve each error-log question after 1 day, 1 week and 3 weeks. A question you solve correctly three times in a row can leave the log.

33.5 Exam-day preparation

33.5.1 Online test environment

Check the system and internet the day before, keep rough sheets and two pens ready, and read the on-screen instructions about navigation and marking. Start with the section you are strongest in if the test allows it, keep an eye on the timer, and stay calm if a question looks unfamiliar: skip it and come back.

33.6 Mixed mock test

33.6.1 Basic level

Q1 What is 35% of 640?

(a) 214 **(b)** 220 **(c)** 240 **(d)** 224

Q2 What is the LCM of 12, 18 and 30?

(a) 90 **(b)** 360 **(c)** 120 **(d)** 180

Q3 A shirt marked ₹1,200 is sold at a 15% discount. What is the selling price?

(a) 1,080 **(b)** 1,000 **(c)** 1,020 **(d)** 1,180

Q4 A can do a job in 12 days and B in 18 days. How long do they take together?

(a) 6 days **(b)** 8 days **(c)** 7.2 days **(d)** 15 days

Q5 What is the simple interest on ₹8,000 at 6% for 2 years?

(a) 480 (b) 1,000 (c) 900 (d) 960

Q6 A train 150 m long passes a pole in 9 seconds. What is its speed in km/h?

(a) 54 km/h (b) 60 km/h (c) 50 km/h (d) 66 km/h

Q7 The average of 6 numbers is 15. What is their sum?

(a) 21 (b) 75 (c) 96 (d) 90

Q8 Divide ₹720 in the ratio 4 : 5. What is the smaller share?

(a) 400 (b) 360 (c) 320 (d) 300

33.6.2 Intermediate level

Q9 What is the compound interest on ₹10,000 at 10% for 3 years?

(a) 3,000 (b) 3,100 (c) 3,300 (d) 3,310

Q10 What is the unit digit of $3^{65} \times 7^{42}$?

(a) 7 (b) 1 (c) 3 (d) 9

Q11 A boat goes 30 km downstream in 2 hours and 18 km upstream in 2 hours. What is the speed of the stream?

(a) 6 km/h (b) 12 km/h (c) 3 km/h (d) 2 km/h

Q12 In how many ways can the letters of LEVEL be arranged?

(a) 60 (b) 120 (c) 20 (d) 30

Q13 Two dice are thrown. What is the probability that the sum is 9?

(a) 1/9 (b) 1/6 (c) 5/36 (d) 1/12

Q14 In what ratio must tea at ₹160/kg be mixed with tea at ₹220/kg to get a mixture worth ₹180/kg?

(a) 1 : 2 (b) 3 : 1 (c) 2 : 1 (d) 3 : 2

Q15 A father is 30 years older than his son. In 5 years he will be 3 times as old as his son. What is the son's present age?

(a) 8 years (b) 12 years (c) 15 years (d) 10 years

Q16 What is the sum of the first 25 terms of the AP 4, 7, 10, ...?

(a) 975 (b) 1,025 (c) 1,000 (d) 1,100

33.6.3 Advanced level

Q17 A sum becomes ₹13,310 in 3 years at 10% compound interest. What is the sum?

(a) 10,000 (b) 11,000 (c) 9,000 (d) 12,100

Q18 How many trailing zeros does 125! have?

(a) 25 (b) 30 (c) 28 (d) 31

Q19 A committee of 4 is chosen from 5 men and 4 women. In how many ways can it include at least 2 women?

(a) 60 (b) 120 (c) 126 (d) 81

Q20 The diagonals of a rhombus are 16 cm and 12 cm. What is its side?

(a) 10 cm **(b)** 8 cm **(c)** 14 cm **(d)** 20 cm

Q21 I. $x^2 - 9x + 20 = 0$ II. $y^2 - 11y + 30 = 0$. Which relation holds?

(a) $x > y$ **(b)** $x \le y$ **(c)** $x < y$ **(d)** $x \ge y$

Q22 A cylinder and a cone have equal radii and heights. If the cylinder's volume is 600 cm³, what is the cone's volume?

(a) 300 cm³ **(b)** 1,800 cm³ **(c)** 200 cm³ **(d)** 400 cm³

Q23 $\log_2 32 + \log_3 81 = ?$

(a) 8 **(b)** 10 **(c)** 9 **(d)** 7

Q24 Of 120 students, 70 play cricket, 50 football and 20 both. How many play neither?

(a) 10 **(b)** 30 **(c)** 0 **(d)** 20

Q25 A man walks at 5 km/h and reaches his office 6 minutes late; at 6 km/h he is 4 minutes early. How far is the office?

(a) 5 km **(b)** 4 km **(c)** 6 km **(d)** 7.5 km

33.7 Solutions

Q1 **Answer: (d) 224**

7/20 × 640 = **224**.

Q2 **Answer: (d) 180**

$2^2 \times 3^2 \times 5$ = **180**.

Q3 **Answer: (c) 1,020**

1200 × 0.85 = **₹1,020**.

Q4 **Answer: (c) 7.2 days**

12 × 18/30 = **7.2 days**.

Q5 **Answer: (d) 960**

8000 × 6 × 2/100 = **₹960**.

Q6 **Answer: (b) 60 km/h**

150/9 m/s × 18/5 = **60 km/h**.

Q7 **Answer: (d) 90**

6 × 15 = **90**.

Q8 **Answer: (c) 320**

720 × 4/9 = **₹320**.

Q9 **Answer: (d) 3,310**

10000 × 1.331 − 10000 = **₹3,310**.

Q10 **Answer: (a) 7**

3^{65}: 65 mod 4 = 1 → unit digit 3. 7^{42}: 42 mod 4 = 2 → unit digit 9. 3 × 9 = 27 → unit digit **7**.

Q11 **Answer: (c) 3 km/h**

D = 15, U = 9 → v = (15 − 9)/2 = **3 km/h**.

Q12 **Answer: (d) 30**

5!/(2! 2!) = **30**.

Q13 **Answer: (a) 1/9**

(3,6), (4,5), (5,4), (6,3): 4/36 = **1/9**.

Q14 **Answer: (c) 2 : 1**

(220 − 180) : (180 − 160) = 40 : 20 = **2 : 1**.

Q15 **Answer: (d) 10 years**

x + 35 = 3(x + 5) → 2x = 20 → x = **10**.

Q16 **Answer: (c) 1,000**

25/2 × (8 + 72) = **1,000**.

Q17 **Answer: (a) 10,000**

13310/1.331 = **₹10,000**.

Q18 **Answer: (d) 31**

25 + 5 + 1 = **31**.

Q19 **Answer: (d) 81**

C(4,2)C(5,2) + C(4,3)C(5,1) + C(4,4) = 60 + 20 + 1 = **81**.

Q20 **Answer: (a) 10 cm**

Half-diagonals 8 and 6: side √(64 + 36) = **10 cm**.

Q21 **Answer: (b) x ≤ y**

x = 4, 5; y = 5, 6. Every x ≤ every y (5 = 5 is possible): **x ≤ y**.

Q22 **Answer: (c) 200 cm³**

Cone = ⅓ cylinder = **200 cm³**.

Q23 **Answer: (c) 9**

5 + 4 = **9**.

Q24 **Answer: (d) 20**

120 − 100 = **20**.

Q25 **Answer: (a) 5 km**

D(1/5 − 1/6) = 10/60 → D/30 = 1/6 → D = **5 km**.

Classify each question in seconds, attempt in three passes, and skip anything that takes twice your time budget. Guess only after eliminating options when there is negative marking. Back-substitute options, check size and unit digits, take weekly mocks, analyse accuracy and speed by topic, and keep revising your formula sheet and error log.